SAS® Model Manager: User’s Guide

2026.03 - 2026.04*

* This document might apply to additional versions of the software. Open this document in SAS Help Center and click on the version in the banner to see all available versions.

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SAS® Model Manager: User's Guide

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Introduction to SAS Model Manager

About Managing Models

SAS Model Manager helps streamline analytical model registration, deployment, monitoring, and management of SAS and open-source models, keeping them in one central place for data scientists and information technology (IT) teams. Using SAS Model Manager, you can store models in a common model repository, and organize them within projects and folders. You can also evaluate models for champion model selection, monitor performance of models, and publish models. All model development and model maintenance personnel, including data modelers, validation testers, scoring officers, and analysts, can use SAS Model Manager.

You can store and manage all types of models within the common model repository. This includes open-source models that are developed using Python and R programming languages, SAS models, as well as model content that is stored in PMML, ONNX, or RDS formats. In addition to models developed with custom Python code, you can use models that are developed with packages such as scikit-learn, TensorFlow, and XGBoost.

Data tables are an integral part of the modeling process. Data tables are used for scoring, publishing validation, and performance monitoring. In addition, data tables are used to record adherence to your modeling methodology for audit compliance. Performance data can be created from your operational data, provided that it has the required structure (for example, the data contains a target variable). For information about preparing and managing your data, see *Getting Started with SAS Data Preparation for the SAS Viya Platform* and "About Managing Data" on page 13.

Here are some of the tasks that you can accomplish with SAS Model Manager:

- Use a single interface to access all of your business modeling projects and models. All models are stored in a common model repository. Models can also be accessed in one place using the **Models** category. Your most recent model and project items are also available on the **Home** page, along with links to What's New and how-to (video) articles in SAS Communities, and links to production documentation.
- Import models that you develop using a SAS application, such as Model Studio, SAS Visual Analytics, and SAS Studio, as well as SAS code, open-source programming language such as Python or R, and PMML models. You can also create a new model with the model's files in a folder or project.
- Compare models to assess candidate models.
- Manage data that is used for scoring, validating published models, and performance monitoring.
- Run scoring tests to validate models.
- Publish SAS models and open-source models to supported publishing destinations. You can then score the published models within the publishing destinations or by using external applications or interfaces.
- Score open-source models within run-time containers.
- Run monitoring jobs to measure the performance of models on historical and current data.
- Use native Python code in Jupyter notebooks to import, export, and manage models.
- View various model and project metrics such as the number of published models within a dashboard.
- Create custom workflow definitions to meet your business requirements and to match your business processes. You can then start a workflow process to track the progress of your project.

Sign In to SAS Model Manager

Note: If you are already signed in to another SAS web application, you can access SAS Model Manager by clicking ⠿ and selecting **Manage Models**.

To sign in to SAS Model Manager:

1 In the address bar of your web browser, enter the URL for SAS Model Manager and press **Enter**. The **Sign In** page appears.

   Note: If you are in a single sign-on environment, you are not prompted to sign in. Contact your system administrator if you need the URL for SAS Model Manager. The default URL is https://*host_name*:*port*/SASModelManager. The last part of the URL (the application name) is case sensitive. If the default port of 80 is used during configuration, you do not need to include the colon and port in the URL.

2 Enter a user ID and password.

3 Click **Sign In**.

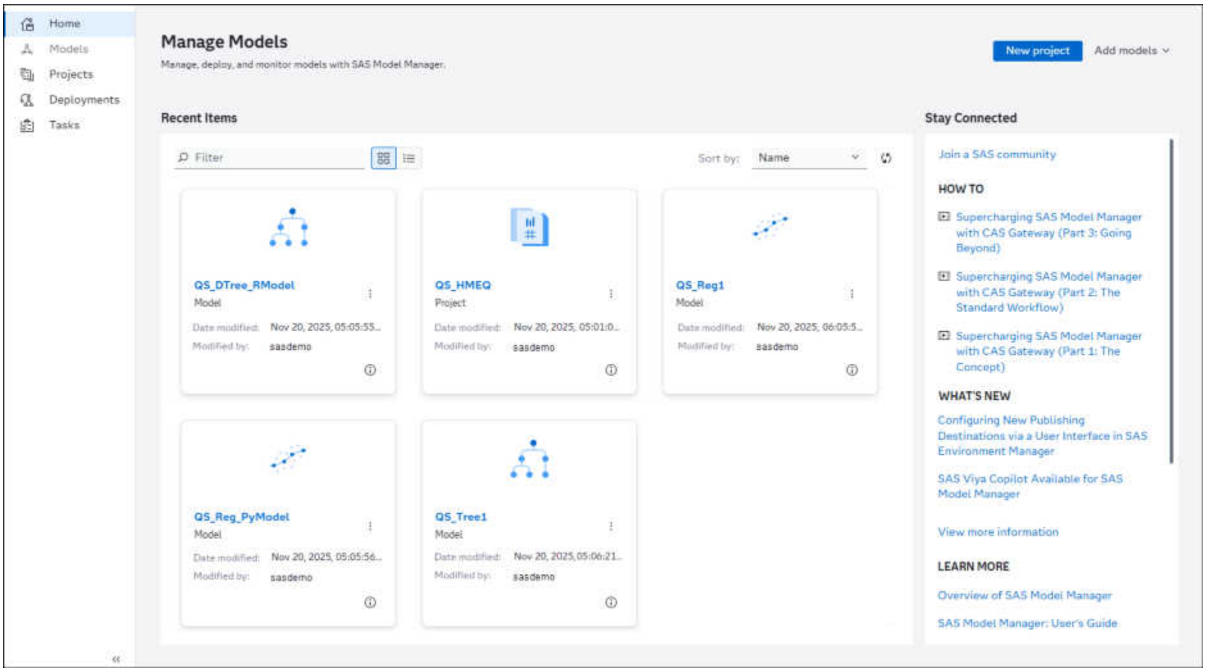
4 (Optional) If this is your first time signing in to SAS Model Manager, the Welcome to SAS window appears. Here you can set up a profile, which enables you to customize some settings. You will then be taken to the SAS Model Manager Home page.

TIP You can install SAS Model Manager as a Progressive Web App (PWA).

SAS Model Manager Home

The SAS Model Manager Home page appears the first time you sign in to the web application. The category page that you are on when you sign out of SAS Model Manager is saved. The next time you sign in, the same category page is displayed.

Note: The left navigation bar is also expanded by default the first time you sign in. The next time you sign in, the navigation bar appears in the state that you left it when you signed out.



The Home page contains the following information:

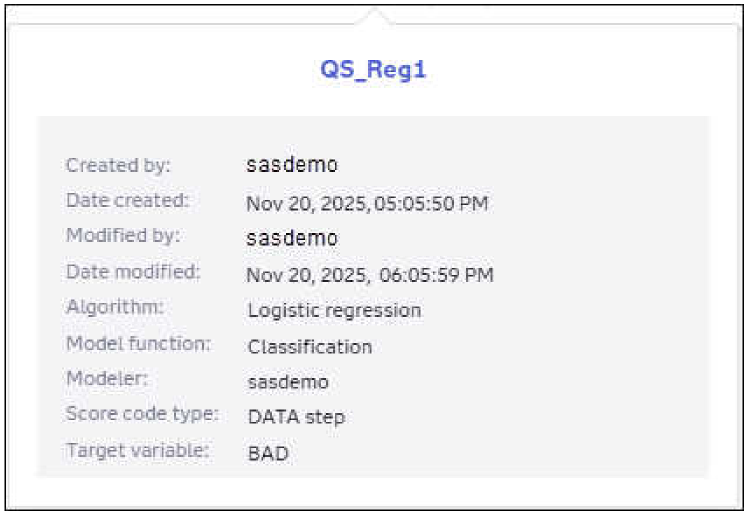
- the most recent items that you opened with the date that they were last modified and by whom
- links to the five most recent how-to videos and What's New articles on SAS Communities
- links to the What's New, overview, and user documentation for SAS Model Manager

Here are some of the actions that you can perform from Home:

- Click to display the items in a list view.
- Click to display the items in a tile view.

  Note: This is the default view.

- Sort the recent items by name or date modified.
- Click the name of an item to open it.
- Click to view additional properties for an item.



- Click ⋮ to open an item, remove a recent item, or share a link to the item.
- Create a new project.
- Add models to a folder.

Note: Only the 40 most recent items appear on the Home page.

Share Links to SAS Model Manager Content

When the icon appears in the toolbar, you can perform the following tasks:

- Copy a link to a category view, object, tab, or page to the clipboard.
- Share a link to a category view, object, tab, or page to Microsoft Teams.

Using SAS Model Manager as a Progressive Web App

Benefits of Using a Progressive Web App

The benefits of using SAS Model Manager as a Progressive Web App (PWA) include the following:

- Application persistence – By default, your session will never time out, so you can restart your work more quickly. Your administrator can control the timing with a configuration property in SAS Environment Manager.
- Performance – When installed as a PWA, SAS Model Manager typically starts faster than when accessed in the browser.
- Desktop experience – Installing SAS Model Manager as a PWA confers all the benefits of a traditional installation. You can launch SAS Model Manager from the **Start** menu or Taskbar, and you do not need to sort through countless tabs to find the correct instance of SAS Model Manager.
- Renaming – You can rename each PWA instance to quickly access different environments, such as development, test, or production servers.

Installing SAS Model Manager as a Progressive Web App

SAS Model Manager must be deployed with TLS and HTTPS enabled.

To install SAS Model Manager as a PWA:

1 Open SAS Model Manager in a Chromium-based web browser and sign in.

2 Open the web browser's **More** menu and select **Install SAS**.

Note: The **Install SAS** option might be located under another menu option. For example, within the Microsoft Edge browser it is located under **Apps**.

SAS Model Manager is now installed as a desktop program named SAS. After one web application has been installed as a PWA, all other SAS applications are accessible in the PWA.

To uninstall the PWA for SAS Model Manager:

1 Open the PWA for SAS Model Manager.

2 In the PWA menu, select **Uninstall SAS**.

Manage Application Settings

Use the Settings window to edit user preferences or customize accessibility settings for all SAS web applications. You can also manage model repositories.

To access the Settings window, click your name in the application bar and select **Settings**.

For information about settings, see the following documentation:

- "Settings" in *SAS Viya Platform: General Usage Help for Web Applications*
- Model Repositories

High-Level Model Support Matrix for Primary Functions

Here is a summary of the primary functions that are supported by SAS Model Manager on SAS Viya. The type of model score code and the assignment of the "Score code" file role can determine which functions you can perform for the different types of models. Additional requirements and restrictions might apply, depending on the function that is being performed. For more information, click the links in the column titles to go to the function-specific section of the user's guide.

> **IMPORTANT** The values in the **Model Score Code Type** column in each table below are associated with the **Score code type** model property. For more information, see "Set Model General Properties" on page 85.

Open-Source Models

You can import models that were created using an open-source programming language such as Python and R and then perform tasks that are associated with the primary functions.

Table 1.1 Support for Primary Functions

Programming Language	Model Score Code Type	Import	Score	Publish	Monitor Performance	Compare	Retrain
Python	Python	Yes	Yes	Yes	Yes	Yes	No
R	R	Yes	Yes	Yes	Yes	Yes	No

Here are some requirements and restrictions for open-source models:

- Models that have a score code type of Python or R can be scored if the score code is in the correct format. The score code must also be in the correct format for publishing and running performance. For more information, see "Scoring Python Models" on page 170 and "Scoring R Models" on page 173.
- Python and R open-source models can be published to CAS, Git, SAS Micro Analytic Service, Amazon Web Services (AWS), Azure, Azure Machine Learning, and Private Docker publishing destinations. For more information, see "Publishing Requirements and Restrictions" on page 143.
- The **Model Card** tab is available for Python models with a model function of classification or prediction. For more information, see "Generating Model Card Content" on page 71.

Predictive Model Markup (PMML) Models

Predictive Model Markup Language (PMML) is an XML-based predictive model interchange format. PMML models that you create using PMML 4.2 support DATA step score code. When you are importing valid PMML models, the score code type model property is set to DATA step, instead of PMML. PMML models with a score code type of DATA step can be scored and published. For more information, see "Import Models" on page 63.

Note: The **Model Card** tab is available for PMML models with a model function of classification or prediction, and a score code type of DATA step. For more information, see "Generating Model Card Content" on page 71.

Table 1.2 Support for Primary Functions

Model Score Code Type	Import	Score	Publish	Monitor Performance	Compare	Retrain
DATA step	Yes	Yes	Yes	Yes	Yes	No
PMML	Yes	No	No	Yes	Yes	No

SAS Viya Models

You can build models with Model Studio, SAS Visual Analytics, or with SAS Viya modeling procedures within SAS Studio. These models can then be registered to the common model repository from those applications or imported into the common model repository using the SAS Model Manager web application.

Note: The **Model Card** tab is available for SAS Viya models with a model function of classification or prediction. For more information, see "Generating Model Card Content" on page 71.

Table 1.3 Support for Primary Functions

<table>
<tr><th>Product or Tool</th><th>Model Score Code Type</th><th>Import</th><th>Score</th><th>Publish</th><th>Monitor Performance</th><th>Compare</th><th>Retrain</th></tr>
<tr><td rowspan="3">SAS Visual Data Mining and Machine Learning</td><td>DATA step</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td></tr>
<tr><td>DS2 multi-type</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td></tr>
<tr><td colspan="7">Note: You can build SAS Visual Data Mining and Machine Learning models with Model Studio, SAS Visual Analytics, or with SAS Viya modeling procedures within SAS Studio. When SAS Viya models that contain one or more analytic store files are registered into the common model repository, their score code type is set to DS2 multi-type, instead of Analytic store.</td></tr>
<tr><td rowspan="2">SAS Visual Statistics</td><td>DATA step</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>No</td></tr>
<tr><td colspan="7">Note: You can build SAS Visual Statistics models with SAS Visual Analytics, or with SAS Viya modeling procedures within SAS Studio.</td></tr>
<tr><td rowspan="2">SAS Visual Text Analytics</td><td>DS2 multi-type</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>No</td></tr>
<tr><td colspan="7">Note: Here are some restrictions:</td></tr>
</table>

Product or Tool	Model Score Code Type	Import	Score	Publish	Monitor Performance	Compare	Retrain
	■ You can build SAS Visual Text Analytics models with Model Studio or with SAS Viya modeling procedures within SAS Studio. ■ Text categories, Text concepts, Text sentiment, and Text topics models have a score code type of DS2 multi-type and can be published to CAS, Git, Hadoop, and SAS Micro Analytic Service destinations. With the 2024.12 stable release, SAS Visual Text Analytics models can also be published to container destinations. ■ Only Text categories models can be monitored for performance						

Here are some restrictions for SAS Viya models:

- Only SAS Visual Data Mining and Machine Learning models that are built and registered using Model Studio can be retrained using SAS Model Manager. For more information, see "Retrain a Project from Model Studio" on page 138.
- Scoring of decisions that contain a model with a score code type of SAS program is not supported by SAS Intelligent Decisioning. For more information about scoring decisions, see *SAS Intelligent Decisioning: User's Guide*.

For more information about SAS Viya models, see the following documentation:

- *SAS Viya: Machine Learning User's Guide*
- *SAS Visual Text Analytics: User's Guide*
- *SAS Visual Analytics: Working with Machine Learning Objects*
- *SAS Visual Analytics: Working with Statistics Objects*
- "SAS Viya: Machine Learning" in *SAS Procedures by Name and Product*
- *SAS Viya: Getting Started with Machine Learning*

SAS 9.4 Models

Models that you create with SAS 9.4 can also be imported into SAS Model Manager on SAS Viya. You can then perform tasks associated with the primary functions.

Note:

- When analytic store models are imported using a SASAST file or are registered into the common model repository from Model Studio, SAS Visual Analytics, or SAS Studio, their score code type is set to DS2 multi-type, instead of Analytic store.
- PMML models that you create with PMML 4.2 support DATA step score code. When importing valid PMML models, set the score code type model property to DATA step, instead of PMML. PMML models with a score code type of DATA step can be scored and published. See "Import Models" on page 63.

- The **Model Card** tab is available for SAS models with a model function of classification or prediction. For more information, see "Generating Model Card Content" on page 71.

Table 1.4 *Support for Primary Functions*

Product or Tool	Model Score Code Type	Import	Score	Publish	Monitor Performance	Compare	Retrain
Base SAS or other code editor	SAS program	Yes	No	No	Yes	Yes	No
SAS Enterprise Miner	DATA step	Yes	Yes	Yes	Yes	Yes	No
	PMML	Yes	No	No	No	Yes	No
	Analytic store	Yes	Yes	Yes	Yes	Yes	No
	Note: Applies only to SAS analytic store (SASAST) files. SAS package (SPK) files are not supported.						
SAS HPFOREST and HPSVM procedures	Analytic store	Yes	Yes	Yes	Yes	Yes	No
SAS/STAT linear model procedures	DATA step	Yes	Yes	Yes	Yes	Yes	No

System-Supplied Score Code Types

Note: A model with the score code type of DS2 multi-type contains code files for a DS2 embedded process and a DS2 package. It can also contain one or more analytic stores. When you import or register analytic store models into the common model repository, the DS2 score code is generated and the score code type is set to DS2 multi-type, instead of Analytic store. When adding a model using the new custom model feature, if an analytic store file is included with the model files but the DS2 score code is not included, the model cannot be scored or published.

Table 1.5 Support for Primary Functions

Model Score Code Type	Model Card	Import	Score	Publish	Monitor Performance	Compare
Analytic store	Yes	Yes	Yes	Yes	Yes	Yes
DATA step	Yes	Yes	Yes	Yes	Yes	Yes
DS2 embedded process	Yes	Yes	Yes	Yes	Yes	Yes
DS2 multi-type	Yes	Yes	Yes	Yes	Yes	Yes
DS2 package	Yes	Yes	Yes	Yes	Yes	Yes
PMML[1]	No	Yes	No	No	Yes	Yes
Python	Yes	Yes	Yes[3]	Yes	Yes	Yes
R	No	Yes	Yes[3]	Yes	Yes	Yes
SAS program	No	Yes	No[2]	No	Yes	Yes
Additional Score Code Types Note: Monitoring performance of models with the following score code types is supported only when the performance definition includes input tables that contain scored data. For more information, see “Monitoring Performance” on page 117.						
C	No	Yes	No	No	Yes	Yes
CAS language	No	Yes	No	No	Yes	Yes
Java	No	Yes	No	No	Yes	Yes
Lua	No	Yes	No	No	Yes	Yes
MATLAB	No	Yes	No	No	Yes	Yes

1 PMML models that you create with PMML 4.2 support DATA step score code. When you are importing valid PMML models, set the score code type model property to DATA step, instead of to PMML. PMML models with a score code type of DATA step can be scored and published. See “Import Models” on page 63.

2 Scoring and publishing of decisions that contain a model with a score code type of SAS program is not supported by SAS Intelligent Decisioning. For more information about scoring or publishing decisions, see *SAS Intelligent Decisioning: User’s Guide*.

3 Models that have a score code type of Python or R can be scored, if the score code is in the correct format. The score code must also be in the correct format for publishing and running performance. See “Scoring Python Models” on page 170 and “Scoring R Models” on page 173.

2

Managing Data

About Managing Data

The Manage Data window is accessible from both the Models and Projects category views in SAS Model Manager. You can view existing data, import data from local files, import data that is stored in SAS folders, and copy data in caslib directories. Data can be imported from a CSV data file or a SAS data set.

CAS data libraries (caslibs) can be created using SAS Data Explorer. To create a new global caslib, see "Making Data Available to CAS" in *SAS Data Explorer: User's Guide*.

Note: You can access the Manage Data window only after a project or model has been added to the common model repository.

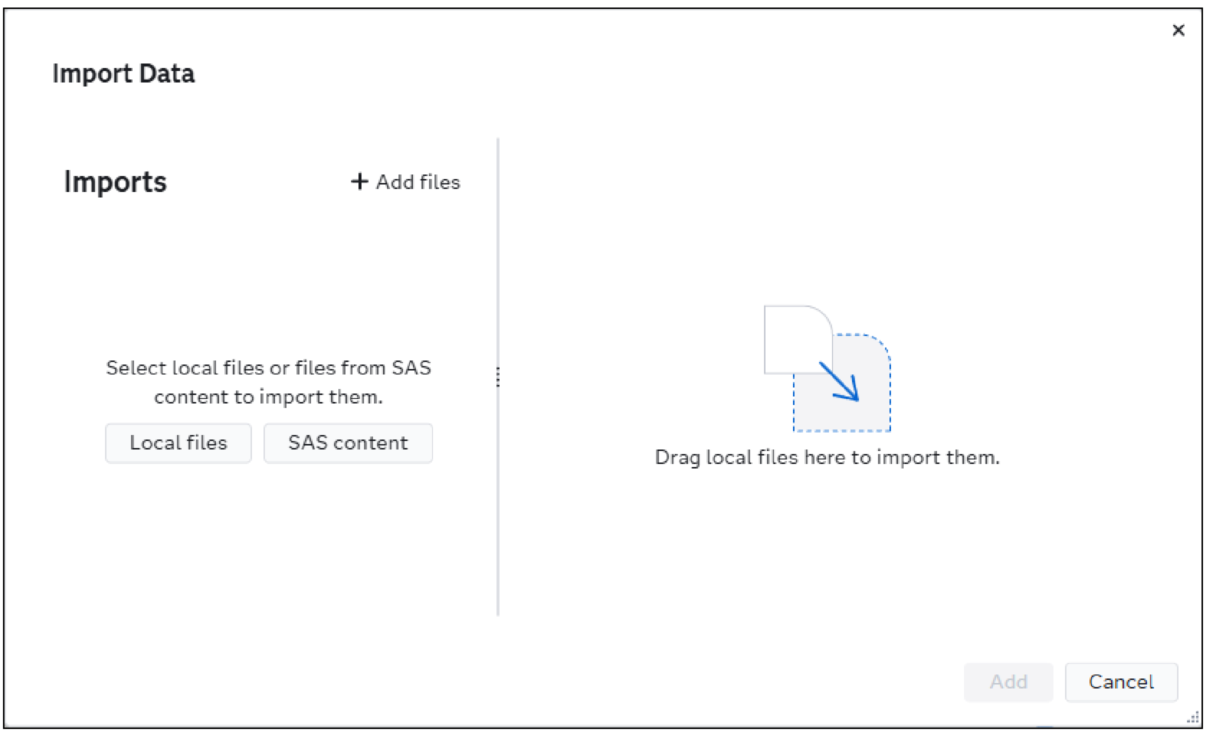
CAUTION
Do not add or include files that contain executable or dynamic content that is not valid or approved for your environment. Examples of executable content are batch (.bat) files or executable (.exe) files. Examples of dynamic content are HTML, Microsoft Office documents, JavaScript, dynamic PDF forms, and other forms of executable content. This type of content, when downloaded, could execute in your local environment and provide a way for malicious content to enter it. The Files service is used when adding data files, model files, and project files. As general guidance, SAS recommends that users be vigilant and cautious with any content that is added using the Files service and accessed within SAS Model Manager.

Import Data

Note: Before importing data tables to use for performance monitoring, ensure that the table names meet the requirements. For more information, see "Naming Requirements for Input Data Tables" on page 119.

1 On the toolbar above the table, click ⋮ and select **Manage data**. The Manage Data window appears.

2 Click **Import data**.



3 Drag your data files into the right pane.

> **TIP** You can also click **Local files** and navigate to the file location.

4 Make sure that the target location is set to the correct library location. The default library location is `cas-shared-default/Public`.

Note: You can accept the default format or select a target format that is appropriate for the source data. Not all source data should be saved to some formats, even if they are supported by the target caslib. For example, the CSV format might be supported, but you cannot save a binary image file to a CSV target file and get useful data. Also, it is recommended to use the SASHDAT format for large data files that you want to use for performing analytic calculations.

5. Click **Import all**.
6. Click **Add** to return to the Manage Data window.
7. Click **Close** in the Manage Data window.

Note: Prior to the 2025.09 release, click **Cancel** to close the Manage Data window, without selecting any of the tables.

For more information, see "Working with Data in CAS" in *SAS Data Explorer: User's Guide*.

3

Managing Model Repositories

About Model Repositories

SAS Model Manager provides a secure common model repository to store all types of models that are versioned. The common model repository enables you to search, query, sort, and filter models by attributes that are associated with models across all accessible organizational folders and projects. You can use model repositories to separate your project and model content, as well as to set permissions for objects within a repository. Some examples are having different repositories for test and production environments, or for different organizations. Model repositories are managed within the Settings window of the SAS Model Manager web application. You can add, delete, and rename a repository. The default repository can be renamed, but it cannot be deleted.

Note: Only SAS administrators and other authorized users can create, update, or delete repositories. In addition, Authenticated Users cannot initially access new custom repositories. A SAS administrator must grant access for a user or group to a new custom repository. For more information, see “Managing Content” in *SAS Model Manager: Administrator’s Guide*.

To access the settings for repositories, click the user button in the application bar and select **Settings** ⇨ **SAS Model Manager** ⇨ **Repositories**.

Create a New Repository

Note: By default, only SAS administrators can create new repositories.

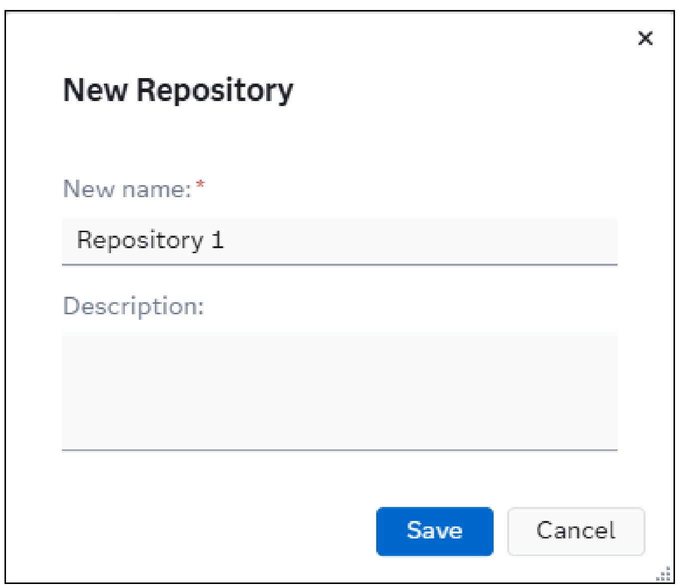
1 Click the user button in the application bar and select **Settings** ⇨ **SAS Model Manager** ⇨ **Repositories**.

2 Click .

3 Enter a name for the repository.

4 (Optional) Enter a description for the repository.

Note: After you save the new repository, the description cannot be edited.



5 Click **Save**.

Rename a Repository

Note: By default, only SAS administrators can rename repositories.

1 Click your name in the application bar and select **Settings** ⇨ **SAS Model Manager** ⇨ **Repositories**.

2 Select a repository, click ⋮ , and select **Rename**.

3 Enter a new name for the repository.

4 Click **Rename**.

Delete a Repository

Note: By default, only SAS administrators can delete repositories. A repository cannot be deleted if it contains model or project content.

1 Click your name in the application bar and select **Settings** ⇨ **SAS Model Manager** ⇨ **Repositories**.

2 Select a repository and click 🗑.

3 In the confirmation message, click **Delete**.

4

Working with Projects

About Projects

Often numerous statisticians, modelers, and data scientists, working in their tool of choice, are asked to solve a business problem by creating an analytical model. The Projects category view enables users to view models grouped by business problem. It also enables a user to perform an apples-to-apples comparison of models in order to decide which model is the best to use as the champion.

You create projects within folders. The models within a project are associated with a project version. A project version enables you to group models based on business requirements. When you group your models by a specific time period, you can see whether a particular business problem has changed. You can also determine how you addressed those changes and use that insight to govern how you will address these changes in the future.

A project consists of the models, variables, tests, and other resources that you use to determine a champion model. For example, a banking project might include models, data, and tests that are used to determine the champion model for a home equity scoring application. The home equity scoring application predicts whether a bank customer is an acceptable risk for granting a home equity loan.

Note: The **History** tab of a project shows events for the primary actions, and the date on which the project was modified and by whom.

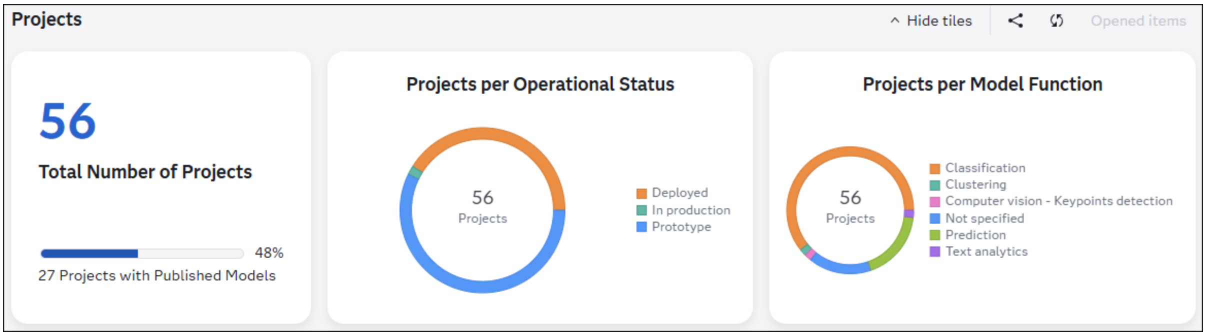
View Dashboard and Filter Projects

View Projects Dashboard

When you first view the Projects category, the dashboard appears at the top of the application window. The dashboard contains a graphical representation of the following metrics:

- the total number of projects
- number of projects with published models
- number of projects per model function
- number of published models per destination

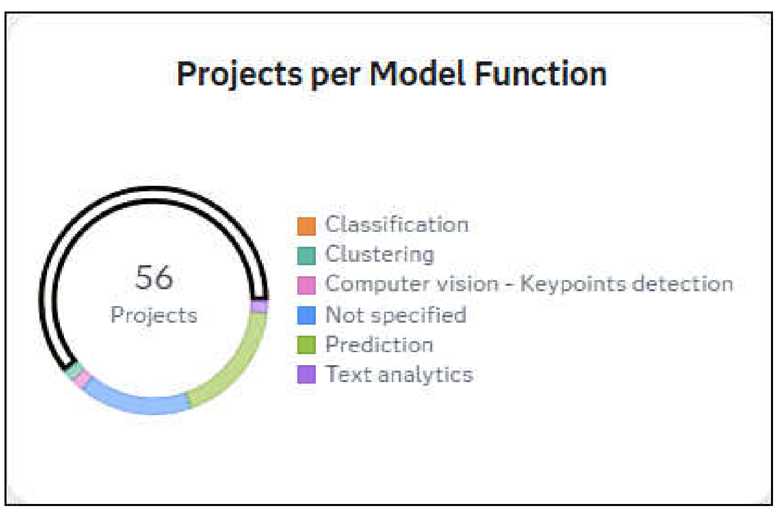
You can click **Hide tiles** or **Show tiles** on the toolbar. You can also share a link to the Projects category by clicking the icon.

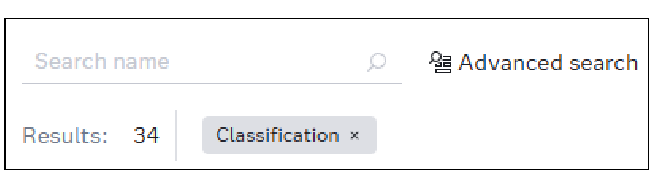


Filter the List of Projects Using the Dashboard Charts

You can click various sections of the different dashboard charts to filter the list of projects. You can apply filters to each of the charts to create a compound query.

Here is an example of filtering the list of projects to display only projects that have a **Classification** model function:





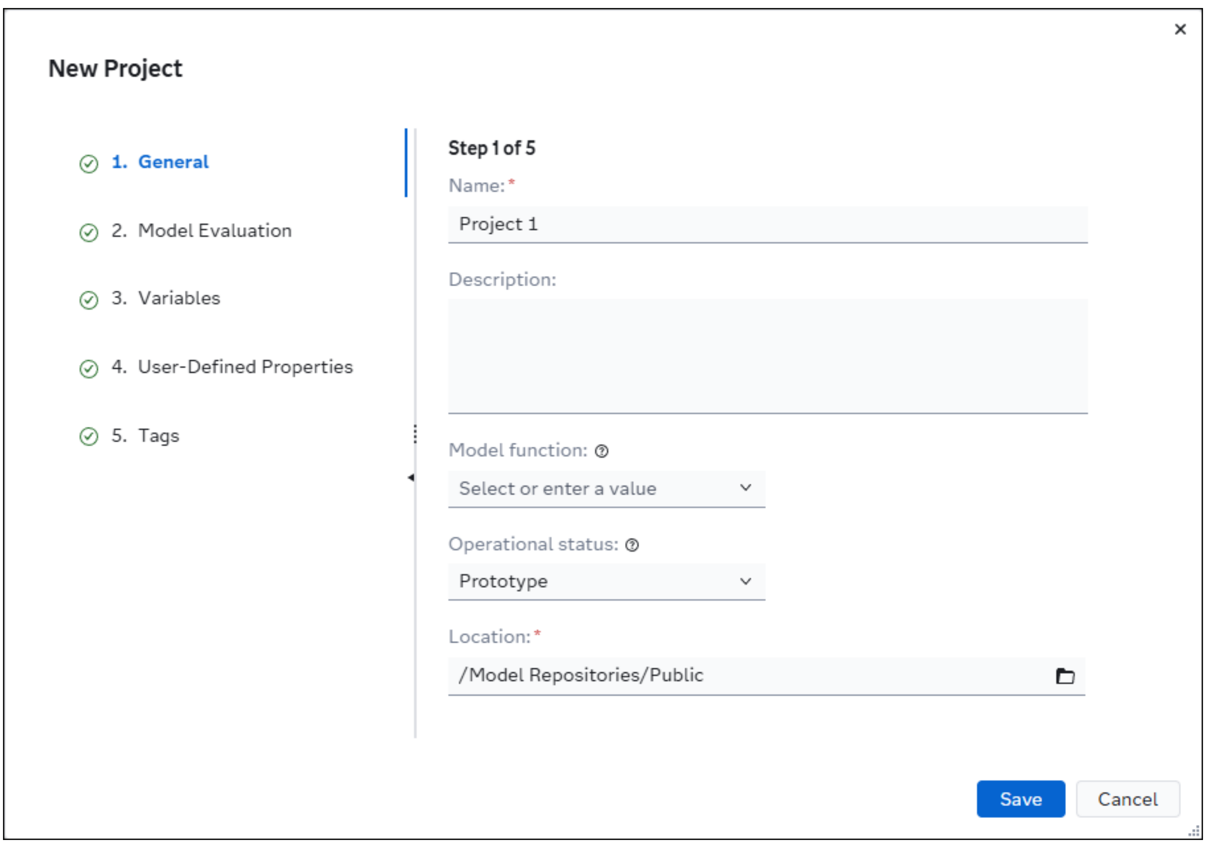
You can also use the Advanced Search for Projects window to specify additional information in order to further filter the list. The filters that you applied using the dashboard charts are reflected in the Advanced Search for Projects window. For more information, see "Advanced Search for Projects" on page 48.

TIP To remove an individual filter from a chart, you can click the empty area beside the chart or you can click the × icon within the tokens that are located above the list of projects. You can also remove all filters by clicking the ✕ icon, which is located in the right side of the toolbar.

Create a New Project

1 Click to navigate to the Projects category view.

2 Click **New project**. The New Project wizard appears.



3 On the **General** page, enter a name for the project.

TIP You can save a project at any time as long as all of the required fields have been specified.

4 (Optional) Enter a description for the project.

5 (Optional) Select a model function from the list or enter your own value. The model function indicates the type of output that your predictive model project generates.

> **IMPORTANT** To monitor the performance of the models within a project, it is recommended to select **Classification**, **Prediction**, or **Text categories** when creating a project. Only projects with those model functions can be monitored for performance.

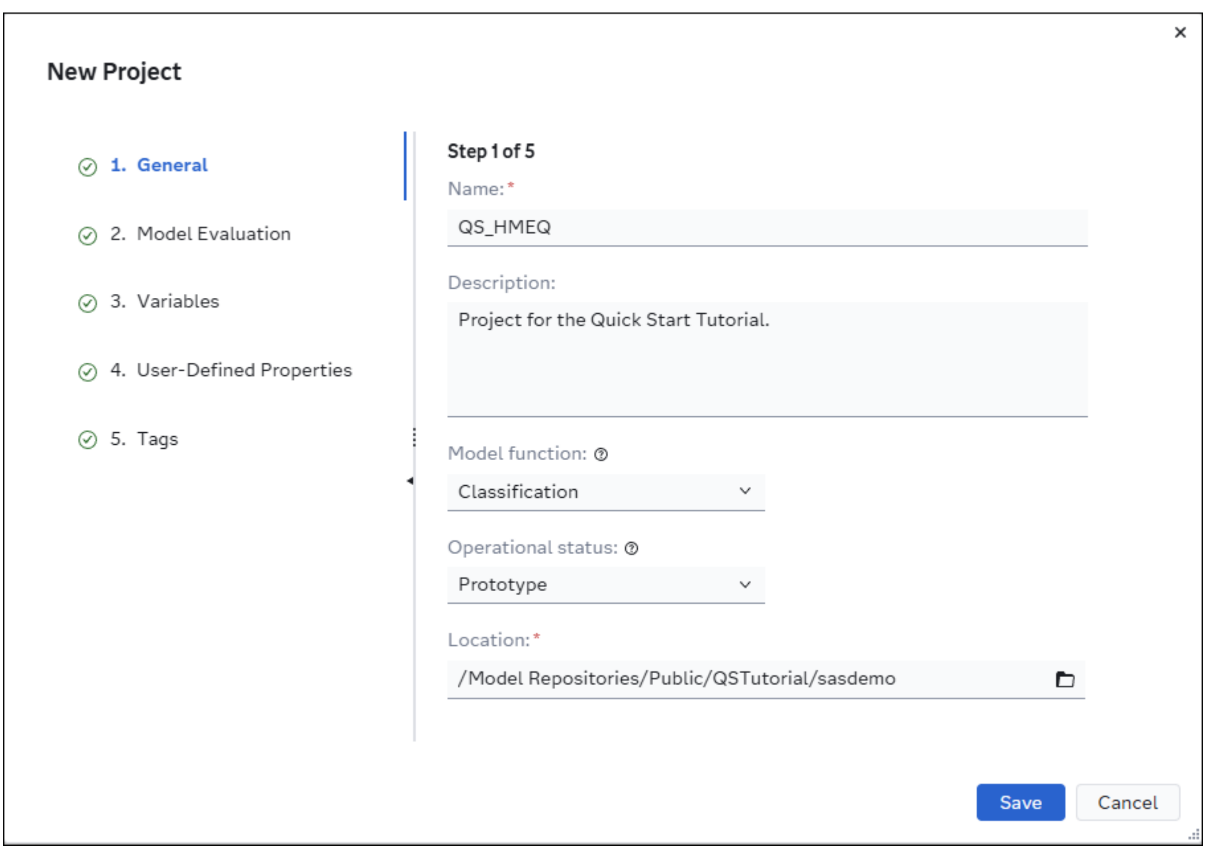
6 Accept the default value of **Prototype** for the operational status.

7 Accept the default location or select a new location.

To select a new location, click , select the desired repository or folder, and then click **OK**.

Note: In the Choose a Location window, you can create a folder within a repository folder to store projects and models for your organization. Repository folders must be created using the Settings window. For more information, see "About Model Repositories" on page 17.

It is recommended that the `DMRepository` and `VTARepository` repository folders should be reserved for projects and models that are registered from Model Studio. Also, the `VARepository` repository folder should be reserved for models that are registered from SAS Visual Analytics. For more information, see "Register Models" in *SAS Viya: Machine Learning User's Guide*.



8 (Optional) Click the **Model Evaluation** page and specify values for each setting:

a Select a default training table. Click , enter the name of the table in the filter box, and select it. Click **OK**.

> **TIP** If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see "Overview of SAS Data Explorer" in *SAS Data Explorer: User's Guide*.

b Specify the values for the properties that are used for model evaluation.

Note: The following properties are used when running a performance definition. If you do not specify values for these properties when creating a project, you must specify them before running a performance definition.

Classification project

- Target variable
- Target level
- Target event value or Target event values
- Output event probability variable or Predicted target variable

Prediction project

- Target variable
- Target level
- Output prediction variable

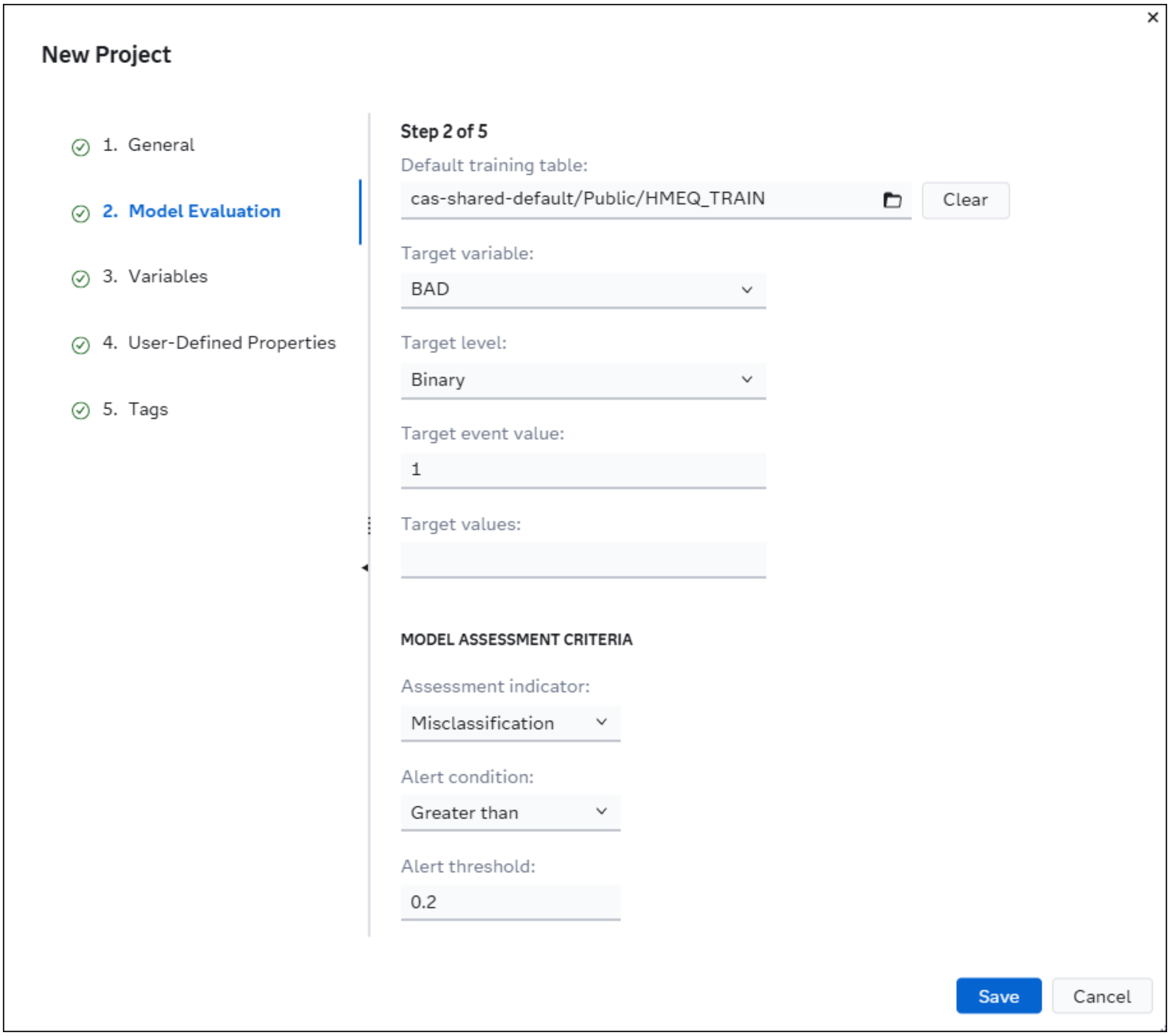
Text categories project

- Actual category variable
- Document ID variable
- Text variable
- Predicted category variable

For information, see "Set Model Evaluation Properties".

c Specify values for the model assessment indicator, alert condition, and alert threshold.

Note: You cannot currently select an assessment indicator when creating a project under these conditions: a model function is not specified, or the model function is Analytical, Text analytics, or Transformation.



9 (Optional) Click the **Variables** page and select input and output variables from the default training table or another input table. You can also add custom variables.

Note: If you do not specify project variables on this page, you are prompted to set them when you are setting a model as the project champion or as a challenger.

a To select input variables, click **Browse** and select one or more variables, and then click **OK**.

If you did not specify a default training table, here are the other ways that you can select variables.

- Select an input table:

1 Click

2 Select a data table and click **OK**.

- Add one or more custom variables:

1 Click **Add custom variables**.

2 Enter a name for the variable.

3 Select a data type.

4 Expand the **Optional** section to specify a length, measurement, and description for the variable.

5 Click **Add**.

6 Repeat steps 1 through 5 for each variable that you want to add.

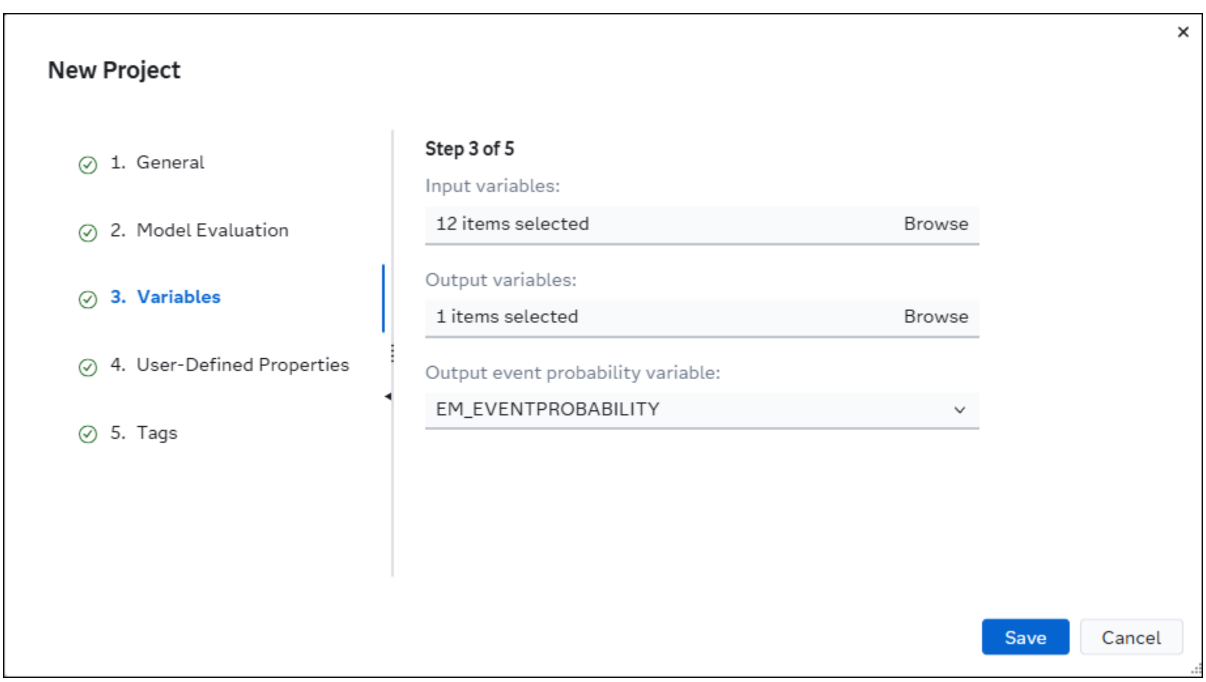
7 Click **OK**.

b To select output variables, **Browse** and select one or more variables, and then click **OK**.

Note: You have the same choices for selecting output variables that you had for input variables. See the previous step.

c Select a value for each of the project-specific output variables. For example, if you selected **Classification** as the model function on the **General** page, select a value for the **Output event probability variable**.

Note: The model function that you specified determines the project-specific output variables that are displayed. If you do not specify values for the output variables when creating a project, you must specify them before running a performance definition.



10 (Optional) Click the **User-Defined Properties** page.

> **TIP** You can also add user-defined properties after you create a project. For more information, see "Add User-Defined Properties" on page 42.

a Click **Add Property**. The Add Property window appears.

b Enter a name for the property.

c Select a data type for the property.

d Enter a value for the property.

e (Optional) Select the **Add another property** check box to add another variable without closing the window.

f Click **Add** to add the property to the list.

g Repeat steps b through f for each property that you want to add.

11 (Optional) Click the **Tags** page.

Enter a value and select an existing tag, or press Enter to add a new tag. Repeat this step to add more tags.

Note: The tag name can contain only alphanumeric characters, double-byte characters, an underscore (_), a hyphen (-), and a period (.). Spaces are not allowed.

> **TIP** You can also add tags after you create a project.For more information, see "Add Tags to a Project" on page 43.

12 Click **Save**.

Adding Models

After you create a project, you add models to a project version on the **Models** tab. A project can contain multiple versions. You can also copy a model from a folder or another project version. You can view and compare models in all versions or in one selected version on the **Models** tab. After model evaluation, you set one of the candidate models as the champion model and can also set one or more models as challengers.

Here are some of the tasks that you can perform on the **Models** tab of a project:

- add a new custom model
- import models

- copy a model
- compare models

Managing Variables

Input variables and output variables can be added to both project and model objects on the **Variables** tab of an object. The same variable name cannot be used for both an input and output variable.

Note: When you add, edit, or delete project variables, you might receive a warning message indicating that the model role will be cleared. This happens when the project champion model and one or more challenger models have been set.

Variable Naming Restrictions

Note: Do not use any of these operators or keywords as variable names: AND, OR, IN, NOT, LIKE, TRUE, or FALSE. Do not use _N_ or any DS2 reserved word as a variable name. For more information, see “Reserved Words in the DS2 Language” in *SAS DS2 Programmer’s Guide*.

Add Variables from a Data Table

1. Click the **Variables** tab.
2. Click **Add variable** and select **Data table**. The Choose Variables window appears.
3. Click 🗀 to select a table. The Choose Data window appears.

   TIP If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see “Overview of SAS Data Explorer” in *SAS Data Explorer: User’s Guide*.

4. Select the data table that you want to import the variables from and click **OK**.

5 In the Choose Variables window, select the variables that you want to add and click +› to add the variables to the input variables or output variables list. You can also click +» to add all of the variables from the available items list.

6 Click **OK**.

7 Click .

Add Custom Variables

1 Click the **Variables** tab.

2 Click **Add variable** and select **Custom variable**. The Add Custom Variables window appears.

3 Enter a name for the variable.

4 Select a data type.

5 Select a variable type.

> **Note:** If you select "Input" or "(none)" as the variable type, you can select the **Assess variable for bias** option. To mark a variable to be assessed for bias, it must also have a measurement value of Binary, Nominal, or Ordinal. When a variable is marked to be assessed for bias, it is used to calculate fairness and bias metrics for a model using the training data. If no variables are marked for bias at the time the model was trained, the fairness and bias metrics are not available.

6 Expand the **Optional** section to specify a length, measurement, and description for the variable.

7 Click **Add**.

8 Repeat steps 3 through 6 for each variable that you want to add.

9 Click **OK**.

Assess Variables for Bias

When a variable is marked to be assessed for bias, it is used to calculate fairness and bias metrics for a model when running performance monitoring in a project that has a model function of classification or prediction. If no variables are marked for bias at the time the model was trained, the fairness assessment using the training data is not available on the model card for a model.

Note: If a model is set as champion, and any of the variables are marked to be assessed for bias, the variable is also marked to be assessed for bias at the project level.

To mark a variable to be assessed for bias:

1 Click the **Variables** tab.

2 Select a variable from the list and click **Assess variable for bias**.

Note: When marking a variable to be assessed for bias, it must have a measurement value of Binary, Nominal, or Ordinal. The **Assess variable for bias** button is enabled for input variables or for variables that are not specified as an input or output variable. Output variables cannot be assessed for bias.

TIP You can also mark a variable to be assessed for bias in the Edit Variable window or when adding custom variables..

Edit Variables

1 Click the **Variables** tab.

2 Click on the name of the variable that you want to edit. The Edit Variable window appears.

3 Edit the properties as needed and click **OK**.

4 Click .

Delete Variables

1 Click the **Variables** tab.

2 Select the check box for the variables that you want to delete, click ⋮, and then select **Delete**.

3 Click .

Modifying Project Properties

Project properties contain the project metadata. Project metadata includes information such as the name of the project, the type of project (model function), the project owner, the project identifier, the project location, and the names of the tables and variables that are used by project processes. The project properties are organized into these types: General, Model Evaluation, Tags, and User-Defined.

Set Project General Properties

The **General Properties** tab of a project contains both system-defined properties that you cannot modify, and project specific properties that can be modified, such as the description of the project. None of the project properties are required, except for the name and location.

To set the project general properties, click the **Properties** tab, modify the property values, and then click .

Table 4.1 *List of General Properties*

Property	Description
Name	Specifies the name of the project. A project can be renamed only from the Projects category view.
Description	Specifies the description of the project.
Model function	Specifies the type of output that your predictive model project generates. Ensure that the types of models that you are going to use in the project fit within the selected model function type. For more information, see Table 4.2 on page 35.
Operational status	Specifies the current state of the project and its models: Prototype indicates that the project has started, but that the project's models have not yet been deployed. This is the default status when you create a new project.

Property	Description
	Note: This status replaces the previous status value of “Under development”. Deployed indicates that at least one model within the current project has been successfully published to a destination. In production indicates that the project contains models that have been successfully deployed and are being used in a production environment. This status must be set manually. **Note:** The status replaces the previous status value of “Active”. Deactivated indicates that the project contains one or more models that were previously in a production environment and have been temporarily or permanently suspended. **Note:** This status replaces the previous status values of “Inactive” and “Retired”.
Location	Specifies the location of the project in the common model repository.
Champion version	Specifies the project version that contains the champion model.
Champion model name	Specifies the name of the model that is set as the project champion.
UUID	Specifies the universally unique identifier for a project object.
External URL	Specifies a user-defined URL to a project object in another application or to documentation related to the project.
External project ID	Specifies the project ID for a project that was registered from an external application, such as Model Studio. **TIP** You can view the relationship between a project in SAS Model Manager and another SAS Viya

Property	Description
	application. For more information, see "View SAS Lineage".

Table 4.2 Types of Model Functions

Model Function	Description
Analytical	Function for any model that is not Prediction, Classification, Clustering, or Segmentation.
Classification	Function for models that have target variables that contain binary, categorical, nominal, or ordinal values.
Clustering	Function for clustering models.
Computer vision - Image classification	Function for computer vision models that are used for image classification.
Computer vision - Object detection	Function for computer vision models that are used for object detection in a classification network.
Computer vision - Keywords detection	Function for computer vision models that are used for keywords detection in neural networks.
Computer vision - Image segmentation	Function for computer vision models that are used for image segmentation.
Prediction	Function for models that have interval targets with continuous values.
Segmentation	Function for segmentation models.
Single-series forecasting	Function for models used to forecast future data based on past data for a single time series.
Text analytics	Function for a SAS Visual Text Analytics modeling project. **Note:** This model function appears on the **Properties** tab of a project only when a SAS Visual Text Analytics project contains different types of text

Model Function	Description
	models. It is not displayed as an option in the model function drop-down list
Text categories	Function for SAS Viya Text Analytics categorization models.
Text concepts	Function for SAS Viya Text Analytics concepts models.
Text sentiment	Function for SAS Viya Text Analytics sentiment models.
Text topics	Function for SAS Viya Text Analytics topics models.
Transformation	Function for models that are used to determine mathematical functions, which can be used to stabilize variances, remove nonlinearity, and correct non-normality in variables to improve the fit of your model.
Text generation	Function for text generation models, which are designed to generate coherent and contextually relevant text based on input prompts. These models can be categorized into several types based on their architecture, training objectives, and use cases. See: Large Language Model (LLM) Properties **Note:** Support added with the 2025.08 release.
Prompt template	Function for prompt template models, which are designed to help translate user input and parameters into instructions for a language model. This can be used to guide a model's response, helping it understand the context and generate relevant and coherent language-based output. See: Large Language Model (LLM) Properties **Note:** Support added with the 2025.08 release.

Set Model Evaluation Properties

This section contains project properties that are used for comparing, assessing, and evaluating the health of models.

Property	Description
Default training table	Specifies the default training table, which can be used to test models on-demand or to validate models after a user publishes them to a publishing destination. This property is optional.
Target variable	Specifies the name of the target variable that was used to train the model.
Target level	Specifies the target level of binary, nominal, ordinal, or interval.
Target event value	Specifies the target variable value that defines the desired target variable event for binary and interval targets. **Note:** When you select an ordinal or nominal target level the Target Event Values section appears to enable you to add multiple target event values on page 40.
Target values	For class binary targets, the set of possible outcome classes, separated by commas. For example, binary class target values might be `1, 0` or `Yes`, `No`.
Predicted target variable	Specifies the variable that contains the predicted target values, when the target level is set to Nominal or Ordinal.
Actual category variable	Specifies the actual target variable , when the **Model function** property is set to **Text categories**.
Document ID variable	Specifies the document ID variable , when the **Model function** property is set to **Text categories**.

Text variable	Specifies the text variable , when the **Model function** property is set to **Text categories**.
Predicted category variable	Specifies the output variable name, when the **Model function** property is set to **Text categories**. **Note:** This property was renamed with the 2022.09 stable release.
Output cluster variable	The output cluster variable name, when the **Model function** property is set to **Clustering**.
Output concept variable	Specifies the output variable name, when the **Model function** property is set to **Text concepts**.
Output event probability variable	Specifies the output event probability variable name, when the **Model function** property is set to **Classification**, **Analytical**, **Computer vision - Image classification**, or **Transformation**. **Note:** This property is also displayed for user-defined model functions.
Output prediction variable	The output prediction variable name, when the **Model function** property is set to **Prediction**, **Analytical**, **Computer vision - Image segmentation**, **Computer vision - Keypoints detection**, **Computer vision - Object detection**, **Single-series forecasting**, or **Transformation**. **Note:** This property is also displayed for user-defined model functions.
Output segmentation variable	The output segmentation variable name, when the **Model function** property is set to **Analytical**, **Segmentation** or **Transformation**. **Note:** This property is also displayed for user-defined model functions.
Output sentiment variable	Specifies the output variable name, when the **Model function** property is set to **Text sentiment**.

Output topic variable	Specifies the output variable name, when the **Model function** property is set to **Text topics**.
Default Thresholds for Training Metrics	This section contains properties that specify the default values for the Accuracy, Generalizability, and Fairness training metrics that are used for comparing and assessing models using the training data. You can use the default values for the training metrics or customize the thresholds for your organizations specific use cases. **Note:** This section is only displayed for projects with a model function of classification or prediction.
Model Assessment Criteria	This section contains the properties that are used for comparing and assessing models. **Note:** This section is only displayed for projects with a model function of classification or prediction.
Assessment indicator	Specifies performance indicators that are used to assess a model. The applicable indicators are displayed in the drop-down list based on the specified model function.
Alert condition	Specifies the operator to use with the alert threshold value to determine whether the model is performing within an acceptable tolerance for the associated assessment indicator. An example is whether the Misclassification rate for the model is below 25%.
Alert threshold	Specifies the threshold value to use with the alert condition operator to determine whether the model is performing within an acceptable tolerance for the associated assessment indicator.
KPI Alert Rules	This section contains the properties that are used for specifying key performance indicator (KPI) alert notifications. You can specify the rules for key performance indicators (KPIs) to use in

order to determine what notifications the system should send to users at the project level. Alert notifications are sent when the associated alert condition and threshold are not met.

You can add, modify, or delete KPI rules. For more information, see “Add KPI Alert Rules” on page 41.

Add Target Event Values

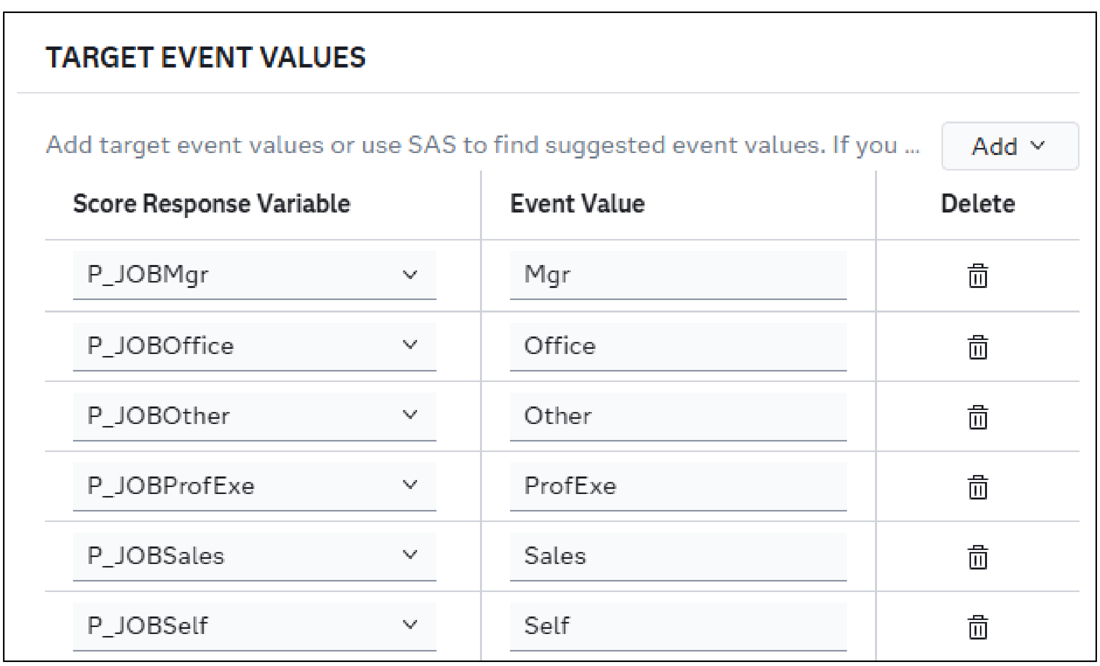
When the target level for a project is set to `Nominal` or `Ordinal` the **Target Event Values** sections appears, instead of the **Target event value** property.

To add target event values:

1 Click **Add** and select **Add event value**, or select **Find event values** to find the suggested event values within the default training table.

Note: If you do not find the desired variables, add them on the **Variables** tab.

2 Map a score response variable to the event values that you added.



3 Click to save the changes to the project.

Add KPI Alert Rules

When you add a new KPI alert rule to the current project, a default rule name is generated. The default rule name is a combination of the KPI name, the variable name, and the values entered in the alert condition and alert threshold fields. The default name can be modified. You can also specify the priority and the number of occurrences that must be reached before sending an alert notification.

1 Click +. The Add KPI Alert Rule window appears.

2 Select a key performance indicator (KPI) category and a specific KPI.

 Note: When selecting a fairness or model drift KPI metric you are also prompted to select the variables to assess. Fairness metrics use variables that are marked to be assessed for bias. Model drift metrics use input variables or output variables depending on which KPI is selected.

 Click to select one or more variables to assess.

3 Select an alert condition.

4 Enter a value for the alert threshold.

5 Select a priority for the KPI alert.

6 Enter a value for the number of occurrences after which you want to set an alert for this KPI rule.

7 Accept the default rule name or enter your own rule name.

8 Click **OK**.

9 Click .

Here are other actions that you can perform:

- Modify the values of existing KPI rules.
- Select one or more KPI rules and click to delete them from the current project.

Note: After modifying or deleting KPI rules, you must click to save the changes to the project.

Specify Model Usage Properties

You can specify the values of the model usage properties at the project level or model level. The values of the model-level properties are inherited from the project-level properties, if the model is within a project. However, model-specific information can be specified for each property value at the model level. The information specified in this section is surfaced on the **Model Card** tab of a model object.

- Model purpose
- Intended use
- Expected benefit
- Out-of-scope use cases
- Limitations
- Model responsible party

Note: The **Model responsible party** property is read-only at the model level, if the model is located within a project.

Add User-Defined Properties

You can add your own project or model properties. The property-value pair is metadata for the project or model, both of which can be searched.

To add user-defined properties:

1. On the **Properties** tab, select **User-Defined**.
2. Click **Add Property**. The Add Property window appears.

   TIP If user-defined properties already exist, click + above the table.

   a. Enter a name for the property.
   b. Select a data type for the property.
   c. Enter a value for the property.
   d. Click **Add** to add the property to the list.
   e. Repeat steps a through d for each property that you want to add.
3. Click .

To edit a property:

1. Click on a property name within the table.
2. Edit the name, the data type, or value of the property.
3. Click **OK**.
4. Click .

To delete properties, select one or more properties in the table, and then click .

Add Tags to a Project

Tags are ideas and topics that indicate what your content is about. You can use tags to indicate the market sector, a company department, or other types of content that you want to associate with your project.

You can add one or more tags to a project. When you add tags to a project, they are also added to a global list of tags that is available to be added to other projects within the common model repository.

Automated tags are applied to a project when specific actions are performed, such as models published from a project or performance monitoring is scheduled. They also show key information about a project, such as an active workflow being associated with the project, performance results available for a project, or relationships that might exist.

Note: Tag property values for a project appear as tokens in the **Tags** column of the Projects category view and can be used for sorting projects. The function for searching projects for tag values is currently not supported.

To add a tag:

1. On the **Properties** tab of a project, select **Tags**.
2. Enter a value and select an existing tag, or press Enter to add a new tag.

   Note: The tag name can contain only alphanumeric characters, double-byte characters, the underscore (_), the hyphen (-), and the period (.). Spaces are not allowed.

   Repeat this step for additional tags.
3. Click .

To delete a tag, click × within the token, and then click . The tag is removed from the project.

Managing Project Files

On the **Files** tab of a project, you can add, modify, and delete custom key performance indicator (KPI) SAS code files. You can also download and run standard and custom KPI SAS code files. The standard KPI file cannot be deleted. The project files are grouped into categories based on their file role and file name. The standard and custom KPI files are used when running performance for models within a project.

Add Files

Note: Currently, when you add a new file to a project, the file role is set to **Custom KPI**.

1 Click +. The **Add Project File** window appears.

2 Click , select a file to add to the project, and then click **Open**.

Note: You can add only those files that have a .sas file extension.

3 Click **Add**.

Delete Project Files

Select the file and click . In the confirmation message, click **Delete**.

Note: The standard KPI file (`ProjectKPI.sas`) cannot be deleted.

Download Project Files

Select the file and click . The project file is downloaded to your local machine.

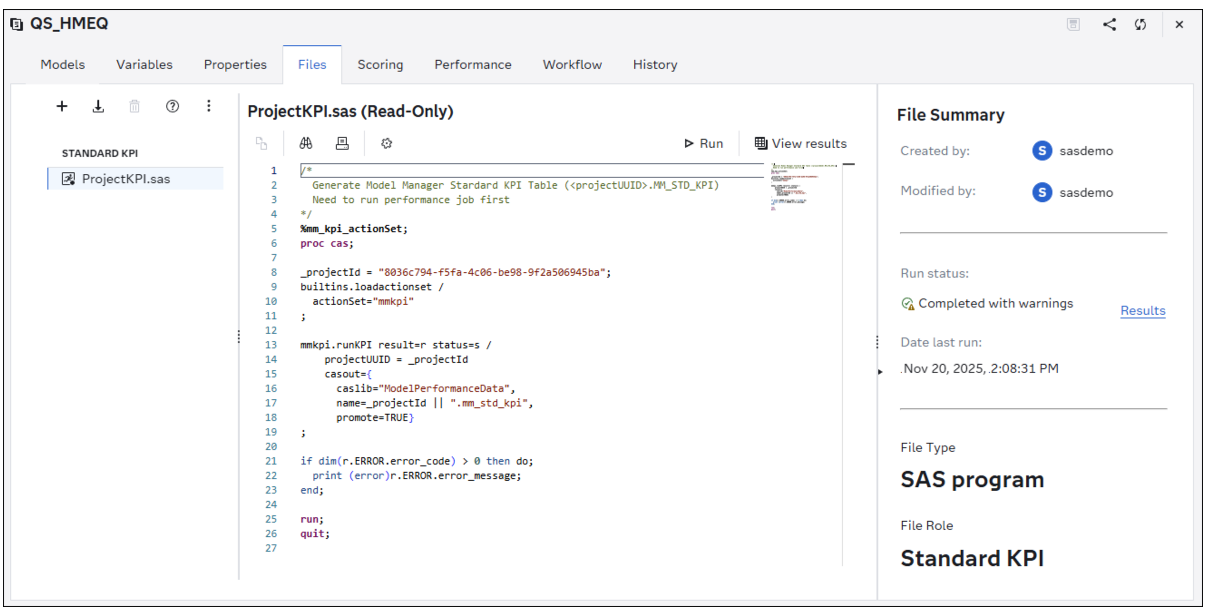
Run a Project Code File

You can run standard KPI and custom KPI code files. However, you cannot edit or delete the standard KPI file (ProjectKPI.sas).

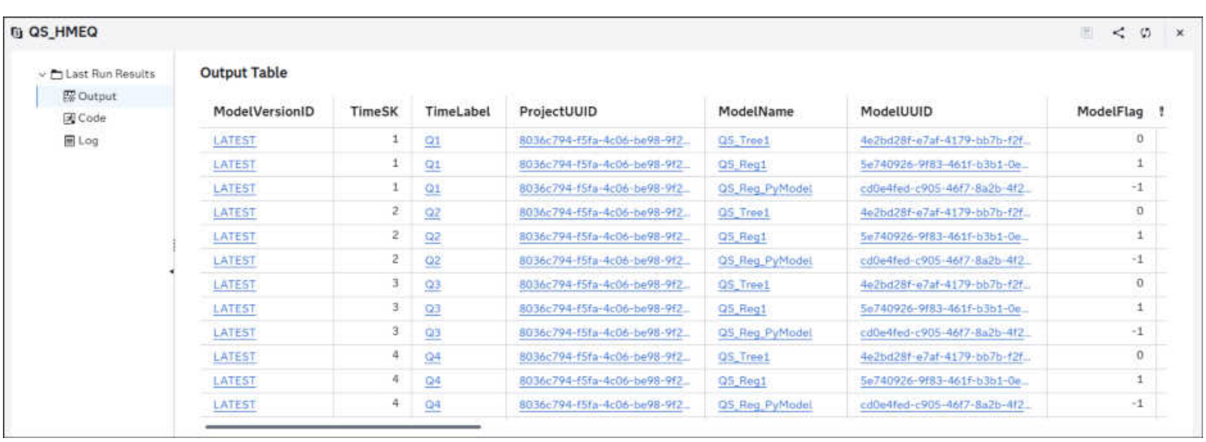
Note: The SAS Model Manager standard KPI table (<projectUUID>.mm_std_kpi) is also generated when running the project's performance definition.

1 Select a file and click ▷ **Run** to execute the code within the file.

Note: You can view the status of the running code within the **File Summary** panel. When the run status has a value of completed, the **Results** link appears in this panel. The **View results** button is available in the toolbar



2 Click **View results** in the toolbar.



3 Click × to return to the project's **Files** tab.

View File Properties

To view the file properties, hover over the ⓘ icon that appears next to a project file or select a file and then click in the toolbar.

Managing Project Versions

A project version is a container of models. An initial version is created automatically when you create a project. You can view a list of the project versions on the **Models** tab in the **Version** drop-down list. The latest version is displayed by default. You can also choose to display all versions, create a new version, or manage existing versions from the **Version** drop-down list. When you create a new project version, you can specify a name and description for the version, such as a time interval for a project cycle.

A version is a sequential number that increments by one each time you add a new version. A project can contain multiple editable versions. A project version is used to differentiate collections of models that are meant to solve the project's problem over time-boundaries. Your version might represent a calendar year, a retail season, or a fiscal quarter. A version contains all of the candidate model resources that you need to determine a champion model as well as all champion model resources. For example, you might develop models for a scoring program that determines whether a customer is eligible for a home equity loan.

Create a New Project Version

1. On the **Models** tab, click ▾ in the **Version** drop-down list, and select **New version**. The **New Project Version** window appears.
2. Enter a name for the version, or accept the default name (for example, Version 2).
3. (Optional) Enter a description for the version.
4. Click **Save**.

Manage Project Versions

In addition to creating a new project version, you can edit the description of a version, rename a version, or delete a version.

1. On the **Models** tab, click ▼ in the **Version** drop-down list, and select **Manage versions**. The Manage Project Versions window appears.
2. (Optional) Edit the description of a version.
3. Create a new project version.
   a. Click . The New Project Version window appears.
   b. Enter a name for the version, or accept the default name (for example, Version 2).
   c. (Optional) Enter a description for the version.
   d. Click **Save**.
4. Rename a project version.
   a. Select a version and click **Rename**. The Rename window appears.
   b. Enter a new name for the version.
   c. Click **Rename**.
5. Delete a project version.

   Note: When only one project version exists, it cannot be deleted. You must also have the appropriate permissions to delete a version.

   a. Select a version and click .
   b. Click **Delete** in the confirmation message.
6. Click **Close** to close the Manage Project Versions window.

Delete a Project

Note: You must also have the appropriate permissions to delete a project.

1. In the Projects category view, select one or more projects.

2 Click ⋮ and select **Delete**.

3 In the confirmation message, click **Delete**.

Rename a Project

1 In the Projects category view, select a project, click ⋮ , and select **Rename**.

Note: Open objects cannot be renamed.

2 Enter a new name for the project.

3 Click **Rename**.

Search for Projects

In the **Projects** category view, you can perform multiple types of searches:

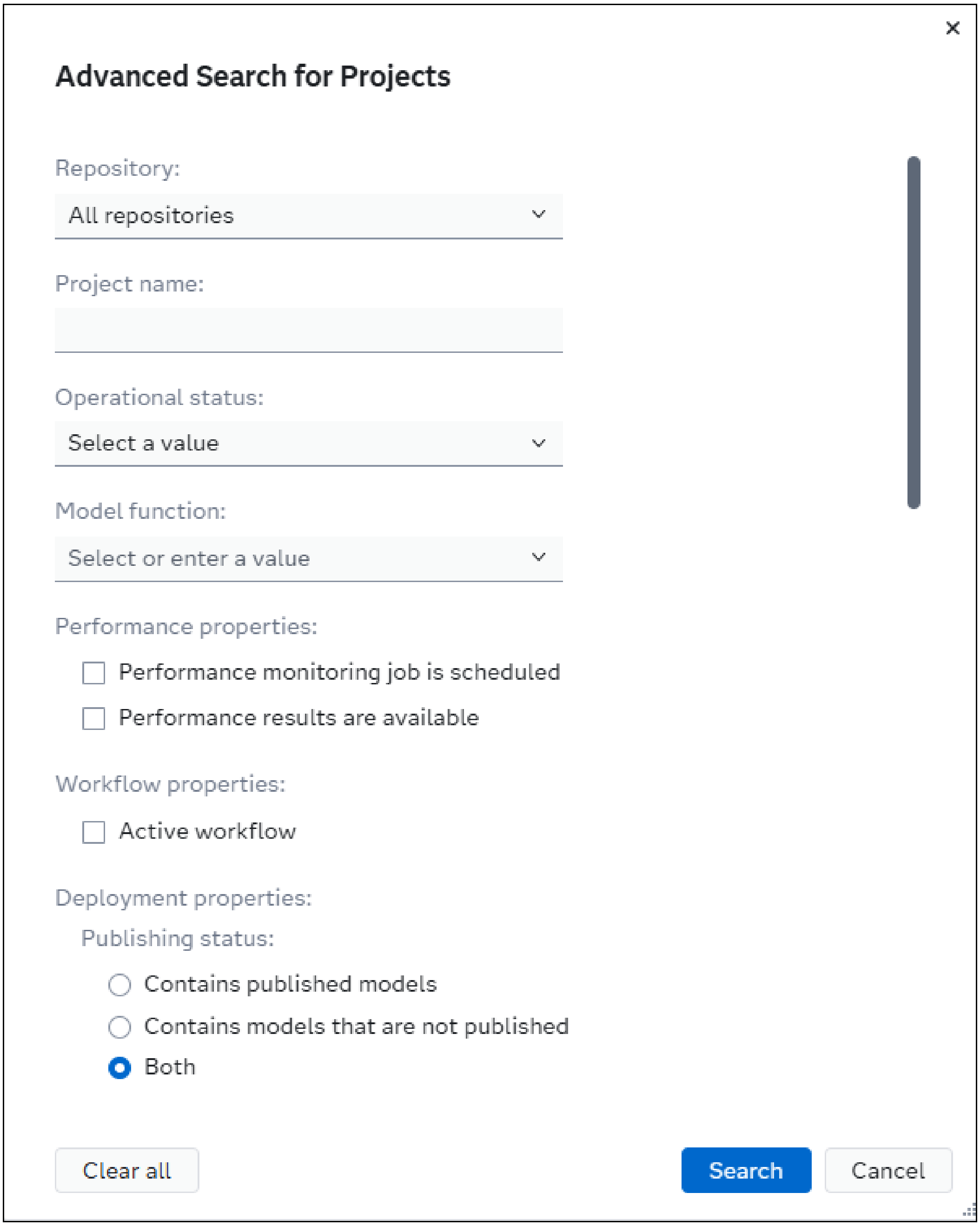
- search for projects by name using the search field above the models list.
- perform an advanced search for projects using project properties, input variables, output variables, or user-defined properties.
- search for projects across applications using the search field in the application bar.

  For more information about searching, see "Search" in *SAS Viya Platform: General Usage Help for Web Applications*.

Advanced Search for Projects

To perform an advanced search for projects:

1 Click **Advanced search**, which is to the right of the search box. The Advanced Search for Projects window appears.



2 To search for a project in a specific repository, click ▼ in the **Repository** drop-down list, and select the desired repository. **All repositories** is selected as the default for the advanced search.

3 Enter a possible value in the **Project name** field to search for a project by name.

4 Specify values for one or more of the following fields to include them in the search criteria:

Note: If you previously used the dashboard charts to filter the list of projects, the filters are applied to the associated fields in this window.

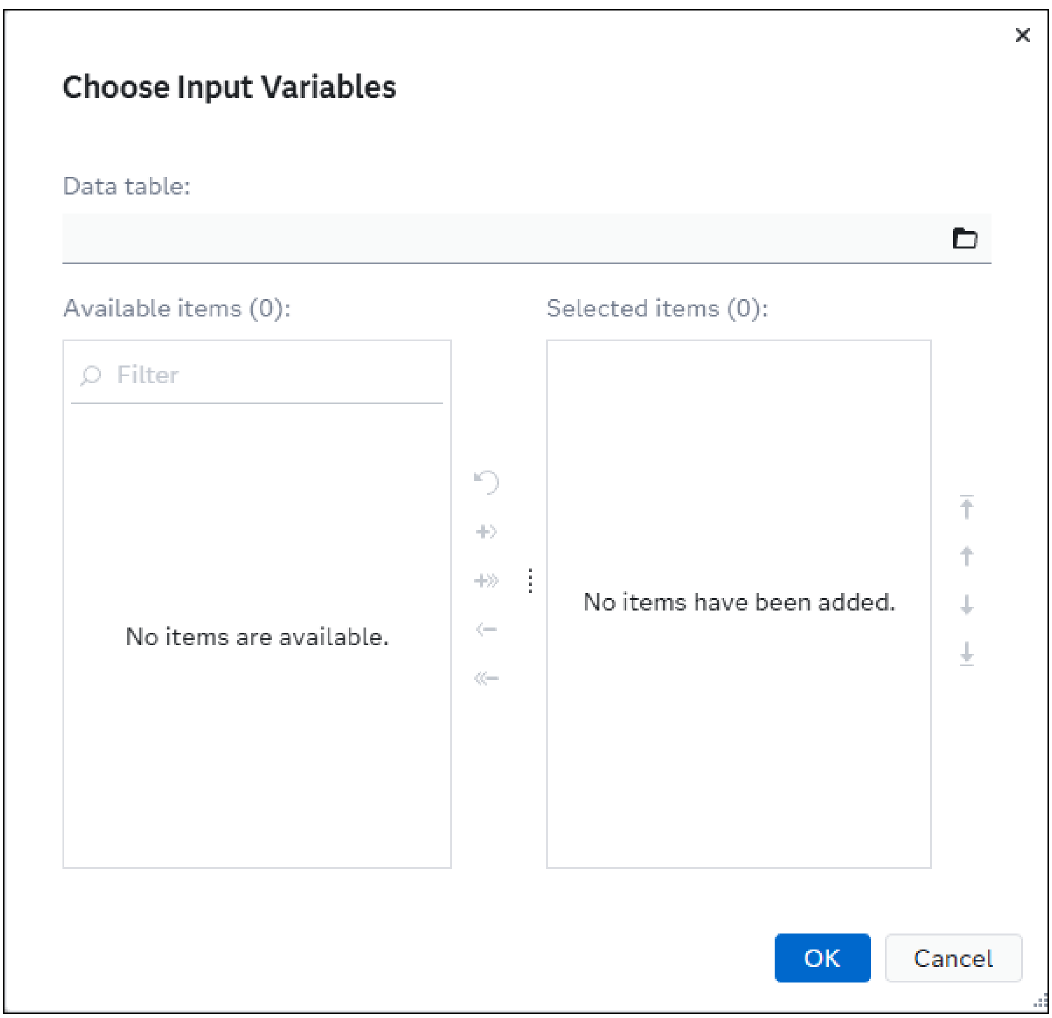
- **Operational status**

- **Model function**
- **Performance properties**
- **Workflow Properties**
- **Deployment properties**
- **Publishing destination**
- **Date created (range)**
- **Created by**

5 To search for a project by **Target variable**, enter a value or click to select a variable from a data table.

6 To search for a project by input variables or output variables, select a data table and choose the variables that are associated with models or project objects.

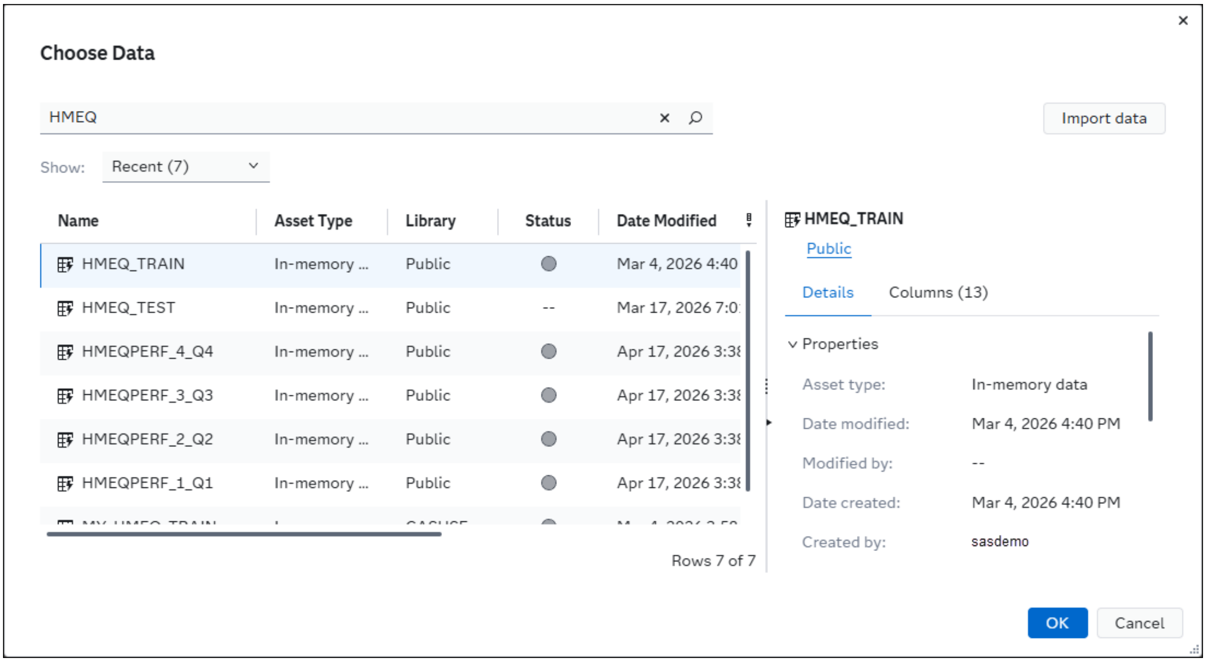
a Click next to the **Input variables**, or **Output variables** field. The Choose Input Variables or Choose Output Variables window appears.



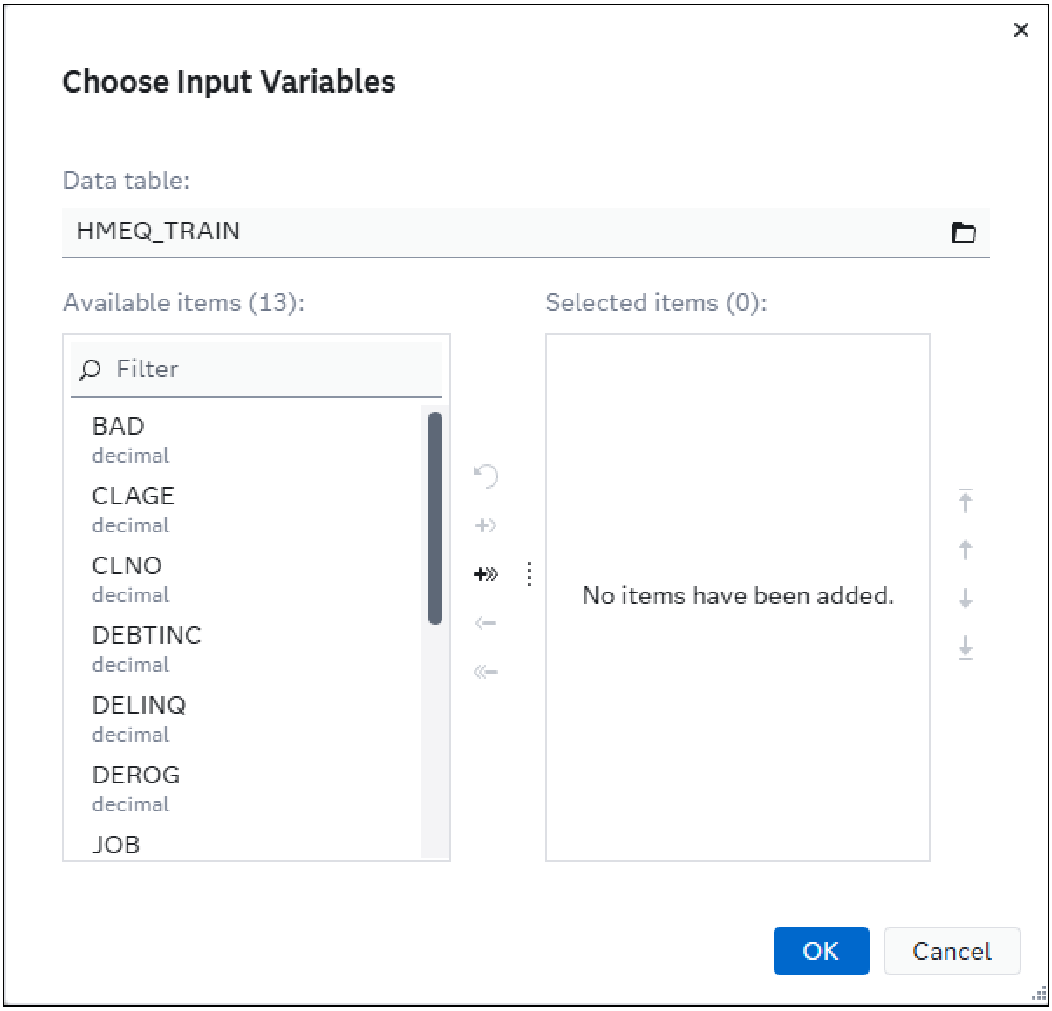
b Click next to the **Data table** field. The Choose Data window appears.

c Select a data table from the list. The details of the selected data appear on the right-hand side.

TIP Use the search bar to locate the desired data table and filter search results.



d When you have selected your data table, click **OK**. You are returned to theChoose Input Variables or Choose Output Variables window.



e Select the variables that you want to add and click +›. You can also click +» to add all of the variables from the available items list.

f To remove variables, select the variables that you want to remove and click ‹–. Click «– to remove all of the variables from the available items list.

> **TIP** Click ⟲ to undo your last action.

g Click **OK** to close the window. The variable or variables that you selected appear in the **Input variables** or **Output variables** fields.

> **TIP** Click |⟲ next to the **Input variables** or **Output variables** fields in the Search for Projects window to clear the selected variables.

7 Select relationships such as **Govern model rist (SAS Model Risk Management)**.

8 To search for user-defined properties that are associated with project objects, specify one or more property name and value pairs, or specify property names only.

a. Click + to add a row to the **User-defined properties** table.

b. Click ⌄ in the **Name** drop-down list, and select a user-defined property.

c. (Optional) Enter the value for the user-defined property in the **Value** column.

   Note: When the data type of a user-defined property is set to date or datetime, the calendar selector icon appears.

d. Select a row and click 🗑 to remove the user-defined property from the search criteria.

9 Click **Search**. The Projects category view opens and the project or projects that meet the advanced search criteria are listed.

> **TIP** Click **Clear All** to clear all previously entered search criteria in the Search for Projects window.

View SAS Lineage

You can view the relationships between objects such as models, projects, and data tables in the SAS Lineage application. SAS Lineage uses the **Search for Subjects** window to find the subjects for lineage diagrams.

Note: If models are registered from a Model Studio project into a SAS Model Manager repository folder and a project with the same name already exists, an underscore with a number is appended to the project name to avoid naming conflicts (for example, HMEQProject_1). This can occur when another user has previously registered models from a Model Studio project and that project has the same name and repository location within the SAS Model Manager repository.

1 Click ⁝⁝⁝ and select **Explore Lineage**.

2 Click **Search for subjects** in the SAS Lineage window.

3 Enter the name of your object (model or model project) in the search box.

4 Select a subject with the name of your object and click **Open.**.

   Note: The relationship between your object (model or model project) in SAS Model Manager and the Model Studio object is shown.

For more information, see the following documentation:

- "Open a SAS Lineage Subject from Another Application" in *SAS Lineage: User's Guide*
- "Searching for a SAS Lineage Subject " in *SAS Lineage: User's Guide*

5

Working with Models

About Models

You can add new custom models or import existing models using the SAS Model Manager web application. Models can be stored within a folder or project version in the SAS Model Manager common model repository.

The Models category view enables you to access all of the models in the common model repository in one place. The models can be located in a folder, or a project version. You can import models, add new custom models, compare models, and export models. You can also view model relationships and related objects, as well as search for models. Models that are within a project version can be published from the Projects category view or when opened from the Models category view. Models within folders cannot be published.

Several options exist for importing open-source models such as Python and R into SAS Model Manager. You can place your model files in a ZIP file and then import the model, or you can add a new custom model and add local files. You can also use native Python code in a Jupyter notebook to submit REST API requests to the Model Repository API. For more information, see model-management-resources GitHub repository.

Note:

Models can also be built using Model Studio, SAS Visual Analytics, or SAS Studio and then registered into the common model repository. You can view the relationship between a model in SAS Model Manager and another SAS Viya application. For more information, see "View SAS Lineage".

For more information, see the following documentation:

- *SAS Viya: Getting Started with Machine Learning*
- *Welcome to SAS Model Studio*
- *SAS Visual Analytics: Getting Started with Analytical Models*
- *SAS Procedures by Name and Product*
- *SAS Studio: Task Reference Guide*

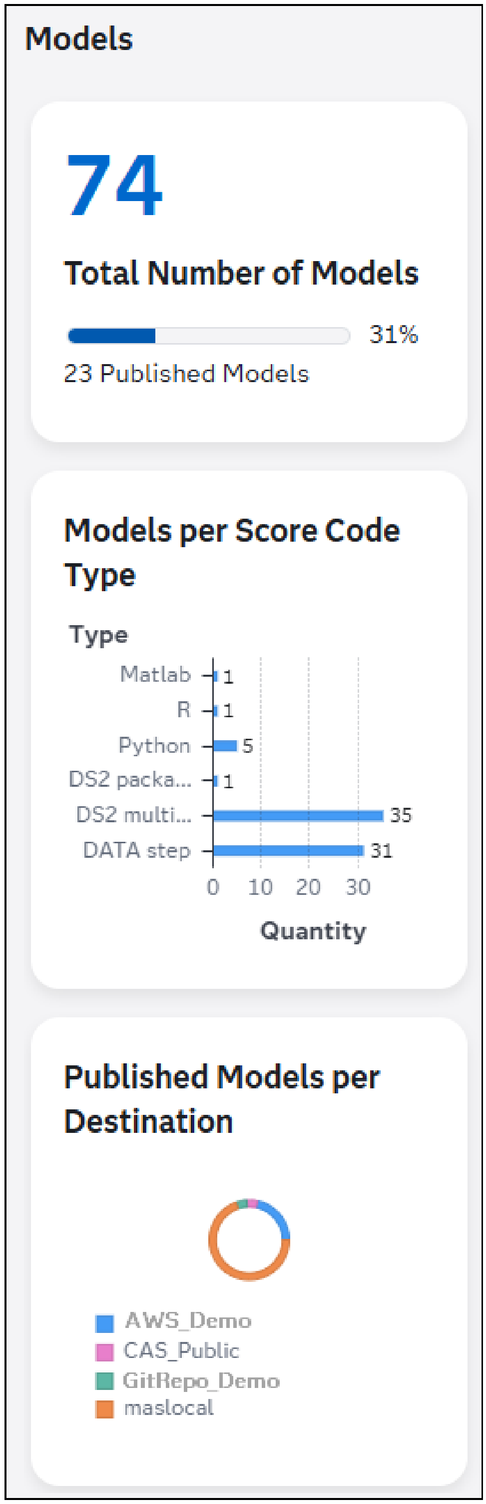
View Dashboard and Filter Models

View Models Dashboard

When you first view the Models category the dashboard appears at the top of the application window. The dashboard contains a graphical representation of the following metrics:

- total number of models
- number of published models
- number of models per score code type
- number of published models per destination

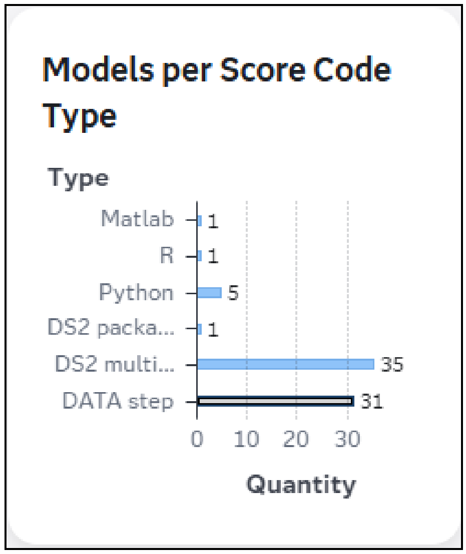
You can click **Hide tiles** or **Show tiles** on the toolbar.

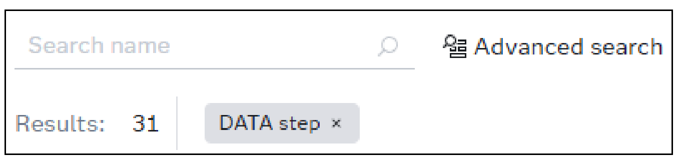


Filter Models List Using the Dashboard Charts

You can click various sections of the different dashboard charts to filter the list of models. You can apply filters for each of the charts to create a compound query.

Here is an example of filtering the list of models to display only models that have a score code type of **DATA step**:





You can also use the Advanced Search for Models window to specify additional information in order to further filter the list. The filters that you applied using the charts are reflected in the Advanced Search for Models window. For more information, see "Advanced Search for Models" on page 99.

TIP To remove an individual filter from a chart, you can click the empty area beside the chart or you can click the × icon within the tokens that are located above the list of models. You can also remove all filters by clicking the ✕ icon, which is located in the right side of the toolbar.

Add a New Custom Model

You can add a new custom model from one or more model files and store it within a folder or within a project version. When you add a new custom model in the Models category view, you can choose a repository and folder to store the new model. When you create a model from the **Models** tab of a project, you can select a project version to store the model.

CAUTION
Do not add or include files that contain executable or dynamic content that is not valid or approved for your environment. Examples of executable content are

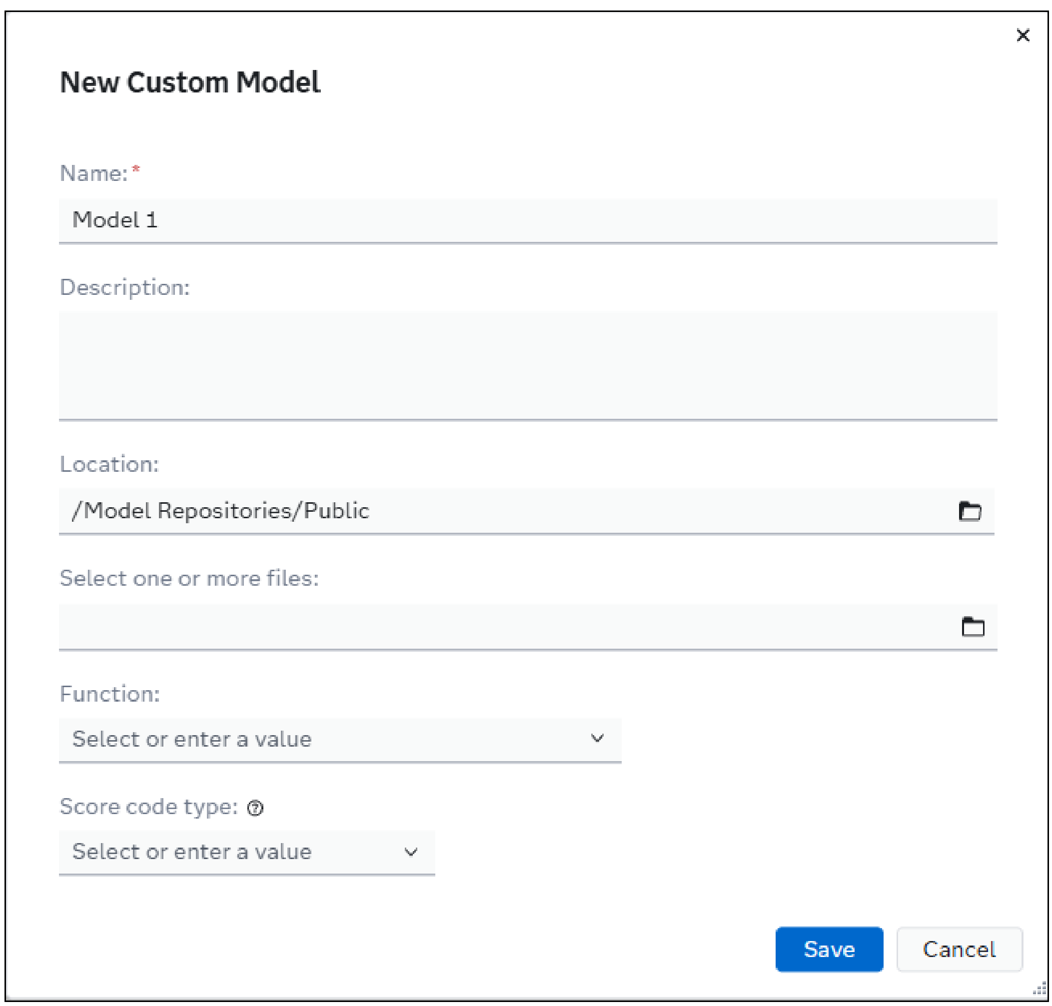
batch (.bat) files or executable (.exe) files. Examples of dynamic content are HTML, Microsoft Office documents, JavaScript, dynamic PDF forms, and other forms of executable content. This type of content, when downloaded, could execute in your local environment and provide a way for malicious content to enter it. The Files service is used when adding data files, model files, and project files. As general guidance, SAS recommends that users be vigilant and cautious with any content that is added using the Files service and accessed within SAS Model Manager.

Add a Model within a Folder

To add a new custom model in a repository or folder:

1 Click to navigate to the Models category view.

2 Click **Add models** and select **New custom model**. The New Model window appears.



3 Enter a name for the new custom model.

Note: Including the following special characters in a model name could cause errors when saving the model or while running a performance definition or job that contains the model:

- a percent sign (%) or an ampersand (&)
- unbalanced braces, brackets, parenthesis, or single or double quotation marks

4 (Optional) Enter a description for the new model.

5 Click 🗀 to select a location to store the new model, and then click **OK**.

Note: In the Choose a Location window, you can create a folder within a repository folder to store projects and models for your organization. Repository folders must be created using the Settings window. For more information, see "About Model Repositories" on page 17.

It is recommended that the `DMRepository` and `VTARepository` repository folders should be reserved for projects and models that are registered from Model Studio. Also, the `VARepository` repository folder should be reserved for models that are registered from SAS Visual Analytics. For more information, see "Register Models" in *SAS Viya: Machine Learning User's Guide*.

6 (Optional) Click 🗀, select one or more files to include within the new model, and then click **Open**.

7 To select a function, enter a value or click ⌄ in the **Function** drop-down list, and select the desired function.

8 To select a score code type, enter a value or click ⌄ in the **Score code type** drop-down list, and select the desired score code type.

9 Click **Save**. The new model object opens.

Add a Custom Model within a Project

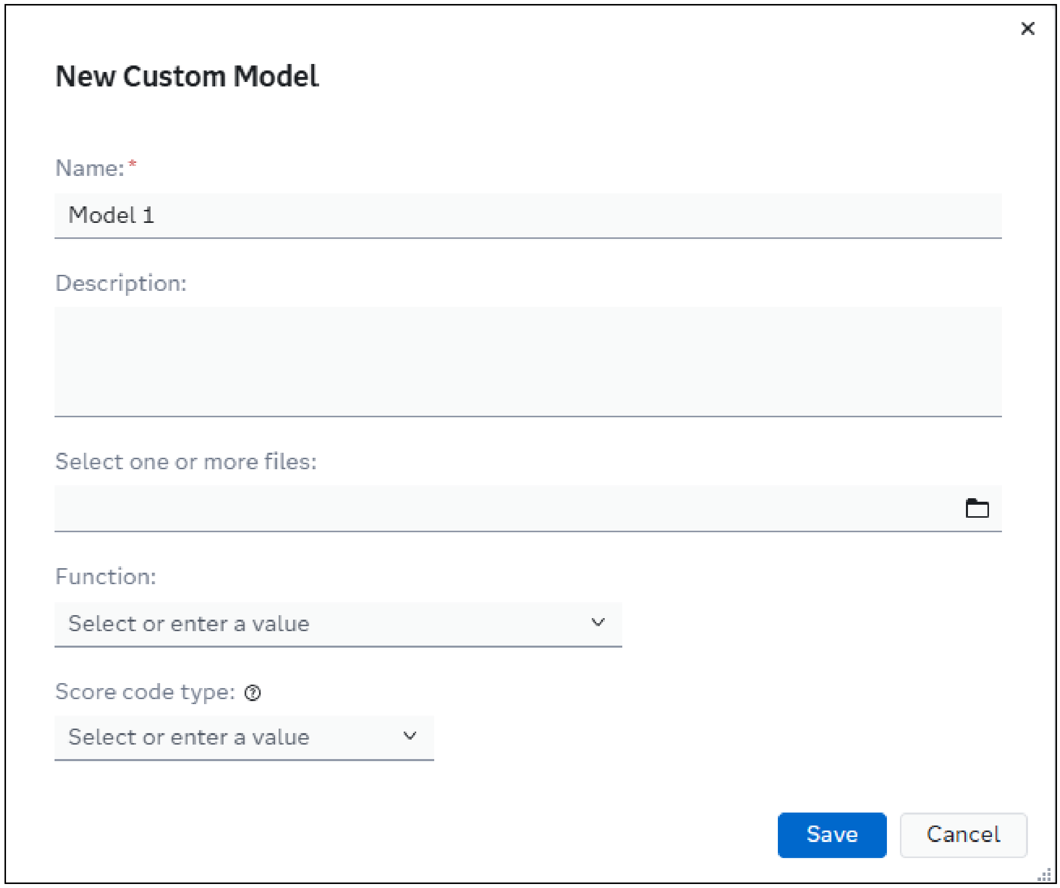
To add a new custom model within a project version:

Note: By default, models are added within the latest project version. You can select a different project version from the **Version** drop-down list.

1 Click ⧉ to navigate to the Projects category view.

2 Open a project.

3 Click **Add models** and select **New custom model**. The New Custom Model window appears.



4 Enter a name for the new model.

Note: Including the following special characters in a model name could cause errors when saving the model or while running a performance definition or job that contains the model:

- a percent sign (%) or an ampersand (&)
- unbalanced braces, brackets, parenthesis, or single or double quotation marks

5 (Optional) Enter a description for the new model.

6 (Optional) Click , select one or more files to include within the new model, and then click **Open**.

7 To select a function, enter a value or click in the **Function** drop-down list, and select the desired function.

8 To select a score code type, enter a value or click in the **Score code type** drop-down list, and select the desired score code type.

9 Click **Save**. The new model object opens.

TIP After you edit the model content, click to return to the project.

Import Models

You can import models into a project or into a folder. However, only specific file types can be used when you import models into a project or folder. Models that are imported into a folder can then later be moved into a folder or project version. If a model already exists in a project version with the same value for the external model ID property, a new model version is created. The model name is updated only when a new name is specified and the external model ID is the same. Otherwise, a model with the same name cannot be imported.

Note: You cannot score, publish, or monitor performance of models that are within a folder. Models must be within a project version in order for you to perform these tasks.

Restrictions

Here are the file types that can be used to import models:

CAUTION
Do not add or include files that contain executable or dynamic content that is not valid or approved for your environment. Examples of executable content are batch (.bat) files or executable (.exe) files. Examples of dynamic content are HTML, Microsoft Office documents, JavaScript, dynamic PDF forms, and other forms of executable content. This type of content, when downloaded, could execute in your local environment and provide a way for malicious content to enter it. The Files service is used when adding data files, model files, and project files. As general guidance, SAS recommends that users be vigilant and cautious with any content that is added using the Files service and accessed within SAS Model Manager.

PMML XML File
: an XML file that contains PMML model information. Predictive Modeling Markup Language (PMML) is an XML-based standard for representing data mining results. You can import PMML models that are produced by other applications. PMML 4.1, 4.2, 4.3, and 4.4 are supported. Some classification and prediction models that are created using PMML are converted to DATA step score code during the model import process. The file extensions can be .xml or .pmml, provided that the file contains valid PMML XML code. When you are importing valid PMML models, the score code type model property is set to DATA step, instead of PMML. PMML models with a score code type of DATA step can be scored and published.

Note: Here are some PMML model file restrictions and limitations:

- The length of the variable names that are included in a PMML model file cannot be greater than 32 characters.
- PMML XML files cannot be edited using the code editor. They are read-only.
- Files with the .pmml file extension cannot be viewed within the editor.

For more information, see “Concepts: PMML Support” on page 220.

SAS analytic store (ASTORE) file
: a file that is created by using the ASTORE procedure, as well as by using some SAS Visual Data Mining and Machine Learning modeling procedures or CAS actions. An ASTORE file is a binary representation of a model that encapsulates in a portable format the scoring logic and metadata such as input and output variables. The file extensions can be .sasast or .astore, provided that the file contains a valid analytic store model.

SAS package (SPK) file
: a compressed container file that contains a mining result and model component files. SPK files can be created using SAS Enterprise Miner or SAS 9.4 modeling procedures and macros.

ZIP file
: an archive file that contains model files. Model files that are associated with a specific model are stored within the ZIP file. The ZIP file can contain model folders at the same level or in a hierarchical folder structure. Each model folder within the ZIP file is imported as a separate model object that contains the contents of the model folder. When you import models from a ZIP file into a project version, the hierarchical folder structure is ignored.

Note: When importing Python and R open-source models, you can place your model files in a ZIP file and then import the model, or you can create a new model and add local files. Also, when creating a ZIP file, you must specify to use Unicode (UTF-8) encoded file names within the ZIP file.

Import Models into a Folder

To import models into a folder:

Note: You can import models only into repository folders within the common model repository (**Model Repositories** folder) or folders within a repository folder.

1. Click to navigate to the Models category view.
2. Click **Add models** and select **Import**. The Import Models window appears.
3. Click to select a location to store the models, and then click **OK**.

Note: In the Choose a Location window, you can create a folder within a repository folder to store projects and models for your organization. Repository folders must be created using the Settings window. For more information, see "About Model Repositories" on page 17.

It is recommended that the `DMRepository` and `VTARepository` repository folders should be reserved for projects and models that are registered from Model Studio. Also, the `VARepository` repository folder should be reserved for models that are registered from SAS Visual Analytics. For more information, see "Register Models" in *SAS Viya: Machine Learning User's Guide*.

4 Click to select a file that contains your model contents. Select only one file at a time in the Open window. The name of the selected file is used as the default model name.

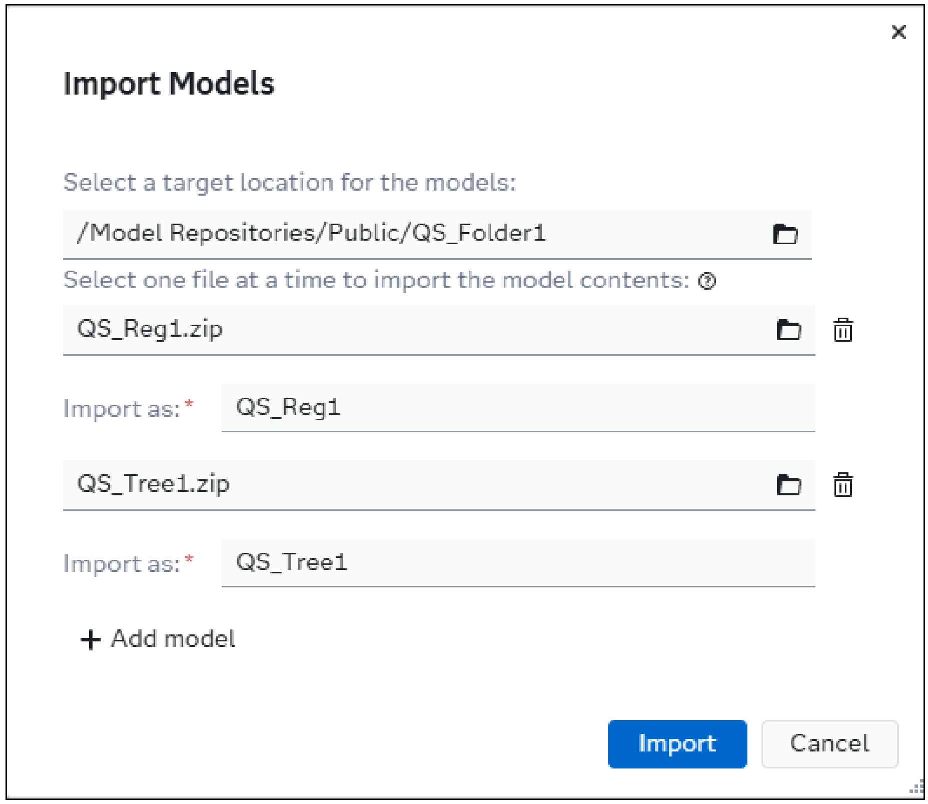
Click **Open**.

Note:

- Including the following special characters in a model name could cause errors when importing the model or while running a performance definition or job that contains the model:
  - a percent sign (%) or an ampersand (&)
  - unbalanced braces, brackets, parenthesis, or single or double quotation marks
- If you previously exported an analytic store model from SAS Model Manager, the analytic store files are included in the ZIP file. When you re-import the analytic store model to the same system or another system, the analytic store files are copied to the ModelStore caslib.

5 Click + **Add model** to add rows so that you can import more models.

6 Repeat steps 4 and 5 until you have selected all of the models that you want to import.



TIP To remove extra lines, click before you click **Import**.

7 Click **Import**.

Import Models into a Project

To import model into a project version:

1 Click to navigate to the Projects category view.

2 Open a project.

3 Click **Add models** and select **Import**. The Import Models window appears.

4 Click to select a file that contains your model contents. Select only one file at a time in the Open window. The name of the selected file is used as the default model name.

Click **Open**.

Note:

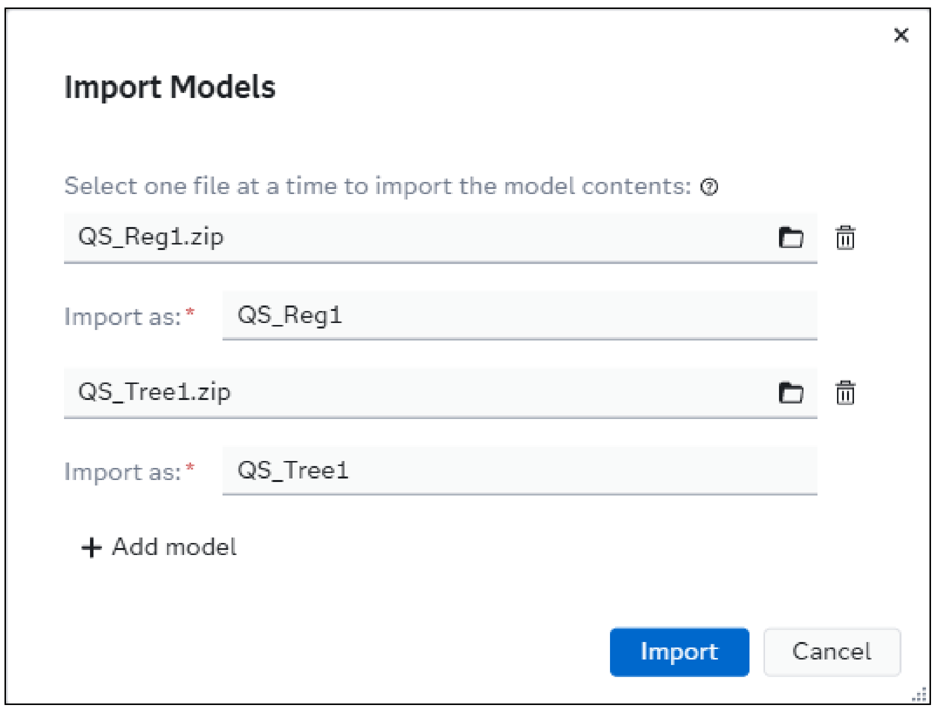
- Including the following special characters in a model name could cause errors when importing the model or while running a performance definition or job that contains the model:

- □ a percent sign (%) or an ampersand (&)
- □ unbalanced braces, brackets, parenthesis, or single or double quotation marks

■ If you previously exported an analytic store model from SAS Model Manager, the analytic store files are included in the ZIP file. When you re-import the analytic store model to the same system or another system, the analytic store files are copied to the ModelStore caslib.

5 Click + **Add model** to add rows so that you can import more models.

6 Repeat steps 4 and 5 until you have selected all of the models that you want to import.



TIP To remove extra lines, click 🗑 before you click **Import**.

7 Click **Import**.

Note: If a model already exists in a project version with the same value for the external model ID property, a new model version is created. The model name is updated only when a new name is specified. However, if the external model ID property values do not match for the model with the same name, the model cannot be imported. For more information about the external model ID property and other model properties, see Table 5.2 on page 85.

See Also

- %MM_IMPORT_MODEL Macro
- %MM_IMPORT_ASTORE_MODEL Macro
- "Register a Model" in *SAS Studio: Task Reference Guide*
- REGISTERMODEL Procedure in *SAS Viya: Machine Learning Procedures*
- Register SAS Model Step
- Register Python Model Step

Export a Model

You can export one model at a time from the Models category view or from the **Models** tab of a project.

Note: When you export an analytic store model, the analytic store files are included in the ZIP file. If you re-import the analytic store model to the same system or another system, the analytic store files are copied to the ModelStore caslib.

To export a model:

1. Select a model from the list.
2. Click ⋮ and select **Export as ZIP**.

   The contents of the model in a ZIP file is downloaded to your local machine.

Move or Copy a Model

You can move a model from a folder to another folder or project version using the Models category view. Only SAS Administrators and users who have Delete permission for the source location where the model resides and Write permission for the target location can move a model. By default, all other users can copy a model from a folder or another project from the **Models** tab of a project only.

For more information, see "Managing Permissions" in *SAS Model Manager: Administrator's Guide*.

Move a Model

To move a model:

Note: Only models located within a folder can be moved.

1. Click to navigate to the Models category view.
2. Select a model from the list.
3. Click ⋮ and select **Move**. The Choose a Location window appears.
4. Navigate to the folder or project version that you want to move the model to.
5. Click **OK**.

Copy a Model

To copy a model from a folder or another project:

1. Open a project and click the **Models** tab.
2. Click **Add models** and select **Copy from** from the drop-down list. The Choose a Model window appears.
3. Click > to navigate to a folder or a project version.
4. Click > for the model folder and select the model object. The model object is indicated by the icon .
5. Click **OK**.

Note: Only the latest version of the source model is copied in to the project as a new model object. The initial version for the model is 1.0.

Delete a Model

You can delete one or more models at a time from the Models category view or from the **Models** tab of a project.

Note: The project champion model and open model objects cannot be deleted.

1 Select one or more models.
2 Click ⋮ and select **Delete**.
3 In the confirmation message, click **Delete**.

Rename a Model

You can rename one model at a time from the Models category view or from the **Models** tab of a project.

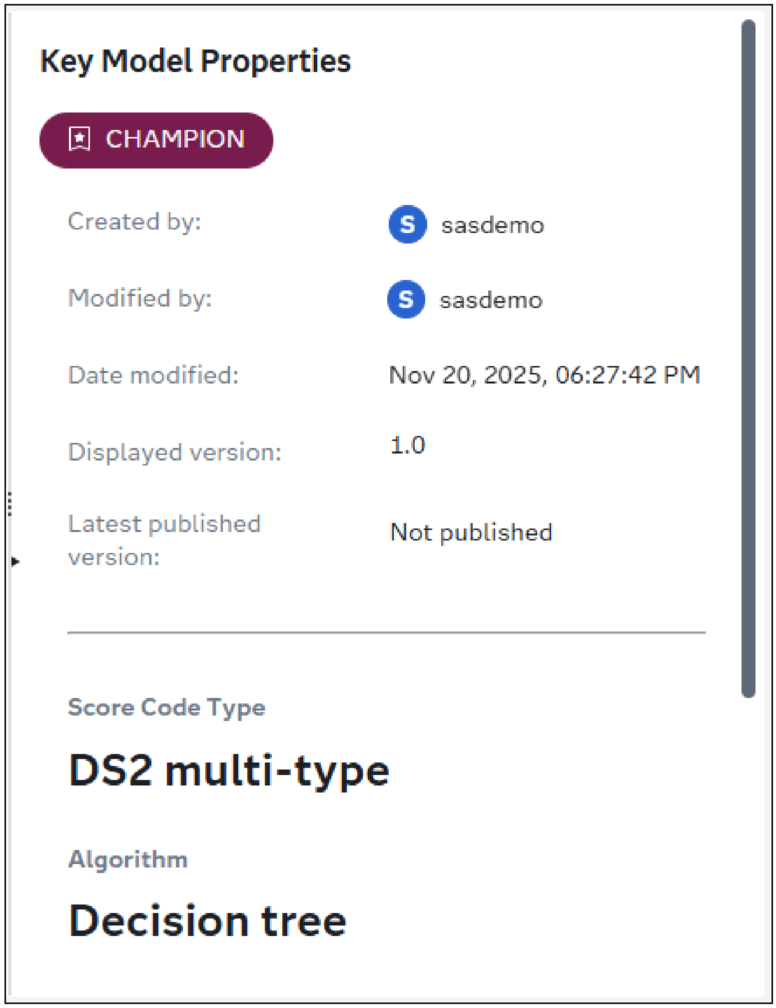
Note: Open objects cannot be renamed.

To rename a model:

1 Select a model, click ⋮, and select **Rename**.
2 Enter a new name for the model.
3 Click **Rename**.

Manage Model Content and Versions

When you open a model, you can manage model files, add model input and output variables, modify the model properties, and add or view model versions. You can also publish a model, if it is located within a project version. Models within a folder cannot be published. You can open a model from the Models category view and from the **Models** tab of a project. The primary information about the displayed version of a model is shown in the **Key Model Properties** pane.



Generating Model Card Content

The first release of the model card feature supports classification and prediction models from various sources. As users and teams develop and manage their models within SAS Viya, the content on the **Model Card** tab of a model object is automatically populated. The model card includes information about the training data, model performance at the time of training, and model performance over time.

Prerequisites

Here are the minimum prerequisites for viewing content on the **Model Card** tab for a model:

- The model must have a function of either classification or prediction.
- The score code type must be DATA step, DS2 embedded process, DS2 multi-type, DS2 package, analytic store, or Python.

- The training table associated with the model must be in a global caslib. Personal caslibs such as CASUSER should not be used.
- Classification model must contain the dmcas_fitstat.json and dmcas_misc.json files. Prediction models must contain the dmcas_fitstat.json and dmcas_lift.json.

For more information, see "Generate Model Files" on page 76 and "Modifying Model Properties" on page 84.

Model Card Examples

You can reference the following resources for examples of using the Model Card feature:

- Creating a Model Nutrition Label: Model Cards for SAS Model Studio Models article in SAS Communities
- Creating a Model Nutrition Label: Model Cards for Python Models article in SAS Communities
- Creating a Complete Model Card for a Python Model using python-sasctl Jupyter notebook on GitHub

Overview Section

The **Overview** section provides an at-a-glance review of the model. It provides visuals of model health that are easy to understand and are supported by other sections of the model card. This section shows the model performance during training and over time. It also shows influential variables, variable privacy classifications, and the completeness of the model card. Most of the data in the **Overview** section is populated automatically for models that are registered from Model Studio, but there are areas that require attention after a model is registered. SAS models and Python open-source models can also be built using macros in SAS Studio or python-sasctl. Models that are imported into the common model repository are populated based on the metrics files that are included within the model.

You can review and update the default thresholds for training metrics in the **Model Evaluation** section of the project **Properties** tab. It is recommended that you update these thresholds to ensure that they are appropriate for your use case, as thresholds might differ from one use case to the next. For the performance monitoring metric results to be available, you must define key performance indicator (KPI) alert rules, and run performance monitoring for the specific model. To ensure that the variable privacy classifications are available, you must analyze the data in the **Data Summary** section of the model card. If you want the "No" to change to a "Yes" in the **Limitations Documented** section, then complete the limitations in the **Model Usage** section of the model card. Finally, to see fairness metrics, you must specify one or more variables to assess for bias before training the model in Model Studio, SAS Studio, or using python-sasctl.

For more information, see:

- Analyze Assets
- Reviewing and Managing Semantic Types
- Set Model Evaluation Properties
- Add KPI Alert Rules
- Assess Variables for Bias
- Monitoring Performance on page 117

Model Usage Section

The **Model Usage** section describes the intended use, expected benefits, out-of-scope use cases, and limitations of the model. The content in this section must be specified on the **Properties** tab at either the project or model level. The values of the model usage model-level properties are inherited from the project-level properties. However, model-specific information can be specified for each property value at the model level.

For more information, see "Specify Model Usage Properties" on page 94.

Data Summary Section

The **Data Summary** section provides information about the training data from SAS Information Catalog. When registering your model from Model Studio, the Default training table model property should already be populated, but you might be prompted to run an analysis of the training data in the **Data Summary** section. Click **Analyze data** to run an analysis of the training data.

> **IMPORTANT** The training table associated with the model must be in a global caslib. Personal caslibs such as CASUSER should not be used.

If you want to view the information privacy for the variables that are marked to be assessed for bias, the information privacy for those variables must be set for the training data asset in SAS Information Catalog before you analyze the data. Only SAS administrators can set the classifications and information privacy in SAS Information Catalog. Other users can only view the classifications and information privacy.

For more information, see "Analyze Assets" in *SAS Information Catalog: User's Guide*.

Model Summary Section

The **Model Summary** section examines the model's performance at the time of training, which corresponds to the training donut charts in the **Overview** section. The information in this section is populated automatically when a model is registered from SAS Model Studio. SAS models and Python open-source models can also be built using macros in SAS Studio or python-sasctl, and then imported into the common model repository. The **Model Summary** section includes information about the model target, algorithm, development tool and version, various accuracy measures across training, testing, and validation partitions, generalizability, and variable importance. If you have marked a variable to be assessed for bias, those metrics appear in this section if the fairness metrics were generated when the model was built or trained. Overall, this section provides information about how well the model was performing during training, which can provide a baseline for monitoring model performance over time.

Model Audit Section

The **Model Audit** section shows the performance of the model over time. It provides more detailed information, then the performance monitoring donut charts in the **Overview** section of the model card.

The **Model Audit** section relies on two capabilities of SAS Model Manager: performance monitoring and key performance indicator (KPI) alert rules. Performance monitoring reviews model performance over time in batch at user-defined time points. To generate performance monitoring results, you need data with the actual or ground-truth values, and SAS Model Manager graphs model performance over time. Without the ground-truth values, you get only metrics on data drift.

For more information, see:

- Add KPI Alert Rules
- Monitoring Performance on page 117

Managing Model Files

On the **Files** tab of a model, you can add, delete, and download files, as well as assign roles to model files. The model files are grouped into a few primary categories based on their file role and file name. You can add most file types to a model. You can also edit supported file types in the code editor. File types that are not supported by the editor or that are read-only can be downloaded, and they can be viewed by another application.

CAUTION

Do not add or include files that contain executable or dynamic content that is not valid or approved for your environment. Examples of executable content are batch (.bat) files or executable (.exe) files. Examples of dynamic content are HTML, Microsoft Office documents, JavaScript, dynamic PDF forms, and other forms of executable content. This type of content, when downloaded, could execute in your local environment and provide a way for malicious content to enter it. The Files service is used when adding data files, model files, and project files. As general guidance, SAS recommends that users be vigilant and cautious with any content that is added using the Files service and accessed within SAS Model Manager.

Add Model Files

1. Click +. The **Add Model Files** window appears.
2. Click , select one or more files to add to the model, and then click **Open**.
3. Click **Add**.

Note: Starting with the 2025.08 release, you can add standard Markdown files to your model. You can also edit the file content and preview it within SAS Model Manager. GitHub-flavored Markdown formats such as tables, strikethroughs, and task lists are supported starting with the 2025.09 release. Embedded images are not supported. For more information, see Markdown Guide - Basic Syntax and GitHub Flavored Markdown Spec.

Assign Model File Roles

Place your pointer over ⓘ next to a model file to view the file properties. Roles might need to be assigned for your model files. To assign a role, select the file and click . Enter a value for the file role or select the file role from the drop-down menu. Click **Save**.

Some roles such as **Score code** and **Score resource** are automatically assigned when you import or create new models based on their file names. The files are also grouped into a few primary categories based on their file role and file name.

Note: The file type for the file that is assigned the file role of **Score code** must be associated with the value that is specified for the score code type model property. For example, if the file name is myScoreCode.py, then the value for the score code type should be Python. This ensures that the correct score code file is used and that valid code is generated for a scoring job.

Delete Model Files

Select the file and click 🗑. In the confirmation message, click **Delete**.

Download Model Files

Select the file and click ⤓. The model file is downloaded to your local machine.

Generate Model Files

This is a list of the model files that are generated when importing or registering a model into the SAS Model Manager common model repository, or when exporting a model from the common model repository. You can also generate some of these files by using SAS modeling procedures or CAS actions.

Table 5.1 *List of Model Files*

File Type	File Name	Purpose	Required or Optional	Notes
File metadata	fileMetadata.json	Contains the list of models files that are associated with specific file roles.	Optional	This file is generated when a model is exported from SAS Model Manager. It is also recommended to manually create this file and include it when importing an open-source model.
Analytic store metadata	AstoreMetadata.json	Contains the metadata about the analytic store when an ASTORE file is generated for a model.	Required	This file is created when building a model.
Score code and scoring resources	See notes for examples of file names.	Enables the scoring of a model within SAS Model Manager.	Required	The content of the score code varies depending on the type of model. File name examples: ■ score.sas

File Type	File Name	Purpose	Required or Optional	Notes
				■ dmcas_ epscorecode.sas ■ dmcas_ packagescorecode.sas ■ dmcas_ packagescorecode.sas ■ dmcas_ scorecode.sas ■ hmeq_classtree_ r.rds ■ hmeq_classtree_r_ score.r ■ dmcas_ scorecode.sas ■ hmeq_ logistic.pickle ■ hmeq_logistic_ score.py
Fit statistics	dmcas_fitstat.json	Contains the fit statistics from training the model using the training data.	Required for **Model Card** tab content and model comparison	This file is generated using the CAS assess action. It is included when registering a model from Model Studio. It can also be generated using python-sasctl.
Lift statistics	dmcas_lift.json	Contains the lift statistics from training the model using the training data.	Required for **Model Card** tab content and model comparison	This file is generated using the CAS assess action. It is included when registering a model from Model Studio. It can also be generated using python-sasctl.
Misclassification statistics	dmcas_misc.json	Contains the misclassification statistics from training the model using the training data.	Required for **Model Card** tab content	This file is generated when building a model in Model Studio. It can also be generated using python-sasctl.
Misclassification statistics table	dmcas_ miscTable.json	Contains misclassification	Optional	This file is generated when building a

File Type	File Name	Purpose	Required or Optional	Notes
		statistics from training, testing, and validation data at different cut-off points.		model in Model Studio.
Model information from Model Studio	dmcas_ modelInfo.json	Contains information such as score type, algorithm, scoring and registration status, and analytic store (ASTORE) name.	Optional	This file is generated when building a model in Model Studio.
Package metadata from Model Studio	dmcas_ packagemetadata.json	Contains information about model inputs and outputs.	Optional	This file is generated when building a model in Model Studio.
Properties from Model Studio	dmcas_ properties.json	Contains model hyper-parameters and other information relevant to building the model.	Optional	This file is generated when building a model in Model Studio.
Relative importance	dmcas_ relativeimportance.json	Contains the variable relative importance	Required for **Model Card** tab content	This file is generated when building a model in Model Studio and the Variable importance is selected under the post-training properties for global interpretability. It can also be generated using python-sasctl.
ROC statistics	dmcas_roc.json	Contains the ROC statistics from training the model using the training data.	Required for model comparison	This file is generated using the CAS assess action. It is included when registering a model from Model Studio.

File Type	File Name	Purpose	Required or Optional	Notes
Input variables	dmcas_ scoreinputs.json or inputVar.json	Contains the input variables for the model.	Required	The dmcas_ scoreinputs.json file is generated by Model Studio and the inputVar.json is created by SAS Model Manager.
Output variables	dmcas_ scoreoutputs.json or outputVar.json	Contains the output variables for the model.	Required	The dmcas_ scoreoutputs.json file is generated by Model Studio and the outputVar.json is created by SAS Model Manager.
ASTORE key	dmcas_ sha1key.json	Contains the analytic store (ASTORE) key and name.	Optional	This file is generated when building a model in Model Studio and registering it into SAS Model Manager.
Target variable information	dmcas_ targetInfo.json	Contains information about the target variable such as variable name, frequency, and so on.	Required for **Model Card** tab content	This file is generated when building a model in Model Studio and registering it into SAS Model Manager.
Model properties	ModelProperties.js on	Contains the general properties for a model.	Required	This file is generated when registering a model from Model Studio or SAS Visual Analytics into SAS Model Manager. It is also generated when exporting a model from SAS Model Manager. python-sasctl contains a function to generate this file as well.
Requirements	requirements.json	Contains the requirements for	Optional	This file is optional, but it recommended

File Type	File Name	Purpose	Required or Optional	Notes
		the Python model, such as the version of the packages that must be installed.		for a Python model in order to be able to score and validate the model.
Assess for bias metadata	assessforbiasmetadata.json	Indicates which variables to assess for bias.	Optional	This file is generated when registering a model from Model Studio.
Group metrics	groupMetrics.json	Contains the fairness metrics from training the model.	Required for **Model Card** tab content	This file is generated using the CAS assessBias action. It is included when registering a model from Model Studio. This file can also be generated using python-sasctl.
Main model properties and information	master.json	Contains a compilation of information about the model including algorithm, inputs, outputs, location, and so on.	Optional	This file is generated when registering a model from Model Studio.
Maximum differences	maxDifferences.json	Contains the maximum difference values for variables that are included as part of the Fairness metrics.	Recommended for **Model Card** tab content	This file is generated using the CAS assessBias action. It is included when registering a model from Model Studio. It can also be generated using python-sasctl.
Parameters	paramests.json	Contains the parameters for regression models.	Optional	This file is generated when registering a model from Model Studio.
Partial dependence	partialdependence1.json	Contains the partial	Optional	This file is generated when registering a model from Model Studio.

File Type	File Name	Purpose	Required or Optional	Notes
		dependence for the model.		
Training code	traincode.sas	Contains the code that was used to train the model.	Optional	This file is generated when registering a model from Model Studio.
Variable importance	varimportance.json	Contains the variable importance for tree-based models.	Optional	This file is generated when registering a model from Model Studio.

Managing Variables

Input variables and output variables can be added to both project and model objects on the **Variables** tab of an object. The same variable name cannot be used for both an input and output variable. You can also map model output variables to project output variables for models with a score code type of DATA step, SAS program, or PMML.

Note: When you add, edit, or delete model variables, you might receive a warning message indicating that the model role will be cleared. This happens when the model is set as the project champion or as a challenger.

Variable Naming Restrictions

Note: Do not use any of these operators or keywords as variable names: AND, OR, IN, NOT, LIKE, TRUE, or FALSE. Do not use _N_ or any DS2 reserved word as a variable name. For more information, see "Reserved Words in the DS2 Language" in *SAS DS2 Programmer's Guide*.

Add Variables from a Data Table

1 Click the **Variables** tab.

2 Click **Add variable** and select **Data table**. The Choose Variables window appears.

3 Click to select a table. The Choose Data window appears.

> **TIP** If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see "Overview of SAS Data Explorer" in *SAS Data Explorer: User's Guide*.

4 Select the data table that you want to import the variables from and click **OK**.

5 In the Choose Variables window, select the variables that you want to add and click to add the variables to the input variables or output variables list. You can also click to add all of the variables from the available items list.

6 Click **OK**.

7 Click .

Add Custom Variables

1 Click the **Variables** tab.

2 Click **Add variable** and select **Custom variable**. The Add Custom Variables window appears.

3 Enter a name for the variable.

4 Select a data type.

5 Select a variable type.

> **Note:** If you select "Input" or "(none)" as the variable type, you can select the **Assess variable for bias** option. To mark a variable to be assessed for bias, it must also have a measurement value of Binary, Nominal, or Ordinal. When a variable is marked to be assessed for bias, it is used to calculate fairness and bias metrics for a model using the training data. If no variables are marked for bias at the time the model was trained, the fairness and bias metrics are not available.

6 Expand the **Optional** section to specify a length, measurement, and description for the variable.

7 Click **Add**.

8 Repeat steps 3 through 6 for each variable that you want to add.

9 Click **OK**.

Assess Variables for Bias

When a variable is marked to be assessed for bias, it is used to calculate fairness and bias metrics for a model when running performance monitoring in a project that has a model function of classification or prediction. If no variables are marked for bias at the time the model was trained, the fairness assessment using the training data is not available on the model card for a model.

Note: If a model is set as champion, and any of the variables are marked to be assessed for bias, the variable is also marked to be assessed for bias at the project level.

To mark a variable to be assessed for bias:

1 Click the **Variables** tab.

2 Select a variable from the list and click **Assess variable for bias**.

   Note: When marking a variable to be assessed for bias, it must have a measurement value of Binary, Nominal, or Ordinal. The **Assess variable for bias** button is enabled for input variables or for variables that are not specified as an input or output variable. Output variables cannot be assessed for bias.

TIP You can also mark a variable to be assessed for bias in the Edit Variable window or when adding custom variables..

Edit Variables

1 Click the **Variables** tab.

2 Click on the name of the variable that you want to edit. The Edit Variable window appears.

3 Edit the properties as needed and click **OK**.

4 Click .

Delete Variables

1 Click the **Variables** tab.

2 Select the check box for the variables that you want to delete, click ⋮, and then select **Delete**.

3 Click .

Map Output Variables

To set a model as the project champion or as a challenger, run performance, or publish a model from a project, you must map the model output variables to the project output variables. If you do not map the output variables after importing a model, you are prompted to map them when setting a model as the project champion or as a challenger.

Note: You can modify the model output variable mappings only for a model with a score code type of DATA step, SAS program, or PMML. The names of model output variables must also match the names of the project output variables for other types of models. Otherwise, you cannot run performance for a model or publish a model.

To map output variables:

1 Click the **Variables** tab of a model.

2 Click ⋮ and select **Map output variables**. The Map Output Variables window appears.

3 Select the model output variables to map with each of the project output variables.

4 Click **OK**.

5 Click .

Modifying Model Properties

Model properties contain the model metadata. Model metadata includes information such as the name of the model, the type of model, the modeler, the model identifier, the name and path of the repository, and of the tables and variables that are used by model processes. The model properties are organized into three types: General, Relationships, and User-Defined.

IMPORTANT The **Relationships** page is available only when SAS Model Risk Management is installed and you have access to it.

Set Model General Properties

General Properties contains both system-defined properties that you cannot modify, and model specific properties that can be modified, such as the description of the project.

To set the model general properties, click the **Properties** tab, modify the property values, and then click .

Table 5.2 *List of General Properties*

Property	Description
Name	Specifies the name of the model. It can be renamed only from the Models category view or the **Models** tab of a project.
Description	Specifies the description of the model.
Location	Specifies the location of the model in the common model repository.
Project name	Specifies the name of the project that contains the model.
Project version	Specifies the project version that contains the model.
Function	Specifies the type of output that your model generates. For more information, see Table 5.3 on page 87.
Score code type	Specifies the type of score code that your model uses. A value must be specified in order for you to be able to publish a model, run a test, or monitor performance for a model. You can select a value from the list or enter your own value. User-defined values are not added to the list. Instead, they are stored within the model properties. **Note:** A model with the score code type of DS2 multi-type can contain code files for a DS2 embedded process, a DS2 package, and one or more analytic stores. Depending on score code type, you can score, monitor, or publish a model. For more information, see “High-Level Model Support Matrix for Primary Functions” on page 7.
Train table	Specifies the **Train table** that is used to validate scoring functions or scoring model files when a user publishes the associated project champion model or challenger models to a database. This property is optional.

Property	Description
Train code type	Specifies the type of train code that your model uses. This property is for informational purposes only. You can select a value from the list or enter your own value. User-defined values are not added to the list. Instead, they are stored within the model properties.
Algorithm	Specifies the computational algorithm that is used for the selected model.
Input variable type	Specifies the type of input variables and whether the variables come from the trainInputVar.json model file or from the inputVar.json model file. Models that are registered from SAS Studio might contain the trainInputVar.json model file. Valid values are `score` and `train`. **Note:** If both files are included with the registered model, the property value is set to `score`.
Target variable	Specifies the name of the target variable.
Target event value	Specifies the target variable value that defines the desired target variable event.
Target level	Specifies the target level of binary, nominal, ordinal, or interval.
Output category variable	Specifies the output variable name, when the **Model function** property is set to **Text categories**.
Output event probability variable	Specifies the output event probability variable name, when the **Model function** property is set to **Classification**, **Analytical**, **Computer vision - Image classification**, or **Transformation**.
Output prediction variable	The output prediction variable name, when the **Model function** property is set to **Prediction**, **Analytical**, **Computer vision - Image segmentation**, **Computer vision - Keypoints detection**, **Computer vision - Object detection**, **Single-series forecasting**, or **Transformation**.
Output segmentation variable	The output segmentation variable name, when the **Model function** property is set to **Clustering**, **Analytical**, **Segmentation**, or **Transformation**.
Output sentiment variable	Specifies the output variable name, when the **Model function** property is set to **Text sentiment**.
Output topic variable	Specifies the output variable name, when the **Model function** property is set to **Text topics**.

Property	Description
Modeler	Specifies the user ID for the user that built the model.
Tool	Specifies the tool that was used to build the model. An example is Model Studio.
Tool version	Specifies the version number of the tool that is specified in the **Tool** property.
UUID	Specifies the universally unique identifier for a model object.
External model ID	Specifies the model ID for a model that was registered from an external application, such as Model Studio. **TIP** You can view the relationship between a model in SAS Model Manager and another SAS Viya application. For more information, see "View SAS Lineage".
External URL	Specifies a user-defined URL to a model object in another application or to documentation related to the model.

Table 5.3 *Types of Model Functions*

Model Function (English Display Name)	Key Value	Description
Analytical	analytical	Function for any analytical model that is not associated with another specific model function.
Classification	classification	Function for models that have target variables that contain binary, categorical, or ordinal values.
Clustering	cluster	Function for segmentation or clustering models.
Computer vision - Image classification	image classification	Function for computer vision models that are used for image classification. See: Computer Vision Model Properties
Computer vision - Keywords detection	keypoints detection	Function for computer vision models that are used for keywords detection in neural networks. See: Computer Vision Model Properties

Model Function (English Display Name)	Key Value	Description
Computer vision - Multi-task learning	multi-task learning	Function for computer vision models that are used for multi-task learning. See: Computer Vision Model Properties **Note:** Support added with the 2025.08 release.
Computer vision - Object detection	object detection	Function for computer vision models that are used for object detection in a classification network. See: Computer Vision Model Properties
Computer vision - Image segmentation	image segmentation	Function for computer vision models that are used for image segmentation. See: Computer Vision Model Properties
Prediction	prediction	Function for models that have interval targets with continuous values.
Segmentation	segmentation	Function for segmentation models.
Single-series forecasting	single-series forecasting	Function for models used to forecast future data based on past data for a single time series.
Text categories	text categorization	Function for SAS Visual Text Analytics categorization models.
Text concepts	text concepts	Function for SAS Visual Text Analytics concepts models.
Text sentiment	text sentiment	Function for SAS Visual Text Analytics sentiment models.
Text topics	text topics	Function for SAS Visual Text Analytics topics models.
Transformation	transformation	Function for models that are used to determine mathematical functions. These functions can be used to stabilize variances, remove nonlinearity, and correct non-normality in variables to improve the fit of your model.
Text generation	text generation	Function for text generation models, which are designed to generate coherent and

Model Function (English Display Name)	Key Value	Description
		contextually relevant text based on input prompts. These models can be categorized into several types based on their architecture, training objectives, and use cases. See: Large Language Model (LLM) Properties **Note:** Support added with the 2025.08 release.
Prompt template	prompt template	Function for prompt template models, which are designed to help translate user input and parameters into instructions for a language model. This can be used to guide a model's response, helping it understand the context and generate relevant and coherent language-based output. See: Large Language Model (LLM) Properties **Note:** Support added with the 2025.08 release.

Table 5.4 Computer Vision Model Properties

Property	Description
Image height	Specifies the height of the image.
Image width	Specifies the width of the image.
Image channels	Specifies the channels of the image.
ESP preferred scoring mechanism	Specifies the preferred scoring mechanism that you want to use when scoring computer vision models with SAS Event Stream Processing.

Table 5.5 Large Language Model (LLM) Properties

Property	Description
Language model type	The type of language model that is used by an AI system to understand and generate human-like text, ranging from

Property	Description
	simple statistical N-grams to complex Transformer-based AI like GPT-4
Language model provider	The provider of the language model.
Language model version	The version of the language model.
Language model endpoint	The endpoint of the language model.
Cost per 1M tokens	The cost of processing for a language model.

Table 5.6 *Types of Algorithms*

Algorithms (English Display Name)	Key Value
Bayesian network	bnet
Binning	binning
Boolean rules	boolrule
Decision tree	tree
Ensemble	ensemble
Factorization machine	factmac
Forest	forest
Generalized additive model	gam
Generalized linear model	glm
Gradient boosting	gradboost
Imputation	impute
Linear regression	linearreg
Logistic regression	logisticreg
Moving window principal component analysis	mwpca

Algorithms (English Display Name)	Key Value
Neural networks	neural
Nonparametric logistic regression	nplr
Partial least squares regression	plsreg
Principal component analysis	pca
Quantile regression	quantreg
Replacement	replace
Robust principal component analysis	rpca
Rule-based	Rule-based
SVD	rotated svd
Support vector data description	svdd
Support vector machine	svm
Text topic (discovery)	texttopic
Other	other

Table 5.7 *Types of Score Code and Training Code*

Code Type (English Display Name)	Key Value	Valid Score Code Type	Valid Training Code Type
Analytic store	analyticStore	Yes	No
DATA step	dataStep	Yes	No
DS2 embedded process	ds2EmbeddedProcess	Yes	No
DS2 multi-type	ds2MultiType	Yes	No
DS2 package	ds2Package	Yes	No
PMML	pmml	Yes	No
Python	python	Yes	Yes

Code Type (English Display Name)	Key Value	Valid Score Code Type	Valid Training Code Type
R	r	Yes	Yes
SAS program	sasProgram	Yes	Yes
C	c	Yes	Yes
CAS language	cas	Yes	Yes
Java	java	Yes	Yes
Lua	lua	Yes	Yes
Matlab	matlab	Yes	Yes
ONNX	onnx	Yes	No

Establish Model Relationships

You can use model relationships to establish an association with objects in other product solutions. When the **Govern model risk** property setting is enabled on the **Relationships** page of a model object, SAS Model Risk Management users can then establish a relationship with the model by linking to it.

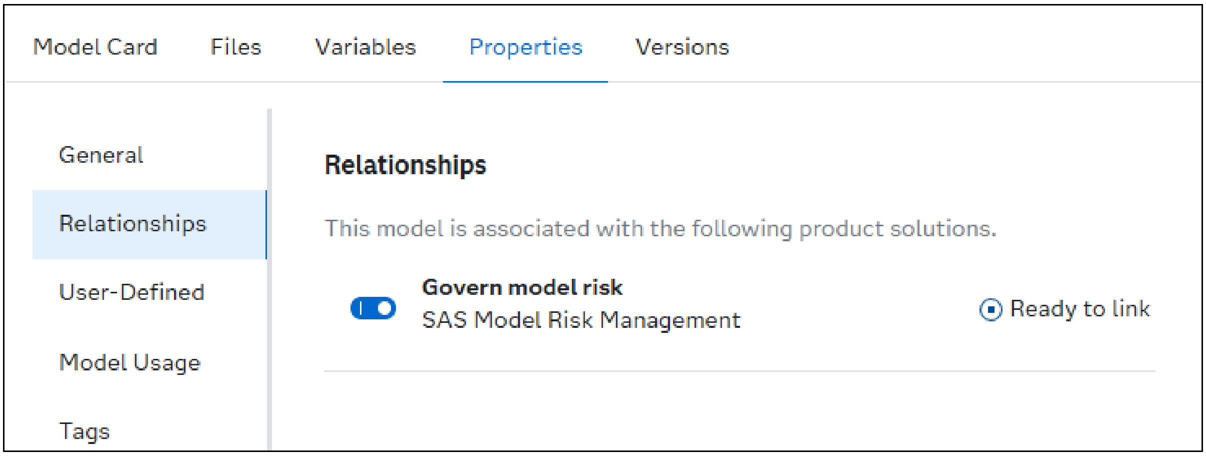
> **IMPORTANT** The **Relationships** page is available only if your company has licensed SAS Model Risk Management and you have the appropriate permissions to it.

Here is an example user scenario:

1 Open a model.

2 On the **Properties** tab, select **Relationships**.

3 Enable the **Govern model risk** toggle button and click . The model is now ready to be linked to from a model object in SAS Model Risk Management.



4 Click and select **Model Risk Management** to switch to SAS Model Risk Management.

5 Click and open the model that you want to establish a relationship with.

> **TIP** For information about the SAS Model Risk Management model object properties and content, see *SAS Model Risk Management: User's Guide*.

6 Click the **Repository Details** tab of the model object.

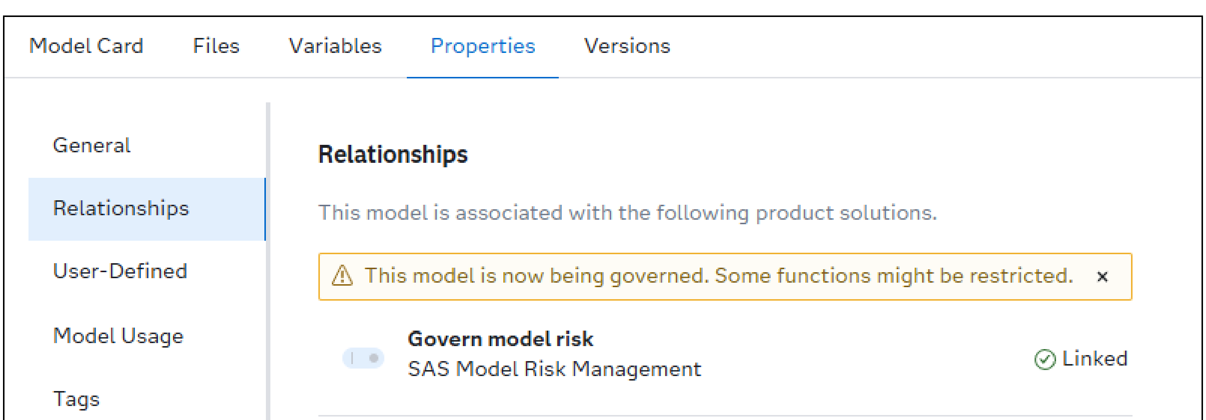
7 Click **Link** and select a source model from the common model repository. Only models that are enabled to govern model risk in SAS Model Manager and that are not already linked are available in the models list.

8 Click **Save**.

9 Click and select **Manage Models** to return to SAS Model Manager.

10 In the Models category view, click the name of the model that you previously linked to from the SAS Model Risk Management model object. The model object opens

11 Return to the **Relationships** page of the model **Properties** tab. The relationship status now indicates that the model is linked.



12 Click **Close**.

13 In the list of models, click to the right of a model to view the model risk card for the related object that the model is associated with.

The **Model Risk Card for Related Object** window appears.

14 Click **Close**.

Add User-Defined Properties

You can add your own project or model properties. The property-value pair is metadata for the project or model, both of which can be searched.

To add user-defined properties:

1 On the **Properties** tab, select **User-Defined**.

2 Click **Add Property**. The Add Property window appears.

> **TIP** If user-defined properties already exist, click + above the table.

a Enter a name for the property.

b Select a data type for the property.

c Enter a value for the property.

d Click **Add** to add the property to the list.

e Repeat steps a through d for each property that you want to add.

3 Click .

To edit a property:

1 Click on a property name within the table.

2 Edit the name, the data type, or value of the property.

3 Click **OK**.

4 Click .

To delete properties, select one or more properties in the table, and then click .

Specify Model Usage Properties

You can specify the values of the model usage properties at the project level or model level. The values of the model-level properties are inherited from the project-level properties, if the model is within a project. However, model-specific information can be specified for each property value at the model level. The information specified in this section is surfaced on the **Model Card** tab of a model object.

- Model purpose
- Intended use
- Expected benefit
- Out-of-scope use cases
- Limitations
- Model responsible party

Note: The **Model responsible party** property is read-only at the model level, if the model is located within a project.

Add Tags to a Model

Tags are ideas and topics that indicate what your content is about. You can use tags to indicate the market sector, a company department, or other types of content that you want to associate with your model.

You can add one or more tags to a model. When you add tags to a model, they are also added to a global list of tags that is available to be added to other models within the common model repository.

Automated tags are applied to a model when specific actions are performed, such as publishing a model. They also show key information about a model, such as the score code type, or relationships that might exist.

Note: Tag property values for a model appear as tokens in the **Tags** column of the Models category view and Models tab of a project. The tag values can be used for sorting models. The function for searching models for tag values is currently not supported.

To add a tag:

1. On the **Properties** tab of a model, select **Tags**.
2. Enter a value and select an existing tag, or press Enter to add a new tag.

   Note: The tag name can contain only alphanumeric characters, double-byte characters, the underscore (_), the hyphen (-), and the period (.). Spaces are not allowed.

   Repeat this step for additional tags.
3. Click .

To delete a tag, click × within the token, and then click . The tag is removed from the model.

Managing Model Versions

The current version of a model is the latest version in which the model properties and file contents are editable. If you add a new model version manually or perform an action that automatically creates a new model version, a snapshot of the model's contents is taken and a version number is assigned. For example, a new model version can be automatically created when you set a model as the champion model or publish a champion model from the project level. However, the contents of the model version that was published or set as the champion is locked and cannot be edited. You can view the contents of the locked model version only. To edit a model when the current model version is locked, you must create a new model version. Model versions cannot be deleted.

Set the Displayed Version

The displayed version is the version whose information is displayed on the other tabs, such as the **Files**, **Variables**, and **Properties** tabs. The version number for the displayed version appears next to the model name in the object title bar. On the **Versions** tab, a ✓ indicates the displayed version. To change the displayed version, select the version that you want to view, and click **Set Version**.

Note: Here are a few restrictions when creating a new model version.

- The current version of an object is the version that has the highest version number. When you create a new version, SAS Model Manager locks the current version before it creates the new version.
- You cannot modify a locked version.
- You cannot unlock a version.

Create a New Version

To create a new version:

1. On the **Versions** tab, click **New Version**. The New Version window appears.
2. Select the version type: **Minor** or **Major**. Version numbers follow the format *Major.Minor*. If you select **Major**, the number to the left of the period is incremented. If you select **Minor**, the number to the right of the period is incremented.

3 Click **Save**.

Publish an Open Model Object

When you open a model that is located within a project version, you can publish it from the object toolbar. Models that are located directly within a folder and that are not members of a project cannot be published. For information about publishing models from the **Models** tab of a project or from the Projects category view, see "About Publishing Models" on page 141.

> **IMPORTANT** Before you can publish a Python model or an R model, the source code must be in the correct format. For more information, see "Concepts: Open-Source Models" on page 169.

1 Open a model.

2 Click **Publish**. The Publish Models window appears.

3 Select a destination.

Note: If you have Read and Write permissions to the caslib that is specified in a publishing destination, the destination is shown in the list.

For more information, see *SAS Viya: CAS Authorization Window* and "Configuring Publishing Destinations" in *SAS Model Manager: Administrator's Guide*.

> **TIP** Click **Details** to view the destination details.

4 (Optional) Assign tags to the published model by entering a tag value. Press Enter after each tag.

> **TIP**
>
> - Tag names cannot start with a period or a hyphen, and cannot contain more than 128 characters. They can contain only alphanumeric characters, the underscore (_), the hyphen (-), and the period (.).
> - When publishing a model to a container destination the model version is automatically added as a tag for the published model within the target destination. You have the option of removing the `latest` tag before publishing your models.

5 Specify a value for any properties that might appear after you select a destination.

Property	Description	Destination Type	Required
Azure Machine Learning workspace	Indicates the workspace in the Azure Machine Learning cloud-based environment to publish the selected items to. Select a value from the drop-down list.	Azure Machine Learning	Yes
Git directory	Indicates the directory path within the Git repository to publish the selected items to. Click and navigate to a folder within the Git repository. Click **OK**. **Note:** If a folder is not selected within the Git repository, the items are published to the root directory.	Git	No

6 In the **Items to Publish** section, select a model version to publish. The default value is the latest version of the model.

7 Choose whether to create a new model version after publishing the model. The option to create a new model version is selected by default.

Note: If you choose to not create a new model version, the content of the model version that was published is locked and cannot be edited.

8 (Optional) Update the published name in the **Items to Publish** section. The maximum length and character restrictions differ, depending on your destination.

For more information, see Table 7.2 on page 145.

9 (Optional) If you have previously published the model with the same published name, you can replace the model. In the **Items to Publish** section, enable the **Replace item with the same name** toggle in order to replace the previously published item of the same name in the same destination.

Note: When publishing models to a container, an Azure Machine Learning, or a Git publishing destination, if a published model with the same name already exists within the destination, a sequential number is added as a tag to indicate the version of the published model.

10 Click **Publish**. The Publishing Results window appears. The status of the publishing request is displayed in the **Status** column.

Note: When you select a CAS destination and click **Publish**, the CAS destination table is automatically reloaded and the newly published item is made available to other applications.

When you are publishing to a SAS Micro Analytic Service destination, the **Micro Analytic Module** column is also displayed with a URL to the published model.

11 When the status changes to `Published successfully`, click **Close**.

Note: The following is true for models that are within a project version: after a model has been published, a new version of the model is created and a publishing validation test is also created. For more information, see "Managing Model Versions" on page 96 and "Validate Published Models" on page 112.

Search for Models

In the **Models** category view, you can perform multiple types of searches:

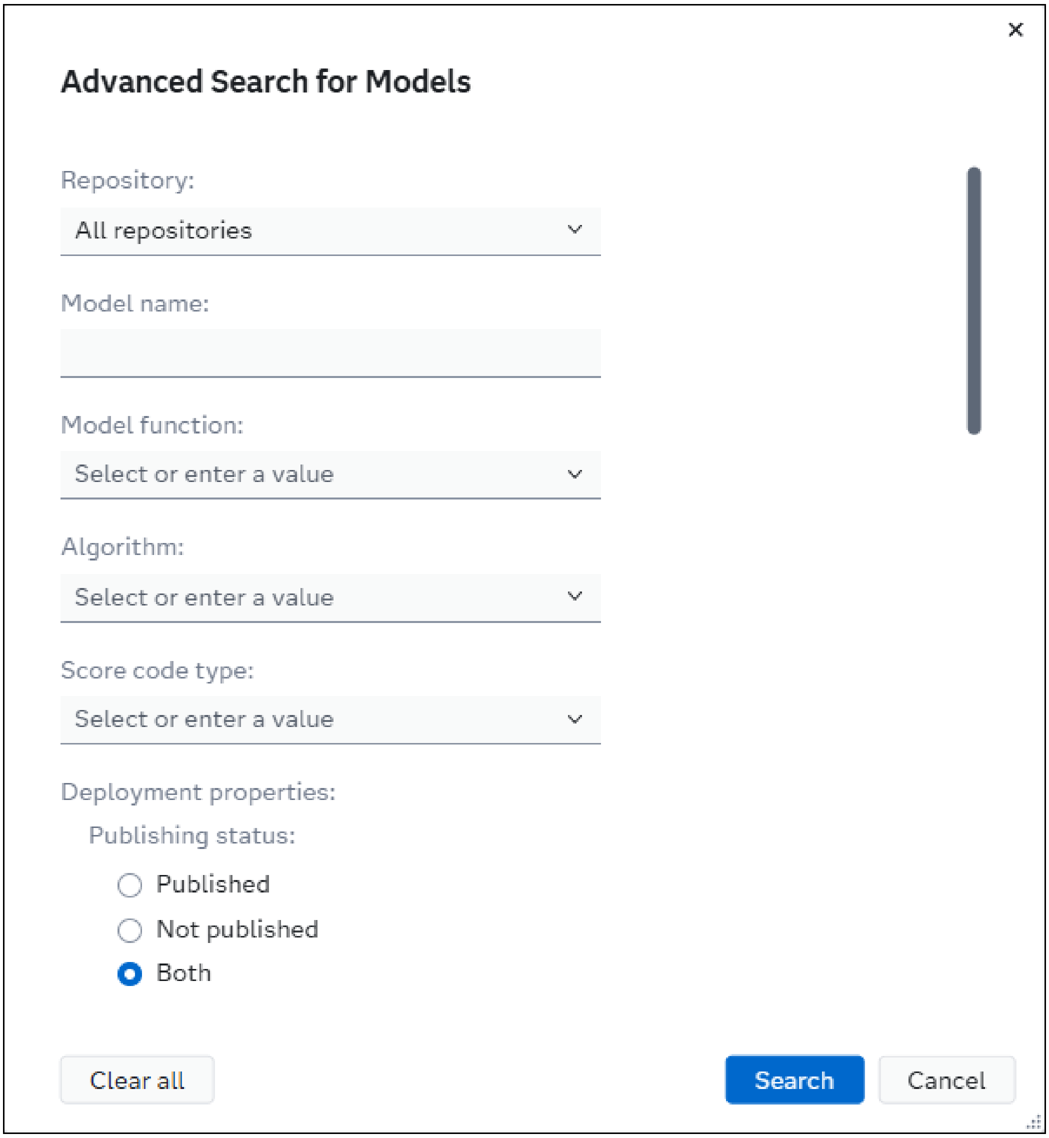
- search for models by name using the search field above the models list.
- perform an advanced search for models using model properties, input variables, output variables, or user-defined properties.
- search for model objects across applications using the search field in the application bar.

For more information about searching, see "Search" in *SAS Viya Platform: General Usage Help for Web Applications*.

Advanced Search for Models

To perform an advanced search for models:

1 Click **Advanced search**, which is to the right of the search box. The Advanced Search for Models window appears.



2 To search for a model in a specific repository, click ▼ in the **Repository** drop-down list, and select the desired repository. **All repositories** is selected as the default for the advanced search.

3 Specify values for one or more of the following fields to include them in your search criteria:

Note: If you previously used the dashboard charts to filter the list of models, the filters are applied to the associated fields in this window.

- **Model name**
- **Model function**
- **Algorithm**
- **Score code type**
- **Deployment properties**
- **Publishing destination**

- **Date created (range)**
- **Modeler**

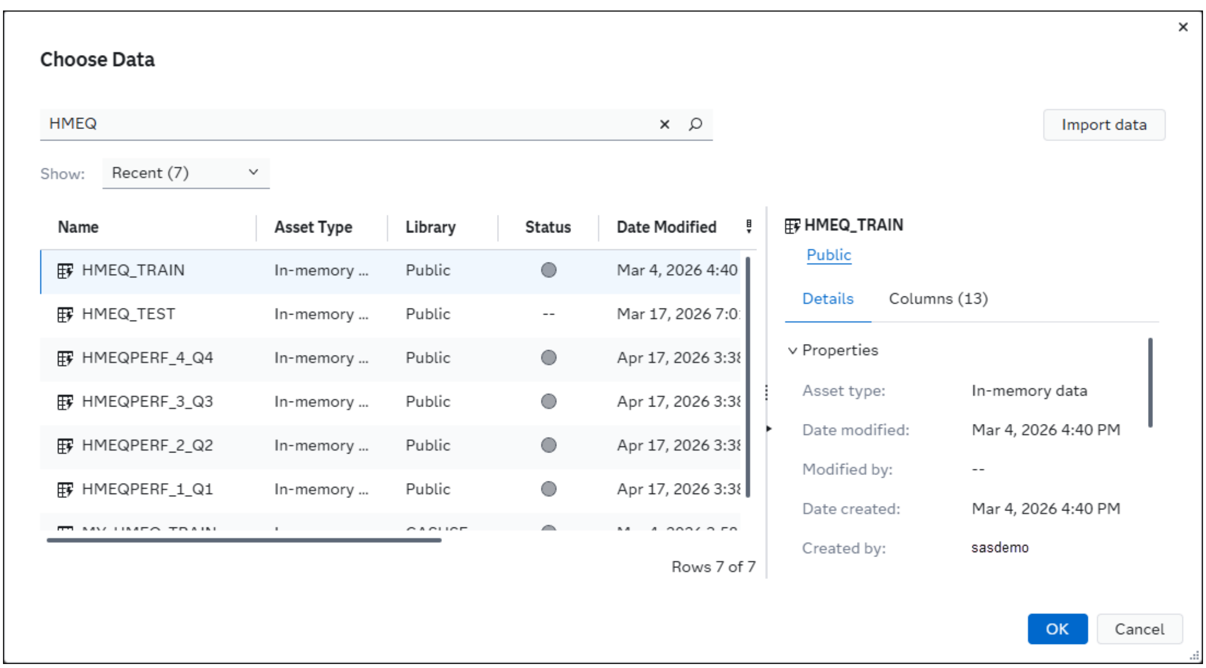
4 To search for a model by **Target variable**, enter a value or click to select a variable from a data table.

5 To search for a model by input or output variables, select a data table and choose the variables that are associated with models or model objects.

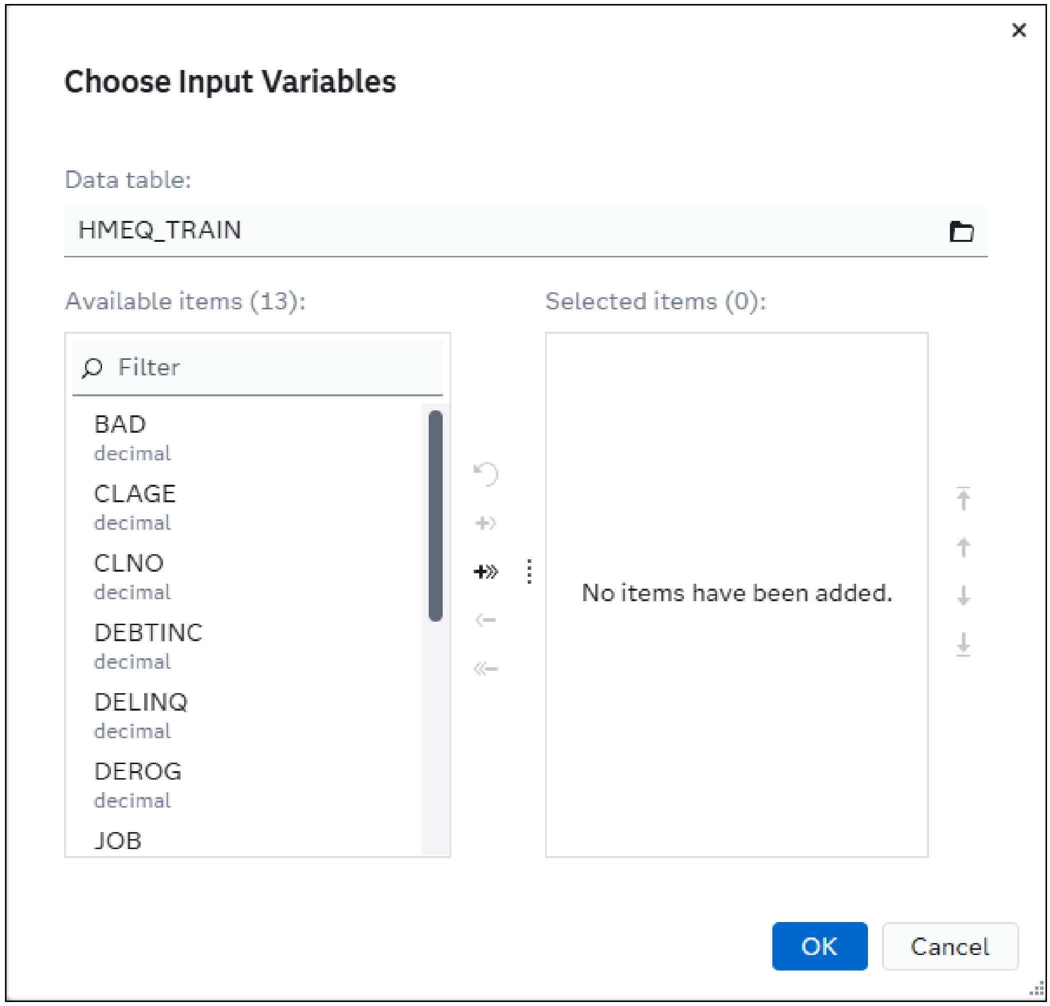
a Click next to the **Input variables** or **Output variables** field. The Choose Data window appears.

b Select a data table from the list on the left-hand side. The details of the selected data appear on the right-hand side. The sample data and available profiles of the data table are also viewable.

> **TIP** Use the search bar to locate the desired data table and filter search results.



c When you have selected your data table, click **OK**. The Choose Variables window appears.



d Select the variables that you want to add and click +›. You can also click +» to add all of the variables from the available items list.

e To remove variables, select the variables that you want to remove and click ‹–. Click «– to remove all of the variables from the available items list.

> **TIP** Click ↶ to undo your last action.

f Click **OK** to close the Choose Variables window. The variables that you selected appear in the **Input variables** and **Output variables** fields.

> **TIP** Click |↶ next to the **Input variables** or **Output variables** fields in the Advanced Search for Models window to clear the selected variables.

6 To search for user-defined properties that are associated with model objects, specify one or more property name and value pairs, or specify property names only.

a Click + to add a row to the **User-defined properties** table.

b Click ▼ in the **Name** drop-down list, and select a user-defined property.

c (Optional) Enter the value for the user-defined property in the **Value** column.

Note: When the data type of a user-defined property is set to date or datetime, the calendar selector icon appears.

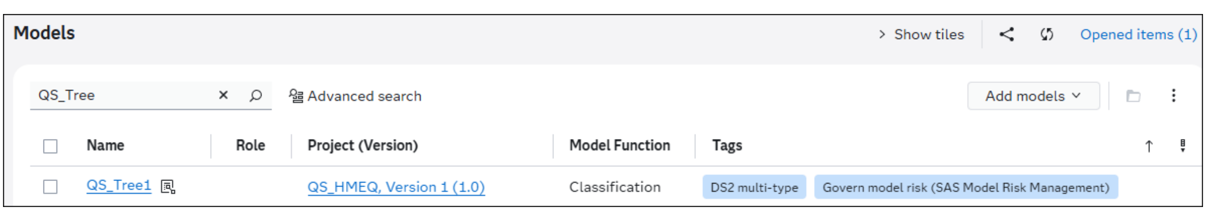
d Select a row and click to remove the user-defined property from the search criteria.

7 Click **Search**. The Models category view displays the list of models that meet the advanced search criteria.

> **TIP** Click **Clear All** to clear all previously entered search criteria in the Advanced Search for Models window.

View Model Relationships and Related Objects

In the Models category view, if model relationships have been established, a **Relationships** token appears in the **Tags** column. Also, the icon appears to the right of the model name.



Here are some tasks that you can perform:

- Click to view the model risk card for the related object.
- Click the **Relationships** token to view the model relationships.

For more information, see "Establish Model Relationships" on page 92.

6

Comparing and Evaluating Models

About Comparing and Evaluating Models

The goal of a modeling project is to identify a champion model that an external scoring application uses to predict an outcome. SAS Model Manager provides tools to evaluate candidate models and declare a project champion model.

Here are the tasks that you can perform to evaluate your models:

- compare and assess models
- create and run a scoring test on a model
- monitor performance of a model
- validate the published models within the destination
- access the published models from external scoring applications

Compare Models

You can compare and assess one or more models. When you compare models, the model comparison output includes model properties, user-defined properties, and variables. The model comparison output might also include fit statistics, and lift and ROC plots for the models if the required model files are available. The fit statistics, as well as plots for lift and ROC, can be produced using the ASSESS procedure or Python packages such as sklearn and scipy.stats. The fit statistics and plots are available by default for the SAS Visual Data Mining and Machine Learning models that are created using Model Studio. On the **Files** tab of a model object, notice that JSON files (dmcas_fitstat.json, dmcas_lift.json, dmcas_roc.json) are included. These JSON files are used to show the fit statistics and plots when comparing models in SAS Model Manager.

Note: You can create the JSON files that are needed to display the fit statistics as well as Lift and ROC plots. Here are some methods that you can use to create the JSON files: the ASSESS procedure, the %MM_MODEL ADD_JSONFILES macro, or Python code and packages such as sklearn and scipy.stats.

For more information, see the following documentation and examples:

- The ASSESS Procedure in *SAS Visual Statistics: Procedures*
- "%MM_MODEL ADD_JSONFILES Macro" in *SAS Model Manager: Macro Reference*
- Examples in the model-management-resources GitHub repository

To compare models:

1 Select one or more models.

2 From the Models category view, click ⋮ and select **Compare**.

   On the **Models** tab of project, click **Compare**.

   The **Compare** page appears.

   > **IMPORTANT** The **Show all** and **Show differences** buttons were hidden with the 2024.09 release.

3 Review the differences for the following model information:

   Note: Section titles appear whether or not the section contains content.

   - Model Properties
   - User-Defined Properties
   - Input Variables
   - Output Variables
   - Target Variable
   - Fit Statistics

     Note: This section compares the fitness statistics for the chosen models.

   - Plots

     Note: This section contains both Lift and ROC plots.

   > **TIP** Click ▸ to expand a section, and click ◂ to collapse a section.

4 Click **Close**.

Test Models

About Testing Models

The purpose of a test is to run the score code of a model and produce scoring results that you can use for scoring accuracy and performance analysis. The test uses the input data table to generate the test output table. If your environment has its own means of executing the score code, then your use of the SAS Model Manager scoring tests is mostly limited to testing the score code. Otherwise, you can use the tests both to test your score code and execute it in a production environment.

To be scored, tested, or validated, a model must have a file that is assigned to the score code model file role and have the following score code types: DATA step, DS2 package, DS2 embedded process, DS2 multi-type, Analytic store, Python, or R. For more information, see “Assign Model File Roles” on page 75 and “Set Model General Properties” on page 85.

When you publish a model, the system creates a publishing validation test. You can edit the publishing validation test to select a test data table and output library. This enables you to score and validate models within the publishing destination that they were previously published to. For more information, see “Validate Published Models” on page 112.

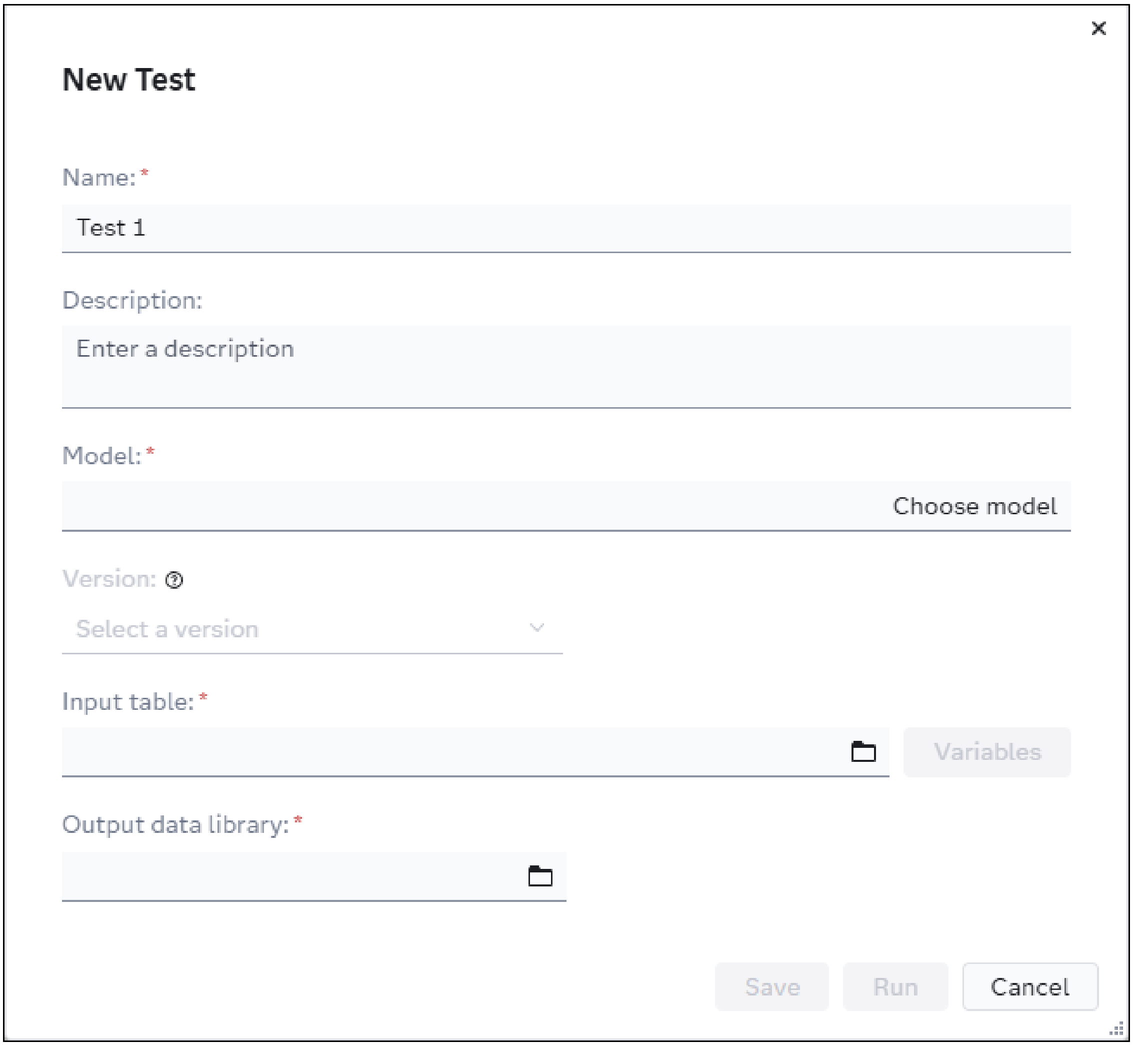
Note: Models that have a score code type of Python or R can be scored, tested, or validated only when the score code is in the correct format. For more information, see “Scoring Python Models” and “Scoring R Models”.

Create and Run a New Test

By default, only the user who creates a test definition can view, update, or delete the test definition, as well as run the test and view the test results. For more information, see “Default Permissions” in *SAS Model Manager: Administrator’s Guide*.

Note: If you have one or more tests, you can select the check box and click **Run**, to run them all at the same time.

1 On the **Scoring** tab of a project, click the **Tests** tab, and then click **New Test**. The New Test window appears.



2 Enter a name for the test if you do not want to use the default name.

3 (Optional) Enter a description for the test.

4 Click **Choose model** and select a model to test.

5 Select the version of the model to use for the test. The default value is **Use latest version**.

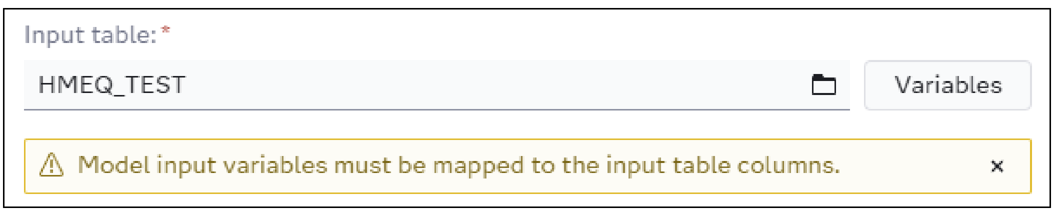
> **TIP** If you select a specific version, that version is always used, even if new versions of the model are available. Select **Use latest version**, if you want to always use the most recent version of the model.

6 Click , select the input table for the test, and click **OK**.

> **TIP** If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see "Overview of SAS Data Explorer" in *SAS Data Explorer: User's Guide*.

7 Map variables.

Note: SAS Model Manager automatically maps model input variables to the columns in the input table when the names and data types of the variables match those of the table columns. If any input variables cannot be mapped automatically, a warning message is displayed. When you are creating a scoring test for a model, if the column names in the input table do not match the model input variable names, a warning message is displayed and the **Variables** button is disabled. The exception is input variables for SAS Visual Text Analytics models with a score code type of SAS program. Those can be mapped.



> **TIP** You can change the automatic variable mappings.

To map variables:

a Click **Variables**. The Variable Mappings window appears.

b For each input variable, select the table column to which the variable should be mapped.

c Click **OK**.

8 (Optional) By default, the library for the output data is the same as that for the input data table. Click to specify a different library to store the new test output table that is created when the test is run. The input data table and output library must be on the same CAS server.

9 Click **Run** to save and run the test. Alternatively, click **Save** to save the test definition without running it.

Note: You can also select a check box for a test and click **Run** in the toolbar to rerun a test.

The status of the test is indicated by the icon in the **Status** column.

Table 6.1 *Test Statuses*

Icon	Status
	The test is not ready to run. The test definition is not complete, or it might contain errors.
	The test is defined correctly and is ready to run.
	The test is running.
	The test completed successfully.
	The test completed, but warnings were issued in the SAS log. The URI to the log file is shown on the **Test Results** page.
	The test did not run successfully. Check the SAS log for information. The URI to the log file is shown on the **Test Results** page.

10 Click in the **Results** column to view the results of the test. The **Output** page is displayed by default. The **Test Results** page displays information about the test, including the URIs for the test definition and test results. It also includes URIs to the SAS code that was run by SAS Model Manager, the output data set, and the SAS log that was generated when the code was run.

You can also click the **Code** or **Log** pages to view the test result details.

You can also work with the output table in other SAS applications to analyze the data, create and compare models, discover relationships hidden in the data, and generate reports based on the data.

Edit a Test

1 On the **Scoring** tab of a project, click the **Tests** tab.

2 Click on a test name. The Edit Test window appears.

3 Edit the test properties as needed, and then click **Save** or **Run**.

Note: You can also select a check box for a test and click **Run** in the toolbar to rerun a test.

Delete a Test

1 On the **Scoring** tab of a project, click the **Tests** tab.

2 Select one or more tests and click 🗑.

Validate Published Models

About Validating Published Models

When you publish a model, the system creates a publishing validation test. You can edit the publishing validation test to select a test data table and output library. The validation of the model runs within the publishing destination that the model was previously published to. The test data table must be loaded in CAS memory in order to run a publishing validation test.

Note: A Git publishing destination is a repository and it has no means of executing code. Therefore, when a model is published to a Git destination, no publishing validation test is created. Publishing validation is also not supported for models that are publishing to an Event Stream Processing (ESP) publishing destination.

Only these models can be validated:

- those that have the score code model file role assigned
- those that have a score code type of DATA step, SAS program, DS2 package, DS2 embedded process, DS2 multi-type, Analytic store, Python, and R.

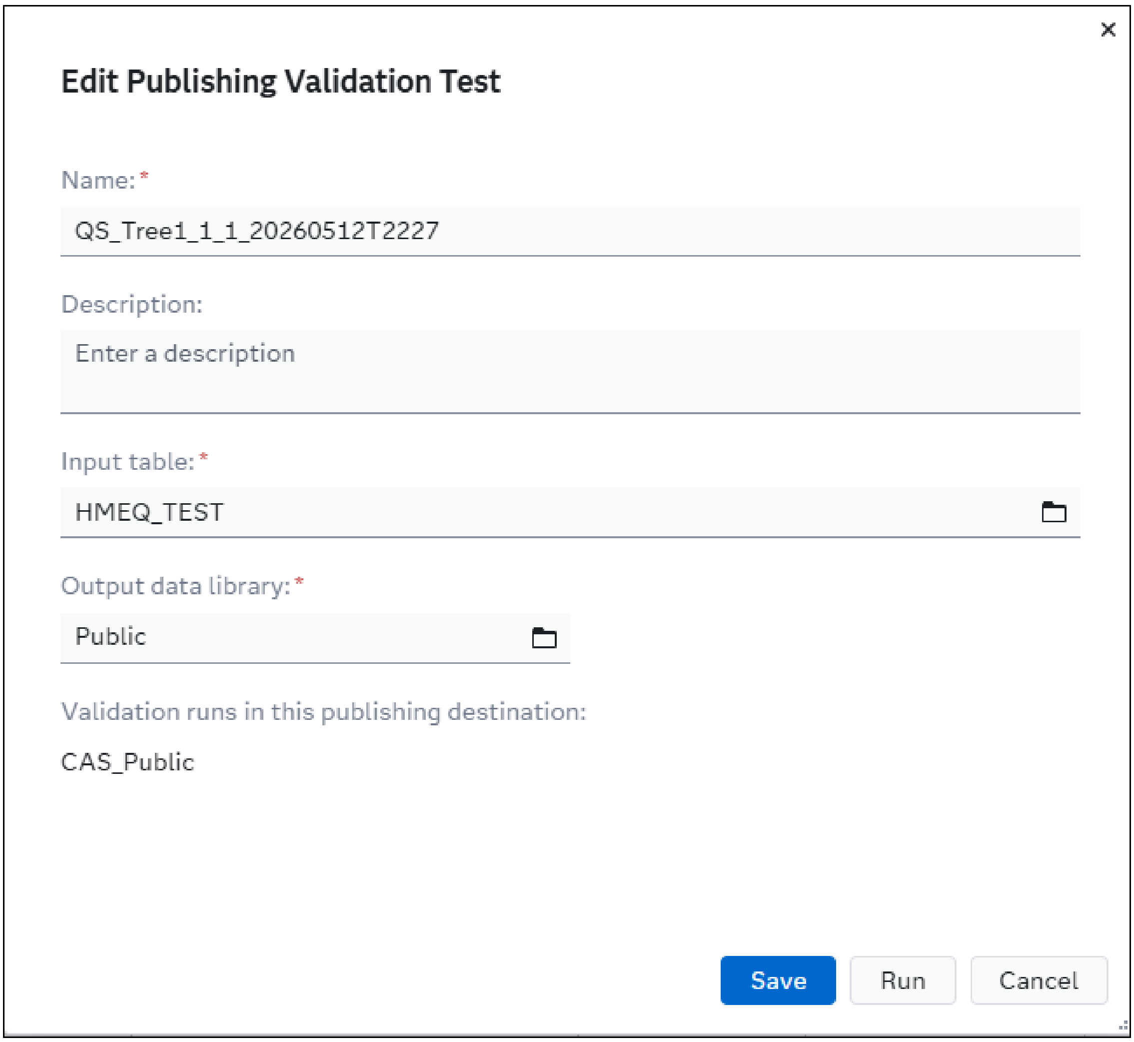
For more information, see "Assign Model File Roles" on page 75 and "Set Model General Properties" on page 85.

Note: Models that have a score code type of Python or R can be validated if the score code is in the correct format. For more information, see "Scoring Python Models" on page 170 and "Scoring R Models" on page 173.

Edit and Run a Publishing Validation Test

1 Click the **Scoring** tab of a project, and then click the **Publishing Validation** tab.

2 Click a test name. The Edit Publishing Validation Test window appears.



3 (Optional) Change the name of the test.

4 (Optional) Enter a description for the test.

5 Click , select the input table for the test, and click **OK**.

> **TIP** If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see "Overview of SAS Data Explorer" in *SAS Data Explorer: User's Guide*.

6 Specify a value for any properties that might appear for the publishing destination and are required for validation. For example, if you select an Azure Machine Learning destination, a property appears for you to select a compute resource.

7 (Optional) By default, the library location is the same as that for the input data table. Click to specify a different library to store the new test output table that is created when the test is run. The input data table and output library must be on the same CAS server.

8 Click **Run** to run the test. Alternatively, click **Save** to save the test definition without running it.

Note: You can also select a check box for a test and click **Run** in the toolbar to rerun a test.

The status of the test is indicated by the icon in the **Status** column. For more information, see Table 6.1 on page 111.

9 Click in the **Results** column to view the results of the test. The **Test Results** page displays information about the test, including the URIs for the test definition and test results. It also includes URIs to the SAS code that was run by SAS Model Manager, the output data set, and the SAS log that was generated when the code was run.

You can click the **Output**, **Code**, or **Log** pages to view the test result details.

You can also work with the output table in other SAS applications to analyze the data, create and compare models, discover relationships hidden in the data, and generate reports based on the data.

Note: When a publishing validation test is run for a model that was published to an Azure Machine Learning destination, the published model is deployed to Azure Kubernetes Service (AKS) in order to perform the validation. After validation is complete, the published model is removed from AKS. For more information about deploying models, see “Deploy a Published Model to Azure Kubernetes Service (AKS)” on page 152.

Delete a Publishing Validation Test

1 On the **Scoring** tab of a project, click the **Publishing Validation** tab.

2 Select one or more publishing validation tests, click ⋮ , and select **Delete**.

3 In the confirmation message, click **Delete**.

Duplicate a Publishing Validation Test

1 On the **Scoring** tab of a project, click the **Publishing Validation** tab.

2 Select one of the publishing validation tests, click ⋮ , and select **Duplicate**.

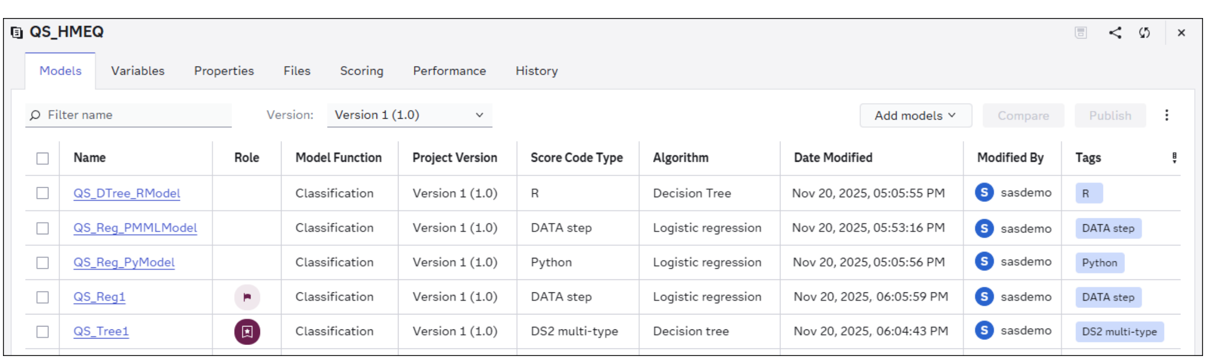
Set Champion and Challenger Models

About Champion and Challenger Models

The champion model is the best predictive model that is chosen from a pool of candidate models. Before you identify the champion model, you can evaluate the structure, performance, and resilience of candidate models. When a champion model is ready for production scoring, you set the model as the champion model. The project version that contains the champion model becomes the champion version for the project. You can publish the champion model to a publishing destination.

You use challenger models to test the strength of champion models. The champion model for a project can have one or more challenger models. A model can be flagged as a challenger model only after a champion model for the project has been selected. A challenger model can be located in any version of a project.

Note: A SAS Visual Data and Machine Learning project champion model is automatically set as a champion within a SAS Model Manager project when you register the models from a Model Studio project, as long as a project champion does not already exist. If a project champion already exists, the new Model Studio project champion is set as a candidate champion.



Set a Champion Model

1. Click the **Models** tab of a project.
2. Select a model, click ⋮, and select **Set as champion**.
3. Choose whether to create a new model version when setting the champion. The option to create a new model version is selected by default.

   Note: Setting the champion locks the current model version. You must create a new version to modify the model content.

4. If the Select Project Output Variables window appears, select the model output variables to use as project level output variables. You can use the same variable names or specify different names for the project output variables.

   Note: You can modify the names of the project output variables only for a model with a score code type of DATA step, SAS program, and PMML. The Map Output Variables window appears only for a model that has a score code type of DATA step, SAS program, PMML, R or Python.

   Click **Save**.

5. If the model input variables are not project input variables, you are prompted to add the input variables to the project.

   In the confirmation message, click **Yes**.

   Note: If you click **No**, the model is not set as the project champion.

6. When the champion model has been set, ★ is displayed in the **Role** column.

Set a Challenger Model

1. Click the **Models** tab of a project.
2. Select a model, click ⋮, and select **Set as challenger**.
3. If the **Select Project Output Variables** window appears, select the model output variables to use as project level output variables. You can use the same variable names or specify different names for the project output variables.

   Note: You can modify the names of the project output variables only for a model with a score code type of DATA step, SAS program, and PMML. The Map

Output Variables window appears only for a model that has a score code type of DATA step, SAS program, PMML, R or Python.

Click **Save**.

4 If the model input variables are not project input variables, you are prompted to add the input variables to the project.

In the confirmation message, click **Yes**.

Note: If you click **No**, the model is not set as a challenger model.

5 When the challenger model has been set, ⚑ is displayed in the **Role** column.

Clear Model Role

1 Click the **Models** tab of a project.

2 Select the champion model or a challenger model, click ⋮, and select **Clear role**.

Note: If you clear the role of the project champion model, all of the associated challenger models are cleared as well. The candidate champion role cannot be cleared.

3 In the confirmation message, click **Yes**.

Monitoring Performance

About Monitoring Performance

Models can begin to degrade the moment that you put them in production. Using SAS Model Manager, you can monitor the performance of your analytical models to see whether they continue to behave as expected after changes occur in market conditions or customer behavior, new data becomes available, or there is concept drift. By monitoring their performance, you can avoid model decay and re-evaluate the business value of your models in production to keep them performing at the highest levels. Comparing and assessing model performance over time is a best practice that preserves the value that your data science and IT teams originally provided.

SAS Model Manager enables you to monitor and evaluate model performance of your champion, challenger, and other models to ensure that your models are performing efficiently. You begin by collecting performance data that has been created by the champion model at intervals. Your organization determines what those intervals are. You then use performance data to assess model prediction accuracy. It includes all of the required variables as well as a target variable. For example, you might want to create performance data tables monthly or quarterly and then use SAS Model Manager to create a performance definition that includes each time interval.

Next, you create a performance definition, and then you run it. The output from running a performance definition includes a summary of performance and model assessment information, performance result tables, and several charts for each model that is specified within the definition. The summary information includes model assessment criteria, scoring observations, performance definition high-level details, and the performance rank of models within the containing project. The results include charts such as Variable Distribution, Characteristic, Stability, Lift, Gini, ROC, Kolmogorov-Smirnov (KS), Average Squared Error (ASE), and Feature Contribution (FCI) charts for each model that was specified within the definition. Additional charts are displayed for prediction models with an interval target, and custom charts can be configured to be displayed as well. Only the charts for each model are displayed on the **Performance** tab of a project. You can explore and visualize the data tables that contain the performance results using SAS Visual Analytics.

Note that model performance can sometimes be improved by tuning or refitting your models, or by using a new champion model.

You can also conduct partial model performance monitoring based on the input data. If your input data contains only the input variables, then the system computes and returns the characteristic analysis. If your input data contains the score output variable, then the system also computes and returns the stability analysis. If your input data contains the response variables, then the system also computes and returns the accuracy measures, such as Gini, ROC, Lift, Kolmogorov-Smirnov (KS), Average Squared Error (ASE), and Feature Contribution Index (FCI). The charts that are displayed depend on the model function and whether there is a binary, nominal, or interval target variable defined within a project. Additional charts are displayed for a classification model with a nominal target, a prediction model with an interval target, and a text categories model.

Alternatively, you can also create performance monitoring output by writing your own SAS program using the performance monitoring macros that are provided with SAS Model Manager. You can then submit your program using SAS Studio. The performance result tables that are produced using the macros can then be viewed in SAS Studio, SAS Environment Manager, or SAS Visual Analytics. For more information, see "Performance Monitoring Macro" in *SAS Model Manager: Macro Reference*.

Note: When performance monitoring is run, the system can score models only with the following score code types: **SAS program**, **DATA step**, **DS2 embedded process**, **DS2 multi-type**, **Analytic store**, **Python**, or **R**. For more information, see "Set Model General Properties" on page 85.

For more information and examples of the performance monitoring charts, see "Concepts: Performance Monitoring" on page 174.

Set Project Properties

1 Click the **Properties** tab of a project.

2 On the **Model Evaluation** page, specify the following properties:

Classification project

- Target variable
- Target level
- Target event value or Target event values
- Output event probability variable or Predicted target variable

Prediction project

- Target variable
- Target level
- Output prediction variable

Text categories project

- Actual category variable
- Document ID variable
- Text variable
- Predicted category variable

3 Click .

Naming Requirements for Input Data Tables

Rules for Data Table Names

Here are rules for data table names:

- They must have at least two levels.
- The value for the *prefix* can contain underscores, but spaces are not recommended in the prefix name or the table name.
- The name can contain alphanumeric characters, double-byte characters, and special characters, except for the following characters: / \ , ; ' "
- The second level must be a number.
- An underscore is treated as a delimiter between the levels within the name.

- The sequence number and time label must be unique across all of the data table names.
- The time label is used to represent the data in the charts. If a time label is not provided, the sequence number is displayed in the charts.
- The time label can contain hyphens. However, it should not contain spaces or underscores.

Supported Formats

Use one of the following formats for the name of the data tables:

- *prefix_sequenceNumber*
- *prefix_sequenceNumber_timeLabel*

Descriptions and Restrictions

Here are the descriptions and restrictions for the different levels of a table name:

prefix
: The prefix is the first level of the table name and specifies which data tables in a data library to use for performance. The prefix can contain underscores, but spaces are not recommended in the prefix name.

sequence number
: The sequence number is the second level of the table name and specifies the order in which the data tables should be used for performance monitoring. The sequence number must be the second level of the table name and unique across all of the data table names. It is recommended that you start with the number *1* and increase from there. However, the sequence number can by any unique numeric value, such as 202306.

time label
: The time label is the third level of the table name and specifies the label to use to represent the data in the performance charts. The time label must be unique across all of the data table names. The label can be a period of time such as Q1 or Q12020, or another meaningful label. Spaces are not recommended in the label name. If a time label is not provided, the sequence number is displayed in the charts.

Create a New Performance Definition

You can create a performance definition only for classification models with a binary or nominal target, for prediction models with an internal target, and for text categories models. The model function and target level properties must be set at both the model level and project level. For more information, see "Modifying Model Properties" on page 84 and "Modifying Project Properties" on page 33.

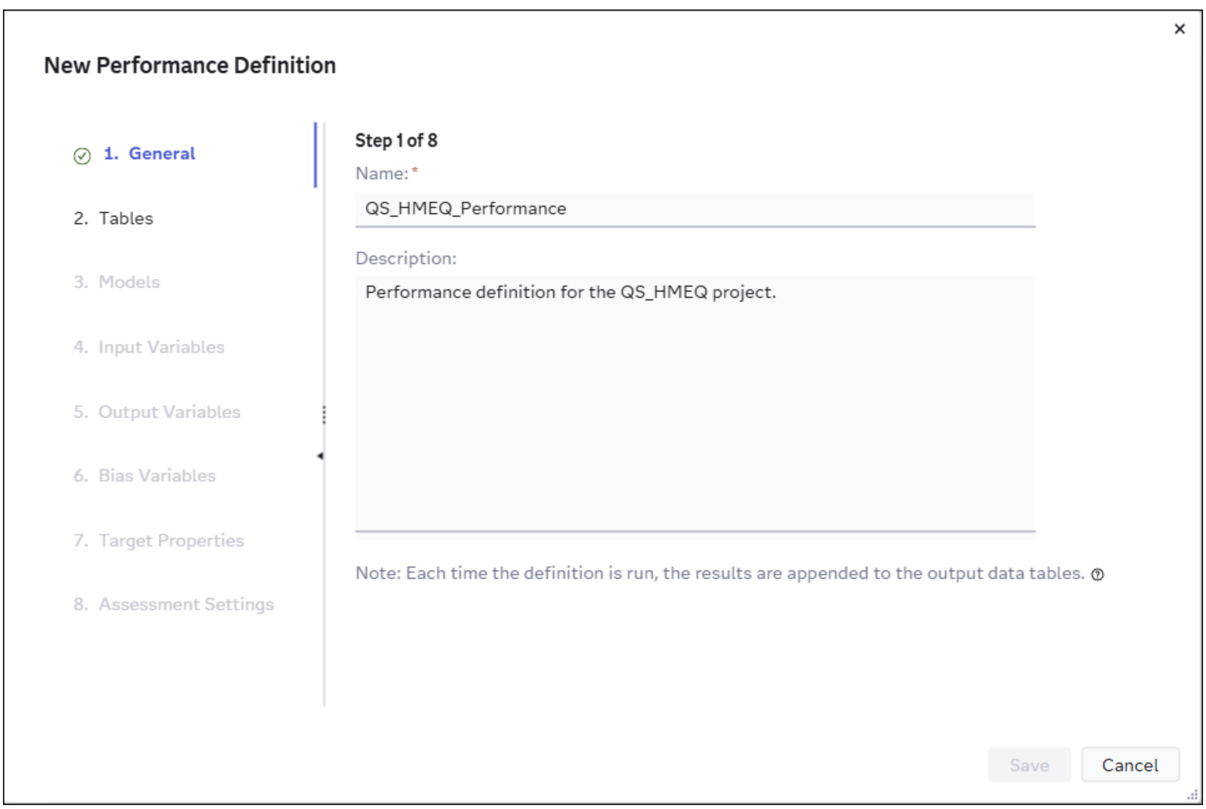
> **IMPORTANT** If the names of the models that are selected as part of the definition contain the following types of special characters, the performance results cannot be generated:
>
> - a percent sign (%) or an ampersand (&)
> - unbalanced braces, brackets, parenthesis, or single or double quotation marks

1 Verify that the required project properties are set.

2 Click the **Performance** tab of a project and then click **New Definition**. The New Performance Definition window appears.

3 On the **General** page, accept the default name for the performance definition or enter a name of your choosing.

4 (Optional) Enter a description.



5 Click the **Tables** page and select an input data method.

> **IMPORTANT** Before selecting an input data method, make sure that the input data tables follow the naming requirements. For more

information, see “Naming Requirements for Input Data Tables” on page 119.

Use a single table

Click 🗀, select a data table and click **OK**.

TIP If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see “Overview of SAS Data Explorer” in *SAS Data Explorer: User’s Guide*.

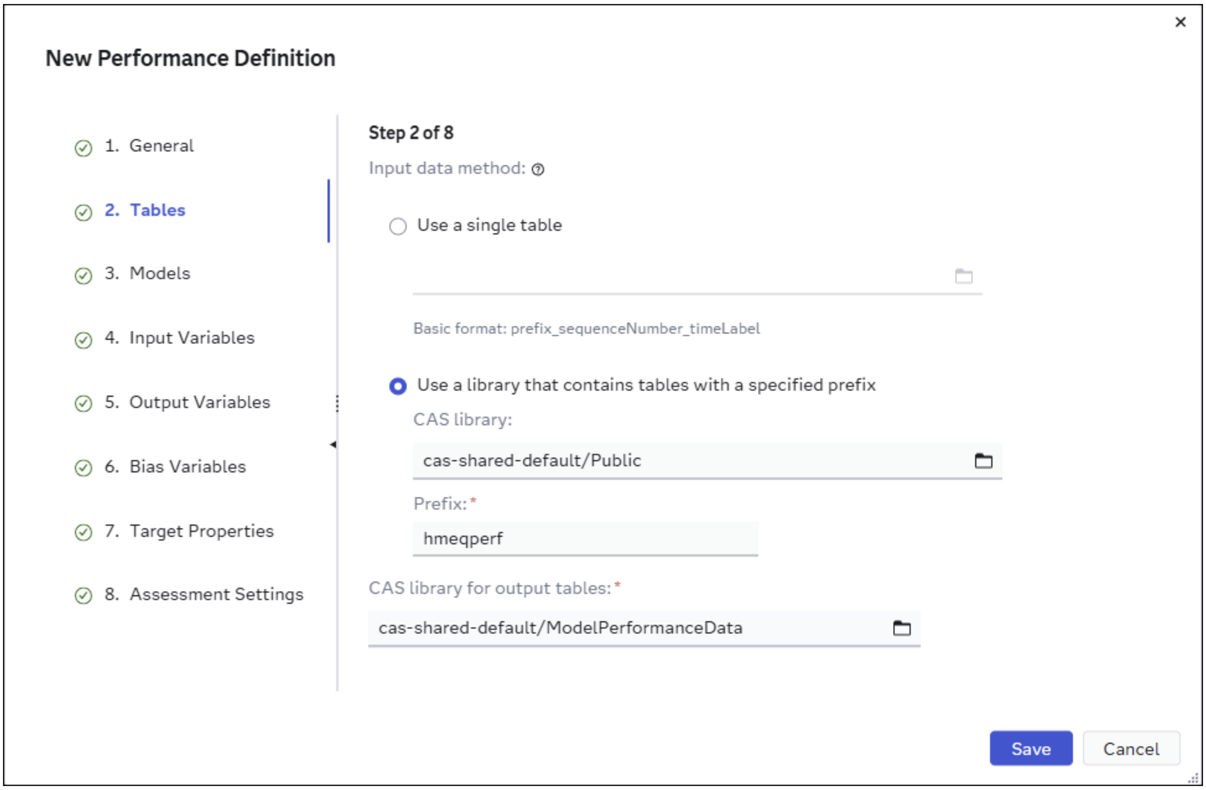
Use a library that contains tables with a specified prefix

- Click 🗀, select the same server as the data table, and select a library. Click **OK**.
- Enter a prefix.

Note: The value for the *prefix* can contain hyphens and underscores, but spaces are not recommended in the prefix name or the table name.

Note: If the data tables that were specified in the performance definition exist in a library, but are not loaded in a CAS session, they are loaded at run time. This condition might occur when the CAS server is restarted.

6 (Optional) Verify that the CAS library for the output tables is set. The default value is `cas-shared-default/ModelPerformanceData`. The input data source library and the output data library must be on the same CAS server.



7 Click the **Models** page and select an option for the data scoring method.

Note: Partial save is supported. After you complete this step of the wizard, you can click **Save**.

System scores data using selected models
When you choose this data method, the model score code is used to score the data before generating the performance results.

Use scored data from input tables
When you choose this data method, the data tables should contain the predicted values for the scored model.

8 Select a model-selection option.

> **TIP** If model roles or selected models are changed after a performance definition is saved, you must edit the performance definition and save it, in order for the changes to take effect.

Note: When selecting models, the function and target level properties are validated at the model level, not the project level. For more information, see “Modifying Model Properties” on page 84.

Use referenced models
The default option to use the current champion. You can also select to include all challenger models.

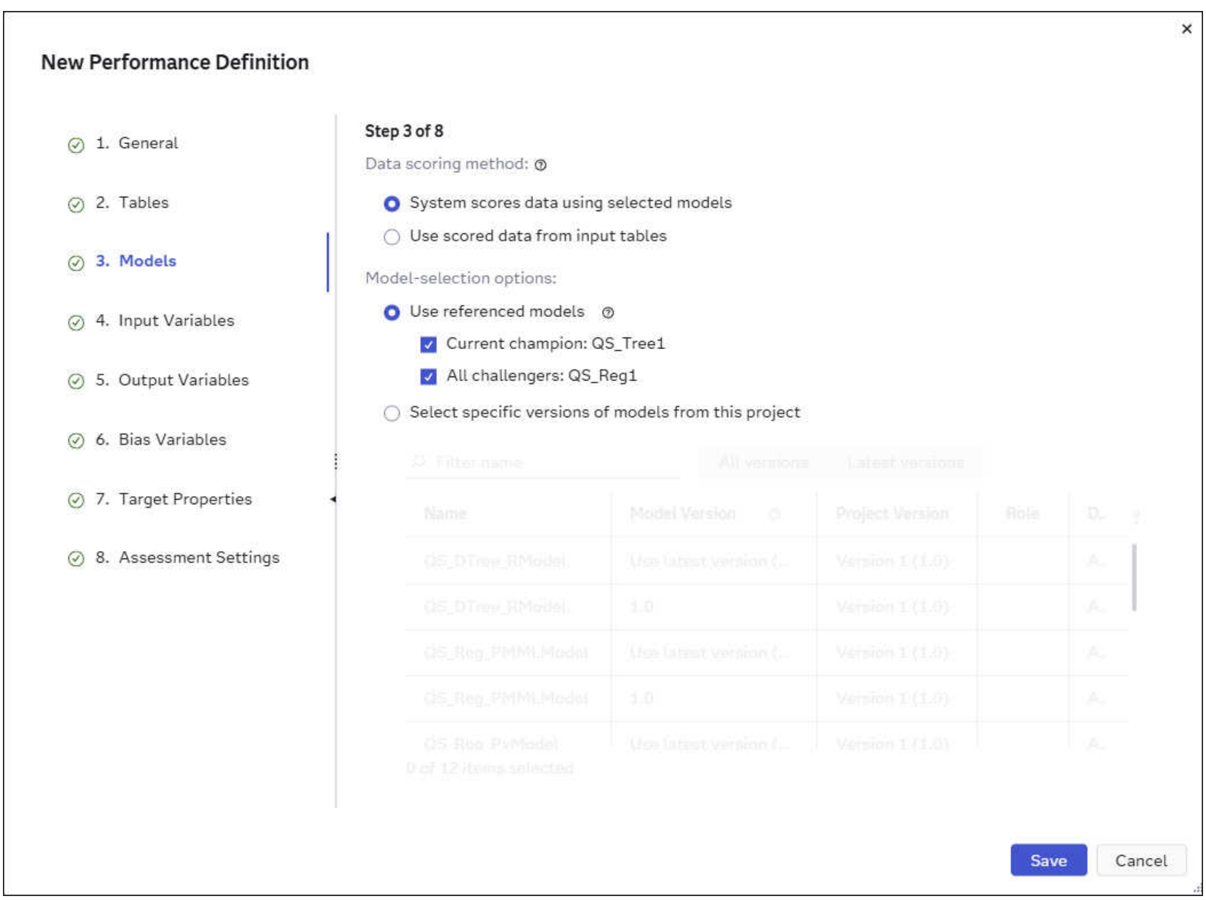
Note: The latest version of the champion and challenger models are used when running performance.

Select specific versions of models from this project

This option enables you to select one or more models and a specific version of each model, or multiple versions of the same model to include in your performance definition for model analysis.

Note: When all model versions are displayed there are two rows for the latest version of the model. In the **Model Version** column, one row contains "Use latest version" and the model version number, and the other contains just the model version number. If you select the row with "Use latest version", the latest version of the model is always used when running performance. If you select the row with a specific version of a model, that model version is always used when running performance, even if newer versions of the model are available.

TIP Click **Latest versions** to display only the current latest version for all of the models within the project.



9 Click the **Input Variables** page to view the input variables to include in the report results.

a Select the variable binning method. The default option is **Adaptive**.

Adaptive
: The variable binning method that recommends bins that have equal width. The common width is a multiple of 1, 2, 2.5, or 5. Therefore, the bin boundaries are positive values that are intuitive and easy to read.

Quantile
: The variable binning method that recommends bins that should have equal count with no special requirements. Because of data distribution, the bin counts might not be exactly equal but similar.

b Select the project input variables that you want to use for analysis when generating the performance results.

> **TIP** By default, all of the project input variables are selected when you are creating a new definition.

c (Optional) You can create custom bins for variables with a decimal data type by adding one or more bin cut-off points. A bin for missing values is created automatically.

The bin cut-off points are used to build bins that cover the range of values for the selected variable. The bins that are built are for missing values, the lower range of values, and the upper range of values, without creating overlapping bins or gaps between bins. Here are some examples:

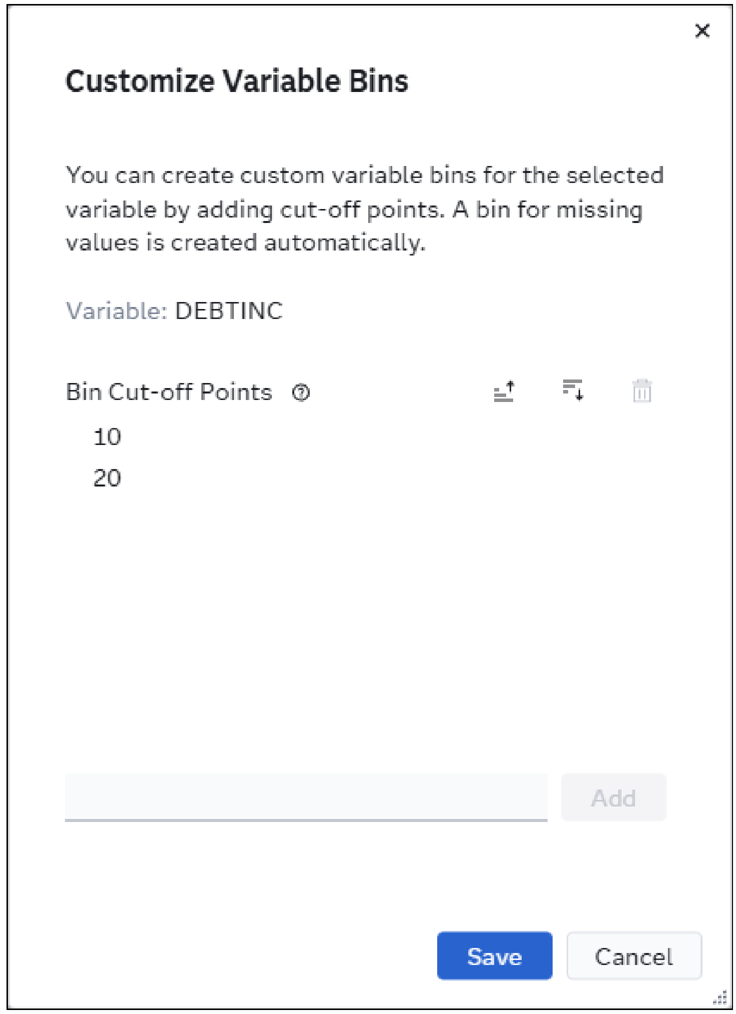
- If a bin cut-off point of X is added, the first bin contains values <= X and the second bin contains values > X. A third bin is also created for missing values, if applicable.
- If two bin cut-off points, X and Y are added, where X < Y, the first bin contains values <= X, the second bin contains values > X and <=Y, and the third bin contains values > Y. A fourth bin is also created for missing values, if applicable.

Create custom bins for a variable:

1 Click in the **Variable Bin Type** column. The Customize Variable Bins window appears.

2 Click **Add cut-off point**.

3 Enter a numeric value, and then click **Add**. Repeat this step to add more bin cut-off points.



4 Click **Save**.

Note: When you save the bin cut-off points, they are automatically sorted in ascending order.

10 Click the **Output Variables** page to view the output variables to include in the report results. Select the project output variables that you want to use for analysis when generating the performance results.

> **TIP** By default, all of the project output variables are selected when you are creating a new definition.

11 (Optional) Click the **Bias Variables** page to view the variables that are marked for bias assessment and will be included in the performance results.

> **TIP** You can modify the variables that are marked for bias assessment on the **Variables** tab of a project.

12 (Optional) Click the **Assessment Settings** page to customize the options that are used for assessment calculations.

Note: Specifying a higher number than the default value for the assessment settings can result in a longer run time, especially when the performance input tables are large. Also, changing the default number of bins in the lift calculation can cause some lift-based KPIs to be unavailable.

13 (Optional) Click the **Target Properties** page to view the properties that are being used to compute stability and accuracy measures to assess model prediction accuracy.

> **TIP** You can modify the values of the target properties on the **Properties** tab of this project. If the values are changed after a performance definition is created, you must edit the performance definition and save it, in order for the changes to take effect.

14 Click **Save**. You are returned to the **Performance** tab.

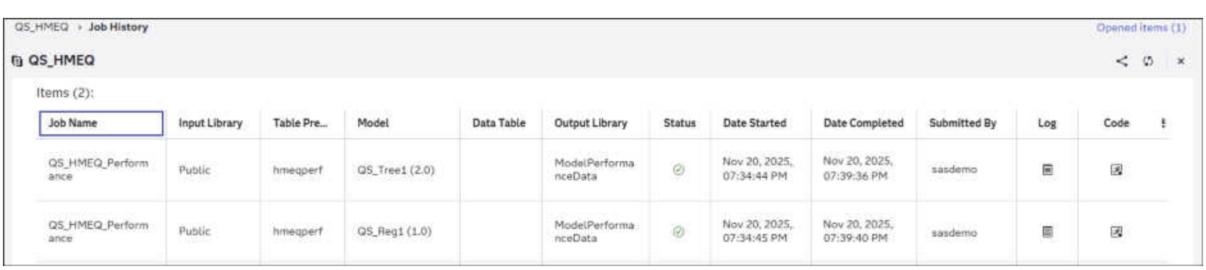
Run Performance and View Results

Note: Data tables are loaded at run time, if necessary, when they are referenced within a performance definition and are not available in a CAS session.

1 On the **Performance** tab, click **Run** and select **Run now**.

> **TIP** You can also schedule performance jobs to run at a specific frequency such as daily, weekly, monthly, or quarterly. Other types of frequencies are also available.For more information, see "Schedule Performance Monitoring" on page 135.

2 Click ⋮ and select **View job history** to view the current status of the performance job.

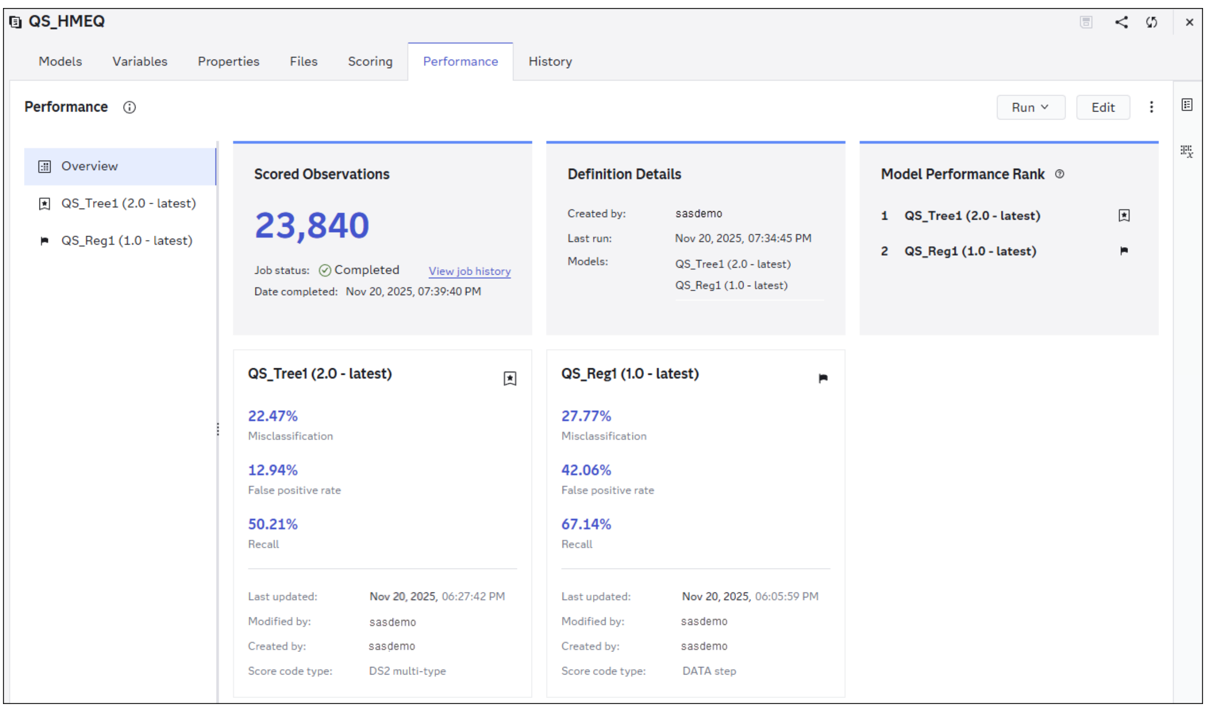


3 When the job is complete, click **Close** to return to the **Performance** tab and view the results.

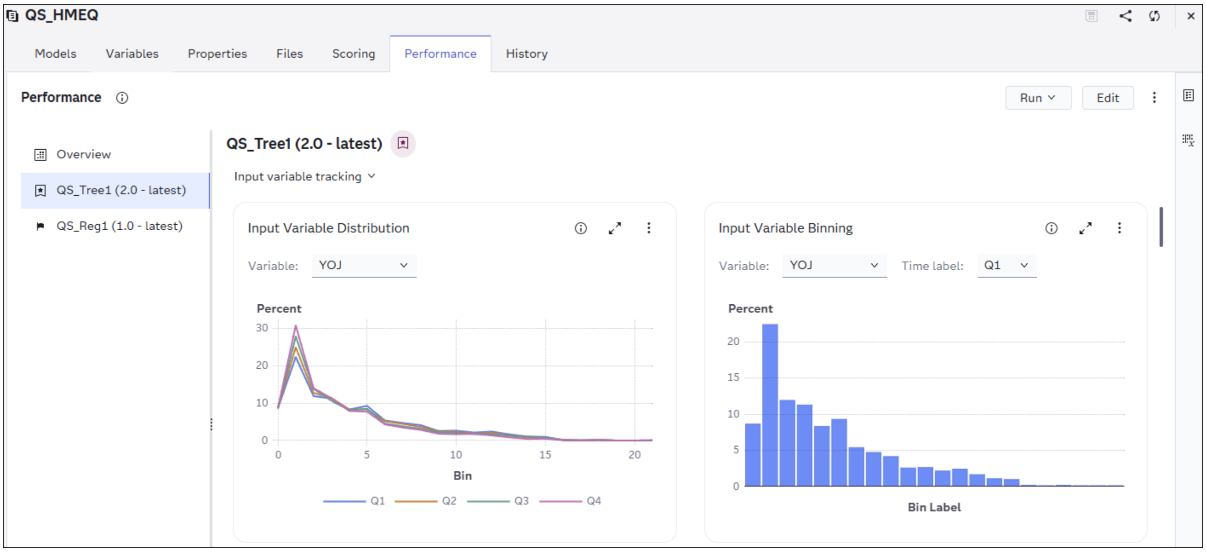
The **Summary** page contains assessment information about the models that are included in the project's performance results. It also includes scoring observations, performance definition high-level details, and the performance rank of models within the containing project.

Note:

The model performance rank is based on the specified model assessment criteria in the project's properties. If the assessment criteria are not specified or user-defined criteria is specified in the project's properties, the misclassification KPI is used to determine the rank of models in a project with a classification model function. Likewise, the average squared error (ASE) KPI is used to determine the rank of models in a project with a prediction model function.



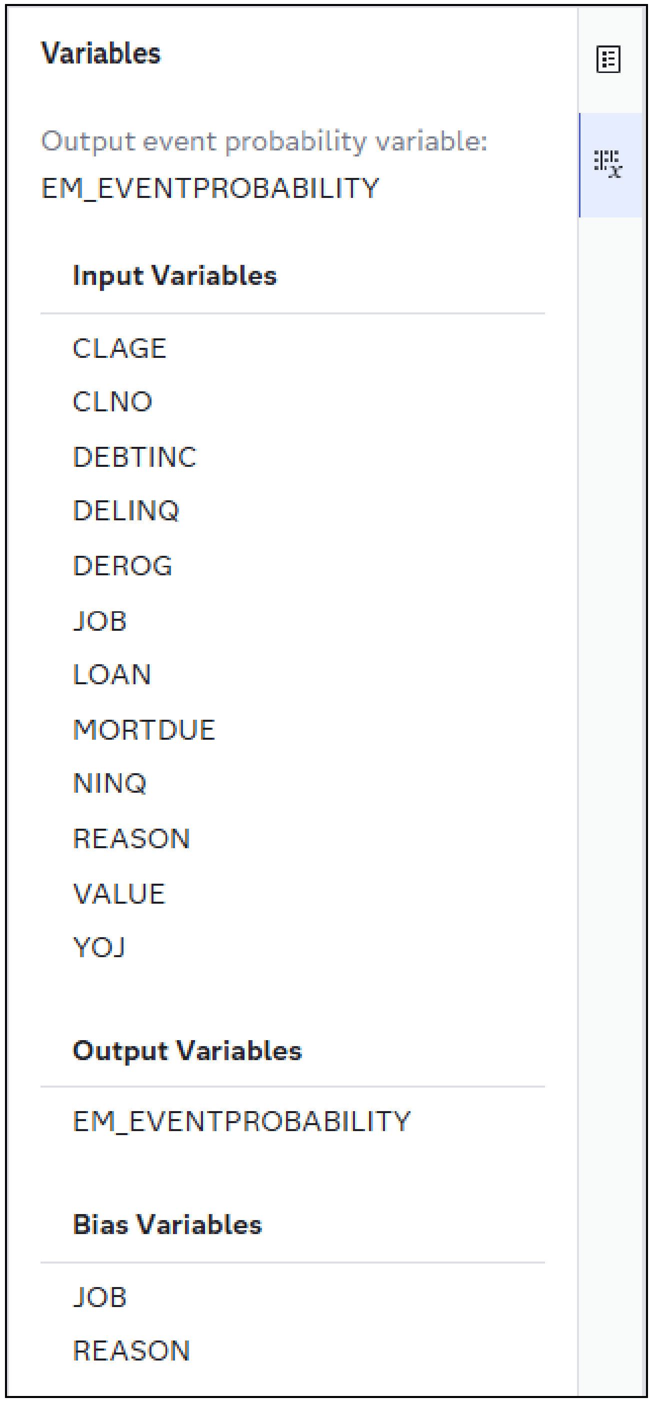
a Click a model object tab to view the performance results for that specific model.



Note: For the Input Variable Distribution and Feature Contribution charts, you can select variables from the drop-down list to change the content of the chart. This is an example of the YOJ variable distribution.

b Click ☷ to view the performance definition details.

c Click to view the variables that are included in the performance definition.



Note: You can conduct partial model performance monitoring based on the input data. If your input data contains only the input variables, then the system computes only the characteristic analysis. If your input data contains the score output variable, then the system also computes the stability analysis. If your input data contains the response variables, then the system also computes the accuracy measures, such as Gini, ROC, Lift, Kolmogorov-Smirnov (KS), Average Squared Error (ASE), and Feature Contribution Index (FCI). Additional charts are available for running performance for a prediction model with an interval target. For more information, see "Model Monitoring" on page 181.

4 (Optional) Click ⋮ and select **Explore and visualize**.

Note: SAS Visual Analytics opens within the same browser window, and the performance data for the associated chart is shown. For more information, see *SAS Visual Analytics: Working with Report Content*.

To return to your project, click ⁝⁝⁝ and select **Manage Models**.

5 Click **Close**.

Edit Performance Definition

Here are some reasons to edit the performance definition content:

- The model role has changed or the model you previously selected is no longer available.
- The project properties or variables have changed.

To edit the definition:

1 Click **Edit**.

2 Modify the performance definition content.

3 Click **Save**.

For information about the definition content, see “Create a New Performance Definition” on page 120.

Clear Performance Results

You can clear all of the performance results and delete the data tables from the CAS output library.

1 On the **Performance** tab, click ⋮ and select **Clear all**.

2 Click **Clear** in the confirmation message.

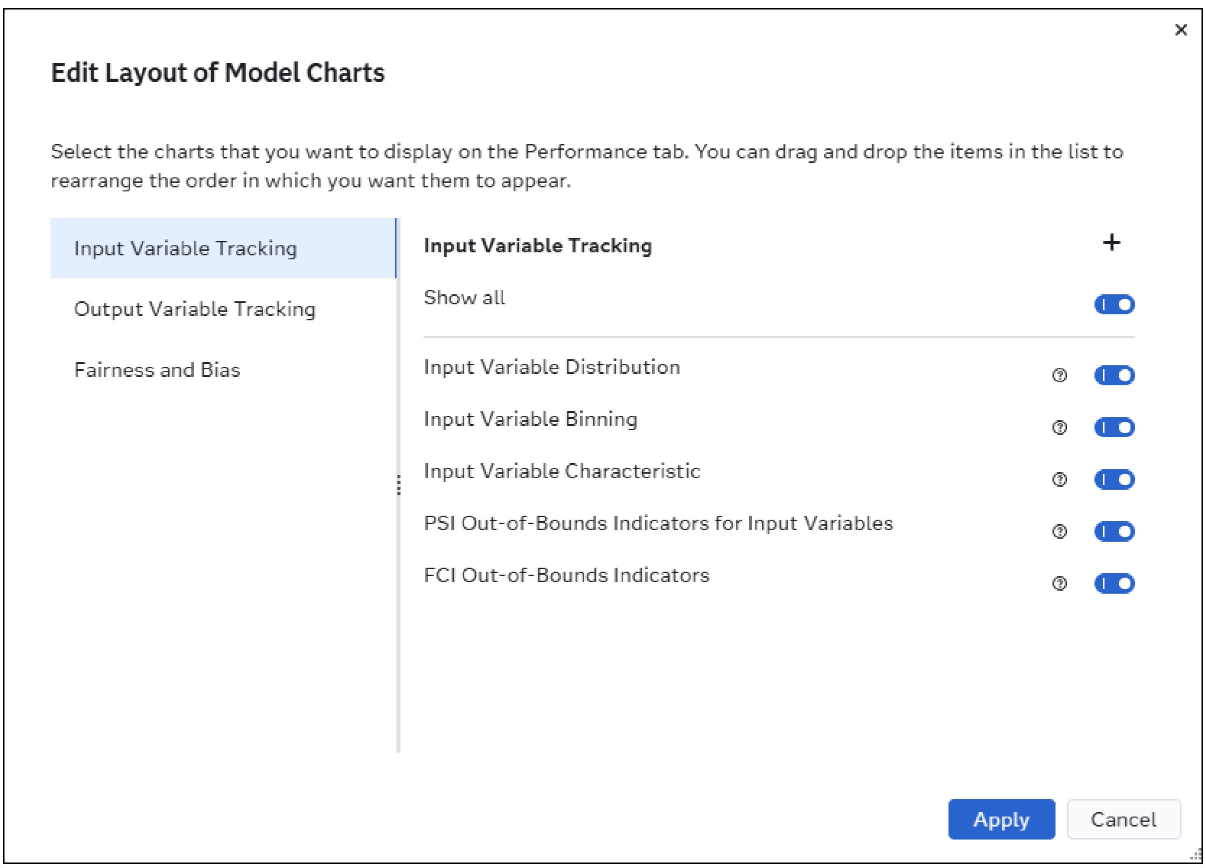
After the performance results are cleared and the data tables are deleted, an information message is displayed indicating that you can edit the definition. Otherwise, click **Run** and select **Run now** to generate the results for the existing definition content. See: “Edit Performance Definition”.

Edit Layout of Performance Results

Edit Layout of Charts

1 On the **Performance** tab, click ⋮ and select **Edit layout**.

Note: Currently, the layout of the **Summary** page cannot be modified. The **Summary**, **Comparison charts**, and **Model charts** submenu options were removed with the 2025.01 release. Also, the **Fairness and Bias** section was added with the 2025.03 release.



2 Select or deselect the charts to be displayed on the **Performance** tab.

3 Arrange the charts to appear in the order that you want by using drag and drop in the **Charts** panel.

TIP Click the Help icon ⓘ next to a chart to view information about that specific chart in the **Charts** panel.

Add Custom Charts

1 Copy an object link from a report in SAS Visual Analytics.

   a Click and select **Explore and Visualize**.

   b Open a report.

   c Click on the object toolbar, and then select **Copy embeddable markup**. The Copy Embeddable Markup window appears.

   Note: If you are using SAS Visual Analytics for the 2025.01 release or an earlier release, then you should select **Copy link**. For more information, see "Copy a Link to an Object" in *SAS Visual Analytics: Designing Reports*.

   d Click **Manage Permissions** to specify who can use the report link. It is recommended to select the **People with existing access** option.

   e Click **Copy markup** and paste the link into a text file so that you can use the link when creating a custom chart in SAS Model Manager.

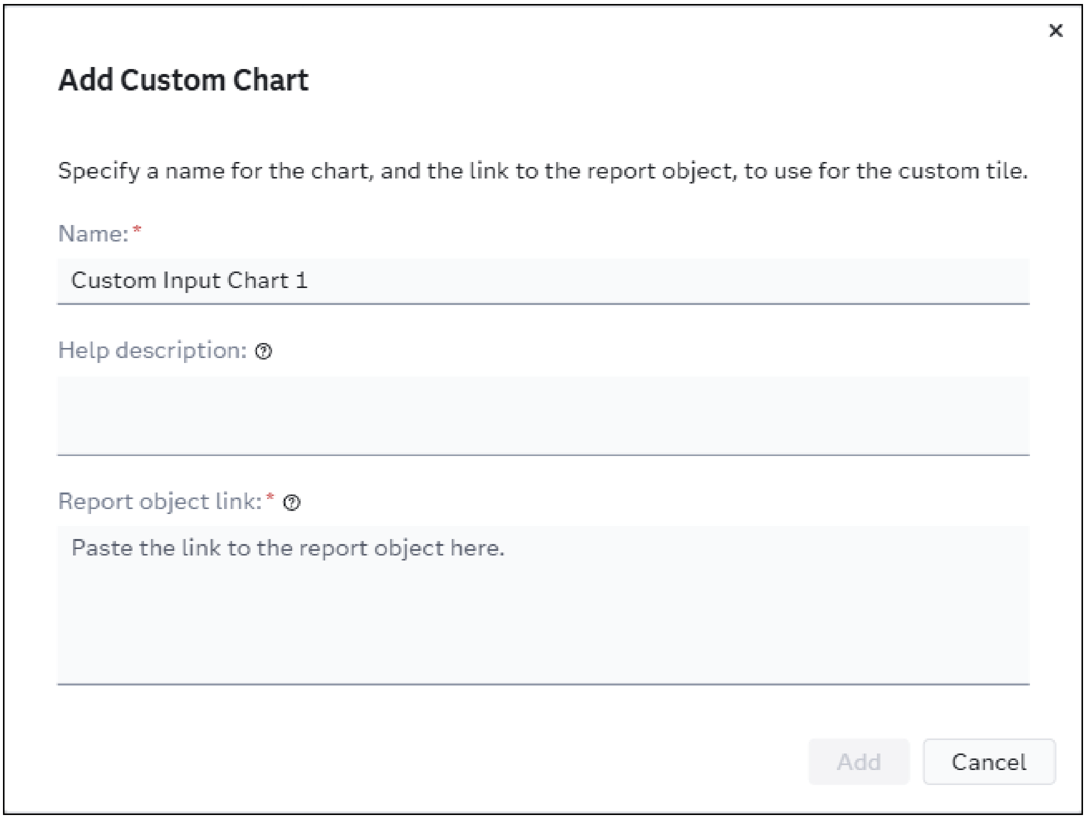
   Note: The SAS Visual Analytics report object that you want to link to must be on the same host as SAS Model Manager.

   For more information, see "Copy Embeddable Markup for an Object" in *SAS Visual Analytics: Designing Reports*.

2 Click and select **Manage Models** to return to SAS Model Manager.

3 On the **Performance** tab of your project, click and select **Edit layout**. The Edit Layout of Model Charts window appears.

4 Click + **Add Custom Chart**. The Add Custom Chart window appears.



5 Paste the report object link in the **Report object link** box.

6 Enter a name for the new custom chart.

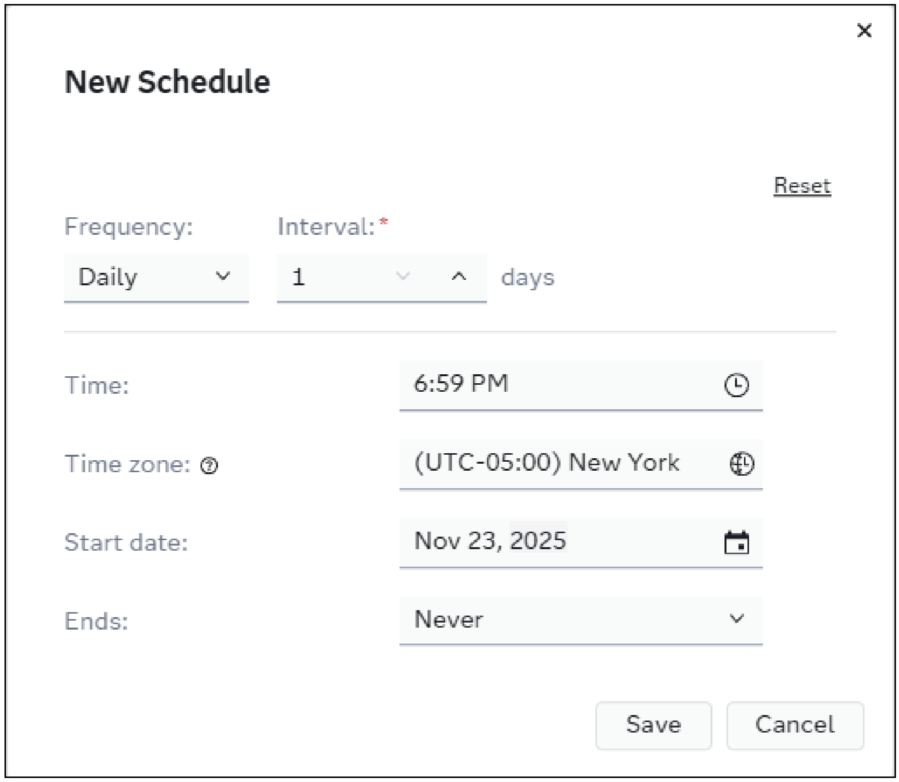
7 Enter a Help description to provide information about your custom chart.

8 Click **Add**.

Schedule Performance Monitoring

After you create a performance definition on the **Performance** tab of a project, you can schedule the performance job to run at a specific frequency such as daily, weekly, monthly, or quarterly. Other types of frequencies are also available.

1 Click **Run** and select **Schedule for later**. The Schedule Job window appears.



2 Select an option from the **Frequency** drop-down list to specify how often to trigger the job to run (such as a specified number of minutes, hourly, or dailly).

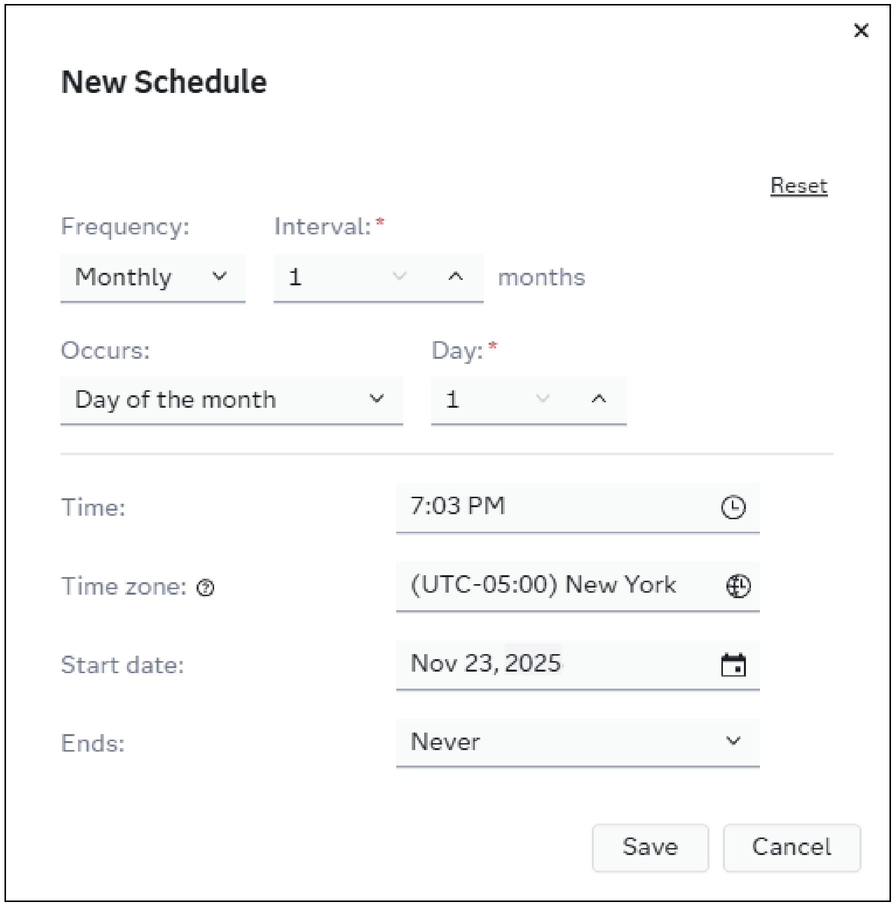
3 Depending on your choice for the frequency interval, different fields appear in the window to enable you to specify a frequency for running the job. For example, if you select **Yearly** in the **Frequency** field, you can specify a day of the month (such as the first of January), the last day of a month, or a specific weekday in a month (such as the third Thursday in February). If you specify **Minutes** in the **Frequency** field, you can specify that the job runs every 5, 10, 15, 20, or 30 minutes. Use these fields to specify the criteria for the trigger interval.

4 In the **Time** field, specify when the job schedule should start. Times are specified in 24-hour format.

5 Specify the time zone to use when evaluating the time for the job schedule and the date on which it starts.

6 Click and select a start date.

7 Specify when the job schedule ends. You can specify that the job schedule never ends, that it ends after a certain number of times, or that it ends on a specific date.



8 Click **Save**.

> **TIP** Only one job schedule can exist for each project performance definition. To view or edit an existing performance job schedule, click **Run** and select **Schedule for later**. Make your changes and click **Save**.

You can also edit or delete a job schedule using SAS Environment Manager. Additional triggers can be added to an existing job schedule only by using SAS Environment Manager. To access SAS Environment Manager, click ⠿ and select **Manage Environment**. For more information, see “Schedule Jobs” in *SAS Environment Manager: User’s Guide*.

CAUTION

Do not delete job requests in SAS Environment Manager. If you delete a job request in SAS Environment Manager, the performance monitoring feature does not work as designed.

Retrain a Project from Model Studio

Only SAS Visual Data Mining and Machine Learning projects that have been registered from Model Studio into the SAS Model Manager common model repository can be retrained. Model studio projects and their models are stored in the `DMRepository` repository folder when they are registered.

To send a retrain request for a Model Studio project:

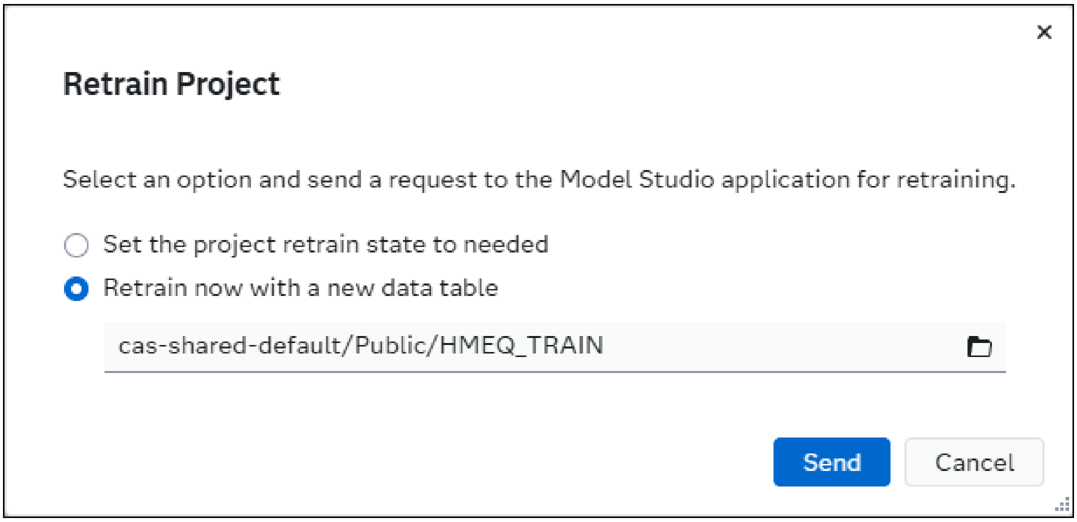
1. Click to navigate to the Projects category view.
2. Open a project that was registered from Model Studio.

   Note: If a project has been registered from Model Studio, an actions menu button appears to the left of the open items icon.

3. Click ⋮ and select **Retrain**.
4. Select an option to send a retrain request to Model Studio.
   - Select the **Set the project retrain state to needed** option if you want to indicate that the project and its models are ready for retrain.
   - Select the **Retrain now with a new data table** option, if you want to select a new data table and send a request to Model Studio to retrain the project now. Click to choose a data table.

     TIP If the desired data table is not available, you can import the table from the Choose Data window. The table is then listed within the results in the Choose Data window and can be selected as the data table for the project. For more information, see "Overview of SAS Data Explorer" in *SAS Data Explorer: User's Guide*.



5 Click **Send**.

Note: When you select the **Retrain now with a new data table** option, all pipelines in the associated Model Studio project are rerun. The previous pipeline results are overwritten. When the pipelines have finished running in Model Studio, a new champion model is selected during pipeline comparison. The new champion model is then auto-registered into a new project version within the associated SAS Model Manager project.

7

Publishing Models

About Publishing Models

You can publish models to a publishing destination so that you can validate the models. You can also access published models from other applications to perform tasks such as scoring. Models can be published to destinations that are defined for CAS, Databricks, Event Stream Processing, Git repository, Hadoop, SAS Micro Analytic Service, SingleStore, and Teradata. They can also be published to an Azure Machine Learning (AML) cloud-based provider, and containers such as Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), and Private Docker. For information about defining publishing destinations, see “Configuring Publishing Destinations” in *SAS Model Manager: Administrator’s Guide*.

Here are the different locations that models can be published from within the SAS Model Manager web application:

- The project champion model and its challengers can be published from the Projects category view.
- The **Models** tab of a project.
- An open model object, if the model is located within a project version.

Note: For information about what types of models are supported for publishing, see "High-Level Model Support Matrix for Primary Functions" on page 7 and "Publishing Requirements and Restrictions" on page 143.

The **SAS Micro Analytic Service (maslocal)** publishing destination is available by default. Other publishing destinations can be defined by a SAS administrator. A global caslib is required for CAS, Hadoop, and Teradata publishing destinations. For more information, see "Configuring Publishing Destinations" in *SAS Model Manager: Administrator's Guide*.

Here are several things to keep in mind:

- When you publish a model from a project version, the system creates a publishing validation test.

  Note: A Git publishing destination is a repository, and thus it has no means of executing code. Therefore, when a model is published to a Git destination, no publishing validation test is created.

- You can edit the publishing validation test to select a test data table and output library.
- Model validation runs within the publishing destination that the model was published to.

For more information, see "Validate Published Models" on page 112 and "Schedule Jobs" in *SAS Environment Manager: User's Guide*.

See Also

- "Publishing Python Models"
- "Publishing R Models"

Publishing Requirements and Restrictions

Model Score Code Types and Publishing Destinations

Before you can publish a model, you must first set its score code type in the model's properties and make sure that the score code file role is set. Only models with a score code type of DATA step, DS2 package, DS2 embedded process, DS2 multi-type, Analytic store, Python, and R and that contain a valid score code file can be published.

Here are a few things to be aware of:

- The model score code type can affect which destinations you can publish your models to.
- A model with the score code type of DS2 multi-type can contain code files for a DS2 embedded process, a DS2 package, and one or more analytic stores. For example, when SAS Visual Data Mining and Machining Learning models that contain an analytic store are registered into the common model repository, their score code type is set to DS2 multi-type, instead of Analytic store.
- Models that have a score code type of Python or R can be published if the score code is in the correct format. For more information, see "Scoring Python Models" on page 170 and "Scoring R Models" on page 173.
- In order to validate models with a score code type of Python within a container publishing destination, you must first add a requirements.json file to the model before you publish it. The requirements.json file should include the install statements for the Python packages, which contain the modules that are used in the Python score code file and its score resource files. For more information, see "Scoring Python Models" on page 170.
- Predictive Model Markup Language (PMML) is an XML-based predictive model interchange format. Some classification and prediction models that you create using PMML 4.2 are converted to DATA step score code during the model import process. When you are importing valid PMML models, the score code type model property is set to DATA step, instead of PMML. PMML models with a score code type of DATA step can be scored and published. For more information, see "Import Models" on page 63 and "Concepts: PMML Support" on page 220.

Here is a list of the model score code types and the types of destinations that you can publish your models to:

Note: The supported container publishing destinations are Amazon Web Services, Azure, Google Cloud Platform (GCP), and Private Docker. When open-source (Python or R) models are published to a container publishing destination, a runtime container is created using an open-source base image. When SAS Viya models are published to a container publishing destination, a runtime container is created using the SAS Container Runtime base image. For more information, see "Container Base Images for Scoring" in *SAS Viya Platform: Publishing Destinations*.

Model Score Code Type	CAS	Git	Hadoop and Teradata	SAS Micro Analytic Service	AML	Container Destinations	ESP[2]
Analytic store	No	Yes	No	No	No	Yes	No
DATA step	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DS2 embedded process	Yes	Yes	Yes	No	Yes	No	No
DS2 multi-type	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DS2 package	No	Yes	No	Yes	Yes	Yes	Yes
PMML	No	No	No	No	No	No	No
Python	Yes	Yes	No	Yes	Yes	Yes[1]	No
R	Yes	Yes	No	No	No	Yes[1]	No
SAS program	No	No	No	No	No	No	No

1 SAS Viya models that are built using pipelines by combining SAS procedures, CAS actions, and open-source code (Python or R) cannot be published to a container destination.

2 Event Stream Processing (ESP) publishing destinations are supported starting with the 2025.12 stable release.

Table 7.1 Publishing Destinations that Use PROC ACCELERATOR

Model Score Code Type	Databricks	SingleStore[1]
Analytic store	No	No
DATA step	Yes	Yes
DS2 embedded process	Yes	Yes
DS2 multi-type	Yes	Yes
DS2 package	No	No
PMML	No	No

Model Score Code Type	Databricks	SingleStore[1]
Python	No	No
R	No	No
SAS program	No	No

1 SingleStore publishing destinations are supported starting with the 2025.12 stable release.

Note: Here are the types of SAS models that are currently supported for publishing to container destinations:

- SAS Visual Data Mining and Machine Learning models that are registered from Model Studio, SAS Visual Analytics, or SAS Studio
- SAS Visual Statistics models that are registered from SAS Visual Analytics or SAS Studio
- SAS Visual Text Analytics models that are registered from Model Studio or SAS Studio

  Note: This is supported with the 2024.12 stable release.
- DATA step models in SAS package files that are created using SAS Enterprise Miner and then imported into SAS Model Manager

For more information, see "Set Model General Properties" on page 85.

Published Name

The maximum length for all published names is 100 characters. Character restrictions differ depending on your destination. See Table 7.2.

Table 7.2 Requirements and Restrictions for Published Names

Destination	Requirements and Restrictions
Container destinations	The published name must start with a letter or an underscore. It cannot contain spaces, multi-byte characters, or special characters other than the underscore. **Note:** ▪ The supported container destinations are Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), and Private Docker. ▪ When publishing a model to an AWS container destination, the published name can contain only alphanumeric characters.

<table>
<tr><th>Destination</th><th>Requirements and Restrictions</th></tr>
<tr><td></td><td>■ When publishing SAS models, the published name that you enter is prepended with a forward slash (/) and then is assigned to the SAS_SCR_APP_PATH environment variable. The value of this variable determines the module name. For more information, see “Changing the Endpoint Name for a Container” in SAS Container Runtime: Programming and Administration Guide.</td></tr>
<tr><td>Azure Machine Learning</td><td>The published name must start with a letter or an underscore. It cannot contain spaces, multi-byte characters, or special characters other than the underscore.</td></tr>
<tr><td>Git</td><td>The published name cannot contain colons (:) or double quotation marks.</td></tr>
<tr><td>SAS Cloud Analytic Services (CAS)</td><td>The published name cannot contain single or double quotation marks.</td></tr>
<tr><td>SAS Micro Analytic Service</td><td>The published name cannot contain the following characters: ! @ # $ % ^ & * ( ) | = ~ ` \ / . { } " ' ;</td></tr>
<tr><td>Databricks</td><td>The published name cannot contain the following characters: ! @ # $ % ^ & * ( ) | = ~ ` \ / . { } " ' ;</td></tr>
<tr><td>Event Stream Processing</td><td>The published name cannot contain the following characters: ! @ # $ % ^ & * ( ) | = ~ ` \ / . { } " ' ;</td></tr>
<tr><td>Hadoop</td><td>The published name cannot contain colons (:) or double quotation marks.</td></tr>
<tr><td>SingleStore</td><td>The published name cannot contain the following characters: ! @ # $ % ^ & * ( ) | = ~ ` \ / . { } " ' ;</td></tr>
<tr><td>Teradata</td><td>The published name must start with a letter or an underscore. It cannot contain spaces, multi-byte characters, or special characters other than the underscore.</td></tr>
</table>

Publish a Project Champion Model

1 Click to navigate to the Projects category view.

2 Select a project, click ⋮, and select **Publish**. The **Publish Models** window appears.

Note: The project champion model and its challengers appear in the **Items to Publish** section.

3 Select a destination.

Note: If you have Read and Write permissions to the caslib that is specified in a publishing destination, the destination is shown in the list.

For more information, see *SAS Viya: CAS Authorization Window* and "Configuring Publishing Destinations" in *SAS Model Manager: Administrator's Guide*.

TIP Click **Details** to view the destination details.

4 (Optional) Assign tags to the published models by entering a tag value. Press Enter after each tag.

TIP

- Tag names cannot start with a period or a hyphen, and cannot contain more than 128 characters. They can contain only alphanumeric characters, the underscore (_), the hyphen (-), and the period (.).
- When publishing a model to a container destination the model version is automatically added as a tag for the published model within the target destination. You have the option of removing the `latest` tag before publishing your models.

5 Specify a value for any properties that might appear after you select a destination.

Property	Description	Destination Type	Required
Azure Machine Learning workspace	Indicates the workspace in the Azure Machine Learning environment to publish the selected items to. Select a value from the drop-down list.	Azure Machine Learning	Yes
Git directory	Indicates the directory path within the Git repository to publish the selected items to.	Git	No

Property	Description	Destination Type	Required
	Click and navigate to a folder within the Git repository. Click **OK**. **Note:** If a folder is not selected within the Git repository, the items are published to the root directory.		

6 For each model in the **Items to Publish** section, perform these steps:

a Select a model version to publish. The default value is the latest version of the model.

b Choose whether to create a new model version after publishing the model. The option to create a new model version is selected by default.

Note: If you choose to not create a new model version, the content of the model version that was published is locked and cannot be edited.

c (Optional) Edit the **Published name** if you do not want to use the default name for the published model. The maximum length and character restrictions differ, depending on your destination.

For more information, see Table 7.2 on page 145.

d (Optional) If you have previously published the project champion model or its challengers, enable the **Replace item with the same name** toggle in order to replace the previously published item of the same name in the same destination. Otherwise, an error message appears, indicating that the published name must be unique, when publishing a model to a CAS, SAS Micro Analytic Service, Hadoop, or Teradata publishing destination.

Note: When publishing models to a container, an Azure Machine Learning, or a Git publishing destination, if a published model with the same name already exists within the destination, a sequential number is added as a tag to indicate the version of the published model.

7 Click **Publish**. The Publishing Results window appears. The status of the publishing request is displayed in the **Status** column.

Note: When you select a CAS destination and click **Publish**, the CAS destination table is automatically reloaded and the newly published item is made available to other applications.

When you are publishing to a SAS Micro Analytic Service destination, the **Micro Analytic Module** column is also displayed with a URL to the published model.

8 When the status changes to `Published successfully`, click **Close**.

> **TIP** If you close the Publishing Results window before publishing has completed, you can still see whether the model was published successfully on the **History** tab of the project.

Note: The following is true for models that are within a project version: after a model has been published, a new version of the model is created and a publishing validation test is also created. For more information, see "Managing Model Versions" on page 96 and "Validate Published Models" on page 112.

Publish Models from a Project Version

You can publish models from the **Models** tab of a project, including the champion model.

> **IMPORTANT** Before you can publish a Python model or an R model, the source code must be in the correct format. For more information, see "Concepts: Open-Source Models" on page 169.

1 Click to navigate to the Projects category view.

2 Open a project.

3 (Optional) Select a version from the **Versions** drop-down list. By default, the models within the current displayed version appear in the list.

4 Select one or more models on the **Models** tab, click ⋮ , and select **Publish**. The Publish Models window appears.

5 Select a destination.

Note: If you have Read and Write permissions to the caslib that is specified in a publishing destination, the destination is shown in the list.

For more information, see *SAS Viya: CAS Authorization Window* and "Configuring Publishing Destinations" in *SAS Model Manager: Administrator's Guide*.

> **TIP** Click **Details** to view the destination details.

6 (Optional) Assign tags to the published models by entering a tag value. Press Enter after each tag.

> **TIP**
>
> - Tag names cannot start with a period or a hyphen, and cannot contain more than 128 characters. They can contain only alphanumeric characters, the underscore (_), the hyphen (-), and the period (.).
> - When publishing a model to a container destination the model version is automatically added as a tag for the published model within the target destination. You have the option of removing the `latest` tag before publishing your models.

7 Specify a value for any properties that might appear after you select a destination.

Property	Description	Destination Type	Required
Azure Machine Learning workspace	Indicates the workspace in the Azure Machine Learning environment to publish the selected items to. Select a value from the drop-down list.	Azure Machine Learning	Yes
Git directory	Indicates the directory path within the Git repository to publish the selected items to. Click 📁 and navigate to a folder within the Git repository. Click **OK**. **Note:** If a folder is not selected within the Git repository, the items are published to the root directory.	Git	No

8 For each model in the **Items to Publish** section, perform these steps:

a Select a model version to publish. The default value is the latest version of the model.

b Choose whether to create a new model version after publishing the model. The option to create a new model version is selected by default.

Note: If you choose to not create a new model version, the content of the model version that was published is locked and cannot be edited.

c (Optional) Edit the **Published name** if you do not want to use the default name for the published model. The maximum length and character restrictions differ, depending on your destination.

For more information, see Table 7.2 on page 145.

d (Optional) If you have previously published the project champion model or its challengers, enable the **Replace item with the same name** toggle in order to replace the previously published item of the same name in the same destination. Otherwise, an error message appears, indicating that the published name must be unique, when publishing a model to a CAS, SAS Micro Analytic Service, Hadoop, or Teradata publishing destination.

Note: When publishing models to a container, an Azure Machine Learning, or a Git publishing destination, if a published model with the same name already exists within the destination, a sequential number is added as a tag to indicate the version of the published model.

9 Click **Publish**. The Publishing Results window appears. The status of the publishing request is displayed in the **Status** column.

Note: When you select a CAS destination and click **Publish**, the CAS destination table is automatically reloaded and the newly published item is made available to other applications.

When you are publishing to a SAS Micro Analytic Service destination, the **Micro Analytic Module** column is also displayed with a URL to the published model.

10 When the status changes to `Published successfully`, click **Close**.

Note: The following is true for models that are within a project version: after a model has been published, a new version of the model is created and a publishing validation test is also created. For more information, see "Managing Model Versions" on page 96 and "Validate Published Models" on page 112.

View Model Publish History

When a user publishes a model, an event is added to the **History** tab of the project.

Note: Only models that are associated with a project can record an event on the **History** tab.

1 Click to navigate to the Projects category view.

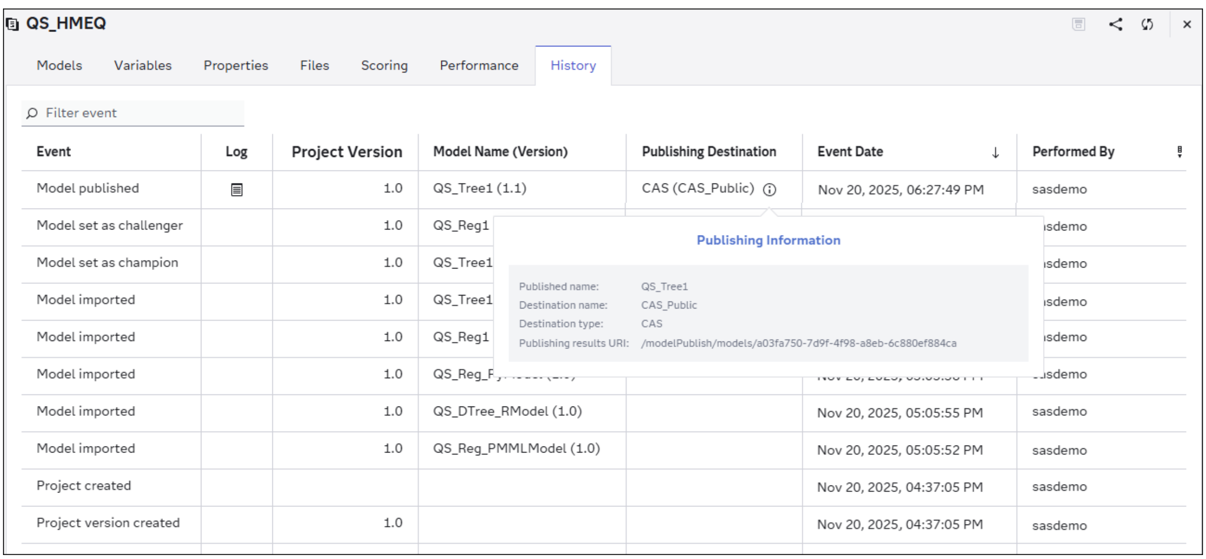
2 Open a project.

3 Click the **History** tab of the project.

4 (Optional) Enter `published` in the **Search event** box to search for models that have been published.

5 Click in the **Publishing Destination** column to view the publishing information for a model.

Figure 7.1 *Example of the Publishing Information for a CAS Publishing Destination*



6 Click anywhere on the screen to close the information pop-up window.

7 Click to view the Model Publish API response log in JSON format.

Deploy a Published Model to Azure Kubernetes Service (AKS)

After you publish a model to an Azure Machine Learning publishing destination you can deploy the published model to AKS. This enables you to make the published model available so that you score it using custom code or another application outside the SAS Viya environment. It also enables you to deploy the published model in a production environment.

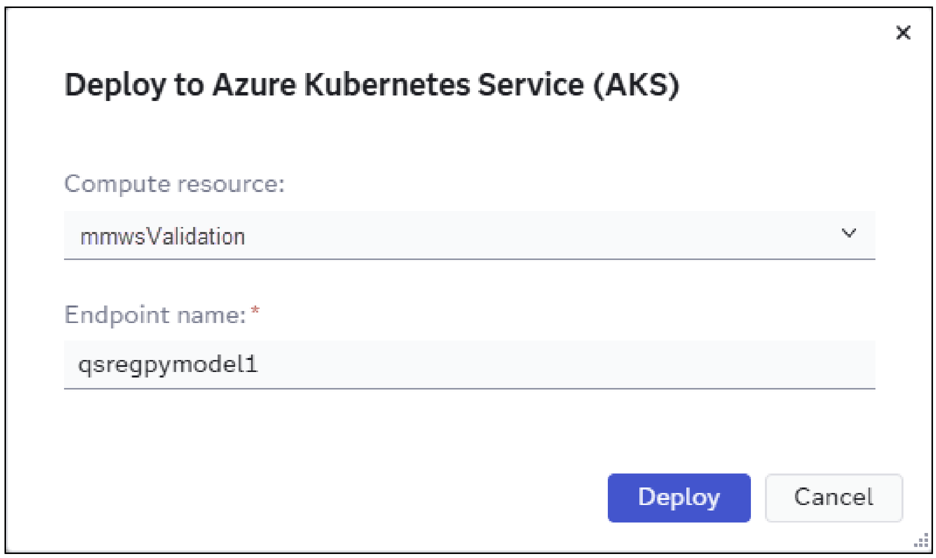
1 Click the **Scoring** tab of a project, and then click the **Publishing Validation** tab.

2 Select a publishing validation test, click ⋮, and select **Deploy to AKS**. The Deploy to Azure Kubernetes Service (AKS) window appears.

3 Select a compute resource that you have access to.

4 Enter a value for the endpoint name.

Note: The endpoint name can contain only alphanumeric characters, and letters must be lowercase.



5 Click **Deploy**.

Note: A toast message is displayed indicating that the published model is being deployed from the Azure Machine Learning to Azure Kubernetes Service (AKS). You can check the deployment status within the Azure Machine Learning destination.

Deleting Published Models

Models that have been published cannot be deleted or removed from a publishing destination using the SAS Model Manager web application. However, you can submit a Model Publish API request to delete a published model from a CAS, SAS Micro Analytic Service, Hadoop or Teradata publishing destination.

Here is a high-level summary of what is deleted for specific destination types:

- The published model is deleted from the CAS model table for CAS and Teradata publishing destinations.

- The published model is deleted from the HDFS directory for a Hadoop publishing destination.
- The micro analytic module that contains the published model is deleted from the SAS Micro Analytic Service publishing destination.

Note: Models that are published to container destinations cannot be deleted from SAS Model Manager or using the Model Publish API service. Users must manually delete the models from their container registry that is associated with the specific container publishing destination.

For information about publishing destinations, see *SAS Viya Platform: Publishing Destinations*.

8

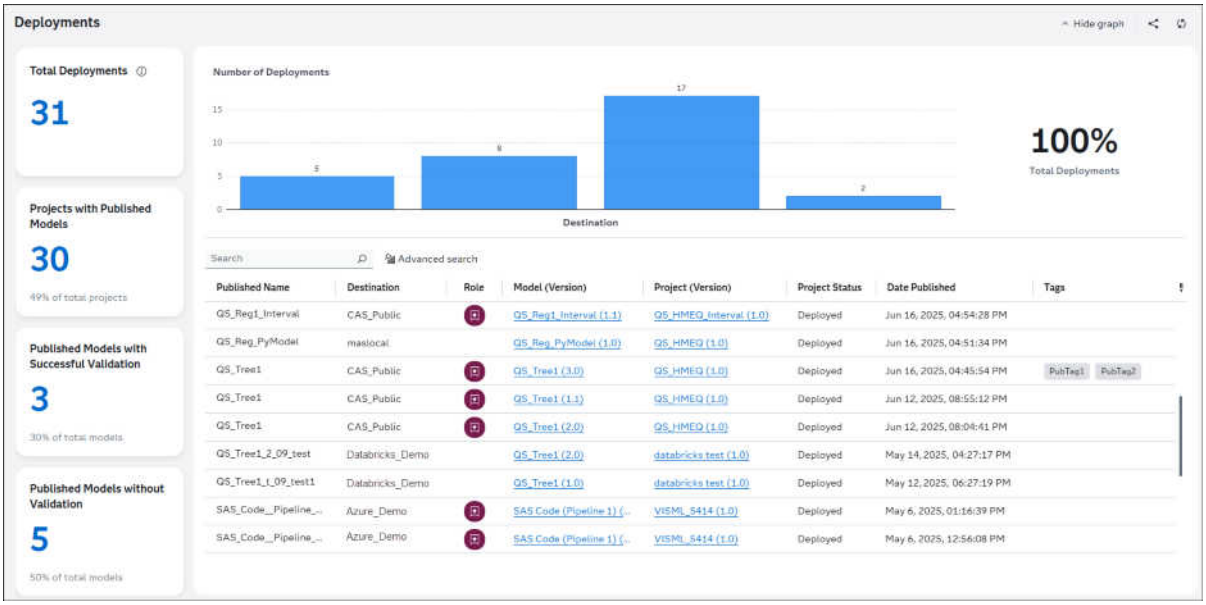
Tracking Deployments

About Deployments

The goal of a modeling project is to identify models that can predict the best data-driven outcomes and deploy them in critical business systems. SAS Model Manager provides tools to evaluate candidate models, declare champion models, and inform your scoring officer that a predictive model is ready for validation or production.

The Deployments category view enables you to access all of your model deployments in one place, view key metrics about your organization's model deployments, and track their status.

Note: Deployments are not available until after you have published models from a project.



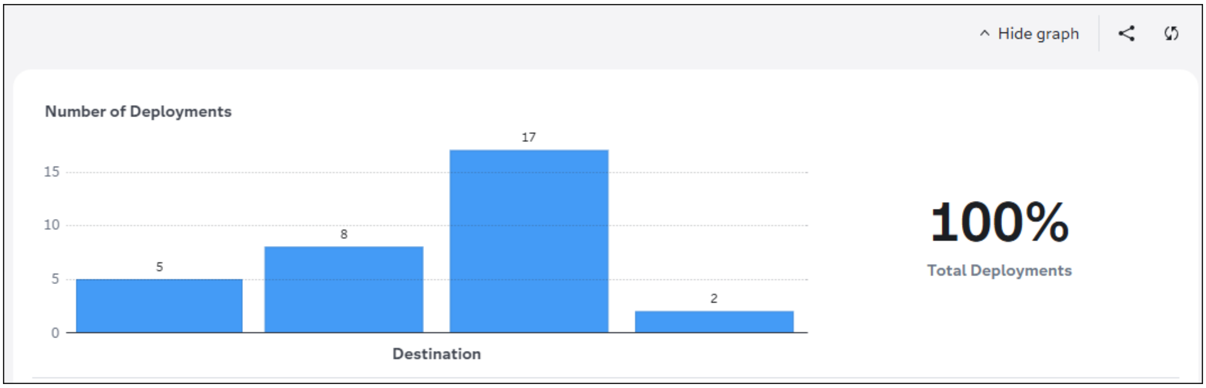
View and Filter Deployments

View Deployments

When you view the Deployments category, a graph appears at the top of the application window and tiles with key information are displayed in the left panel. The graphs and tiles provide the following metrics:

- total deployments
- number of deployments for each destination
- percentage of total deployments
- number of projects and percentage of projects with published models
- number of published models and percentage of published models with successful validation
- number of published models and percentage of published models without validation

You can click **Hide graph** or **Show graph** on the toolbar. You can also share a link to the Deployments category by clicking the icon.

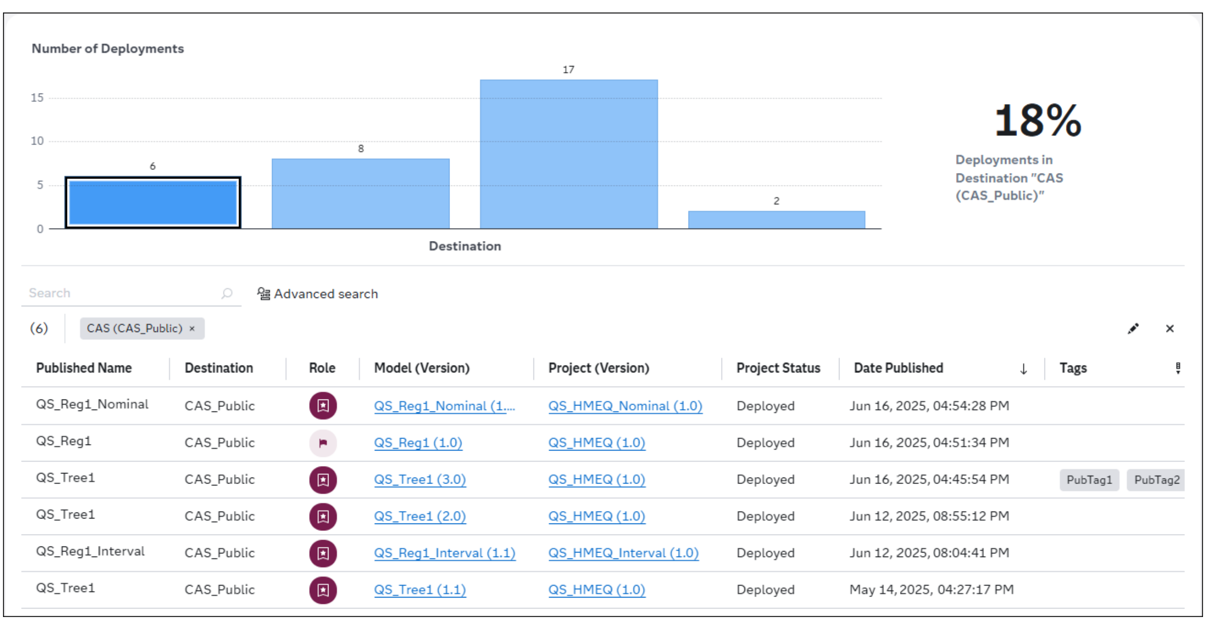


Filter Deployments Using the Graph

You can click sections of the graph to filter the list of models. Tokens are added to the search bar above the list of items.

> **TIP** To select more than one bar in the graph select CTRL+

Here is an example of filtering the list to display only deployments for a specific destination.



You can also use the Advanced Search for Deployment Models window to specify additional information in order to further filter the list. The filters that you applied using the graph are reflected in this window. For more information, see "Advanced Search for Models" on page 99.

TIP To remove an individual filter from a graph, click the empty area beside the graph or click the × icon within the tokens that are located above the list of deployments. To remove all filters, click the × icon, which is located in the right side of the toolbar.

View Deployment Related Content

Values in some of the columns are links that you can click to display the associated content. Here is the content that is displayed for each type of link:

Column Name	Content Displayed
Model (Version)	Opens the associated model object.
Project (Version)	Opens the associated project and displays the **Models** tab.

Search for Deployments

In the **Deployments** category view, you can perform multiple types of searches:

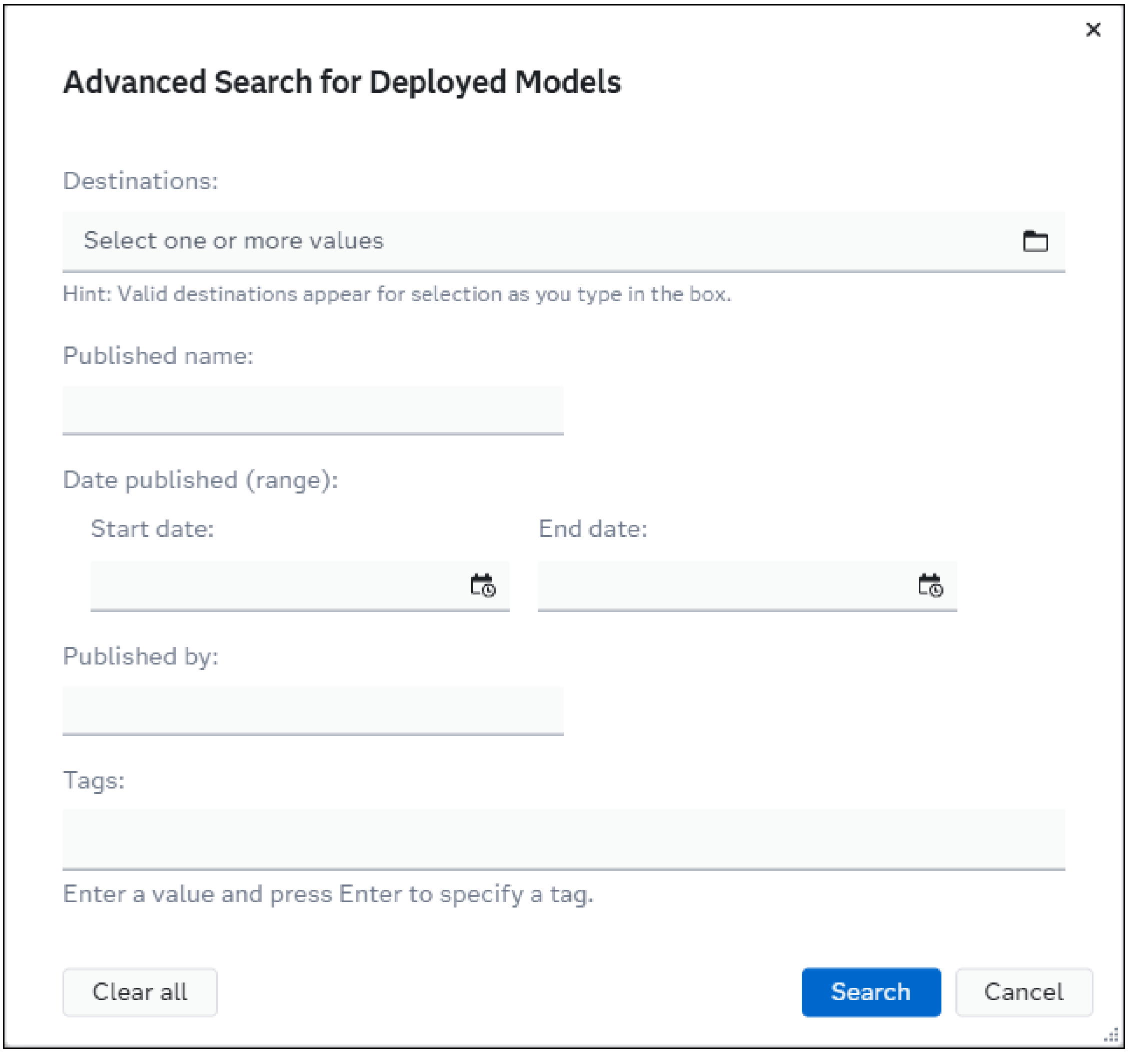
- search for deployments by destination and published name using the search field above the list.
- perform an advanced search for deployments using destinations, published name, date published (range), or published by properties.
- search for objects across applications using the search field in the application bar.

  For more information about searching, see "Search" in *SAS Viya Platform: General Usage Help for Web Applications*.

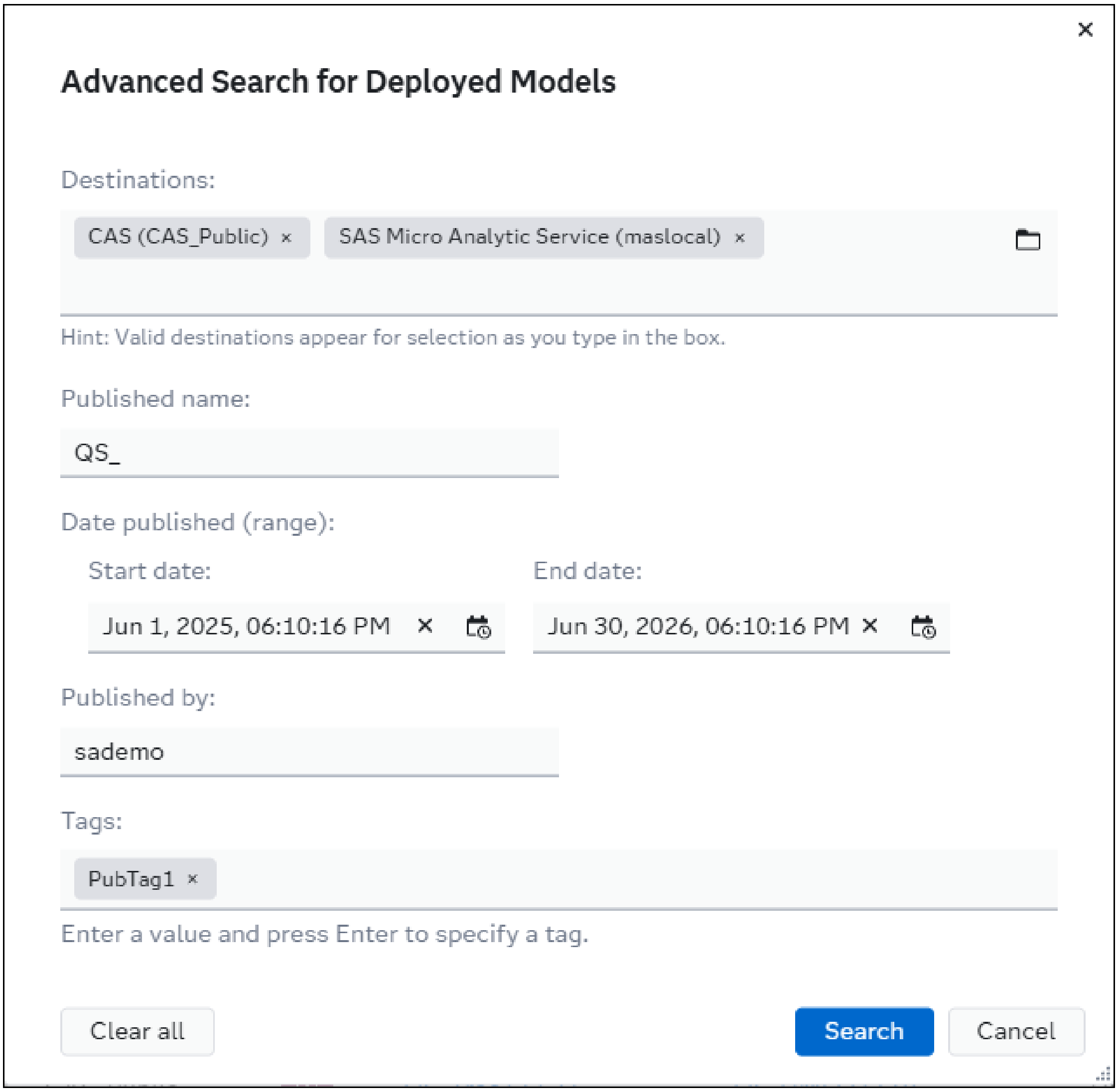
Advanced Search for Deployed Models

To perform an advanced search for models:

1. Click **Advanced search**, which is to the right of the search box. The Advanced Search for Deployed Models window appears.

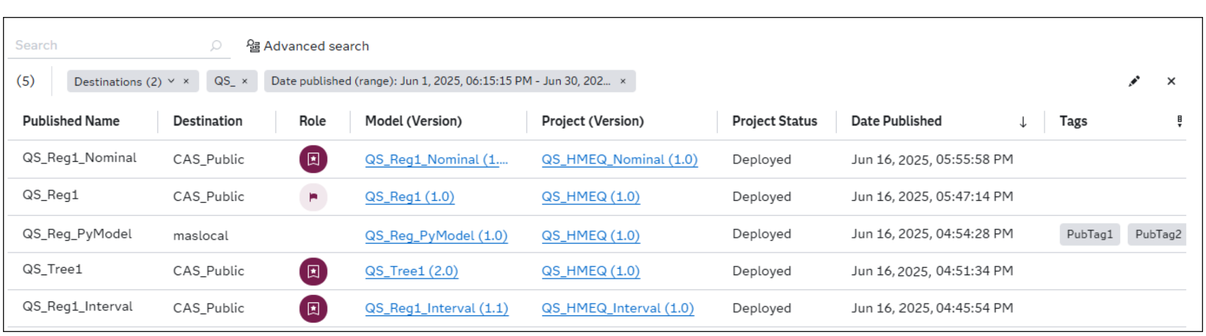


2. To search for deployed models within specific destinations, enter a destination in the **Destinations** box and select it from the list that appears.
3. Enter a published name or part of a published name to include it in the search criteria.
4. Specify a range for the date published in order to limit the list of deployments to specific dates and times.
5. Enter a user ID to include the user that the models were published by in the search criteria.
6. Enter a tag value and press Enter, to search for a tag that is assigned to a published model.



7 Click **Search**. The Deployments category view is displayed with the list of deployments that meet the advanced search criteria.

> **TIP** Click **Clear All** and then click **Search** to clear all previously entered search criteria in the Advanced Search for Deployed Models window.



9

Using SAS Workflow with SAS Model Manager

About Using Workflows

SAS Model Manager uses the **Workflow** tab within a project and the Tasks category view to interface with SAS Workflow. A workflow is an instance of a workflow definition. A workflow can be used to track the progress of objects, such as projects. An authorized user can use SAS Workflow Manager to create workflow definitions and to make them available to SAS Model Manager for use. Workflow definitions contain the set of tasks, participants, and data objects that comprise a business task. The status that you select when completing a task determines the next task in the workflow. All users can access the Tasks category view.

Note: Workflow definitions that are used by SAS Model Manager must contain at least one task.

> **TIP** There is a sample workflow definition available with the *SAS Model Manager: Quick Start Tutorial* on support.sas.com and additional workflow definition samples in the Model Management Resources GitHub repository.

For information about creating workflow definitions, see *SAS Workflow Manager: User's Guide*.

Requirements

Before users can use SAS Workflow with SAS Model Manager, a system administrator must complete the following tasks:

- Configure a user account for a workflow client to have service execution privileges. The Workflow service uses this account to make service task calls when running a workflow instance.
- Grant users or a user group workflow administrative privileges.
- Specify the SAS Model Manager client identifier and enable a workflow definition to make it available to SAS Model Manager.

  Note: The developer of a workflow definition can also complete this task.

For more information, see "Configuring SAS Workflow" in *SAS Model Manager: Administrator's Guide*.

Prompts and Attributes

You can add prompts and attributes to the Start element and user tasks in order to customize fields within workflow tasks. SAS Model Manager provides attributes that enable you to prompt users to select a model or select potential owners from a list within the Start element or a user task as part of the workflow process.

Here are the controls and attribute values that are supported:

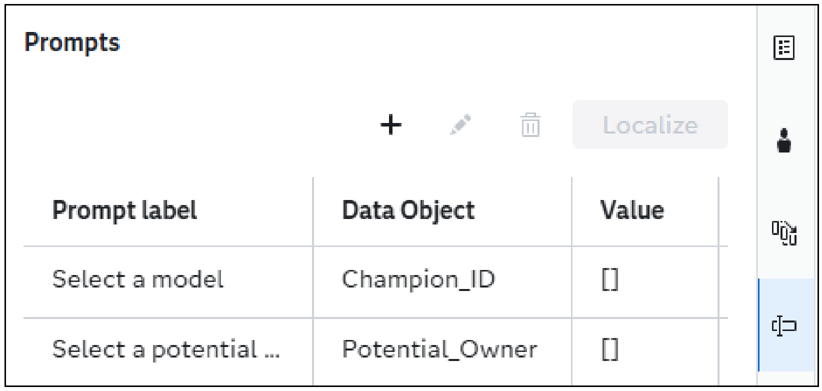
Type of Control	Attribute Value	Description
Choose Models	show_model_list	The show_model_list attribute value and the associated control enable you to configure a prompt as part of a workflow user task

Type of Control	Attribute Value	Description
		that enables a user to select a model from a list. A prompt with the specified name is displayed with the Choose Models control within the associated user task. The **Choose Models** button launches the Choose Models window, which contains a list of all models for the associated project and its containing versions. This enables a user to select a model from a list within a workflow user task.
Choose Members	select_potential_owner	The select_potential_owner attribute value and the associated control enable you to configure a prompt as part of a workflow user task that enables a user to select a user or group as a potential owner during the workflow process. A prompt with the specified name is displayed with the Choose Members control within the associated user task. The user button launches the Choose Members window, which contains a list of users and groups. This enables a user to select a user or group as a potential owner as part of the user task.

Here are the high-level steps for adding attributes to a workflow user task to utilize the controls that SAS Model Manager provides:

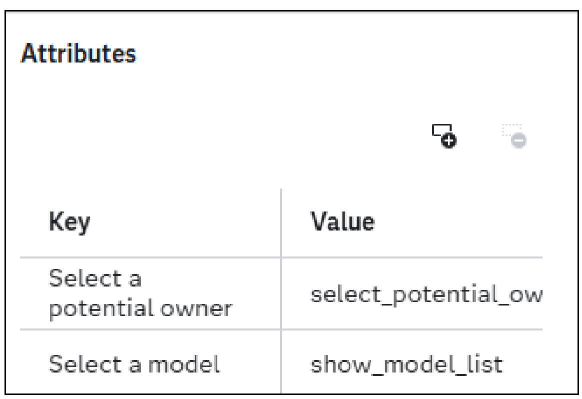
1 Add a user task to your workflow diagram.

2 Add a prompt and associate it with a data object.

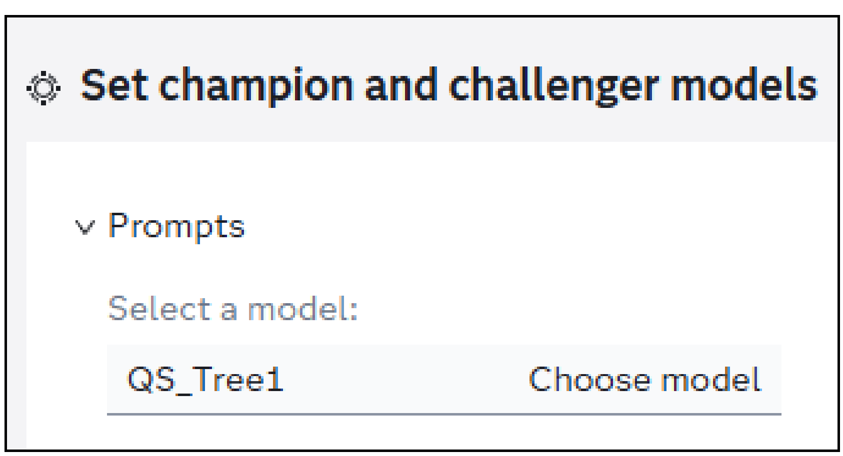


3 Add an attribute with the same name as the prompt and specify an attribute value that SAS Model Manager supports.

> **IMPORTANT** The name of the attribute and the prompt must match, in order for the control that is associated with the attribute to appear in the workflow task.



Here is an example of a user task with the **Select a model** prompt.



Note: Prompts are disabled until a user claims a task.

For an example of how to add a prompt as part of a workflow user task and associate the attribute with the prompt, see “Add the Set Champion and Challenger Models User Task” in *SAS Workflow Manager: Quick Start Tutorial*. You can also use the workflow definition BPMN file that is included in the QuickStartTutorial.zip file and follow the steps in “Add the Set Champion and Challenger Models User Task” in *SAS Workflow Manager: Quick Start Tutorial*.

For more information about adding attributes and prompts, see “Managing Attributes” in *SAS Workflow Manager: User’s Guide*.

Start a New Workflow

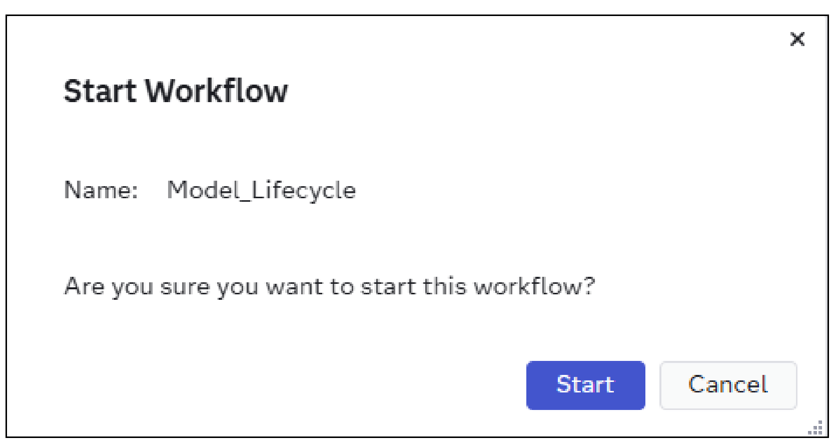
When you start a new workflow, it is associated with the project. For a specific project, only one workflow can be in progress at a time. The tasks within a workflow must be completed or the in-progress workflow process must be terminated, before a new workflow can be started.

To start a workflow:

1 Open a project and click the **Workflow** tab.

Note: The **Workflow** tab appears only for users who have workflow administrative privileges when workflow definitions are available. For more information, see “Requirements” on page 162.

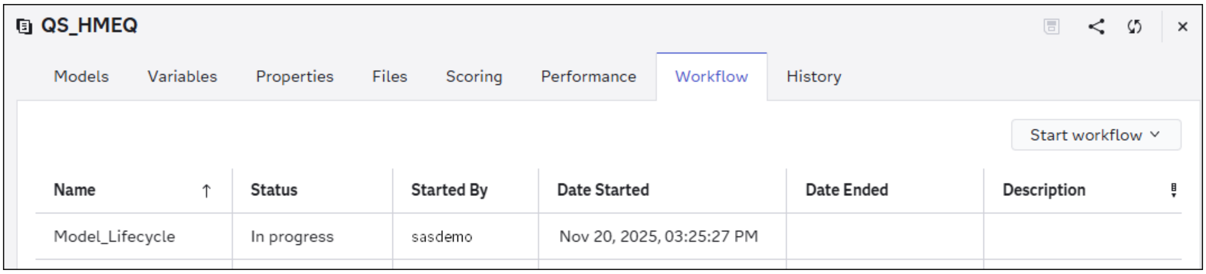
2 Click **Start Workflow** and select a workflow definition from the list. The Start Workflow window appears.



3 Specify values for any prompts that are displayed.

Note: What is displayed in the **Start Workflow** window depends on what is configured in the workflow definition start node. If prompts are not configured for the start node, the default text is “Are you sure you want to start this workflow?”.

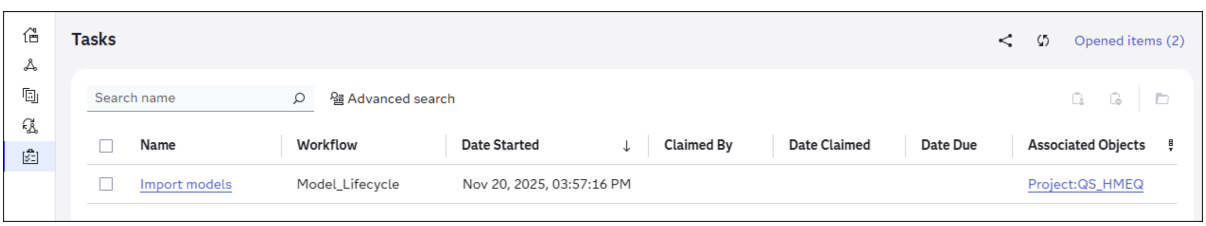
4 Click **Start**. The workflow is added to the list with a status of “In progress”.



Working with Tasks

About Tasks

The Tasks category view displays the tasks for workflows that are in progress, and that you have been assigned to as a potential owner or that have been claimed by you.



In the Tasks category view, you can perform the following:

- claim a task
- open a task
- release a task
- view the object that is associated with a task

Complete a Task

1 Click to view the **Tasks** category.

2 Click on a task to open it.

3 Click to claim the task.

4 Specify values for any prompts that are displayed on the **Prompts** tab.

Note: Prompts are disabled until you claim the task.

5 Click the **Properties** tab to view task properties, including the associated object.

6 Click on the name of the associated object to open it.

7 Complete any actions that are associated with the task. An example is importing models into a project.

8 Click **Complete**.

9 Click **Complete** in the confirmation message. You are returned to the task category view.

Note: Alternatively, you can select one or more tasks and can click to open them, or click to claim them.

Release a Task

1 Select one or more tasks.

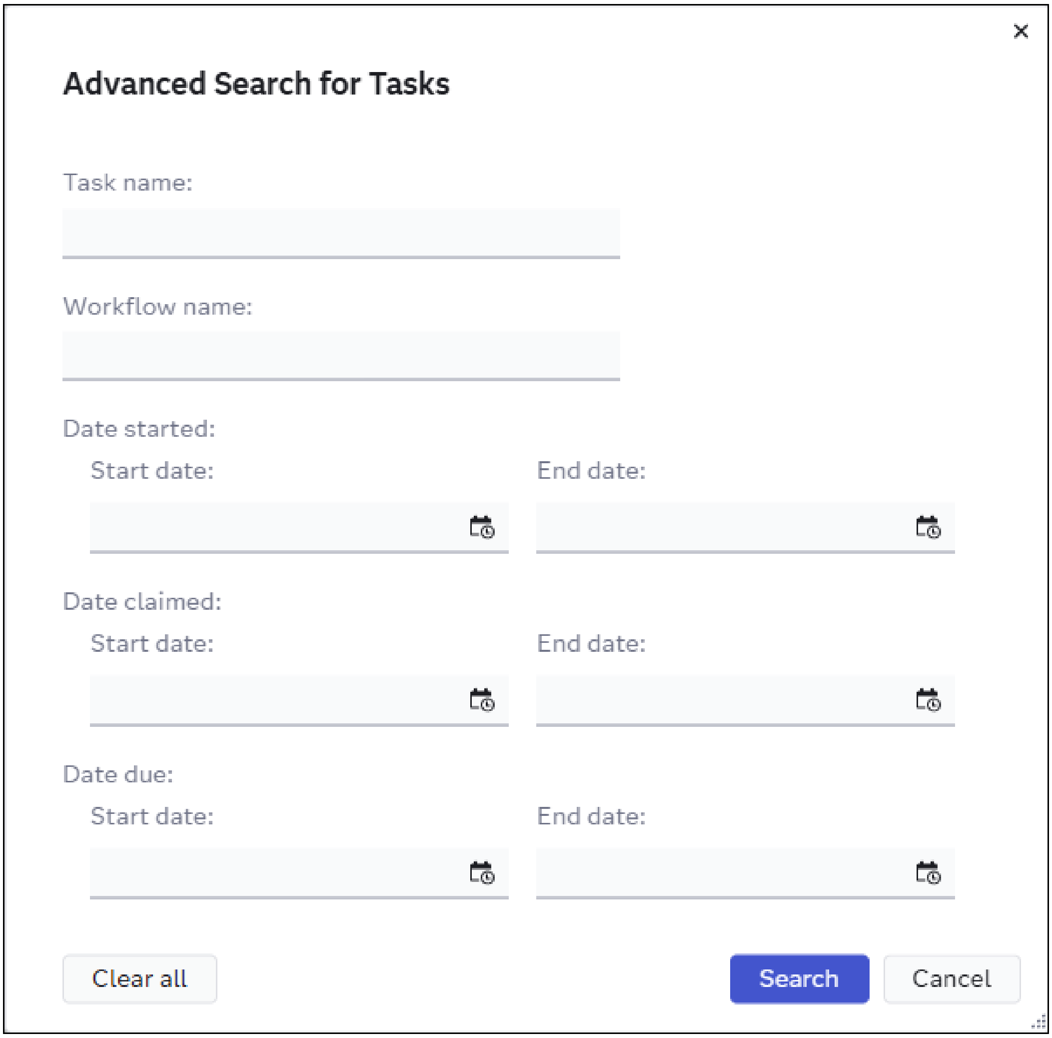
2 Click .

Note: Alternatively, if you already have the task open, you can click .

Search Tasks

In the Tasks category view, you can search the tasks that are displayed in the list. Here are the two options available for searching tasks:

- Enter a value in the **Search** field above the list to filter the list by task name.
- Click **Advanced search**, which is to the right of the search box, to search the list by task name, workflow name, date started, date claimed, or date due.


Advanced Search for Tasks
Task name:
Workflow name:
Date started:
Start date:
End date:
Date claimed:
Start date:
End date:
Date due:
Start date:
End date:
Clear all
Search
Cancel

10

Concepts

Concepts: Open-Source Models

You can use the Python and R open-source programming languages to develop analytic models. This includes model content that is stored in PMML, ONNX, or RDS formats and Python models that are developed with packages such as scikit-learn, TensorFlow, and XGBoost. You can then import those models into SAS Model Manager web applications to compare, test, and evaluate the performance of the models before publishing them to a test or production environment. When you publish open-source models to a container publishing destination, they can then be scored within run-time containers.

Several options exist for importing open-source models into SAS Model Manager. You can place your model files in a ZIP file and then import the model, or you can create a new model and add local files. For more information, see "Add a New Custom Model" on page 59 and "Import Models" on page 63. You can also create models using the Model Repository API. The API can be called in several ways, including using Python programs or using Python in a Jupyter notebook. For more information, see model-management-resources GitHub repository and Model Repository API documentation.

Python Models

Scoring Python Models

After you import a Python model, you can run an on-demand scoring test for the model. You can also publish a model to a publishing destination to score and validate it. When you publish a model from a project version to a CAS, SAS Micro Analytic Service, Azure Machine Learning, or a container publishing destination such as Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), or Private Docker, a publishing validation test is created. The publishing validation test can then be run to validate the model within the publishing destination.

In order to run an on-demand scoring test for a model from within the SAS Model Manager web application, the Python score code must be in the correct format. Here are the format requirements for what must be included in the Python score code:

- import statements for any modules that are needed.
- a score function definition with a list of the input variables.

  Note: Variable names cannot be more than 32 characters long.

- an Output statement enclosed in quotation marks that lists the output variables and directly follows the score function definition. List all output parameters as comma-separated values. Do not specify None if there is no output parameter.
- the lists of variables are case insensitive and must be comma separated.

Here is a Python code example:

```
import numpy
import pandas
import pickle
import settings

def scoreModel(CLAGE, DEBTINC, NINQ,VALUE, LOAN, MORTDUE, YOJ):
   "Output: EM_EVENTPROBABILITY, EM_CLASSIFICATION"
```

If your Python model includes a pickle file, you must also include the following code, which specifies the name of the pickle file that should be loaded:

IMPORTANT The method to load the pickle file was updated in this code example with the 2024.08 stable release. The `pickle.load` function was changed to `pd.read_pickle`.

```
with open(settings.pickle_path + "hmeq_logistic.pickle", "rb") as _pickle_file:
    _thisModelFit = pd.read_pickle(_pickle_file)

def scoreModel(CLAGE, DEBTINC, NINQ,VALUE, LOAN, MORTDUE, YOJ):
   "Output: EM_EVENTPROBABILITY, EM_CLASSIFICATION"

    try:
        global _thisModelFit
    except NameError:
        with open(settings.pickle_path + "hmeq_logistic.pickle", "rb") as _pickle_file:
            _thisModelFit = pd.read_pickle(_pickle_file)
```

After you publish a model to a container destination, a publishing validation test is created. The settings.py file and DS2 package wrapper files are also generated and added to the model. You can use the publishing validation test to score and validate your model within a run-time container.

IMPORTANT In order to validate Python models within a container publishing destination, the following requirements must be met:

- The tool version property must be set for your Python model in SAS Model Manager to a supported Python version.The supported Python versions for validating published models within container destinations are 3.9, 3.10, 3.11, 3.12, and 3.13.
- The Python packages, which contain the modules that are used in the Python score code file and its score resource files must be installed in the run-time container. You can install the packages when you publish a Python model or decision that contains a Python model to a container publishing destination by adding a **`requirements.json`** file that includes the package install statements to your model.

Here is an example of the requirements.json file:

```
[
    {   "step":"install pandas",
        "command":"pip3 install pandas==2.3.1"
    }
    {
        "step":"install statsmodels ",
        "command":"pip3 install statsmodels==0.14.5"
    }
]
```

Note: A **`requirements.txt`** file is also needed when consuming Python models in SAS Event Stream Processing, which requires a specific format for package installation in their virtual environment. For more information, see “Specify Virtual

Environments" in *SAS Event Stream Processing: Using SAS Event Stream Processing Studio*.

Here is an example of the requirements.txt file:

```
pandas==2.3.1
statsmodels==0.14.5
```

TIP The pzmm_generate_requirements_json.ipynb example in the python-sasctl GitHub repository can be used to generate the requirements.json and requirements.txt files. You can also manually create the files in a text editor.

For information about adding files, see "Managing Model Files" on page 74 and Add a Model File in the devsascom-rest-api-samples GitHub repository.

For additional examples of how to format your Python score code file and the requirements.json file, see the Python model samples in the model-management-resources GitHub repository.

See Also

- "Test Models" on page 108
- "DS2 Interface to Python" in *SAS Micro Analytic Service: Programming and Administration Guide*

Publishing Python Models

Python models can be published to CAS, SAS Micro Analytic Service, Git, Azure Machine Learning, or a container publishing destination such as Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), or Private Docker. When you import a Python model, the score code type is set to Python.

See Also

- "About Publishing Models" on page 141
- "Validate Published Models" on page 112

R Models

Scoring R Models

After you import an R model, you can run an on-demand scoring test for the model. You can also publish a model to a publishing destination to score and validate it. When you publish a model from a project version to a CAS or a container publishing destination such as Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), or Private Docker, a publishing validation test is created. The publishing validation test can then be run to validate the model within the publishing destination. Models that are published to a container publishing destination can be scored within run-time containers.

In order to run an on-demand scoring test for a model from within the SAS Model Manager web application, the R score code must be in the correct format.

Note: If an R model is transformed to PMML 4.2 format and the PMML XML file is imported into SAS Model Manager, the score code type is set to DATA step. PMML models with a DATA step score code can be scored using a scoring test within the SAS Model Manager web application.

Here are the format requirements for what must be included in the R score code:

- import statements for any libraries that are needed.
- a score function definition with a list of the input variables.

  Note: Variable names cannot be more than 32 characters long.

- an Output statement that lists the output variables must be declared on the first line within the body of the score function definition. List all output parameters as comma-separated values. Do not specify None if there is no output parameter. Example: `#output: outvar1, outvar2`
- the lists of variables are case insensitive and must be comma separated.
- the path to the RDA or RDS file must be specified. Example: `file = file.path(rdsPath, 'hmeq_classtree_r.rds')`

Here is an R code example:

```
scoreFunctionName <- function(var1, var2, var3)
{
   #output: outvar1, outvar2

   if (!exists("myClassTree"))
   {
      myClassTree <<- readRDS(file = file.path(rdsPath,
'hmeq_classtree_r.rds'))
```

```
    }

    # Include scoring logic here to get a list of the output variables.

    output_list <- list('outvar1' = outvar1, 'outvar2' = outvar2)
    return(output_list)
}
```

Publishing R Models

R models can be published to CAS, a Git repository, or a container publishing destination such as Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), or Private Docker. When you import an R model, the score code type is set to R. To be published to CAS or container publishing destinations, R models must have a score code type of R and an R score code file that contains a scoring function in the correct format.

See Also

- "About Publishing Models" on page 141
- "Validate Published Models" on page 112

Concepts: Performance Monitoring

About Performance Monitoring

You can monitor the effectiveness of models by analyzing the performance results. You can create a performance definition on the **Performance** tab of a project. A performance job is then created for each model when you save the definition. When you run a performance job, the summary information and the charts for each model that appear on the **Performance** tab represent the performance results data that is generated. The key performance indicators (KPIs) help you to assess the health and effectiveness of your models within the project.

The following types of charts are included in the performance results:

Data Composition
: The Variable Distribution charts show you the distributions for an input variable or output variable in one or more time periods, this enables you to see the differences and changes over time.

The Variable Binning charts show you a graphical representation of binning over a period of time for the selected input or output variable.

The Characteristic and Stability charts detect and quantify shifts in the distribution of variable values that occur in input data and scored output data over time. By analyzing these shifts, you can gain insights on scoring input and output variables.

Model Monitoring
: The model monitoring charts are a collection of performance assessment charts that evaluate the predicted and actual target values. The model monitoring charts create several charts:

- Lift
- Gini
- ROC (Receiver Operating Characteristic) on page 182
- KS
- Average Squared Error (ASE) of the Interval Target
- Mean of the Predicted Interval Target
- Standard Deviation of the Predicted Interval Target
- Proportion of Predicted Interval Target Values

Feature Contribution Index
: The feature contribution index (FCI) measures the relationships between input and output variables. This enables you to evaluate the importance of predictors within that model.

Population Stability Index
: The population stability index (PSI) indicates the degree of changes of a distribution from a baseline.

Standard Key Performance Indicator (KPI)
: The Standard KPI Trend chart shows the trend of the selected standard key performance indicator (KPI) that is calculated over a period of time.

Multinomial Target Metrics
: The multinomial target metrics charts show the distribution of predicted and actual target categories, as well as a collection of performance assessment charts that evaluate the predicted and actual target categories.

Text Categorization Metrics
: The text categorization metrics charts appear when you generate performance results for a definition that contains a text categories model.

Fairness and Bias Metrics
: The fairness and bias metrics charts appear when you generate performance results for a definition that contains a classification or prediction model, and variables have been marked for bias assessment.

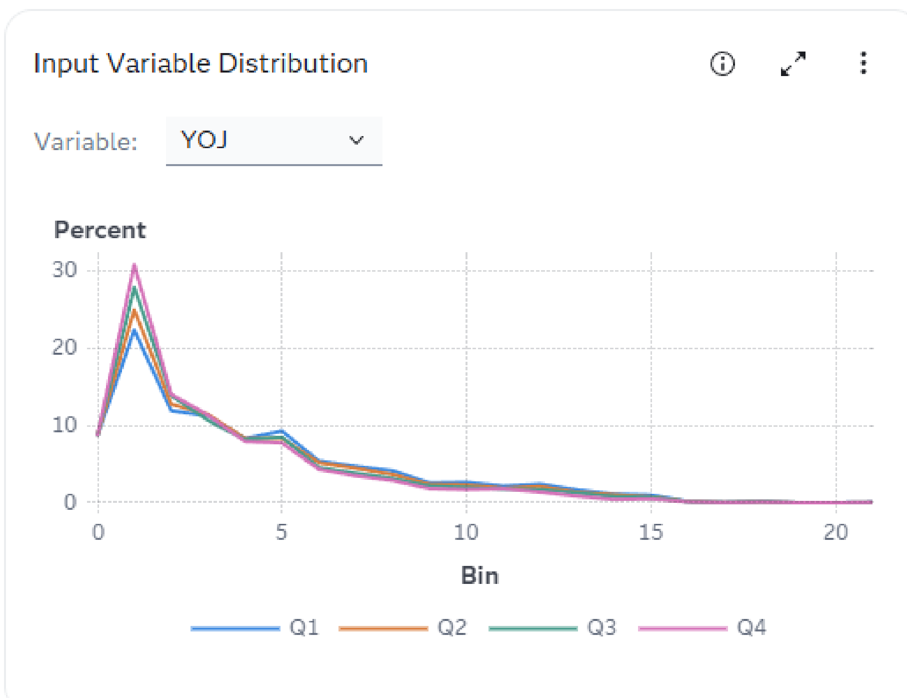
Data Composition

Variable Distribution

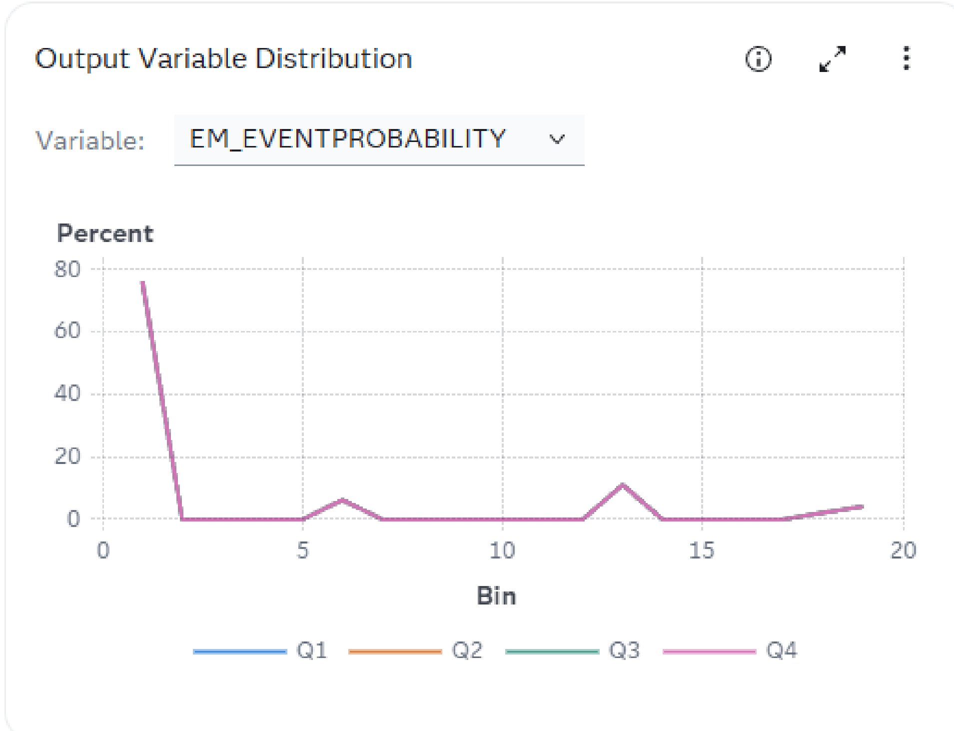
On the **Performance** tab of a project, you can view the variable distribution of a model. The Input Variable Distribution and Output Variable Distribution charts are a graphical representation of distributions over a period of time for the selected variable. Each line plot represents the data for a specific period of time. The Y axis is the percentage of observations in a bin that is proportional to the total count.

To change the variable that appears in the chart, select a variable from the drop-down list.

Here is an example of an Input Variable Distribution chart.



Here is an example of an Output Variable Distribution chart.

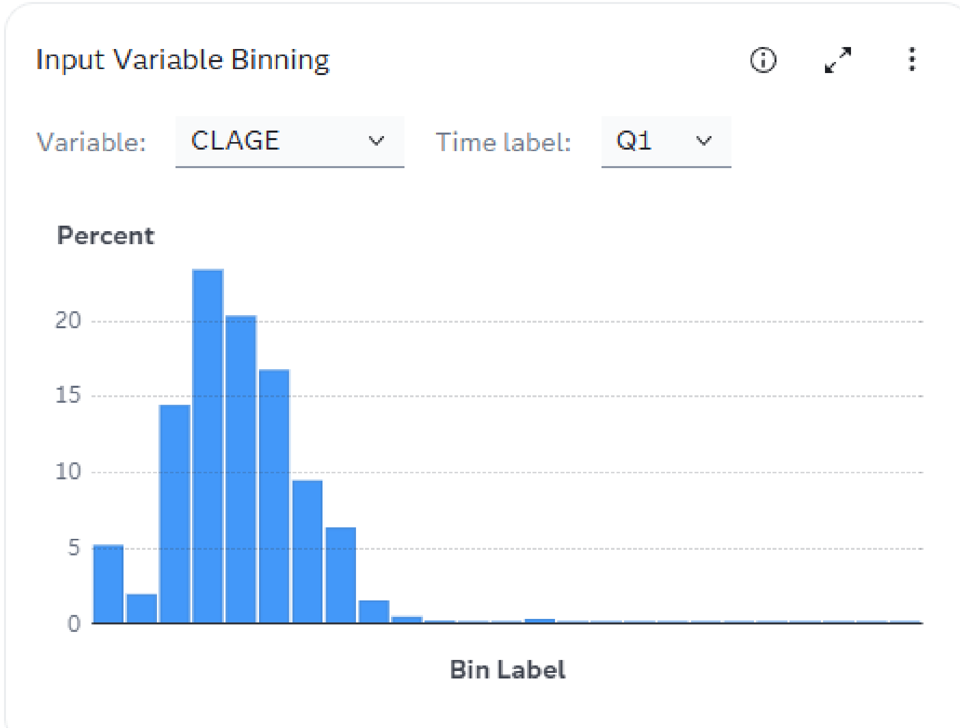


Variable Binning

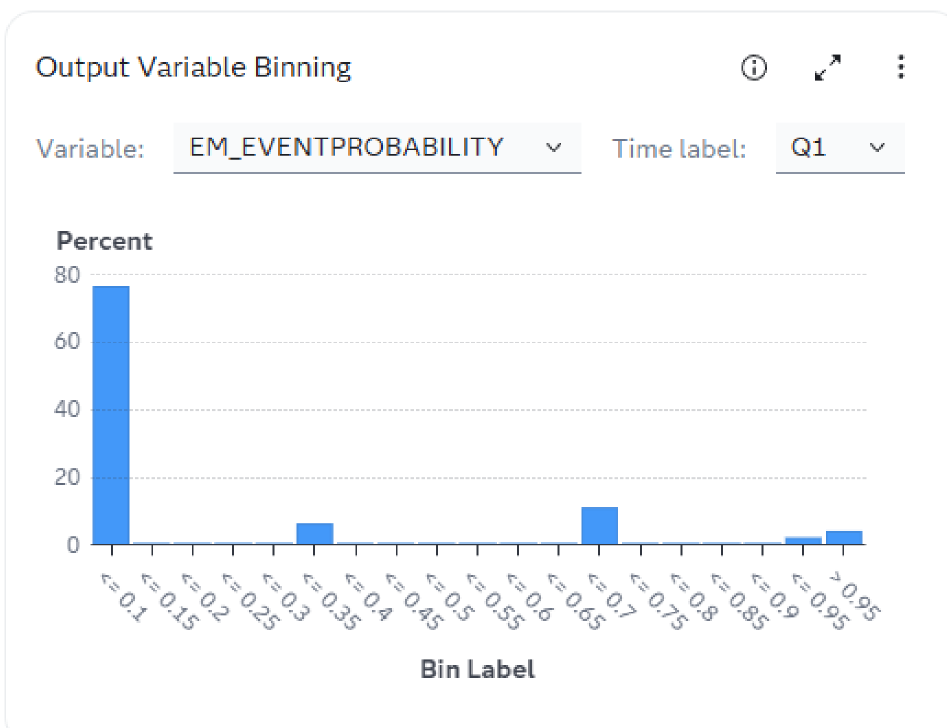
The Input Variable Binning and Output Variable Binning charts show a graphical representation of binning over a period of time for the selected input or output variable. Each bar plot represents the data for a specific bin. The Y axis is the percentage of observations in a bin that is proportional to the total count.

To change the data that appears in the chart, select a variable and a time label from the drop-down lists.

Here is an example of an Input Variable Binning chart.



Here is an example of an Output Variable Binning chart.



Characteristic and Stability

Together, the Input Variable Characteristic and Output Variable Stability charts detect and quantify shifts that can occur in the distribution of model performance data, scoring input data, and the scored output data that a model produces.

Note: For each time period that you run performance, SAS Model Manager creates a new point on the charts. Line segments between points in time do not appear on the charts unless you specify at least three data sources and collection dates as part of the performance definition.

Characteristic

The Input Variable Characteristic chart detects and quantifies the shifts in the distribution of variable values in the input data over time. These shifts can point to significant changes in customer behavior that are due to new technology, competition, marketing promotions, new laws, or other influences.

To find shifts, the first time point in the input data is set as the baseline and the distributions of the variables in subsequent data time points are compared to the baseline data time point.

If large enough shifts occur in the distribution of variable values over time, the original model might not be the best predictive or classification tool to use with the current data.

The Characteristic chart uses a population stability index (PSI) to quantify the shifts in a variable's values distribution that can occur between the baseline input data set and the subsequent input data sets. The PSI is computed for each predictor variable in the data set, using this equation:

$$PSI = \sum \left(\%\,Actual - \%\,Expected\right) \times ln\left(\frac{\%\,Actual}{\%\,Expected}\right)$$

Numeric predictor variable values are placed into bins for frequency analysis. Outlier values are removed to facilitate better placement of values and to avoid scenarios that can aggregate most observations into a single bin.

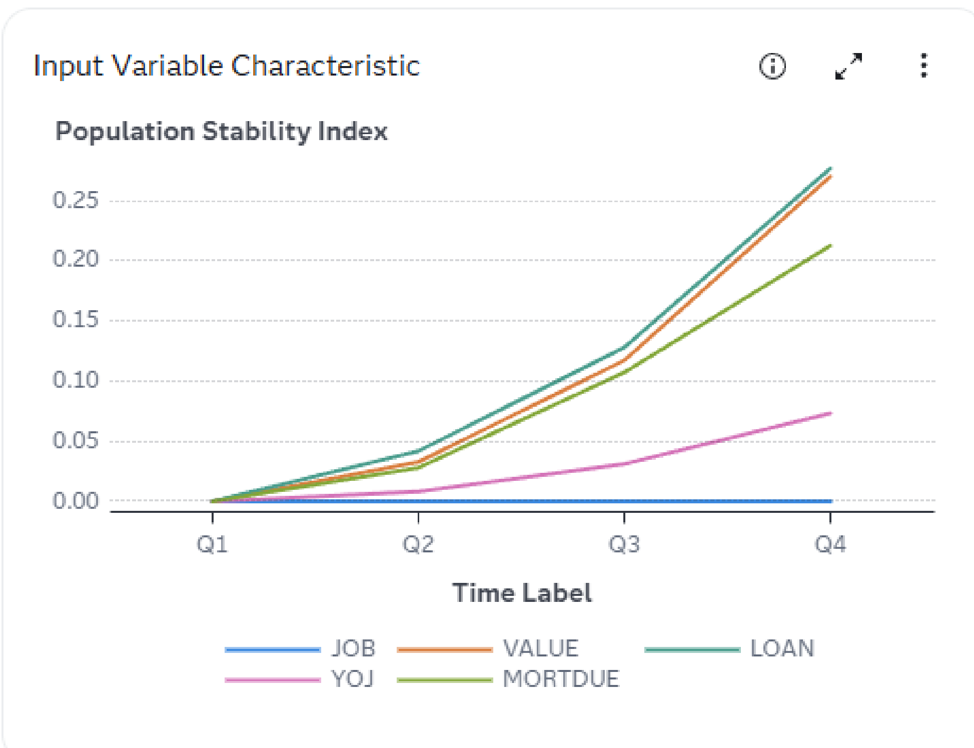
If the baseline input data set and the subsequent input data sets have identical distributions for a variable, the variable's population stability index (PSI) is equal to 0. A variable with a PSI value that is P1>2 is classified as having a moderate change in distribution. The Characteristic chart uses the performance measure P1 to count the number of variables that receive a PSI value that is greater than 0.1.

A variable that has a PSI value that is P1>5 or P25>0 is classified as having a significant change in distribution. A performance measure P25 is used to count the number of variables that have significant changes in distribution, or the number of input variables that receive a PSI score value that is greater than or equal to 0.25.

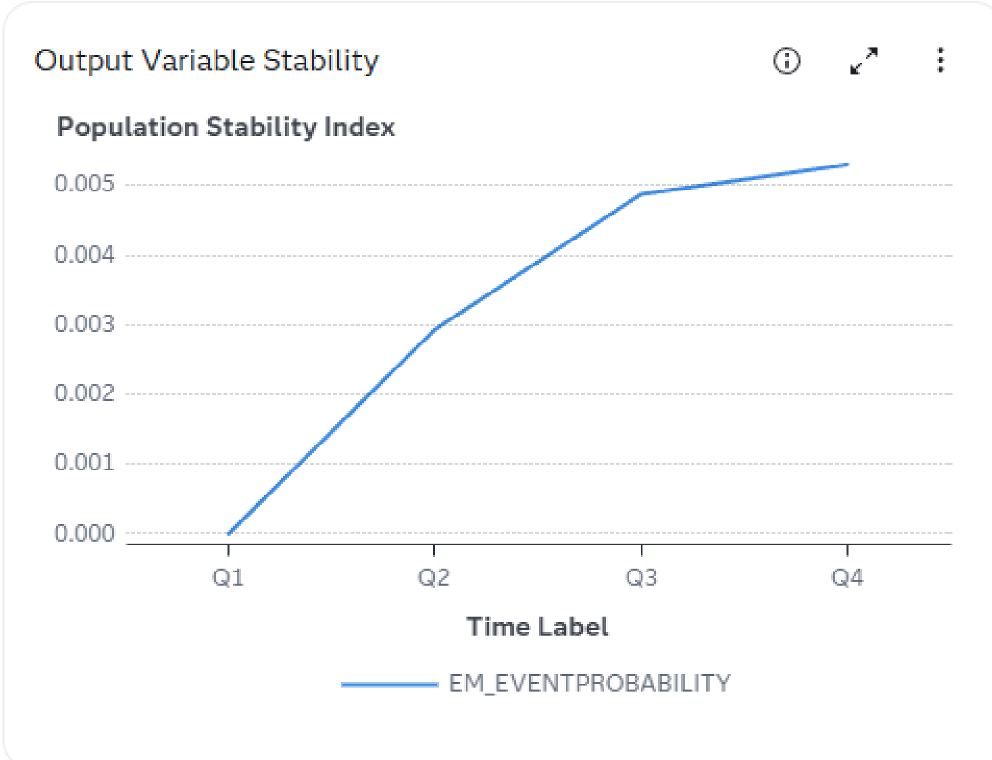
Stability

The Output Variable Stability chart evaluates changes in the distribution of scored output variable values as models score data over time, and detects and quantifies shifts in the distribution of output variable values in the data that is produced by the models. If an output variable from the first time point in the input data and the output variable from the subsequent data time points have identical distributions, then that output variable's population stability index (PSI) is equal to 0. An output variable with a PSI value that is greater than 0.10 and less than 0.25 is classified as having a moderate change in distribution. A variable that has a PSI value that is greater than 0.30 is classified as having a significant change in distribution Too much of a shift in predictive variable output can indicate that model tuning, retraining, or replacement might be necessary.

Here is an example of the Input Variable Characteristic chart.



Here is an example of the Output Variable Stability chart.



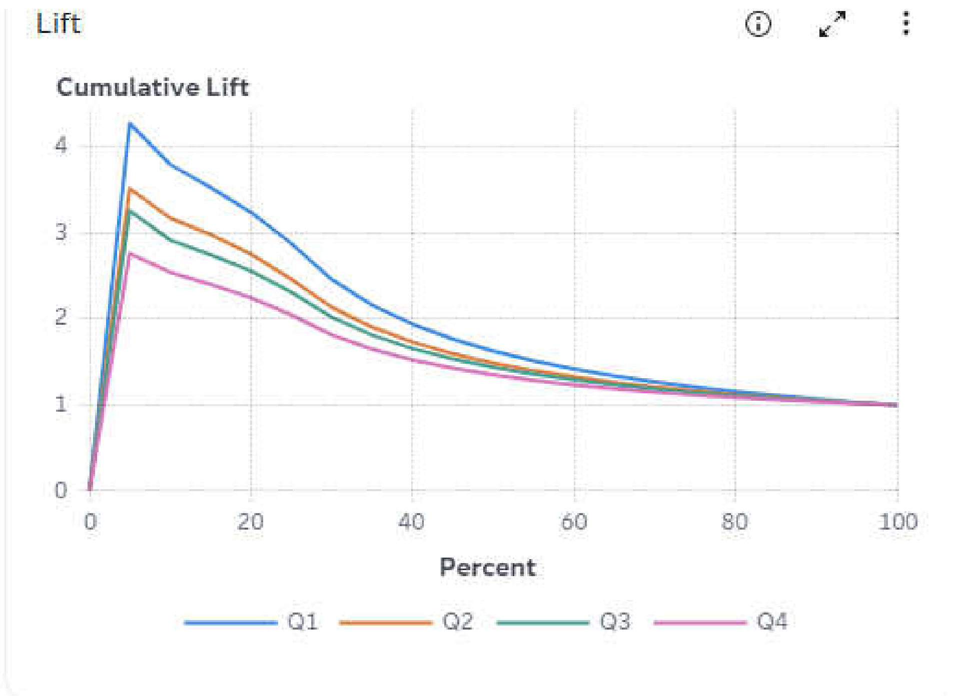
Model Monitoring

Lift

The Lift chart provides a visual summary of the usefulness of the information that is provided by a classification model for predicting a binary outcome variable. Specifically, the report summarizes the utility that you can expect by using the champion model as compared to using baseline information only. Baseline information is the prediction accuracy performance of the initial performance monitoring definition or batch program using operational data.

A monitoring Lift chart can show a model's cumulative lift at a given point in time or the sequential lift performance of a model's lift over time. The Lift performance indexes Lift5Decay, Lift10Decay, Lift15Decay, and Lift20Decay are used to detect model performance degradation. The performance indexes are not displayed in the Lift chart, but are available in the mm_model_indicator performance results data table. The data that underlies the Lift chart is contained in the mm_lift performance results data table.

Here is an example of a monitoring Lift chart.



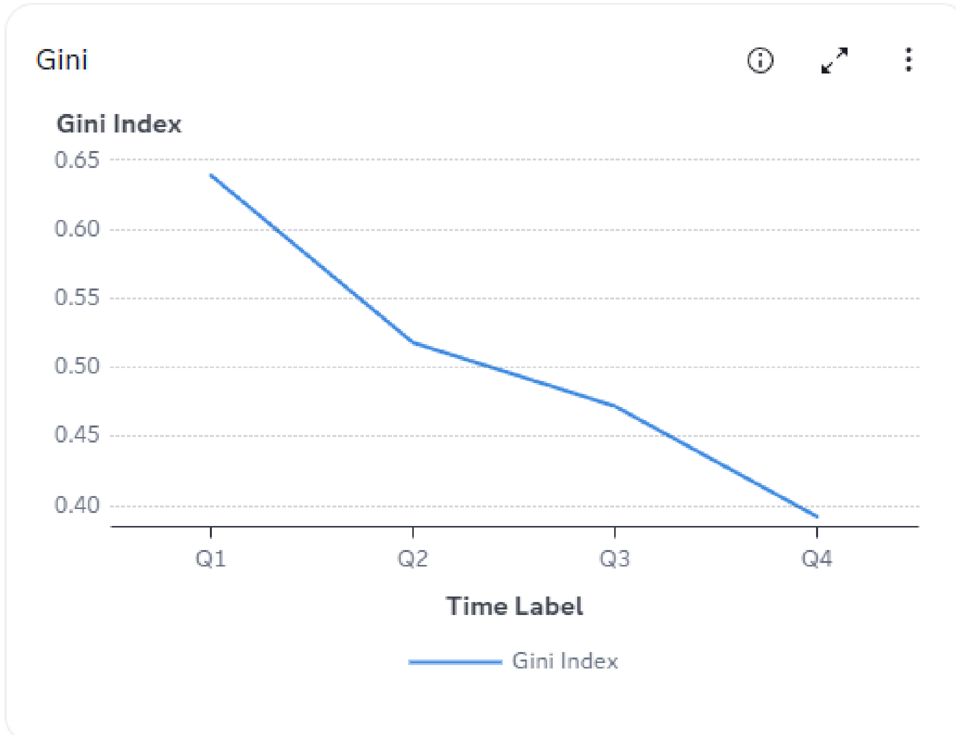
Gini and ROC

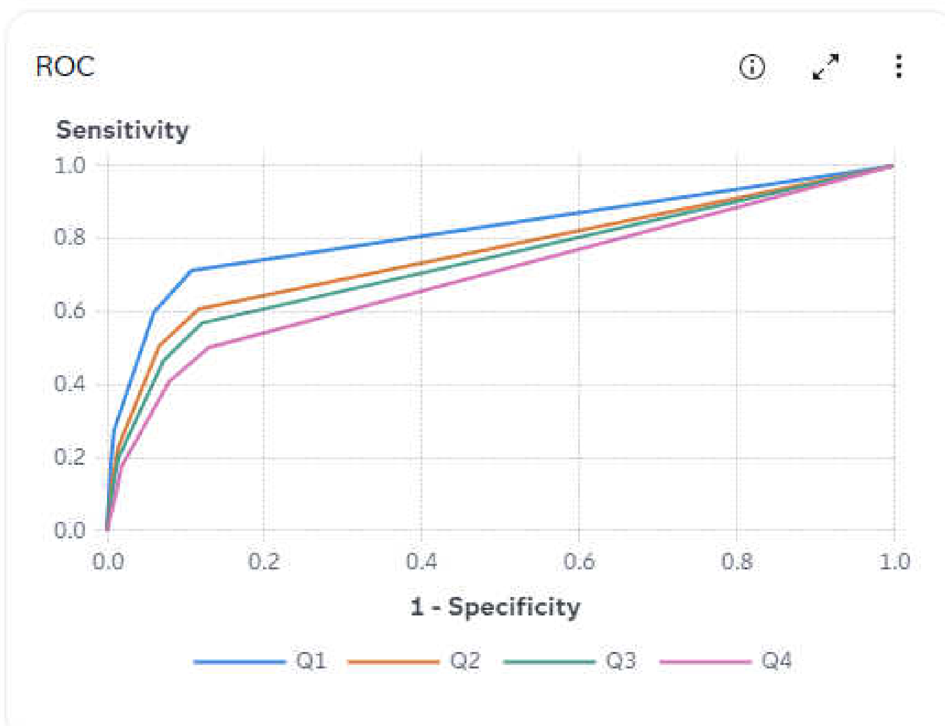
The Gini and ROC charts show you the predictive accuracy of a classification model that has a binary target. The plot displays sensitivity information about the Y axis and 1-Specificity information about the X axis. Sensitivity is the proportion of true positive events. Specificity is the proportion of true negative events. The Gini index is calculated for each ROC curve. The Gini coefficient is a benchmark statistic that can be used to summarize the predictive accuracy of a model, and is directly related to the area under the ROC curve (2*AUC-1).

Use the monitoring Gini chart to detect degradations in the predictive power of a model.

The data that underlies the monitoring Gini and ROC charts are contained in the mm_roc performance results data table.

Here are examples of the monitoring **Gini** and **ROC** charts.


Gini
Gini Index
0.65
0.60
0.55
0.50
0.45
0.40
Q1
Q2
Q3
Q4
Time Label
Gini Index


ROC
Sensitivity
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
1 - Specificity
Q1
Q2
Q3
Q4

KS

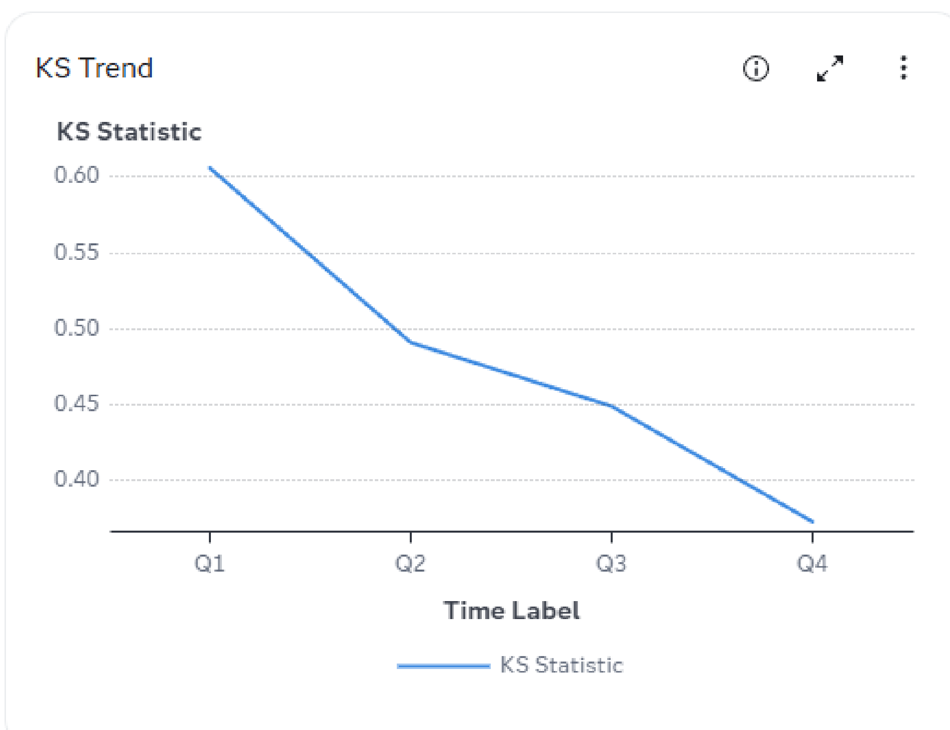
The KS charts contains the Kolmogorov-Smirnov (KS) test plots for models with a binary target. The KS statistic measures the maximum vertical separation, or deviation between the cumulative distributions of events and non-events. This trend chart uses a summary data set that plots the KS statistic and the KS probability cutoff values over time.

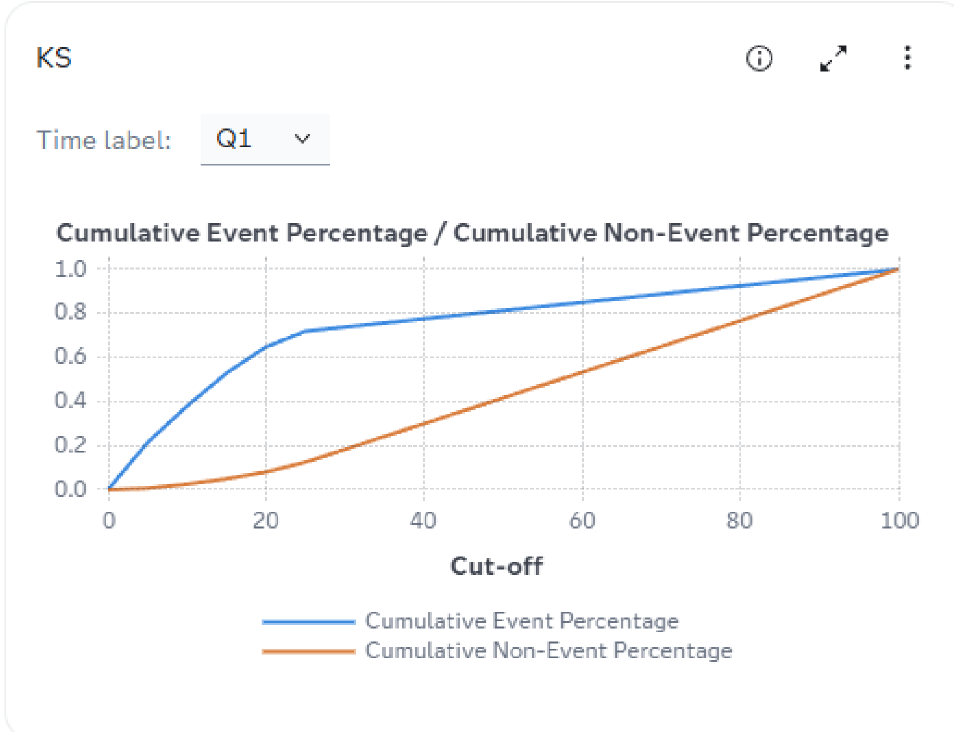
Use the KS chart to detect degradations in the predictive power of a model. To scroll through a successive series of KS performance depictions, select a time interval from the **Time Interval** list box. If model performance is declining, it is indicated by the decreasing distances between the KS plot lines.

The ksDecay performance index detects model performance degradation. The ksDecay performance index is not displayed on the KS chart, but is available in the mm_model_indicator performance results data table.

The data that underlies the KS chart is contained in the mm_ks performance results data table.

Here are examples of the KS charts.

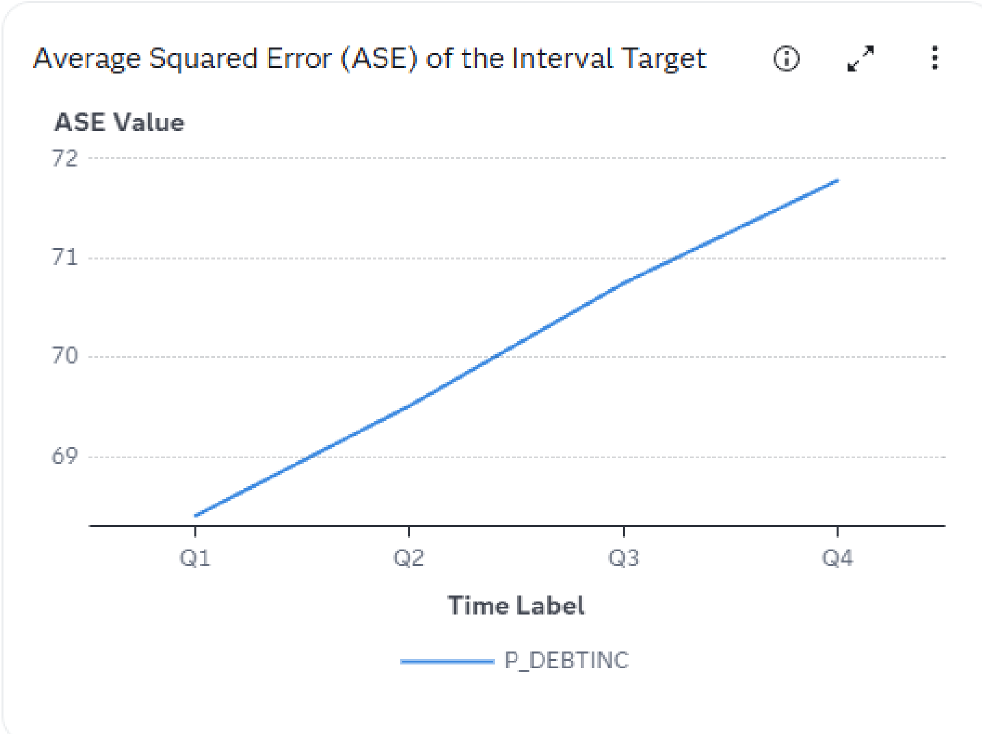




Average Squared Error (ASE) of the Interval Target

The Average Squared Error (ASE) chart checks the accuracy of a prediction model with an interval target by comparing the estimation derived from the test data and the actual outcomes that are associated with the test data for different time periods.

Here is an example of the chart.



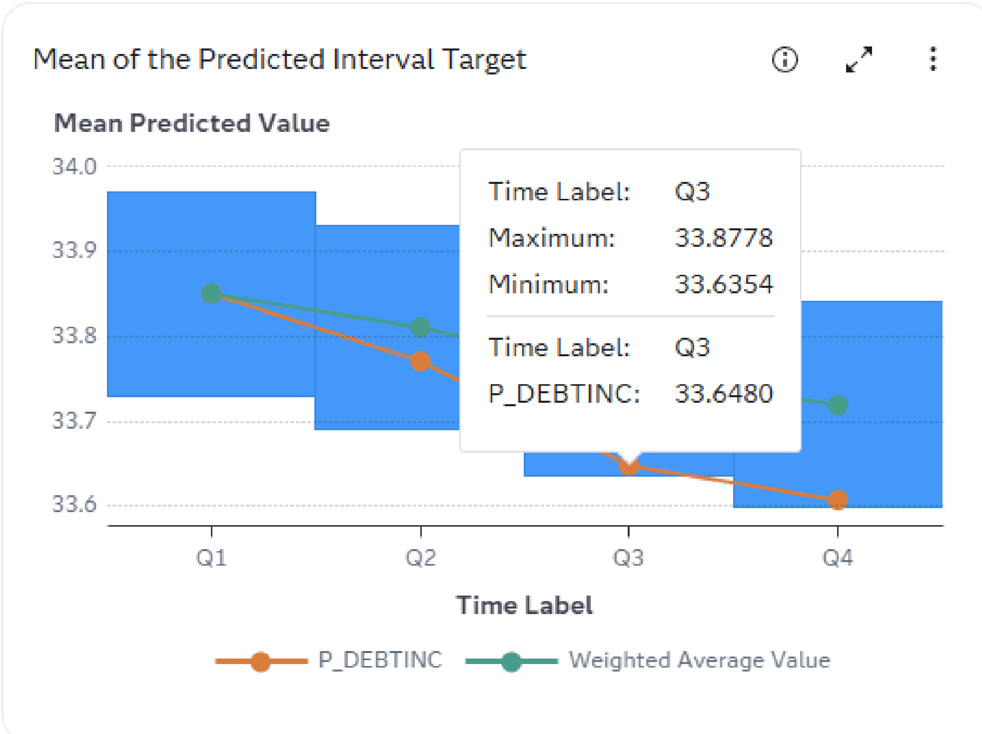
Mean of the Predicted Interval Target

The Mean of the Predicted Interval Target chart shows the mean of values for a prediction model with an interval target for different time periods. The central horizontal line represents the weighted average of the means. The shaded band represents the control limits. If a point in the line chart is outside the shaded band, it indicates that the predicted interval target values at the corresponding time period have deviated significantly. You can then further scrutinize the model outcomes at that time period.

If the underlying data distribution has not changed over time and the current model is still effective in predicting the interval target, you should not see any significant changes in the overall spread of the predicted interval target values.

Concepts from the statistical process control (SPC) are used along with the XSCHART statement in the SHEWHART procedure to monitor the means and the standard deviation of the predicted interval target over time. For more information, see The SHEWHART Procedure – Constructing Charts for Means and Standard Deviations in *SAS/QC User's Guide* and Statistical Process Control Action Set in *SAS Visual Statistics: Programming Guide*.

Here is an example of the Mean of the Predicted Interval Target chart.

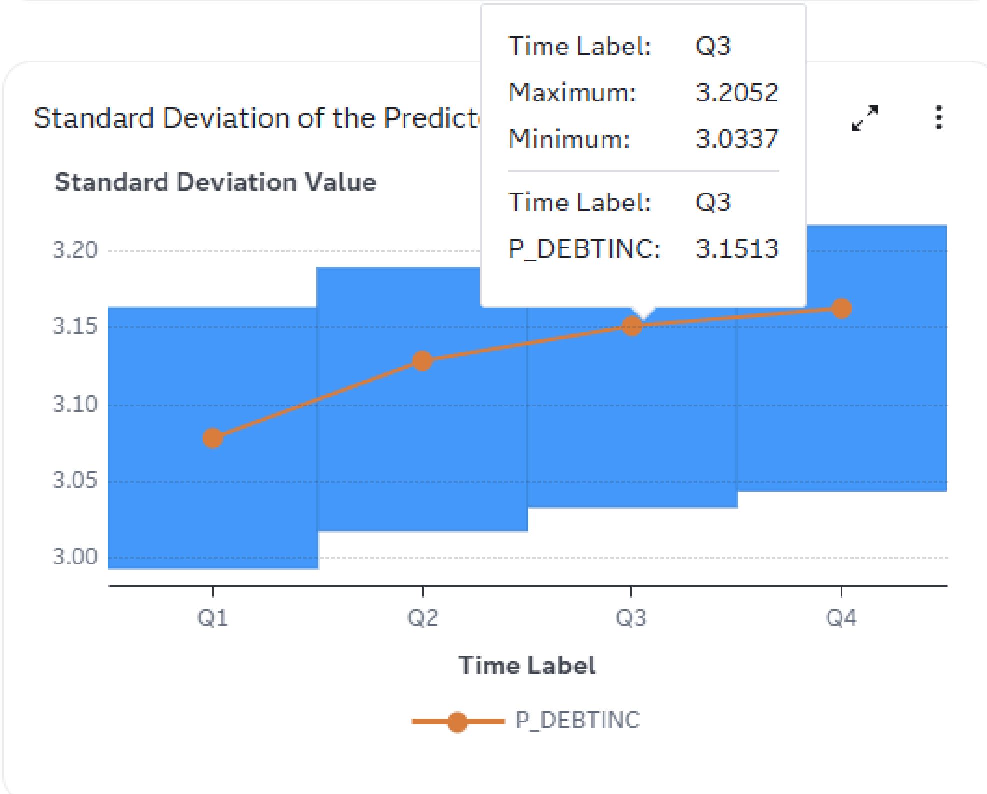


Standard Deviation of the Predicted Interval Target

The Standard Deviation of the Predicted Interval Target chart shows the standard deviation of values for a prediction model with an interval target for different time periods. The central horizontal line represents the weighted average of the standard deviations. The shaded band represents the control limits. If a point in the line chart is outside the shaded band, it indicates that the spread of the predicted interval target values at the corresponding time period has deviated significantly. You can then further scrutinize the model outcomes at that time period.

Concepts from the statistical process control (SPC) are used along with the XSCHART statement in the SHEWHART procedure to monitor the standard deviation of the predicted interval target over time. For more information, see The SHEWHART Procedure - Methods for Estimating the Standard Deviation in *SAS/QC User's Guide*.

Here is an example of the Standard Deviation of the Predicted Interval Target chart.



Proportion of Predicted Interval Target Values

The Proportion of Predicted Interval Target Values charts represent the proportion of values for a prediction model with an interval target that are above the 25th, 50th, and 75th percentiles, over a period of time. When the underlying data distribution has changed over time, you need to know how the distribution changed. For example, the distribution might shift its location, the spread might become smaller or larger, the skewness might change, or the kurtosis might change. Therefore, you can use these charts to monitor the proportion of the predicted interval target values that are above specified thresholds. Without loss of generality, the 25th, the 50th, and the 75th percentiles of the predicted interval target values are represented at the first monitoring time (commonly known as the Time 0).

Concepts of the statistical process control (SPC) are used with the *p* chart to monitor the proportions over time. For more information, see The SHEWHART Procedure - Constructing Charts for Proportion Nonconforming (p Charts) in *SAS/QC User's Guide*

Here are the percentiles that are represented in the Proportion of Predicted Interval Target Values charts:

- Above the 25th Percentile (first quartile)
- Above the 50th Percentile (median)
- Above the 75th Percentile (third quartile)

It is recommended that you look at the *p* charts together. For example, if the 25th percentile *p* chart at time period Q3 is out of bounds, but the 50th and the 75th percentile *p* charts at time period Q3 are not, then you can infer that the distribution has moved the values that are below the 50th percentile, but not those that are above the 50th percentile.

Note: When there is a small sample size, the proportion might not reach the expected value.

Here are examples of the Proportion of Predicted Interval Target Values charts.

Figure 10.1 *Proportion of Predicted Interval Target Values above the 25th Percentile*

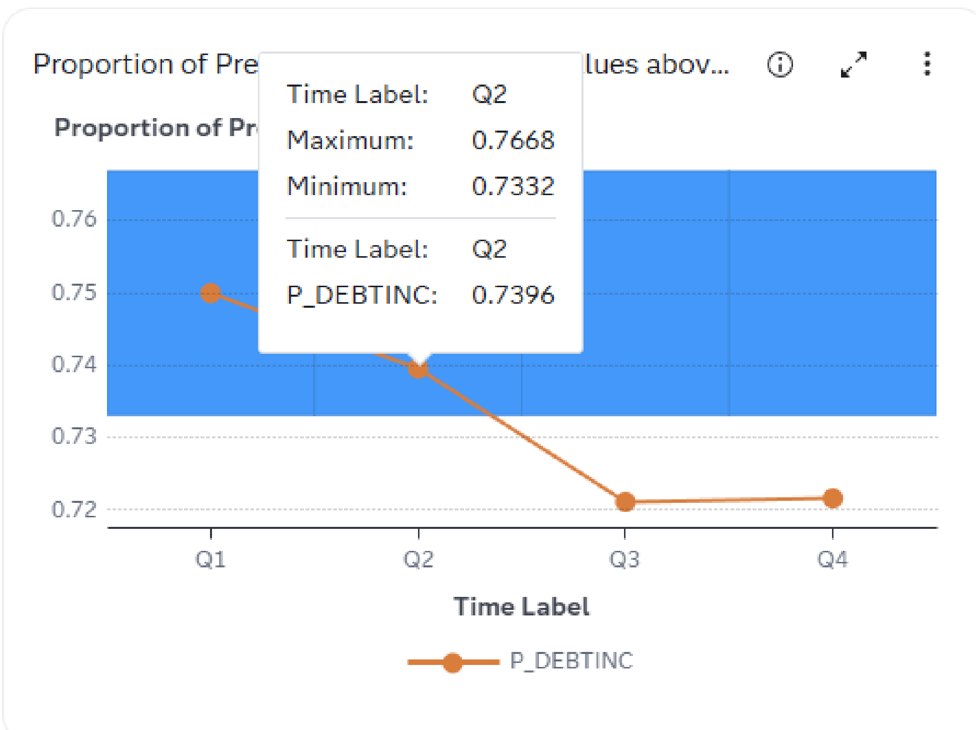


Figure 10.2 Proportion of Predicted Interval Target Values above the 50th Percentile

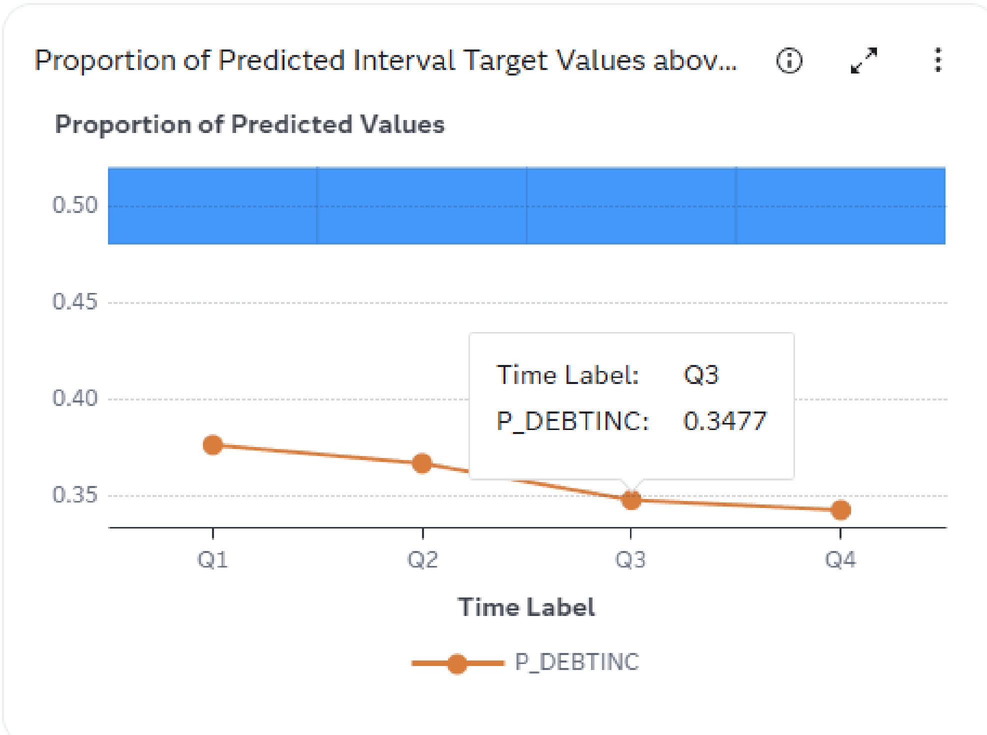
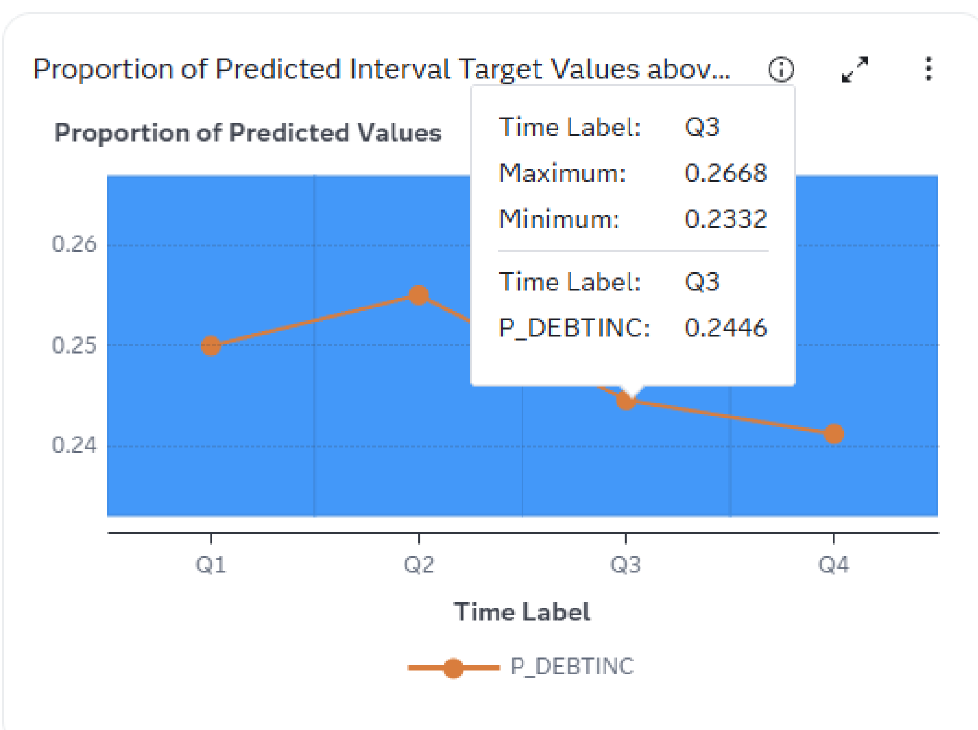


Figure 10.3 Proportion of Predicted Interval Target Values above the 75th Percentile



Feature Contribution

The feature contribution index (FCI) measures the strength of the relationship between each input variable and the predicted numeric output variables. When you are running performance, the FCI macros are used to compute the feature contribution indices for interval and nominal predictors and return the indices in the output data. The output data is then used to create the Feature Contribution chart. The top nine variables with the maximum absolute contribution for first data point are displayed in the drop-down list in descending order. The input data must contain the scored model data or else you must score the model when running performance, in order for the feature contribution index to be computed for the model variables. Otherwise, the Feature Contribution Index chart is not displayed.

The Feature Contribution Index Out-of-Bounds Indicators chart enables you to identify the features that need additional scrutiny.

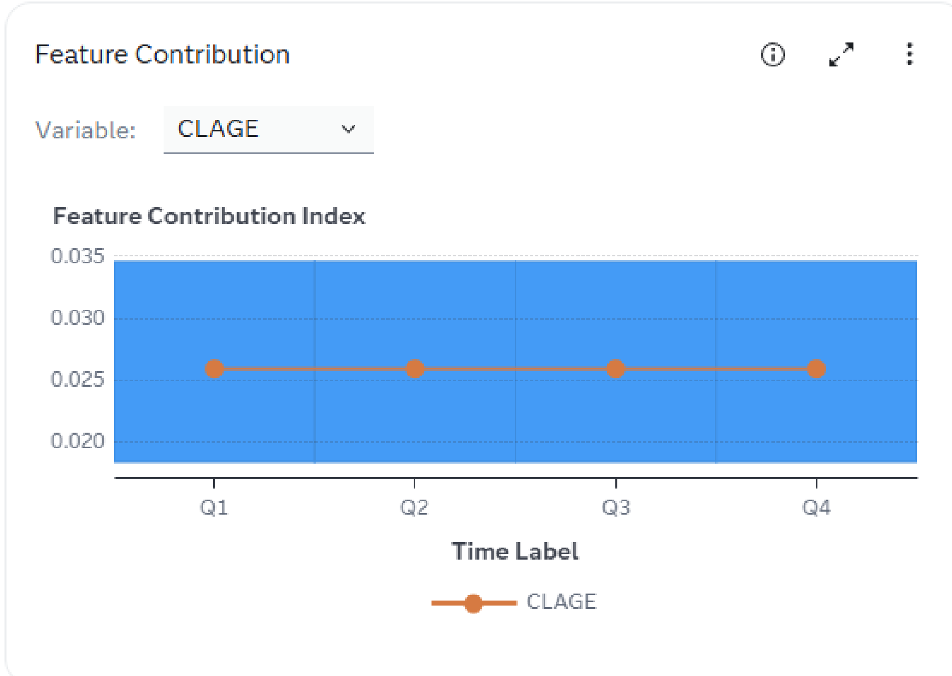
The 95% confidence limits for an FCI are calculated by applying the interval inversion method (Kromrey and Bell 2010 and Steiger 2004). The limits are computed based on the number of levels of the predictor (one for an interval predictor), the number of observations, and the FCI statistic at the baseline time period. If the FCI statistic at the current time period is above (or below) the upper (or lower) confidence limit, then you have reasons to suspect that the current FCI statistic is statistically different from that at the baseline time period. To put it another way, the feature at the current time period contributes more (or less) to the prediction than that at the baseline time period.

The percent out-of-bounds value, which is the ratio of two differences is also calculated. The numerator is the difference from the current FCI statistic to the upper (or lower) confidence limit. The denominator is the difference from the upper (or lower) confidence limit to the baseline FCI statistic. The ratio indicates the urgency of the out-of-bound situation.

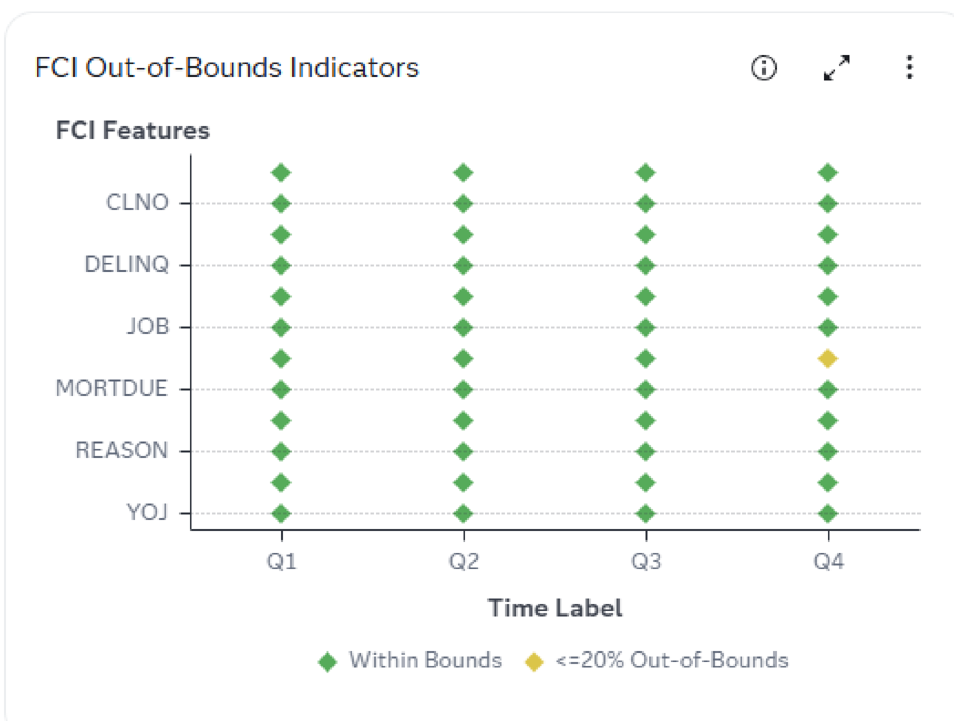
References:

- Kromrey, J. D., and Bell, B. A. (2010). “ES_ANOVA: A SAS Macro for Computing Point and Interval Estimates of Effect Sizes Associated with Analysis of Variance Models,” Proceedings of the SouthEast SAS Users Group Conference (SESUG 2010), paper PO-05. Cary, NC: SAS Institute Inc. Available at https://analytics.ncsu.edu/sesug/2010/PO05.Kromrey.pdf.
- Steiger, J. H. (2004). “Beyond the F Test: Effect Size Confidence Intervals and Tests of Close Fit in the Analysis of Variance and Contrast Analysis,” Psychological Methods, 9(2), 164-182. http://dx.doi.org/10.1037/1082-989X.9.2.164.

Here is an example of the Feature Contribution chart.



Here is an example of the Feature Contribution Index Out-of-Bounds Indicators chart.



See Also

"Feature Contribution Index Macros" in *SAS Model Manager: Macro Reference*

Population Stability

The population stability index (PSI) indicates the degree of changes of a distribution from a baseline. The Population Stability Index Out-of-Bounds Indicators charts enable you to identify the features that need additional scrutiny.

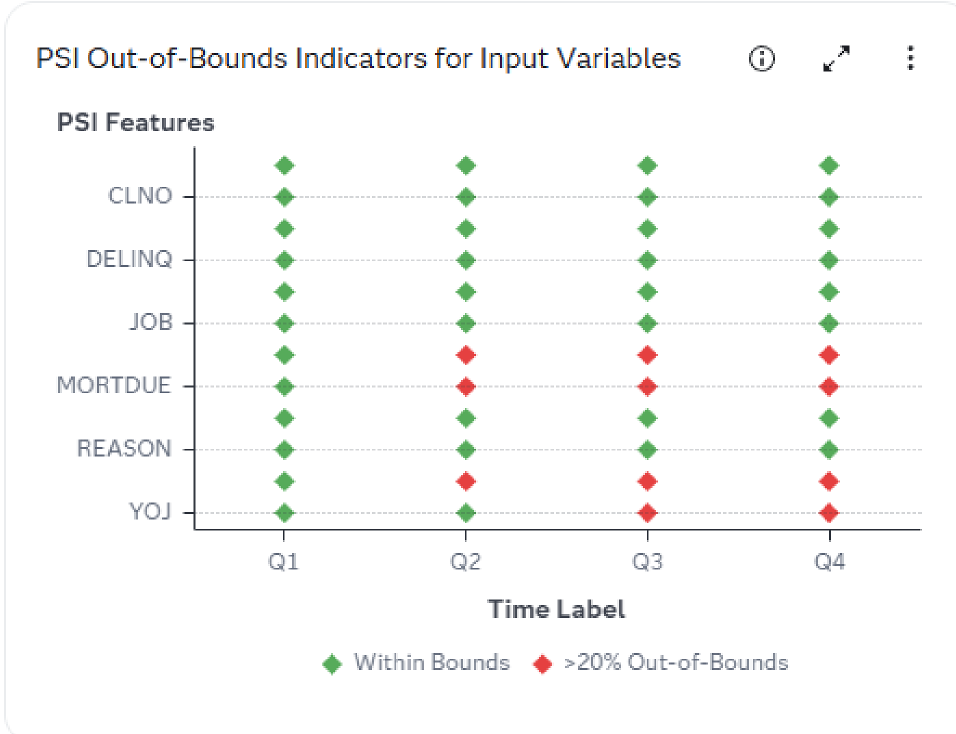
PSI measures the extent a variable has shifted in distribution over time. For a continuous variable, the PSI is calculated on the binned histogram. For a categorical variable, the PSI is calculated on the frequency table. Suppose p0 is the proportion of observations found in a bin of a continuous variable (or a category of a categorical variable) at time t0. Similarly, p1 is the proportion in the same bin (or category) at time t1. The PSI of this bin is (p1 - p0) * ln(p1 / p0). By summing over all the bins, the PSI metric is obtained.

The one-sided 95% confidence limits for the PSI is calculated by applying a method in Bilal Yurdakul and Joshua Naranjo (2020). The baseline PSI is always zero. The upper limit is computed based on the number of bins, and the numbers of observations at both times. If the PSI statistic at the current time period is above the upper confidence limit, then you have reasons to suspect that the current PSI statistic is statistically greater than zero. To put it another way, the feature's distribution at the current time period has shifted from that at the baseline time period.

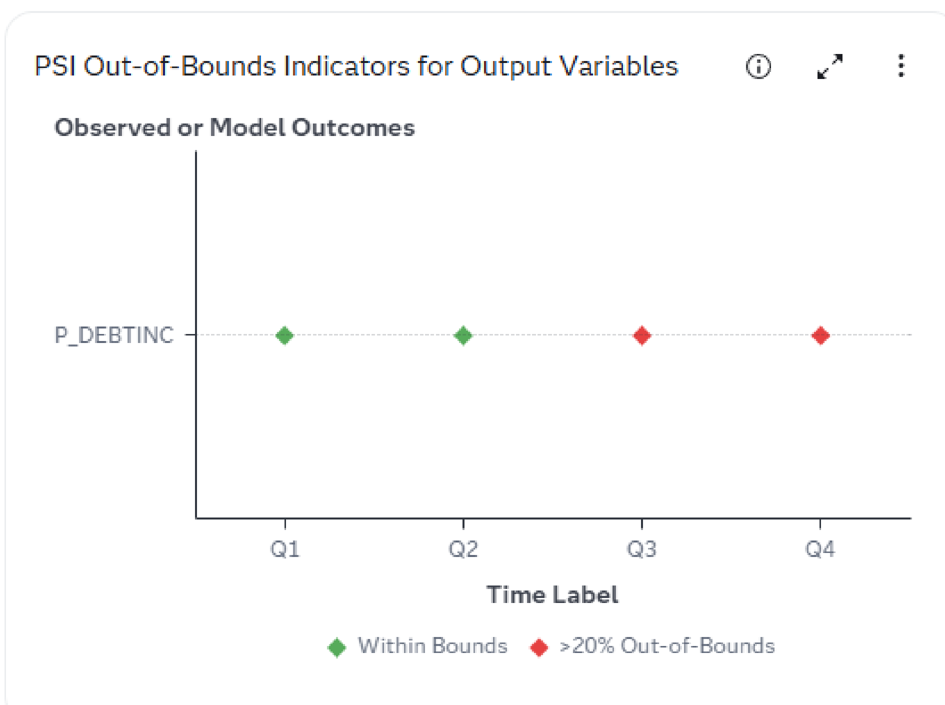
The percent out-of-bounds value, which is the ratio of two differences is also calculated. The numerator is the difference from the current PSI statistic to the upper confidence limit. The denominator is the difference from the upper confidence limit to the baseline PSI (for example, zero). The ratio indicates the urgency of the out-of-bound situation (Bilal Yurdakul and Joshua Naranjo (2020) Statistical properties of the population stability index, Journal of Risk Model Validation, Volume 14, Number 4, Page 89-100).

The same method is used in calculating the PSI statistic and its confidence limit for the output variables. However, since the output variables are often used in making decisions, any shifts in their distributions might affect the decisions made. Therefore, you should scrutinize the output variables methodically for early signs of model deterioration.

Here is an example of the Population Stability Index Out-of-Bounds Indicators for Input Variables chart. Reference:



Here is an example of the Population Stability Index Out-of-Bounds Indicators for Output Variables chart.



Standard KPI Trend

The Standard KPI Trend chart shows the trend of the selected standard key performance indicator (KPI) that is calculated over a period of time.

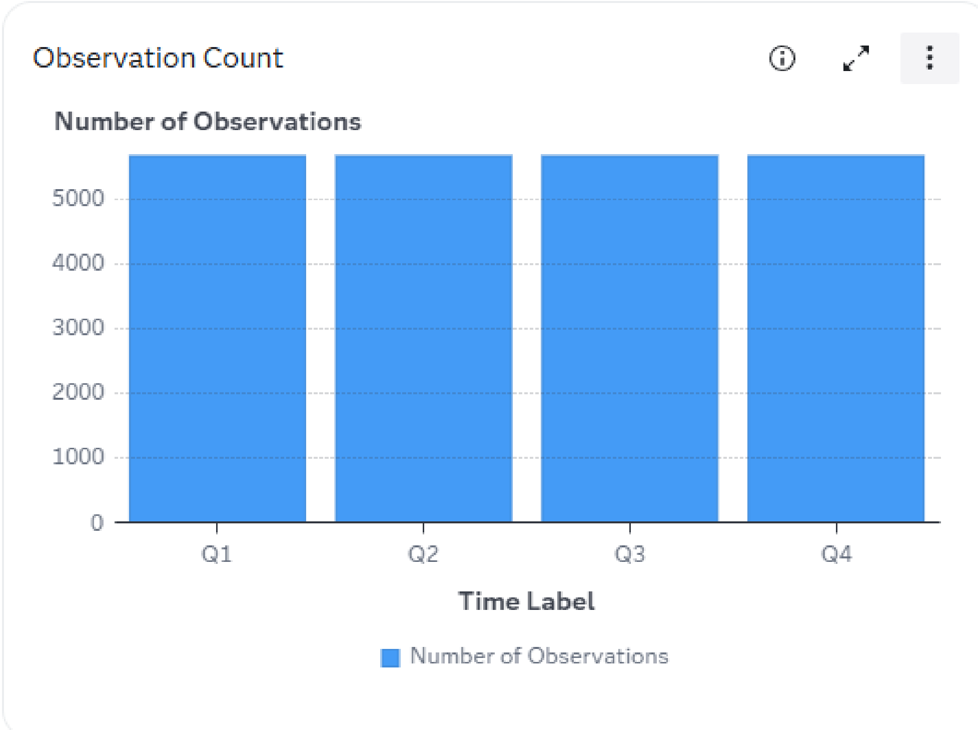
Here is an example of the Standard KPI Trend chart for the Misclassification KPI

Multinomial Target Metrics

Observation Count

The Observation Count chart shows the number of observations monitored at each time period. Large or unexpected changes in observation counts across time periods can be indicative of changes in behavior as well as technical or process-level issues. These counts suggest the relative credibility of the sample at the current time period.

Here is an example of the chart:

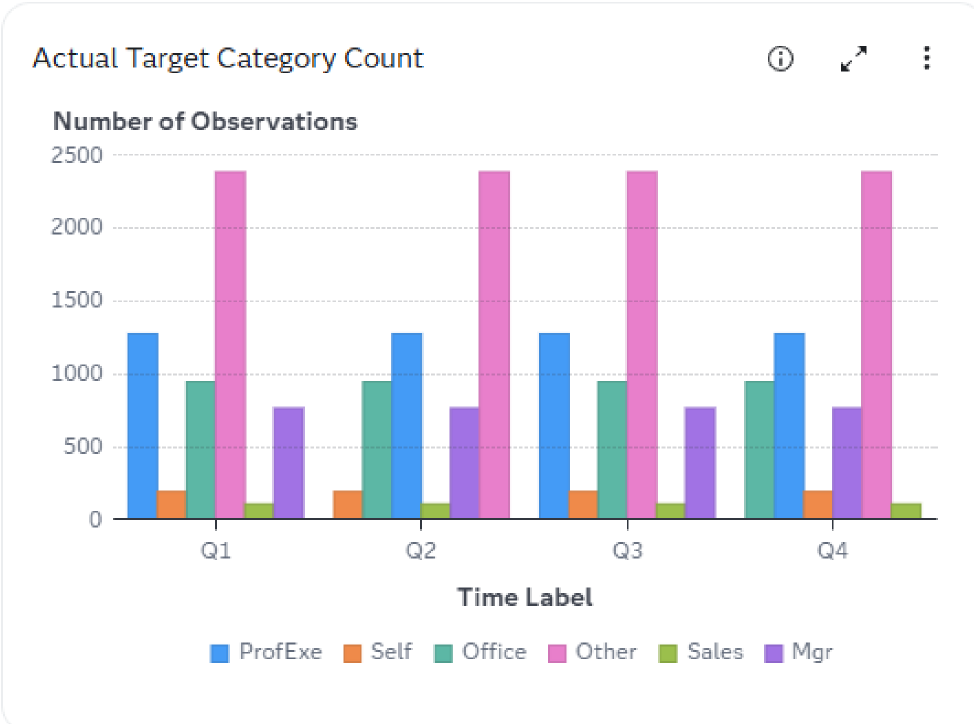


Actual Target Category Count

The Actual Target Category Count chart shows the number of observations for each actual target category at each time period. Large or unexpected changes in counts across time periods can be indicative of changes in behavior as well as technical or process-level issues.

These counts suggest the relative credibility of the categories at the current time period. Particularly, you should look for any extraneous or omitted categories.

Here is an example of the chart:

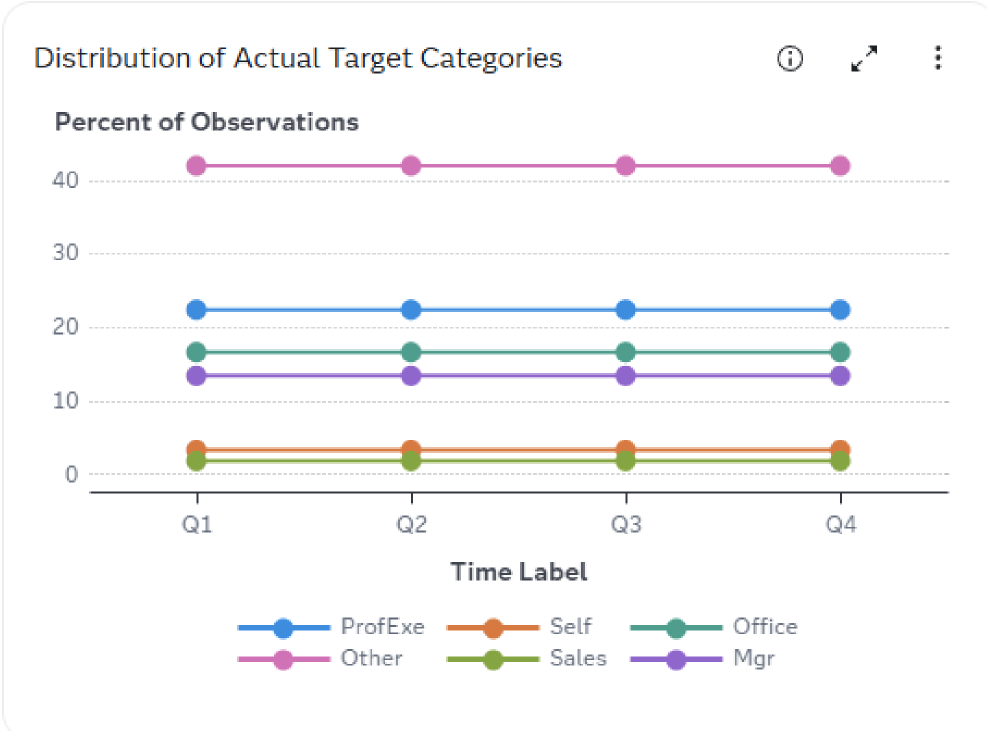


Distribution of Actual Target Categories

The Distribution of Actual Target Categories chart shows the percentage of observations for each target category at each time period. Large or unexpected changes in percentages across time periods can be indicative of changes in behavior as well as technical or process-level issues.

The percentages are calculated based on the number of observations at the current time period. This chart can tell you whether the target variable's distribution has changed. Subsequently, you can evaluate how applicable the model is to the current data.

Here is an example of the chart:

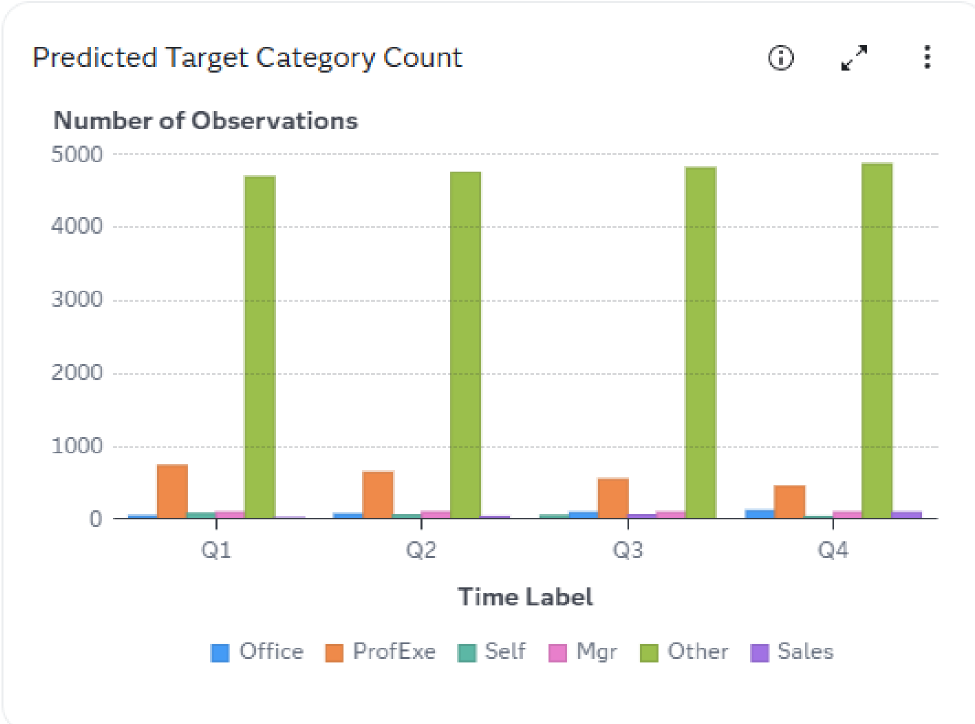


Predicted Target Category Count

The Predicted Target Category Count chart shows the number of observations for each predicted target category at each time period. Large or unexpected differences between the predicted target category count and the actual target category count can indicate reduced model accuracy.

This chart can tell you whether the model prediction's distribution has changed. Subsequently, you can evaluate how applicable the model is to the current data. Also, you should look for any omitted or overly dominant categories.

Here is an example of the chart:

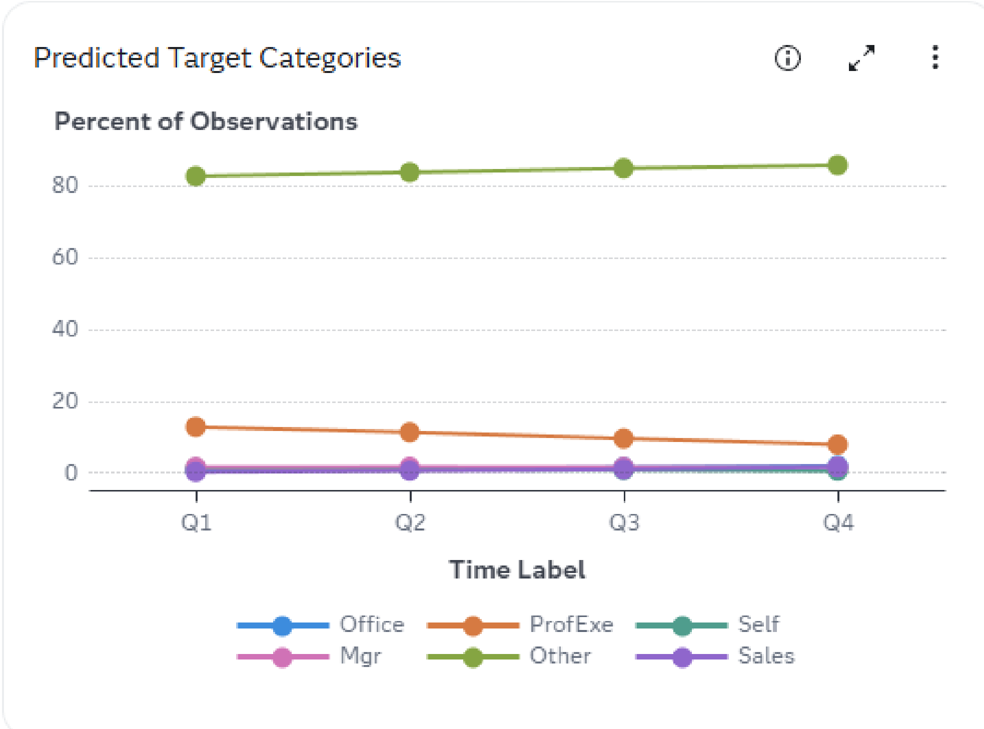


Predicted Target Categories

The Predicted Target Categories chart shows the percentages of each predicted target category at each time period. Large or unexpected differences between the predicted target categories and the actual target categories can indicate model performance changes.

The percentages are calculated based on the number of observations at the current time period. This chart can tell you whether the output variables' distributions have changed. Subsequently, you can evaluate how applicable the model is to the current data. Also, you should look for any omitted or overly dominant categories.

Here is an example of the chart:



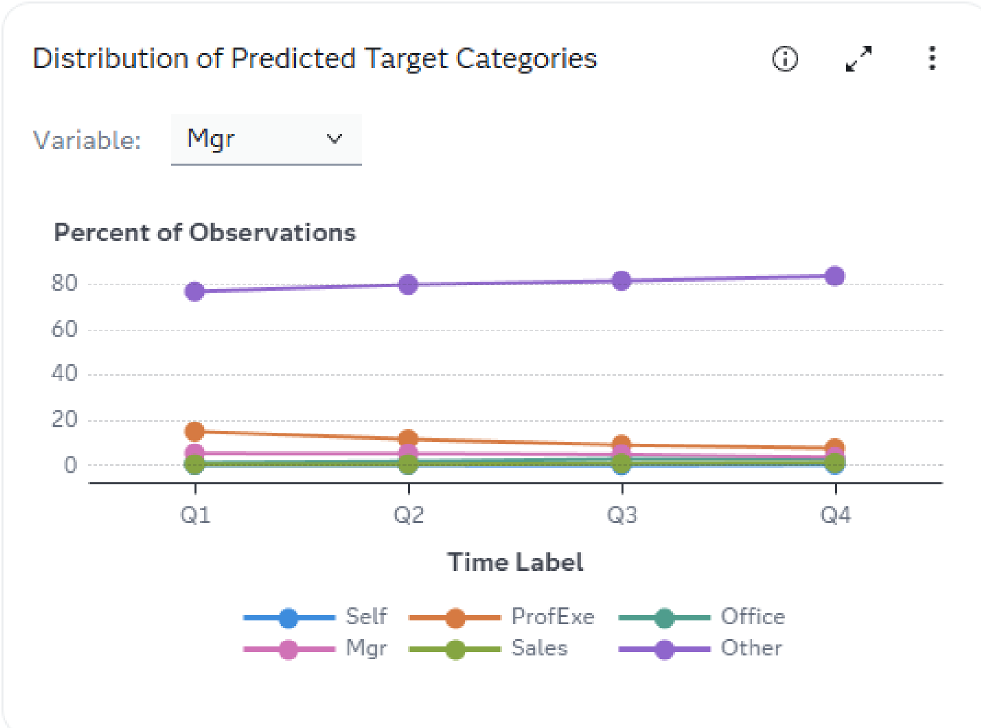
Distribution of Predicted Target Categories

The Distribution of Predicted Target Categories chart shows the percentages of predicted target categories for the selected target category at each time period. Large movements or extraneous trends across time periods can be indicative of changes in distributions of input features. Those incidental changes might be due to behavior shifts as well as systemic or process-level issues

The percentages are calculated based on the number of observations in the observed category at the current time period. This chart can tell you whether the model's classification has changed. To put it another way, this chart is a visual representation of the model confusion matrices across time.

Here is an example of the chart:

> **TIP** To change the target category that appears in the chart, select a value from the drop-down list.

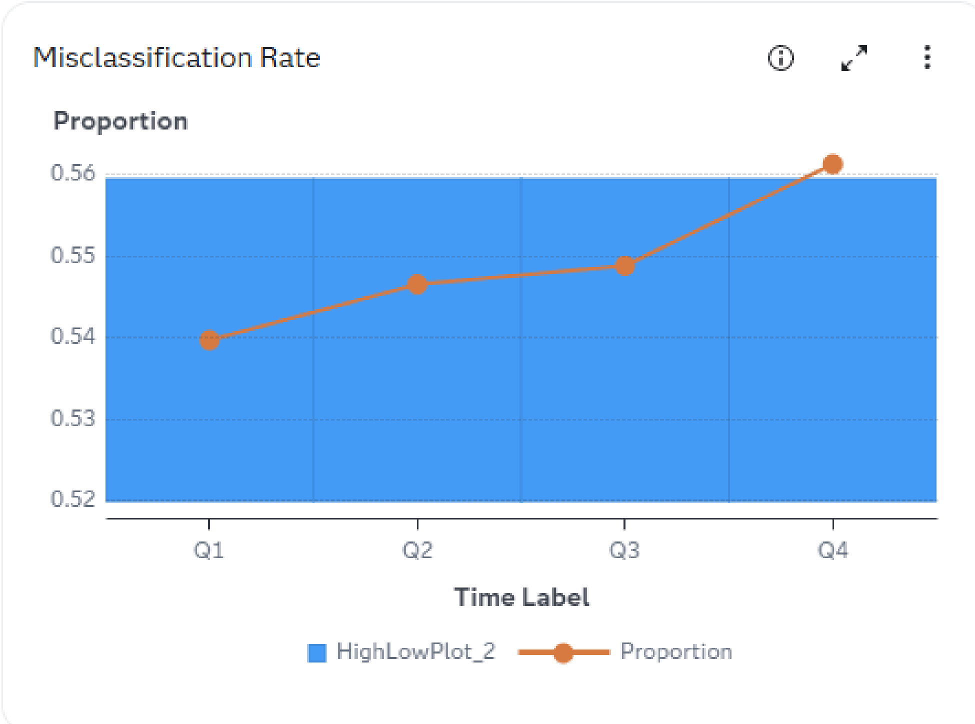


Misclassification Rate

The Misclassification Rate chart shows the proportion of observations where the model was incorrectly classified for each time period. You can use this chart and the bounds within it to detect degradations in the predictive power of a model. The shaded band indicates the uncertainty bounds.

The Shewhart's p-chart algorithm is applied when calculating the uncertainty bounds. The uncertainty bounds are three standard deviations from the misclassification rate at the baseline time period. For more information, see The SHEWHART Procedure in *SAS/QC User's Guide*.

Here is an example of the chart:



Misclassification Rate by Target Category

The Misclassification Rate by Target Category chart shows the proportion of observations where the model was incorrectly classified for the selected target category at each time period. You can use this chart and the bounds within it to detect degradations in the predictive power of a model. The shaded band indicates the uncertainty bounds.

The Shewhart's p-chart algorithm is applied when calculating the uncertainty bounds. The uncertainty bounds are three standard deviations from the respective Misclassification Rate at the baseline time period. For more information, see The SHEWHART Procedure in *SAS/QC User's Guide*.

Here is an example of the chart:

> **TIP** To change the target category that appears in the chart, select a value from the drop-down list.

Predicted Probability by Target Category

The Predicted Probability by Target Category chart shows the predicted probability of the selected target category at each time period. Large or unexpected changes in probabilities across time periods can be indicative of changes in behavior as well as technical or process-level issues.

This chart can tell you whether the distributions of predicted probabilities have changed. Subsequently, you can evaluate how applicable the model is to the current data. Also, you should look for any outliers.

Here are the metrics used for this chart:

Multi-class Area Under Curve (AUC)

The Area Under Curve metric for a multi-class target variable is the average of the AUC values that are calculated using the distinct pairs of target categories. Suppose the target variable has three categories, namely, A, B, and C. Also, let P_A, P_B, and P_C are the corresponding predicted probabilities for the three categories. Then, the AUC values are calculated for the following six pairs of target categories using the designated predicted probability.

```
Pair 1: Event = A, Non-Event = B, exclude category C, and Event Probability is P_A
Pair 2: Event = A, Non-Event = C, exclude category B, and Event Probability is P_A
Pair 3: Event = B, Non-Event = A, exclude category C, and Event Probability is P_B
Pair 4: Event = B, Non-Event = C, exclude category A, and Event Probability is P_B
Pair 5: Event = C, Non-Event = A, exclude category B, and Event Probability is P_C
Pair 6: Event = C, Non-Event = B, exclude category A, and Event Probability is P_C
```

References:

- David J. Hand, and Robert J. Till (2001). "A Simple Generalisation of the Area Under the ROC Curve for Multiple Class Classification Problems", Machine Learning, Volume 45, Pages 171-186.
- SAS Technical Support provided the %MultAUC macro that calculates the multi-class AUC value. Available at: https://support.sas.com/kb/64/029.html and https://support.sas.com/kb/64/addl/fusion_64029_14_multauc.sas.txt

Multi-class Root Average Squared Error (RASE)

Suppose there are N observations, and the target variable has K categories. The Root Average Squared Error is:

```
Square Root ( SUM over i ( SUM over k (Error_i_k )^2 ) / N / K )
where Error_i_k = (Delta_i_k - P_i_k), k = 1,...,K and i = 1,...,N.
The P_i_k is the predicted probability for the k-th target category
of the i-th observation.
The Delta_i_k = 1 if the observed target category of the i-th obervation is the
k-th target category. Otherwise, Delta_i_k = 0.
```

Multi-class Misclassification Error (MCE)

An observation is misclassified if the predicted class is different from the observed target category. The proportion of misclassified observations is the MCE metric.

Here is an example of the chart:

> **TIP** To change the target category that appears in the chart, select a value from the drop-down list.

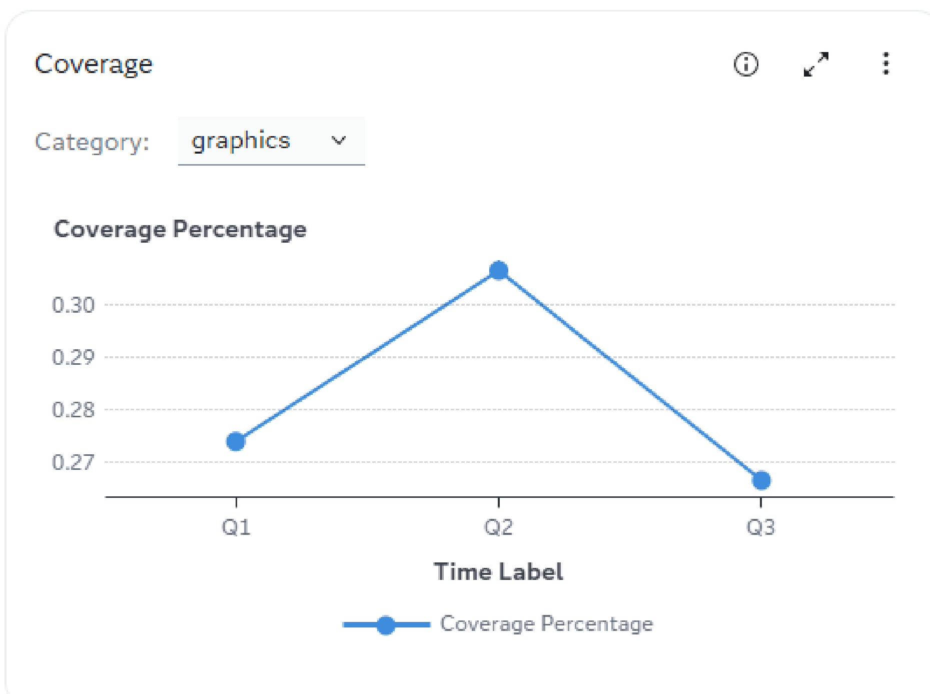
Text Categorization Metrics

Coverage

The Coverage chart shows the percentage of documents for which a model was able to identify any matches and produce a categorical identification. Ideally, every document has a match, and a model should be able to match every document to its respective category.

High coverage is a positive indicator of a healthy model, as it tells that the model is consistently finding matches for the documents passed through it.

Here is an example of a Coverage chart.

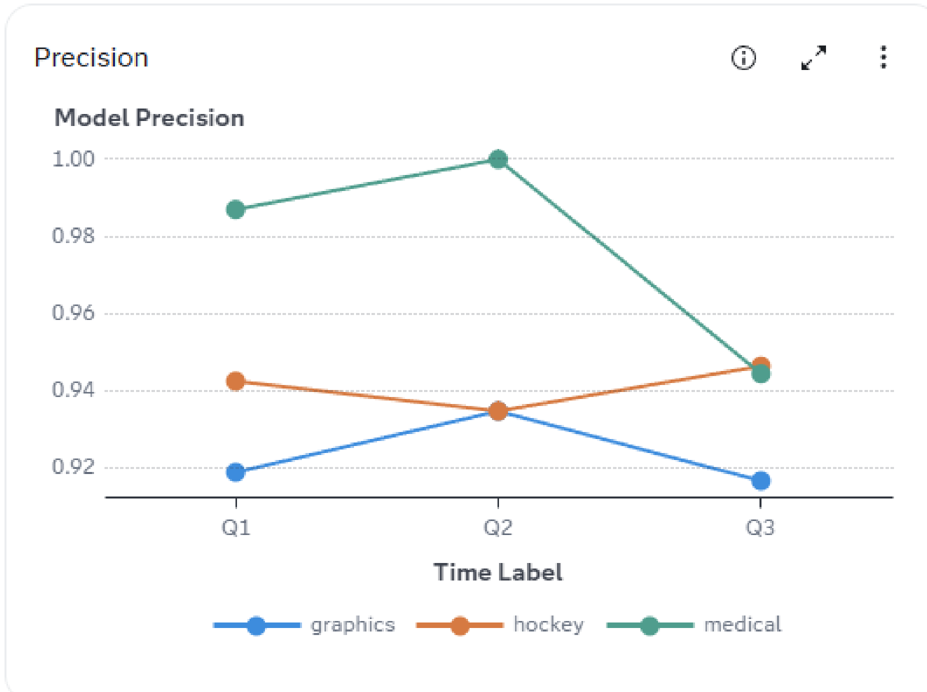


Precision

The Precision chart shows the model's ability to capture documents correctly without creating false matches. Precision is calculated as the ratio of true positives to the sum of true and false positives.

Higher values are preferred for model precision. The maximal value is 100%, which occurs when there are zero false matches made by the model. If precision is low, it is likely that the model's rules are weak, or that the rules are too broad, resulting in false matches.

Here is an example of a Precision chart.

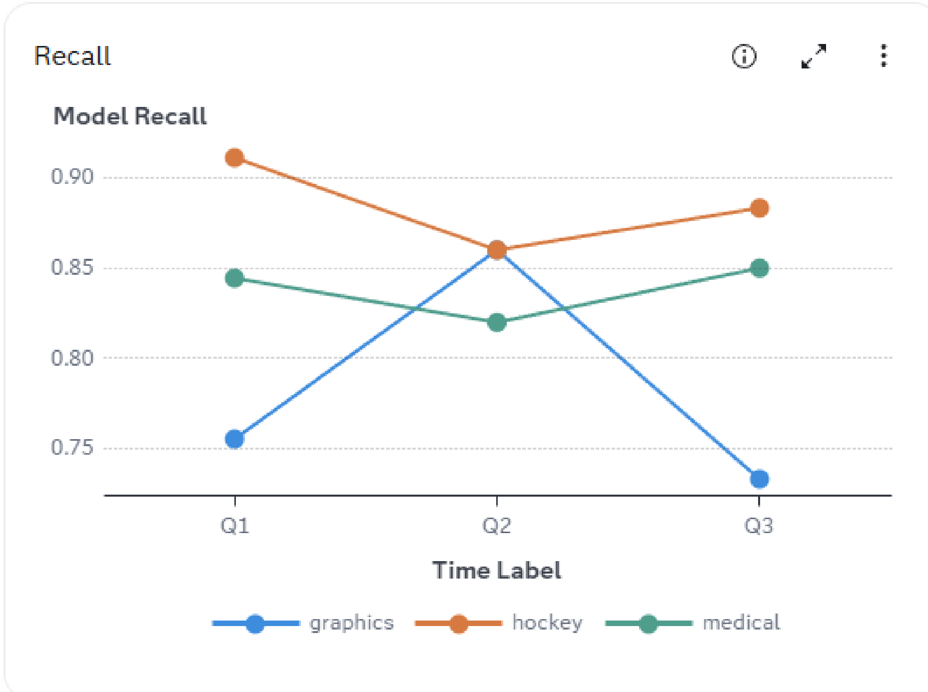


Recall

The Recall chart shows the portion of correct values that are captured by a model. Recall is calculated as the ratio of true positives to the total frequency of positives (sum of true positives and false negatives).

Higher values are preferred for model recall. The maximal value is 100%, which occurs when there are no false negatives, or missed matches. If recall is low, it is recommended to add more variables or rules to the model, as it is failing to capture a significant portion of correct matches.

Here is an example of a Recall chart.



Overall False Positive Rate

The Overall False Positive Rate chart shows the frequency of a model's falsely identified matches. False positive rate (FPR) is calculated as the ratio of false positives to all negatives (false positives + true negatives).

A model's false positive rate should remain relatively steady after its training and deployment. Sudden changes in FPR are likely indicators of external changes in the market, such as the introduction of a new technology, competition, marketing promotions, laws, or other influences.

An increasing FPR is a strong indicator of a degrading model.

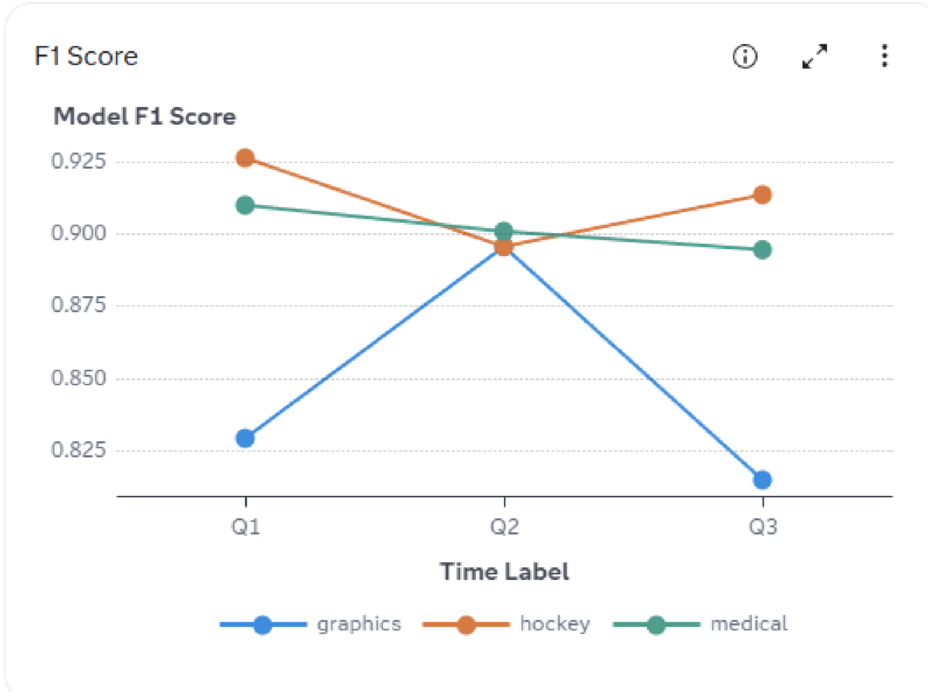
Here is an example of an Overall False Positive Rate chart.

F1 Score

The F1 Score chart shows the precision and recall that serves as a great assessor of model quality. F1 score is calculated as the positive predictive value(PPV) divided by the true positive rate (TPR). PPV is double the product of precision and recall and TPR is the sum of precision and recall.

Higher values are preferred for the F1 Score, and the maximal value is 100%. A low score for this metric is indicative of a model with either a poor precision or a poor recall value, which makes it a great overall indicator.

Here is an example of a F1 Score chart.



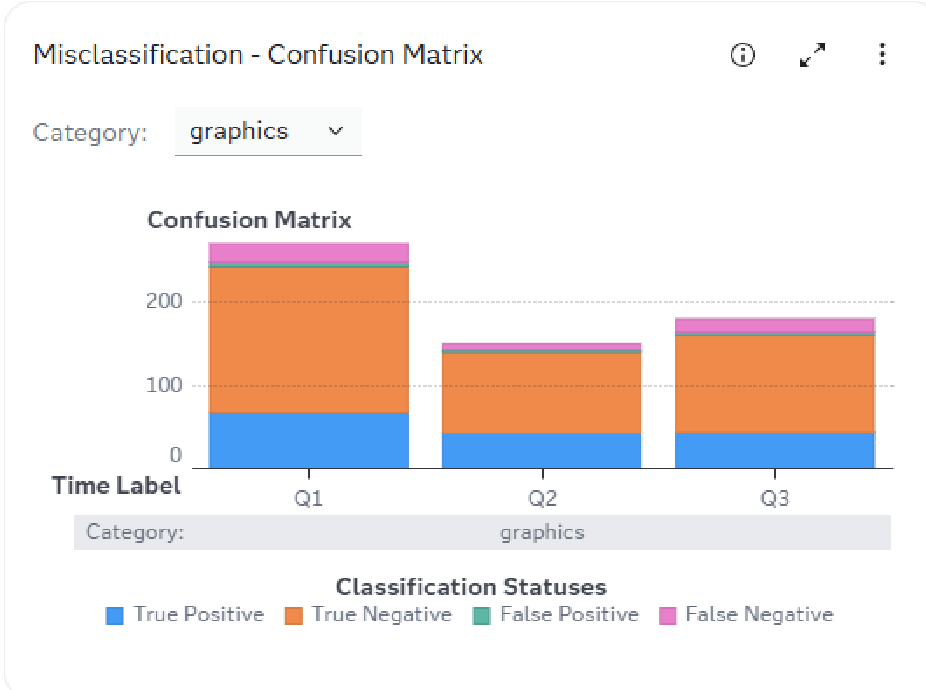
Misclassification - Confusion Matrix

The Misclassification - Confusion Matrix chart shows the percentages of each time period's documents that were scored in each status. The confusion matrix is a numeric or visual tool that separates the data into its classification statuses: true positive, true negative, false positive, and false negative.

When analyzing a confusion matrix, look for increasing frequencies of false positives and false negatives. Increases in false positives are a strong indicator of declining model performance.

Frequencies of true positives and true negatives are useful indicators of the relevance of the model to the sample of documents. A high count of true negatives tells that while the model is accurate, it is not pertinent to the categories of the documents. True positives are indicative of relevance and accuracy.

Here is an example of a Misclassification - Confusion Matrix chart.



Fairness and Bias Metrics

About Fairness and Bias Metrics

Fairness and bias metrics are calculated using the assessBias action that is part of the SAS fair AI tools action set. These charts show how a model's performance for fairness and bias changes over time. This enables you to include measures of model bias when making your decision to refresh, replace, or retrain your production models.

> **IMPORTANT** The fairness and bias metrics are available with the 2025.03 release.

For more information, see the assessBias action in *SAS Viya: Machine Learning Programming Guide*.

Prerequisites

Here are the tasks that must be done before the fairness and bias section of the performance monitoring results appears on the **Performance** tab of your project:

- Mark variables to be assessed for bias on the project's **Variables** tab for classification and prediction models that are written in SAS, Python, or R.
- The bias variables must be available in the input data used by performance monitoring, even if the variables are not used by the model to make a prediction.
- The fairness and bias key performance indicators (KPIs) thresholds are available on the project's **Properties** tab. These KPI thresholds are used to assess model performance in the model card for classification and prediction models.

Parity Metric Trend

The Parity Metric Trend chart shows the parity metric trend monitored at each time period for the selected bias variable and metric when the model function is classification, and the target level is binary or nominal. The available metrics are demographic parity, equal accuracy, equalized odds, equal opportunity, and predictive parity.

Here are the available metrics that can be selected in this chart:

demographic parity
: the maximum pairwise difference for the proportion of observations that are binned into the event level, given the cut-off value.

equal accuracy
: maximum pairwise difference across values of the selected variable for accuracy.

equalized odds
: the larger of the maximum pairwise differences across values of the selected variable for the true positive rate (TPR) and false positive rate (FPR).

equal opportunity
: the maximum pairwise difference across values of the selected variable for true positive rate.

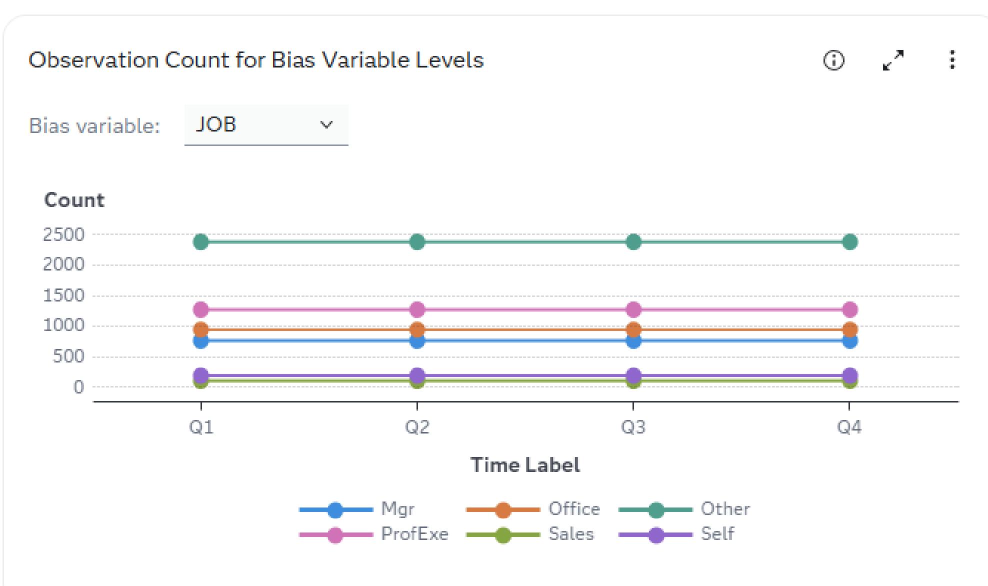
predictive parity
: maximum pairwise difference in the predicted variable corresponding to the event level.

Here is an example of the chart:

Observation Count for Bias Variable Levels

The Observation Count for Bias Variable Levels chart shows the observation count for the different levels of the selected bias variable in the specified time period. This chart is displayed when the project's model function is classification with a target level of binary or nominal, and when the project's model function is prediction with a target level of interval.

Here is an example of the chart:

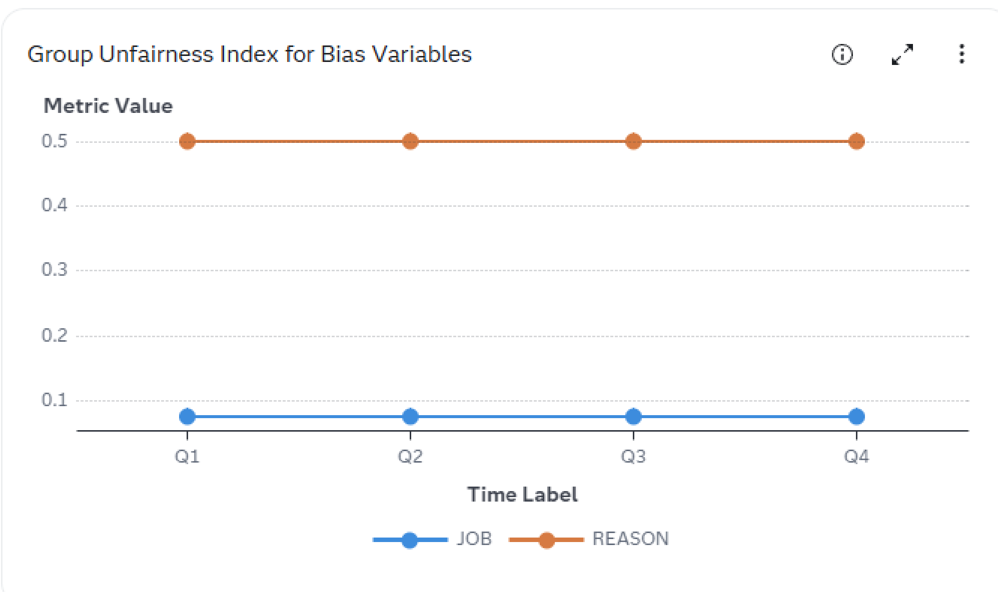


Group Unfairness Index for Bias Variables

The Group Unfairness Index for Bias Variables chart shows the group unfairness index monitored at each time period for bias variables when the model function is classification, and the target level is binary or nominal. The group unfairness index quantifies the degree of group fairness by adapting the population stability index (PSI) measurement to the variables that are selected for bias assessment.

The group unfairness index is calculated using the approach outlined by Szepannek and Lübke. 2021. "Facing the Challenges of Developing Fair Risk Scoring Models." *Frontiers in Artificial Intelligence* (article). https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2021.681915/full.

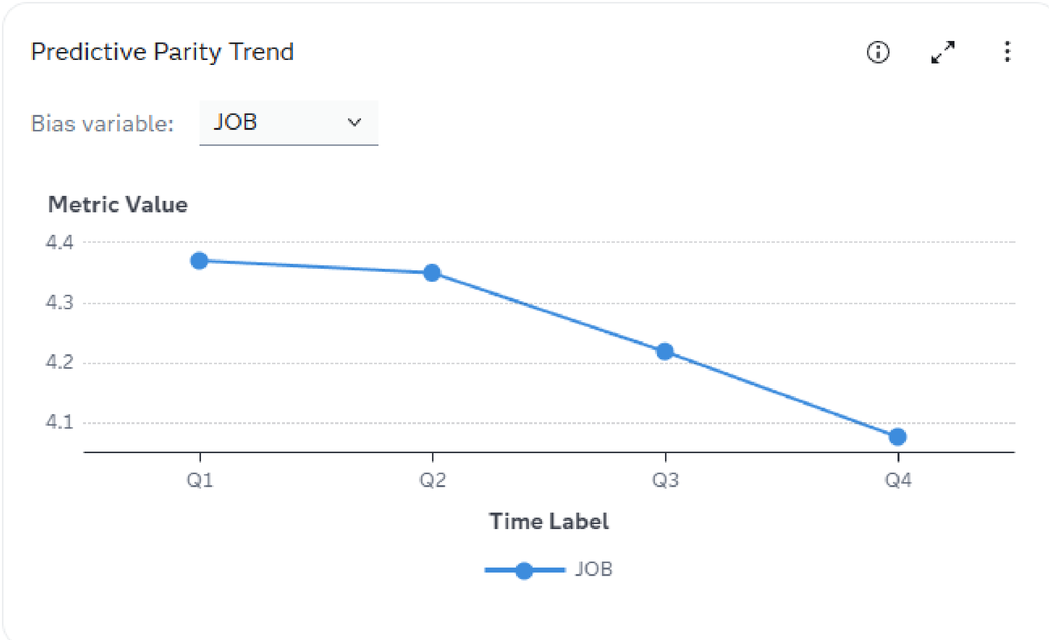
Here is an example of the chart:



Predictive Parity Trend

The Predictive Parity Trend chart shows the predictive parity trend at each time period for the selected bias variable when the model function is prediction, and the target level is interval. Predictive parity is the maximum difference in the predicted variable corresponding to the variable level.

Here is an example of the chart:



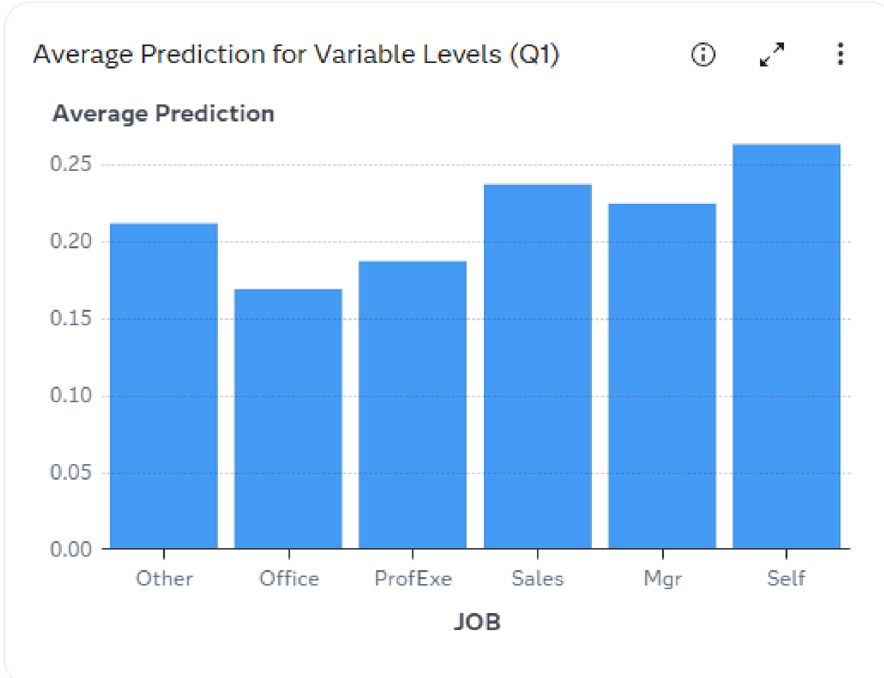
Event Metrics

In the **Event Metrics** section of the performance results, you can select the period of time (time label) and the bias variable for which to display the results for each of the event metrics.

Average Prediction for Variable Levels

The Average Prediction for Variable Levels chart shows the average model prediction for the different levels of the selected bias variable in the specified time period. This chart is displayed when the project's model function is classification with a target level of binary or nominal, and when the project's model function is prediction with a target level of interval.

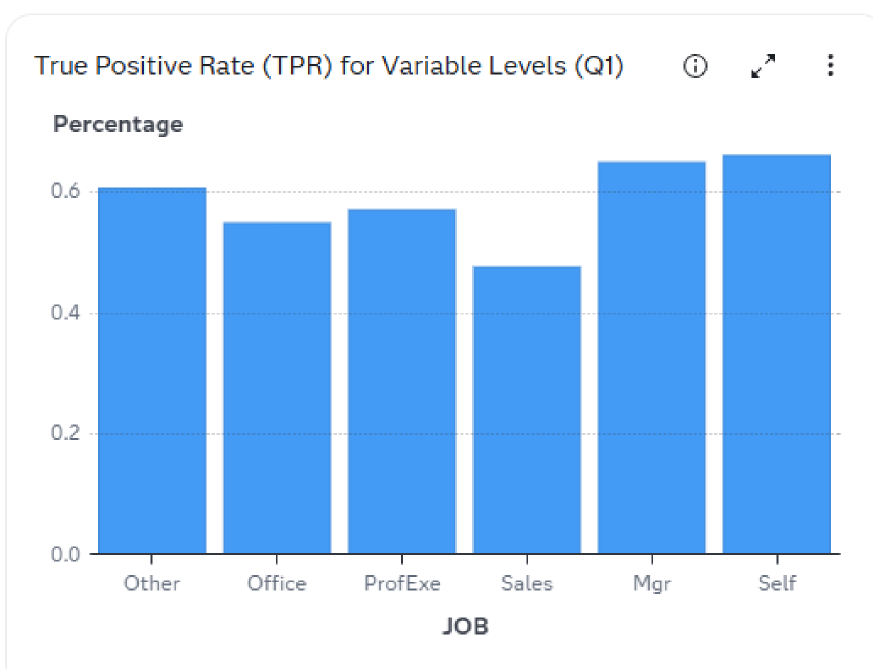
Here is an example of the chart:



True Positive Rate (TPR) for Variable Levels

The True Positive Rate (TPR) for Variable Levels chart shows the true positive rate (TPR) for the different levels of the selected bias variable in the specified time period. TPR is the number of true positives divided by the total number of actual positives (true positives and false negatives). This chart is displayed when the project's model function is classification with a target level of binary or nominal.

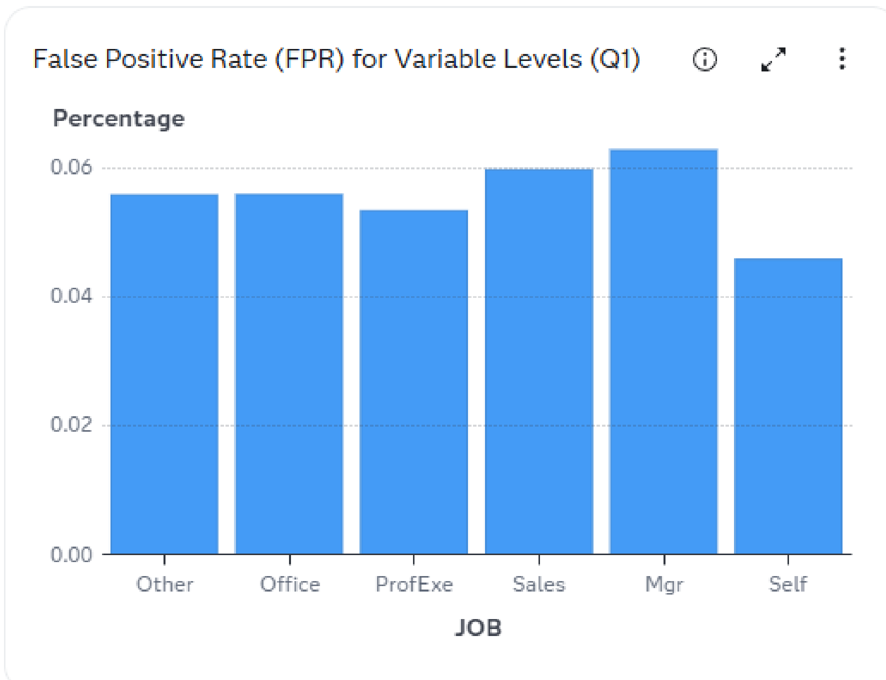
Here is an example of the chart:



False Positive Rate (FPR) for Variable Levels

The False Positive Rate (FPR) for Variable Levels chart shows the false positive rate (FPR) for the different levels of the selected bias variable in the specified time period. FPR is the number of false positives divided by the total number of actual negative events (false positives and true negatives). This chart is displayed when the project's model function is classification with a target level of binary or nominal.

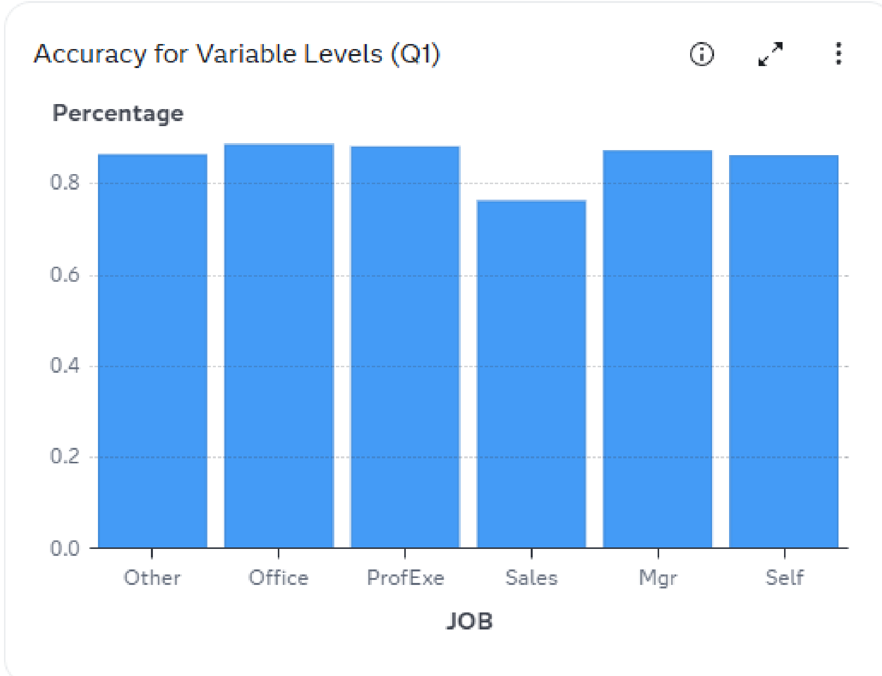
Here is an example of the chart:



Accuracy for Variable Levels

The Accuracy for Variable Levels chart shows the accuracy for the different levels of the selected bias variable in the specified time period. Accuracy is the proportion of events correctly identified. This chart is displayed when the project's model function is classification with a target level of binary or nominal.

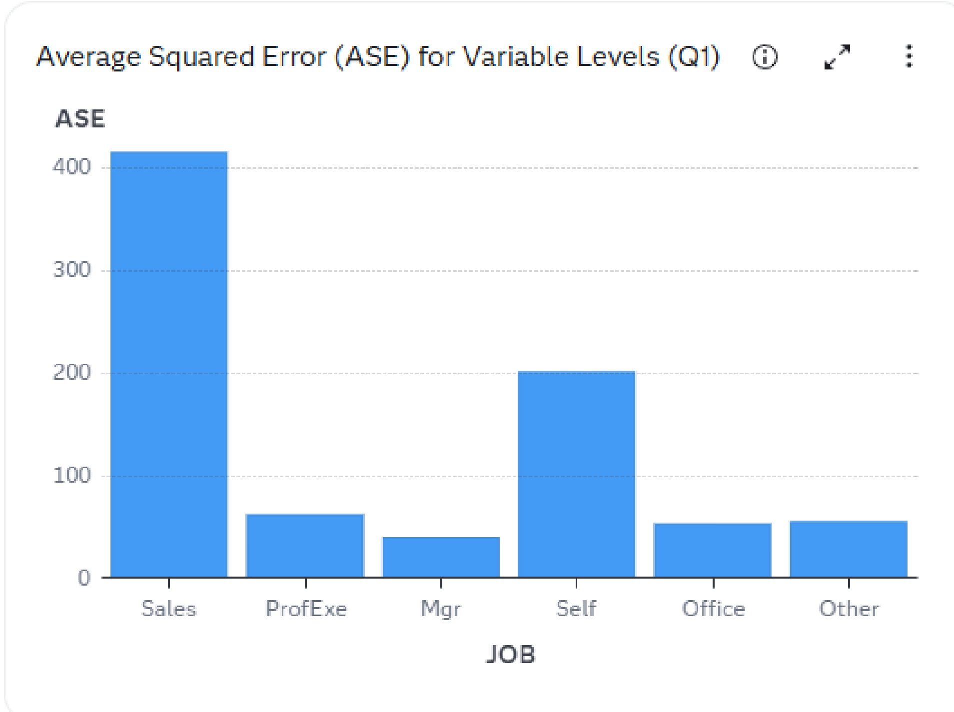
Here is an example of the chart:



Average Squared Error (ASE) for Variable Levels

The Average Squared Error (ASE) for Variable Levels chart shows the average squared error (ASE) for the different levels of the selected bias variable in the specified time period. ASE evaluates the accuracy of a prediction model with an interval target by comparing the estimation derived from the predicted outcome and the actual outcomes. This chart is displayed when the project's model function is prediction with a target level of interval.

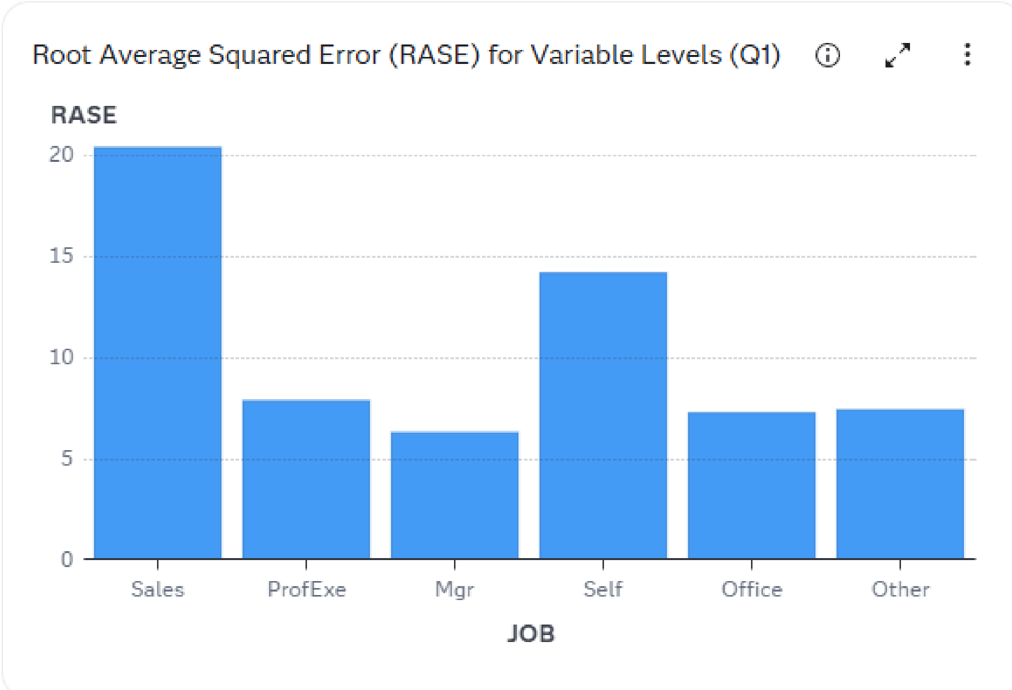
Here is an example of the chart:



Root Average Squared Error (RASE) for Variable Levels

The Root Average Squared Error (RASE) for Variable Levels chart shows the root average squared error (RASE) for the different levels of the selected bias variable in the specified time period. RASE is the square root of the average square difference between the model's predicted value, and the actual value. This chart is displayed when the project's model function is prediction with a target level of interval.

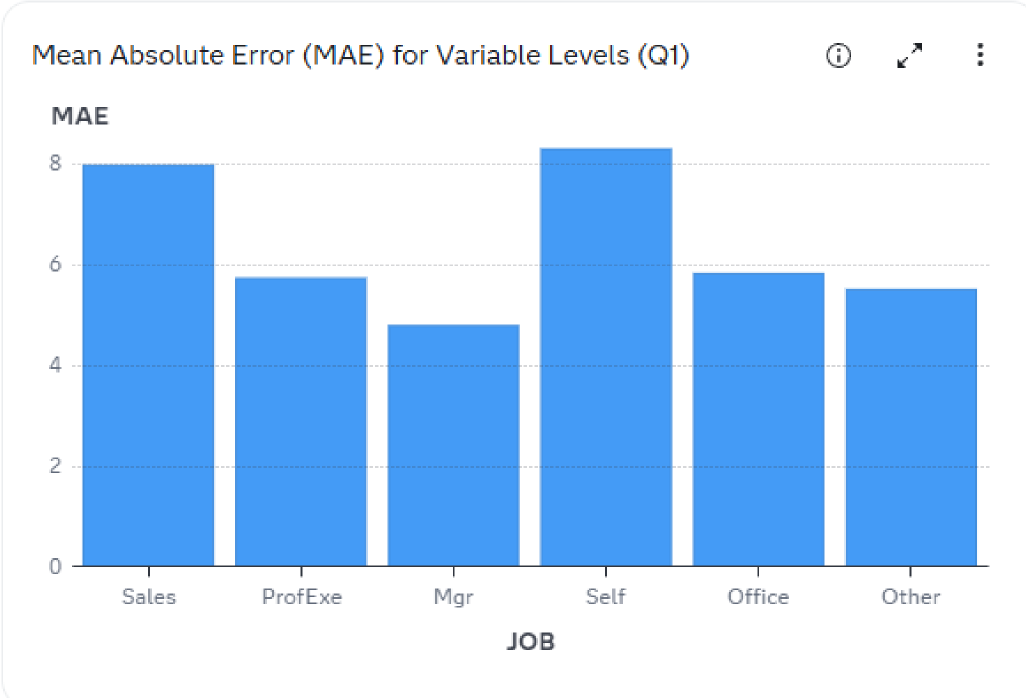
Here is an example of the chart:



Mean Absolute Error (MAE) for Variable Levels

The Mean Absolute Error (MAE) for Variable Levels chart shows the mean absolute error (MAE) for the different levels of the selected bias variable in the specified time period. MAE calculates the average magnitude of the absolute errors between the predicted and actual values. This chart is displayed when the project's model function is prediction with a target level of interval.

Here is an example of the chart:



Concepts: PMML Support

Overview

PMML is an XML markup language that was developed to exchange predictive and statistical models between modeling systems and scoring platforms. Users can import the majority of standard-compliant PMML models and score them within a SAS environment via the SAS PSCORE procedure.

PROC PSCORE Functionality

The SAS PSCORE procedure generates SAS DATA step score code that is functionally equivalent to the PMML model. The generated score code can be executed on all platforms that are supported by SAS to score the data sets. You can submit the score code in SAS Enterprise Miner via the Program Editor, SAS Enterprise Miner Project code, or within a SAS Enterprise Miner Process Flow Diagram, via the SAS Code node. However, the SAS Enterprise Miner UI environment is not necessary to run the score code.

Note: The PSCORE procedure generates both DATA step code and DS2 code. However, only DATA step model score code is generated when you are importing or registering a PMML model into SAS Model Manager.

Supported Versions

PROC PSCORE currently supports the use of PMML 4.1, 4.2, 4.3, and 4.4. Other versions of PMML are not supported for use with PROC PSCORE.

Supported PMML Models

SAS PROC PSCORE supports the following types of PMML models:

- Association rules
- Baseline
- Clustering
- General regression
- Mining
- Naive bayes
- Neural network
- Regression
- Rule set
- Scorecard
- Support vector machine (SVM)
- Time series
- Tree

For a list of valid key values that can be used for the function name and algorithm name in the PMML model file, see Types of Model Functions and Types of Algorithms.

Requirements for PROC PSCORE

In order to use PROC PSCORE, you must have a well formed PMML modeling file, and Write access to the output directory for the DATA step score file. A SAS Enterprise Miner license is not necessary to run PROC PSCORE. SAS 9.4 procedures can also be run on SAS Viya using SAS Studio.

Here are some requirements and restrictions for the PMML XML file:

- The length of the variable names cannot be greater than 32 characters.
- The output section must be included in order to score a model and to run performance monitoring.
- The key values for the function name and algorithm name must be valid in order to score a model and to run performance monitoring.

PROC PSCORE Usage

```
PROC PSCORE PMML FILE = "<full-pathname-of-PMML-file>"
     DS FILE = "<full-pathname-of-output-DS-file>"
```

PROC PSCORE Example

/*Run the PSCORE procedure on a generated PMML file*/

```
PROC PSCORE PMML FILE = "C:\temp\QS_Reg_PMMLModel_v4.4.xml"
     DS FILE = "C:\temp\score.sas";
  run;
```

PMML XML File Example

This is an example of a valid PMML XML model file before it is imported into SAS Model Manager.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PMML xmlns="http://www.dmg.org/PMML-4_4" xmlns:data="http://jpmml.org/
jpmml-model/InlineTable" version="4.4">
    <Header copyright="Copyright(c) 2025 SAS Institute Inc., Cary, NC,
USA. All Rights Reserved." description="Generalized Linear Regression
Model">
        <Application name="SkLearn2PMML package" version="0.122.2"/>
        <Timestamp>2025-10-07T20:46:49Z</Timestamp>
    </Header>
    <DataDictionary>
        <DataField name="BAD" optype="categorical" dataType="integer">
            <Value value="0"/>
            <Value value="1"/>
        </DataField>
        <DataField name="JOB" optype="categorical" dataType="string">
            <Value value="Mgr"/>
            <Value value="Office"/>
            <Value value="Other"/>
            <Value value="ProfExe"/>
```

```
            <Value value="Sales"/>
            <Value value="Self"/>
            <Value value="nan"/>
        </DataField>
        <DataField name="REASON" optype="categorical"
dataType="string">
            <Value value="DebtCon"/>
            <Value value="HomeImp"/>
            <Value value="nan"/>
        </DataField>
        <DataField name="CLAGE" optype="continuous" dataType="double"/>
        <DataField name="CLNO" optype="continuous" dataType="double"/>
        <DataField name="DEBTINC" optype="continuous"
dataType="double"/>
        <DataField name="DELINQ" optype="continuous"
dataType="double"/>
        <DataField name="DEROG" optype="continuous" dataType="double"/>
        <DataField name="NINQ" optype="continuous" dataType="double"/>
    </DataDictionary>
    <RegressionModel functionName="classification"
algorithmName="logisticreg" normalizationMethod="logit">
        <MiningSchema>
            <MiningField name="BAD" usageType="target"/>
            <MiningField name="JOB" invalidValueTreatment="asIs"/>
            <MiningField name="REASON" invalidValueTreatment="asIs"/>
            <MiningField name="CLAGE"/>
            <MiningField name="CLNO"/>
            <MiningField name="DEBTINC"/>
            <MiningField name="DELINQ"/>
            <MiningField name="DEROG"/>
            <MiningField name="NINQ"/>
        </MiningSchema>
        <Output>
            <OutputField name="Predicted_BAD" feature="predictedValue"
optype="categorical" dataType="string"/>
            <OutputField name="Probability_1" optype="continuous"
dataType="double" feature="probability" value="1"/>
        </Output>
        <RegressionTable intercept="-2.012903047898022"
targetCategory="1">
            <NumericPredictor name="CLAGE"
coefficient="-0.0068170080938668295"/>
            <NumericPredictor name="CLNO"
coefficient="-0.012734145498981893"/>
            <NumericPredictor name="DEBTINC"
coefficient="0.032980959962740905"/>
            <NumericPredictor name="DELINQ"
coefficient="0.7147032635793742"/>
            <NumericPredictor name="DEROG"
coefficient="0.6479627319463933"/>
            <NumericPredictor name="NINQ"
coefficient="0.1491968002786953"/>
            <CategoricalPredictor name="JOB" value="Office"
coefficient="-0.27650791665941776"/>
            <CategoricalPredictor name="JOB" value="Other"
coefficient="0.05133658187485258"/>
```

```
            <CategoricalPredictor name="JOB" value="ProfExe"
coefficient="0.11294416787836044"/>
            <CategoricalPredictor name="JOB" value="Sales"
coefficient="1.0016328845570628"/>
            <CategoricalPredictor name="JOB" value="Self"
coefficient="0.1639307501742985"/>
            <CategoricalPredictor name="JOB" value="nan"
coefficient="-0.9966510704192605"/>
            <CategoricalPredictor name="REASON" value="HomeImp"
coefficient="0.30942011677801096"/>
            <CategoricalPredictor name="REASON" value="nan"
coefficient="0.6038953802580702"/>
        </RegressionTable>
        <RegressionTable intercept="0.0" targetCategory="0"/>
    </RegressionModel>
</PMML>
```


Note: Sample PMML model files for PMML 4.2 and 4.4 are available in the QuickStartTutorial.zip file on support.sas.com.

See Also

- "SAS Enterprise Miner 15.4 PMML Support" in *SAS Enterprise Miner: Reference Help*
- PMML Standard
- Machine Learning Algorithms
- *SAS Viya: Machine Learning Node Reference*